

Supplementary Information

An Efficient C-glycoside Production Platform Enabled by Rationally Tuning the Chemoselectivity of Glycosyltransferases

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Supplementary Tables

Supplementary Table 1 | Strains and plasmids in this study.

Strain or plasmid	Characteristics	Source
Strains		
<i>E. coli</i> TOP 10	General cloning host strain	Invitrogen
<i>E. coli</i> ROSETTA (DE3)	Protein production host strain	Novagen
Plasmids		
pET28a	Km ^r , vector for protein expression	Previous work ¹
pET28a- <i>micgt</i>	Km ^r , plasmid for MiCGT-WT expression	Previous work ¹
pET28a- <i>micgt</i> (M21A)	Km ^r , plasmid for MiCGT-M21A expression	Previous work ¹
pET28a- <i>micgt</i> (H23A)	Km ^r , plasmid for MiCGT-H23A expression	Previous work ¹
pET28a- <i>micgt</i> (F89A)	Km ^r , plasmid for MiCGT-F89A expression	Previous work ¹
pET28a- <i>micgt</i> (R92A)	Km ^r , plasmid for MiCGT-R92A expression	Previous work ¹
pET28a- <i>micgt</i> (W93A)	Km ^r , plasmid for MiCGT-W93A expression	Previous work ¹
pET28a- <i>micgt</i> (S122A)	Km ^r , plasmid for MiCGT-S122A expression	Previous work ¹
pET28a- <i>micgt</i> (L123A)	Km ^r , plasmid for MiCGT-L123A expression	Previous work ¹
pET28a- <i>micgt</i> (F142A)	Km ^r , plasmid for MiCGT-F142A expression	Previous work ¹
pET28a- <i>micgt</i> (T143A)	Km ^r , plasmid for MiCGT-T143A expression	Previous work ¹
pET28a- <i>micgt</i> (S144A)	Km ^r , plasmid for MiCGT-S144A expression	Previous work ¹
pET28a- <i>micgt</i> (M148A)	Km ^r , plasmid for MiCGT-M148A expression	Previous work ¹
pET28a- <i>micgt</i> (E152A)	Km ^r , plasmid for MiCGT-E152A expression	Previous work ¹
pET28a- <i>micgt</i> (P188A)	Km ^r , plasmid for MiCGT-P188A expression	Previous work ¹
pET28a- <i>micgt</i> (V190A)	Km ^r , plasmid for MiCGT-V190A expression	Previous work ¹
pET28a- <i>micgt</i> (F191A)	Km ^r , plasmid for MiCGT-F191A expression	Previous work ¹
pET28a- <i>micgt</i> (F198A)	Km ^r , plasmid for MiCGT-F198A expression	Previous work ¹
pET28a- <i>micgt</i> (L202A)	Km ^r , plasmid for MiCGT-L202A expression	Previous work ¹
pET28a- <i>micgt</i> (W364A)	Km ^r , plasmid for MiCGT-W364A expression	Previous work ¹
pET28a- <i>micgt</i> (F383A)	Km ^r , plasmid for MiCGT-F383A expression	Previous work ¹
pET28a- <i>micgt</i> (D385A)	Km ^r , plasmid for MiCGT-D385A expression	Previous work ¹
pET28a- <i>micgt</i> (Q386A)	Km ^r , plasmid for MiCGT-Q386A expression	Previous work ¹
pET28a- <i>micgt</i> (E152Q/V190D/S122P)	Km ^r , plasmid for MiCGT-E152Q/V190D/S122P expression	This study
pET28a- <i>micgt</i> (M148A/V190T/S121D)	Km ^r , plasmid for MiCGT-M148A/V190T/S121D expression	This study
pET28a- <i>micgt</i> (E152Q)	Km ^r , plasmid for MiCGT-E152Q expression	This study
pET28a- <i>micgt</i> (V190D)	Km ^r , plasmid for MiCGT-V190D expression	This study
pET28a- <i>micgt</i> (S122P)	Km ^r , plasmid for MiCGT-S122P expression	This study
pET28a- <i>micgt</i> (E152Q/V190D)	Km ^r , plasmid for MiCGT-E152Q/V190D expression	This study
pET28a- <i>micgt</i> (E152Q/S122P)	Km ^r , plasmid for MiCGT-E152Q/S122P expression	This study
pET28a- <i>micgt</i> (V190D/S122P)	Km ^r , plasmid for MiCGT-V190D/S122P expression	This study
pET28a- <i>micgt</i> (V190T)	Km ^r , plasmid for MiCGT-V190T expression	This study

pET28a- <i>micgt</i> (S121D)	Km ^r , plasmid for MiCGT-S121D expression	This study
pET28a- <i>micgt</i> (M148A/V190T)	Km ^r , plasmid for MiCGT-M148A/V190T expression	This study
pET28a- <i>micgt</i> (M148A/S121D)	Km ^r , plasmid for MiCGT-M148A/S121D expression	This study
pET28a- <i>micgt</i> (V190T/S121D)	Km ^r , plasmid for MiCGT-V190T/S121D expression	This study
pET28a- <i>fccgt</i>	Km ^r , plasmid for FcCGT-WT expression	This study
pET28a- <i>fccgt</i> (T149Q/A189D)	Km ^r , plasmid for FcCGT-T149Q/A189D expression	This study
pET28a- <i>fccgt</i> (V118D/M145A)	Km ^r , plasmid for FcCGT-V118D/M145A expression	This study
pET28a- <i>fccgt</i> (V118S)	Km ^r , plasmid for FcCGT-V118S expression	This study
pET28a- <i>nngt</i>	Km ^r , plasmid for NnGT-WT expression	This study
pET28a- <i>nngt</i> (S126P/C156Q/L197D)	Km ^r , plasmid for NnGT-S126P/C156Q/L197D expression	This study
pET28a- <i>gjgta</i>	Km ^r , plasmid for GjGTa expression	This study
pET28a- <i>gjgta</i> (F116S/F117P/A142M)	Km ^r , plasmid for GjGTa- F116S/F117P/A142M expression	This study
pET28a- <i>gjgtc</i>	Km ^r , plasmid for GjGTc expression	This study
pET28a- <i>gjgtc</i> (M118S/F119P/M148Q/R189D)	Km ^r , plasmid for GjGTc- M118S/F119P/M148Q/R189D expression	This study

Supplementary Table 2 | Primers used in this study.

Primer	Sequence (5'-3')	Description
MiCGT-F	GTGATGCTTTAAACAGCTGC	Universal primer
MiCGT-R	CTTCTTCCAGTTCTCGATCAG	Universal primer
M148NNK-F	GCTGTTTACCAGCAGCACCCGC <u>NNK</u> TGCT CTTTAGAGGAAAC	For construction of 2nd and 3rd round mutation library
E152NNK-F	GCACCCGCATGTGCTCTTTAN <u>NNK</u> GAAACA TTTCCGGCCTTCG	For construction of 2nd and 3rd round mutation library
V190NNK-F	CGGTGAGTAGTGTGCCGCCG <u>NNK</u> TTTCTG AATTAAACC	For construction of 2nd and 3rd round mutation library
F90NNK-F	CAAACACAACCGATCCGTT <u>CNNK</u> CTGCGC TGGGAAGCAATTGC	For construction of 4th round mutation library
S121NNK-F	GAGTGCACCTGGTGATCGAT <u>NNK</u> TCTTTAGT TAGCAGTTTTGT	For construction of 4th round mutation library
S122NNK-F	GTGCACTGGTGATCGATAGC <u>NNK</u> TTAGTT AGCAGTTTTGTGCC	For construction of 4th round mutation library
V124NNK-F	GGTGATCGATAGCTCTTTAN <u>NNK</u> AGCAGTTT TGTGCCGGTTGC	For construction of 4th round mutation library
V190D- V187NNK-F	GCCCGGTTCCGGTGAGTAGT <u>NNK</u> CCGCCG <u>GATTTTCTGAATTTAAAC</u>	For construction of 4th round mutation library
V190T- V187NNK-F	GCCCGGTTCCGGTGAGTAGT <u>NNK</u> CCGCCG <u>ACATTTCTGAATTTAAAC</u>	For construction of 4th round mutation library
V190D- F191NNK-F	GAGTAGTGTGCCGCCG <u>GATNNK</u> CTGAATT TAAACCATTTATTTAC	For construction of 4th round mutation library
V190T- F191NNK-F	GAGTAGTGTGCCGCCG <u>ACANNK</u> CTGAATT TAAACCATTTATTTAC	For construction of 4th round mutation library
N205NNK-F	CCACCATGCTGATCCAG <u>NNK</u> GGCCAGAGC TTTCGCAAAGC	For construction of 4th round mutation library
F209NNK-F	CTGATCCAGAATGGCCAGAGC <u>NNK</u> CGCAA AGCCAATGGCATCTTAATC	For construction of 4th round mutation library
R282NNK-F	GTACGTGAGCTTTGGCAGT <u>NNK</u> TTTGCTTT AAGCAGCGAAC	For construction of 4th round mutation library
W364NNK-F	GCTTTGTGACCCATGGCGGC <u>NNK</u> AATAGC AGTATGGAGGCCG	For construction of 4th round mutation library
G384NNK-F	GCTGGTTTGGCCGCAGTTT <u>NNK</u> GACCAGA AGATCAACGCCG	For construction of 4th round mutation library
D385NNK-F	GGTTTGGCCGCAGTTTGGC <u>NNK</u> CAGAAGA TCAACGCCGAAG	For construction of 4th round mutation library
V190D-F	CGGTGAGTAGTGTGCCGCCG <u>GATTTTCTGA</u> ATTTAAACC	Plasmid construction for deconvolutants expression
S122P-F	GTGCACTGGTGATCGATAGC <u>CCGTTAGTTA</u> GCAGTTTTGTGCC	Plasmid construction for deconvolutants expression
E152Q-V190D- F	GTGCACTGGTGATCGATAGCT <u>CTTTAGTTA</u> GCAGTTTTGTGCC	Plasmid construction for deconvolutants expression
S122P-E152Q-F	CGGTGAGTAGTGTGCCGCCG <u>GTTTTTCTGA</u> ATTTAAACC	Plasmid construction for deconvolutants expression

Primer	Sequence (5'-3')	Description
S122P-V190D-F	GCACCCGCATGTGCTCTTTAGAGGAAACA TTTCCGGCCTTCG	Plasmid construction for deconvolutants expression
V190T-F	CGGTGAGTAGTGTGCCGCCGACATTTCTG AATTTAAACC	Plasmid construction for deconvolutants expression
S121D-F	GAGTGCACCTGGTGATCGATGATTCTTTAGT TAGCAGTTTTG	Plasmid construction for deconvolutants expression
S121D-M148A-F	CGGTGAGTAGTGTGCCGCCGGTTTTCTGA ATTTAAACC	Plasmid construction for deconvolutants expression
M148A-V190T-F	GAGTGCACCTGGTGATCGATAGCTCTTTAGT TAGCAGTTTTGTG	Plasmid construction for deconvolutants expression
S121D-V190T-F	GCTGTTTACCAGCAGCACCCGCATGTGCTC TTTAGAGGAAAC	Plasmid construction for deconvolutants expression
FcCGT-R	GGGTGTTATTGTTGC	Universal primer
FcCGT-V118S-F	GAGCGCGCTGATTACCGATAGCACCTGA TTAGCGCGGTGC	Plasmid construction for FcCGT-V118S expression
Fc-V118D-F	CTGAGCGCGCTGATTACCGATGATACCCT GATTAGCGCGGTGCTG	Plasmid construction for FcCGT-V118D/M145A expression
Fc-V118D-R	CAGCACCGCGCTAATCAGGGTATCATCGG TAATCAGCGCGCTCAG	Plasmid construction for FcCGT-V118D/M145A expression
Fc-M145A-F	CTGTTTACCGCGAGCGCGCGCGTTTAGC CTGACCGCGAGCTTTCC	Plasmid construction for FcCGT-V118D/M145A expression
Fc-M145A-R	GGAAAGCTCGCGGTCAGGCTAAACGCGCG CGCGCTCGCGGTAAACAG	Plasmid construction for FcCGT-V118D/M145A expression
FcCGT-T149Q-F	CCGCGAGCGCGCGCATGTTTAGCCTGCAG GCGAGCTTTCCGGCGATTGTG	Plasmid construction for FcCGT-T149Q/A189D expression
FcCGT-T149Q-R	CACAATCGCCGAAAGCTCGCCTGCAGGC TAAACATGCGCGCGCTCGCGG	Plasmid construction for FcCGT-T149Q/A189D expression
FcCGT-A189D-F	CGCTGAGCAGCGTGCCGCCGGATGTGATG GATAGCAAAAGCC	Plasmid construction for FcCGT-T149Q/A189D expression
FcCGT-A189D-R	GGCTTTTGCTATCCATCACATCCGGCGGCA CGCTGCTCAGCG	Plasmid construction for FcCGT-T149Q/A189D expression
NnGT-R	GCTGCCACCGCCCACTTC	Universal primer
Nn-S126P-F	GCGCGCTGATTACCGATGTGCCGCTGGCG AGCGCGTTTATTAGC	Plasmid construction for NnGT-S126P/C156Q/L197D expression
Nn-S126P-R	GCTAATAAACGCGCTCGCCAGCGGCACAT CGGTAATCAGCGCGC	Plasmid construction for NnGT-S126P/C156Q/L197D expression
Nn-C156Q-F	GCGCGTGATGCTGAGCCTGCAACTGAAC TTTCCGACCTTTGTG	Plasmid construction for NnGT-S126P/C156Q/L197D expression
Nn-C156Q-R	CACAAAGGTCGGAAAGTTCAGTTGCAGGC TCAGCATCCACGCGC	Plasmid construction for NnGT-S126P/C156Q/L197D expression
Nn-L197D-F	CCGAAAAGCTGGCTGCCGCCGGATTACT GGATCTGAGCAACCTG	Plasmid construction for NnGT-S126P/C156Q/L197D expression
Nn-L197D-R	CAGGTTGCTCAGATCCAGTAAATCCGGCG GCAGCCAGCTTTTCGG	Plasmid construction for NnGT-S126P/C156Q/L197D expression
Gja-1	GCAGTAGCACCGCGCTGGCG	Universal primer
Gja-2	CGCCAGCGCGGTGCTACTGC	Universal primer

Primer	Sequence (5'-3')	Description
Gja-F116S-F117P-F	GCATTCGCGCGCTGATTATTGATAGCCCGTGCACGAGCGCGATGCCGTATAGC	Plasmid construction for GjGTa-F116S/F117P/A142M expression
Gja-F116S-F117P-R	GCTATACGGCATCGCGCTCGTGACGGGCATCAATAATCAGCGCGCGAATGC	Plasmid construction for GjGTa-F116S/F117P/A142M expression
Gja-A142M-F	GTGTATTATTTTTTTACGAGCGGCGCGGCCATGGTGGCGCTGTTTCTGTATTTTCCG	Plasmid construction for GjGTa-F116S/F117P/A142M expression
Gja-A142M-R	CGGAAAATACAGAAACAGCGCCACCATGGCCGCGCCGCTCGTAAAAAATAATACAC	Plasmid construction for GjGTa-F116S/F117P/A142M expression
GjGTc-R	GCGGCCCGAGGCTCGCGCTGC	Universal primer
Gjc-M118S-F119P-F	CTGGTGGGCGTGGTTGTGGATAGCCCGTG CAGCAGCATGATTGATGTG	Plasmid construction for GjGTc-M118S/F119P/M148Q/R189D expression
Gjc-M118S-F119P-R	CACATCAATCATGCTGCTGCACGGGCTATC CACAACCACGCCCACCAG	Plasmid construction for GjGTc-M118S/F119P/M148Q/R189D expression
Gjc-M148Q-F	CGGCGATGCTGGGCCTGCAACTGCATCTG CAGAGCCTGCG	Plasmid construction for GjGTc-M118S/F119P/M148Q/R189D expression
Gjc-M148Q-R	CGCAGGCTCTGCAGATGCAGTTGCAGGCC CAGCATCGCCG	Plasmid construction for GjGTc-M118S/F119P/M148Q/R189D expression
Gjc-R189D-F	CGGTGAAAGTGCTGCCGAGCGATCTGTTT GATATGGAAGGCGG	Plasmid construction for GjGTc-M118S/F119P/M148Q/R189D expression
Gjc-R189D-R	CCGCCTTCCATATCAAACAGATCGCTCGGC AGCACTTTCACCG	Plasmid construction for GjGTc-M118S/F119P/M148Q/R189D expression

Supplementary Table 3 | General conditions for PCR.

Step	Condition	Cycle index
1	98 °C, 3 min	*1
2-4	98 °C, 10 sec; (T _m -5) °C, 5 sec; 68 °C, 10 sec/kb.	*34
5	68 °C, 10 min	*1

Supplementary Table 4 | HPLC methods used in this study.

Method	Solvent A	Solvent B	Gradient	Analysis substrates
A	Water containing 0.1% formic acid	MeOH	10%-50% B, 0.1-8.0 min; 50%-85% B, 8.1-10.0 min; 85% B, 10.1-13.0 min; 85%-10% B, 13.1-15.0 min; 10% B, 15.1-22.0 min.	1a-1d; 1f-1ad; 1ag-1ah
B	Water containing 0.1% formic acid	MeOH	5% B, 0.1-22.0 min; 5%-85% B, 22.1-25.0 min; 85% B, 25.1-30.0 min; 85%-5% B, 30.1-33.0 min 5% B, 33.1-40.0 min.	1e
C	Water containing 0.1% formic acid	MeOH	20% B, 0.1-22.0 min; 20%-85% B, 22.1-25.0 min; 85% B, 25.1-30.0 min; 85%-20% B, 30.1-33.0 min 20% B, 33.1-40.0 min.	1ae
D	Water containing 0.1% formic acid	MeOH	30 % B, 0.1-6.0 min; 30%-50% B, 6.1-17.0 min; 50%-85% B, 17.1-18.0 min; 85% B, 18.1-22.0 min; 85%-30% B, 22.1-23.0 min; 30% B, 23.1-30.0 min	1af

Supplementary Table 5 | Kinetic test of MiCGT-WT and mutants in Reactions of 1a.

entry	enzyme	k_{cat} (min^{-1})	K_{m} (mM)	$k_{\text{cat}}/K_{\text{m}}$ ($\text{mM}^{-1} \text{min}^{-1}$)
1	MiCGT-WT	0.509±0.088	0.430±0.155	1.184
2	MiCGT-E152Q	0.963±0.02	0.176±0.025	5.472
3	MiCGT-E152Q/V190D	1.128±0.033	0.189±0.021	5.968
4	MiCGT-E152Q/V190D/S122P	9.473±0.103	0.036±0.003	263.139
5	MiCGT-M148A	2.601±0.102	0.269±0.049	9.669
6	MiCGT-M148A/V190T	6.156±0.668	0.317±0.100	19.420
7	MiCGT-M148A/V190T/S121D	30.686±0.942	0.249±0.030	123.237

Enzymatic assays containing 100 mM Kpi buffer, pH 8.0, 4 mM of saturated UDP-Glc, and varying concentrations (50, 100, 200, 400, 600, 800, 1000, 1500, 2000, 2500, 3000 and 3500 μM) of **1a** was performed at 30 °C for 5 or 10 min in a final volume of 100 μL . All the reactions were terminated by adding 100 μL methanol and centrifuged at 12,000 rpm for 15 min.

Supplementary Table 6 | TON Data for 1a Catalyzed by MiCGT-WT and mutants.

entry	reaction	enzyme	TON
1	1a → 2a	MiCGT-WT ^a	136
2	1a → 2a	MiCGT-E152Q/V190D/S122P ^a	10375
3	1a → 3a	MiCGT-M148A/V190T/S121D ^a	9313
4	1a → 2a	MiCGT-WT ^b	27
5	1a → 2a	MiCGT-E152Q/V190D/S122P ^b	4611
6	1a → 3a	MiCGT-M148A/V190T/S121D ^b	1472

Enzymatic assays containing 100 mM Kpi buffer, pH 8.0, 4 mM of saturated UDP-Glc, and varying concentrations (200, 400, 600, 800, 1000, 1500, 2000, 2500, 3000 and 3500 μM) of **1a** was performed at 30^a or 50^b °C for 24 h in a final volume of 100 μL . All the reactions were terminated by adding 100 μL methanol and centrifuged at 12,000 rpm for 15 min.

Supplementary Table 7 | Data collection and refinement statistics.

	MiCGT-QDP 8XFH	MiCGT-ATD 8XFW
Data collection		
Space group	P 2 ₁ 2 ₁ 2	P 2 ₁ 2 ₁ 2
Cell dimensions		
<i>a</i> , <i>b</i> , <i>c</i> (Å)	92.8, 93.8, 109.5	93.2, 103.6, 110.2
α , β , γ (°)	90.00, 90.00, 90.00	90.00, 90.00, 90.00
Resolution (Å)	50.00-2.80(2.90-2.80) ^a	30.00-3.10(3.31-3.10)
<i>R</i> _{merge}	0.182(0.707)	0.287(1.188)
<i>I</i> / σ (<i>I</i>)	14.5(2.7)	10.6(10.9)
<i>CC</i> _{1/2}	0.985(0.899)	0.983(0.760)
Completeness (%)	100.0(100.0)	99.8(99.8)
Redundancy	12.9(13.1)	6.6(6.2)
Refinement		
Resolution (Å)	50.00-2.80	48.75-3.10
No. reflections	26288	19773
<i>R</i> _{work} / <i>R</i> _{free}	0.221/0.244	0.192/0.240
No. atoms		
Protein	6644	6658
Ligand	50	25
B factors		
Protein	57.0	46.5
Ligand	55.8	26.6
R. m. s. deviations		
Bond lengths (Å)	0.003	0.004
Bond angles (°)	0.83	0.82
Ramachandran plot		
favoured (%)	98.6	98.5
allowed (%)	1.4	1.5
outliers (%)	0	0

^athe highest resolution shell

Supplementary Table 8 | MM/GBSA binding free energies of MiCGT-WT, MiCGT-ATD, and MiCGT-QDP.

entry	enzyme	MM/GBSA ΔG (kcal/mol)
1	MiCGT-WT	-29.78 $\pm$ 5.89
2	MiCGT-E152Q	-36.79 $\pm$ 5.78
3	MiCGT-E152Q/V190D	-25.79 $\pm$ 3.98
4	MiCGT-E152Q/V190D/S122P	-45.67 $\pm$ 3.96
5	MiCGT-M148A	-31.62 $\pm$ 3.11
6	MiCGT-M148A/V190T	-34.48 $\pm$ 3.32
7	MiCGT-M148A/V190T/S121D	-39.05 $\pm$ 3.74

Supplementary Table 9 | MM/GBSA energy decomposition.

Res ID	Residue Contribution (kcal/mol)						
	WT	ATD	QDP	M148A	M148A/V190T	E152Q	E152Q/V190D
H23	-1.64 ± 0.54	-0.48 ± 0.30	-1.03 ± 0.46	-2.52 ± 0.97	-2.45 ± 2.74	-1.71 ± 0.43	0.53 ± 0.58
S121	-2.21 ± 1.62	0.22 ± 0.21	-0.83 ± 0.55	0.00 ± 0.03	0.24 ± 2.33	-0.02 ± 0.06	-3.61 ± 0.66
122	-0.39 ± 0.52	-0.17 ± 0.05	-1.55 ± 0.65	-0.10 ± 0.03	0.13 ± 2.08	-0.09 ± 0.07	-3.50 ± 0.85
148	-0.92 ± 0.51	-0.55 ± 0.82	-1.06 ± 0.34	-0.03 ± 0.03	-0.06 ± 0.03	-0.04 ± 0.04	-0.11 ± 0.11
152	0.76 ± 0.69	-3.12 ± 0.68	-1.00 ± 0.84	0.54 ± 0.07	0.48 ± 0.06	-0.03 ± 0.07	-2.92 ± 0.68
190	-0.05 ± 0.04	-0.73 ± 2.99	0.26 ± 0.05	-0.02 ± 0.01	0.03 ± 0.03	0.00 ± 0.01	0.28 ± 0.08
F90	-0.14 ± 0.21	0.14 ± 0.04	0.17 ± 0.08	0.06 ± 1.43	0.11 ± 1.40	0.01 ± 0.05	-0.22 ± 0.35
W93	-1.56 ± 0.51	-0.09 ± 0.08	0.30 ± 0.28	-0.11 ± 0.04	-0.14 ± 0.06	-0.25 ± 0.07	-0.55 ± 0.41
F142	-0.87 ± 0.53	-1.74 ± 0.58	-1.52 ± 0.41	-0.11 ± 0.04	-0.04 ± 0.07	-0.09 ± 0.04	-0.35 ± 0.19
T143	-0.68 ± 0.34	-0.86 ± 0.19	0.43 ± 0.27	-3.71 ± 0.72	-4.66 ± 1.55	-1.37 ± 1.50	0.68 ± 1.38
S144	-1.72 ± 0.89	-0.8 ± 0.22	0.10 ± 0.32	-0.42 ± 0.49	-0.93 ± 0.79	-0.95 ± 3.20	0.01 ± 1.87
C149	-0.06 ± 0.07	-0.21 ± 0.21	0.05 ± 0.10	-0.03 ± 0.02	-0.06 ± 0.03	-0.05 ± 0.03	0.11 ± 0.09
P188	0.01 ± 0.02	-0.77 ± 2.43	0.08 ± 0.03	-0.04 ± 0.02	-0.30 ± 1.83	-0.01 ± 0.02	-0.04 ± 1.34
F198	-0.47 ± 0.86	-1.54 ± 0.45	0.35 ± 0.18	0.05 ± 0.03	0.04 ± 0.07	0.01 ± 0.03	-1.73 ± 0.75
L202	-0.26 ± 0.36	-1.79 ± 0.5	-1.23 ± 0.31	0.28 ± 2.74	-0.02 ± 0.01	-0.02 ± 0.01	-1.35 ± 2.99
N205	-0.75 ± 0.98	-2.36 ± 0.86	-4.52 ± 0.88	-0.02 ± 0.03	-0.01 ± 0.02	-0.08 ± 0.03	-7.19 ± 0.66
F209	-0.11 ± 0.16	-0.26 ± 0.19	-0.24 ± 0.25	0.03 ± 0.02	0.05 ± 0.02	0.13 ± 0.95	-0.32 ± 0.19
R282	-0.25 ± 0.08	-2.19 ± 0.30	-2.13 ± 0.55	-1.48 ± 4.41	-0.82 ± 2.94	-5.98 ± 0.79	-1.87 ± 4.23
F383	-0.02 ± 0.08	-0.07 ± 0.08	-0.39 ± 0.21	0.23 ± 0.22	0.04 ± 0.04	-0.14 ± 0.62	0.08 ± 0.11
G384	-1.38 ± 0.3	-1.37 ± 0.38	-2.13 ± 0.55	-0.54 ± 0.25	-0.01 ± 0.07	0.02 ± 0.08	-0.48 ± 0.28
D385	-1.85 ± 0.78	0.07 ± 0.70	-1.13 ± 1.12	-0.46 ± 5.03	0.41 ± 0.70	0.11 ± 0.58	0.72 ± 5.05
N389	0.24 ± 0.08	-0.11 ± 0.06	-1.29 ± 0.49	0.48 ± 0.22	-0.47 ± 0.48	-0.34 ± 0.30	-0.04 ± 0.04

Supplementary Table 10 | Description of 70 C-glycosyltransferases for sequence alignment.

No.	Protein	Organism	Accession
1	MiCGT	<i>Mangifera indica</i>	A0A0M4KE44.1
2	AaCGT	<i>Anemarrhena asphodeloides</i>	UTY68313.1
3	DcaCGT	<i>Dendrobium catenatum</i>	QOD39011.1
4	AbCGT	<i>Aloe vera</i>	QNN26307.1
5	MiCGTb	<i>Mangifera indica</i>	AMM73095.1
6	MaCGT	<i>Morus alba</i>	QXJ78679.1
7	GgCGT	<i>Glycyrrhiza glabra</i>	QGL05036.1
8	PhCGT1	<i>Phyllostachys edulis</i>	MK616588.1
9	PhCGT2	<i>Phyllostachys edulis</i>	MK616589.1
10	PhCGT3	<i>Phyllostachys edulis</i>	MK616590.1
11	PhCGT4	<i>Phyllostachys edulis</i>	MK616591.1
12	PgCGT1	<i>Phyllostachys glauca</i>	MK616592.1
13	PpCGT1	<i>Phyllostachys prominens</i>	MK616593.1
14	TcCGT1	<i>Trollius chinensis</i>	MK644229.1
15	LaCGT1	<i>Lemna aequinoctialis</i>	UZX50445.1
16	VtCGT1	<i>Viola tricolor</i>	MT433767.1
17	PsCGTa	<i>Pistia stratiotes</i>	MK894453.1
18	LpCGTa	<i>Landoltia punctata</i>	QLF98869.1
19	AeCGTa	<i>Arisaema erubescens</i>	MK894449.1
20	SbCGTa	<i>Scutellaria baicalensis</i>	QLF98861.1
21	PlCGT	<i>Pueraria montana var. lobata</i>	ON630411.1
22	AdCGT	<i>Angelica decursiva</i>	OM468890.1
23	ScCGT4	<i>Odontosoria chusana</i>	ON936075.1
24	ScCGT3	<i>Odontosoria chusana</i>	ON936074.1
25	ScCGT2	<i>Odontosoria chusana</i>	ON936073.1
26	ScCGT1	<i>Odontosoria chusana</i>	ON936072.1
27	FcCGT	<i>Citrus japonica</i>	BBA18062.1
28	OsCGT	<i>Oryza sativa Indica Group</i>	C3W7B0.1
29	ZmUGT	<i>Zea mays</i>	A0A096SRM5.1
30	FeCGTa	<i>Fagopyrum esculentum</i>	AB909375.1
31	FeCGTb	<i>Fagopyrum esculentum</i>	AB909376.1
32	GmCGT	<i>Glycine max</i>	NM_001360312.1
33	GtUF6CGT1	<i>Gentiana triflora</i>	A0A0B6VIJ5

No.	Protein	Organism	Accession
34	CuCGT	<i>Citrus unshiu</i>	LC131334.1
35	PIUGT43	<i>Pueraria montana</i> var. <i>lobata</i>	AMQ26115.2
36	WjGT1	<i>Eutrema japonicum</i>	LC465149.1
37	ZmCGT1	<i>Zea mays</i>	NM_001155464.3
38	N4CGT1	<i>Hypericum perforatum</i>	LC628878.1
39	N4CGT2	<i>Hypericum perforatum</i>	LC628880.1
40	SbCGTb	<i>Scutellaria baicalensis</i>	QLF98862.1
41	GuCGTb	<i>Glycyrrhiza uralensis</i>	QLF98866.1
42	GuCGTa	<i>Glycyrrhiza uralensis</i>	QLF98865.1
43	UGT74E2	<i>Arabidopsis thaliana</i>	OAP17332.1
44	ZmCGT2	<i>Zea mays</i>	NM_001154527.3
45	TaCGT2-D	<i>Triticum aestivum</i>	A0A3B6TNL8
46	AeCGTb	<i>Arisaema erubescens</i>	MK894450.1
47	BdCGT1	<i>Brachypodium distachyon</i>	I1GYZ6
48	BdCGT2	<i>Brachypodium distachyon</i>	I1GZD7
49	LpCGTb	<i>Landoltia punctata</i>	QLF98870.1
50	PsCGTb	<i>Pistia stratiotes</i>	MK894454.1
51	OsCGT2	<i>Oryza sativa</i>	Q5VMG8
52	OsCGT3	<i>Oryza sativa</i>	Q5VME5
53	OsCGT4	<i>Oryza sativa</i>	A2YBW7
54	OsCGT5	<i>Oryza sativa</i>	A2YBW8
55	OsCGT6	<i>Oryza sativa</i>	A2YBX1
56	SiCGT1	<i>Setaria italica</i>	K3XWN4
57	SiCGT2	<i>Setaria italica</i>	K3XWJ1
58	SiCGT3	<i>Setaria italica</i>	K3Y181
59	SobCGT1	<i>Sorghum bicolor</i>	C5Z8Y8
60	SobCGT2	<i>Sorghum bicolor</i>	C5Z8Y5
61	SobCGT3	<i>Sorghum bicolor</i>	C5Z8Y2
62	SobCGT4	<i>Sorghum bicolor</i>	A0A194YGR9
63	TaCGT1-A	<i>Triticum aestivum</i>	A0A3B6RGT4
64	TaCGT1-B	<i>Triticum aestivum</i>	A0A3B6SEV6
65	TaCGT1-D	<i>Triticum aestivum</i>	A0A3B6TPG6
66	TaCGT2-A	<i>Triticum aestivum</i>	A0A3B6RGR9
67	TaCGT2-B	<i>Triticum aestivum</i>	A0A3B6SB95
68	ZmCGT3	<i>Zea mays</i>	A0A1D6LXE6

No.	Protein	Organism	Accession
69	ZmCGT4	<i>Zea mays</i>	B4FAN3
70	ZmCGT5	<i>Zea mays</i>	B4FM13

Supplementary Table 11 | BLASTP MiCGT homologs in NCBI databases.

Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
2-hydroxyflavanone C-glucosyltransferas FcCGT [<i>Citrus japonica</i>]	605	605	98%	0.0	64.61%	A0A224AM54.1
UDP-glycosyltransferase 13-like NnGT [<i>Nelumbo nucifera</i>]	534	534	98%	0.0	58.03%	XP_010258947.1
UDP-glucose glucosyltransferase GjGTa [<i>Gardenia jasminoides</i>]	213	213	95%	2e-65	30.93%	BAK55747.1
UDP-glucose glucosyltransferase GjGTc [<i>Gardenia jasminoides</i>]	179	179	94%	4e-53	29.88%	BAK55746.1

Supplementary Table 12 | MM/GBSA binding free energies of FcCGT-V118D/M145A, FcCGT-T149Q/A198D, NnGT-WT, NnGT-S126P/C156Q/L197D, GjGTa-WT, GjGTa-F116S/F117P/A142M, GjGTc-WT and GjGTc-M118S/F119P/M148Q/R189D.

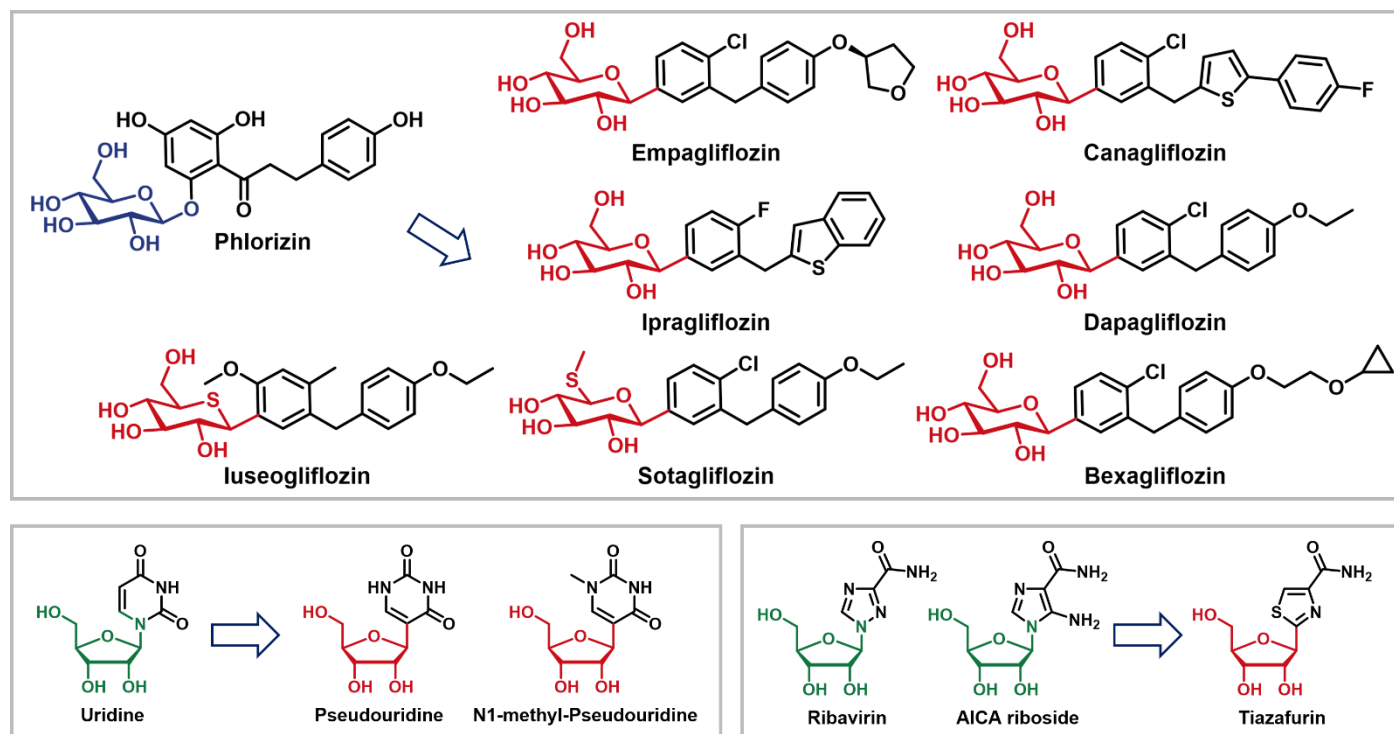
entry	enzyme	MM/GBSA ΔG (kcal/mol)
1	FcCGT-V118D/M145A	-29.90 $\pm$ 3.14
2	FcCGT-T149Q/A198D	-48.54 $\pm$ 3.88
3	NnGT-WT	-34.52 $\pm$ 3.45
4	NnGT-S126P/C156Q/L197D	-25.62 $\pm$ 3.51
5	GjGTa-WT	-42.62 $\pm$ 3.10
6	GjGTa-F116S/F117P/A142M	-38.66 $\pm$ 3.55
7	GjGTc-WT	-45.66 $\pm$ 3.12
8	GjGTc-M118S/F119P/M148Q/R189D	-17.85 $\pm$ 5.17

Supplementary Table 13 | Kinetic test of newly designed UGTs in Reactions of 1a.

entry	enzyme	k_{cat} (min^{-1})	K_m (mM)	k_{cat}/K_m ($\text{mM}^{-1} \text{min}^{-1}$)
1	FcCGT-T149Q/A189D	2.445 $\pm$ 0.096	0.375 $\pm$ 0.072	6.520
2	FcCGT-V118D/M145A	3.194 $\pm$ 0.131	1.064 $\pm$ 0.151	3.002
3	NnGT-S126P/C156Q/L197D	0.523 $\pm$ 0.034	0.319 $\pm$ 0.061	1.639
4	GjGTa-F116S/F117P/A142M	0.021 $\pm$ 0.00007	0.012 $\pm$ 0.0005	17.75

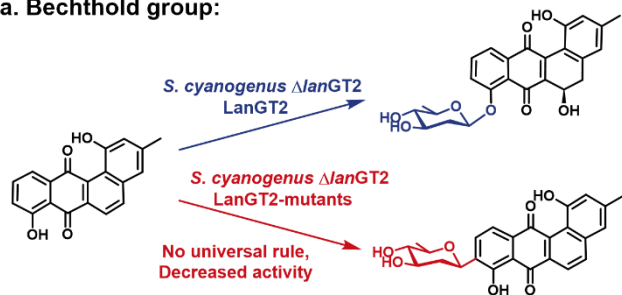
Enzymatic assays containing 100 mM Kpi buffer, pH 8.0, 10 μM enzyme, 4 mM of saturated UDP-Glc, and varying concentrations (50, 100, 200, 400, 600, 800, 1000, 1500, 2000, 2500, 3000 and 3500 μM) of **1a** was performed at 30 $^{\circ}\text{C}$ for 10 or 30 min in a final volume of 100 μL . All the reactions were terminated by adding 100 μL methanol and centrifuged at 12,000 rpm for 15 min.

Supplementary Figures

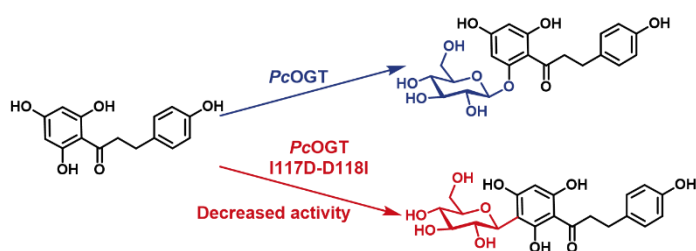


Supplementary Fig. 1 | Development of metabolically stable carbohydrate drugs by replacing other types of glucoside bonds with C-glucoside bonds. AICA riboside: 5-aminoimidazole-4-carboxamide-1- β -D-ribofuranoside).

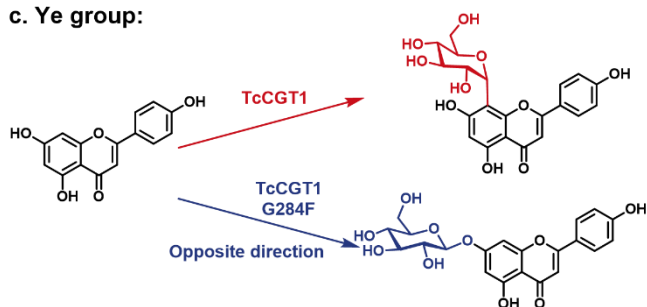
a. Bechthold group:



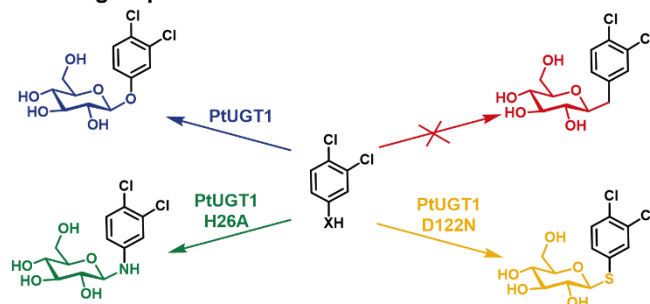
b. Nidetzky group:



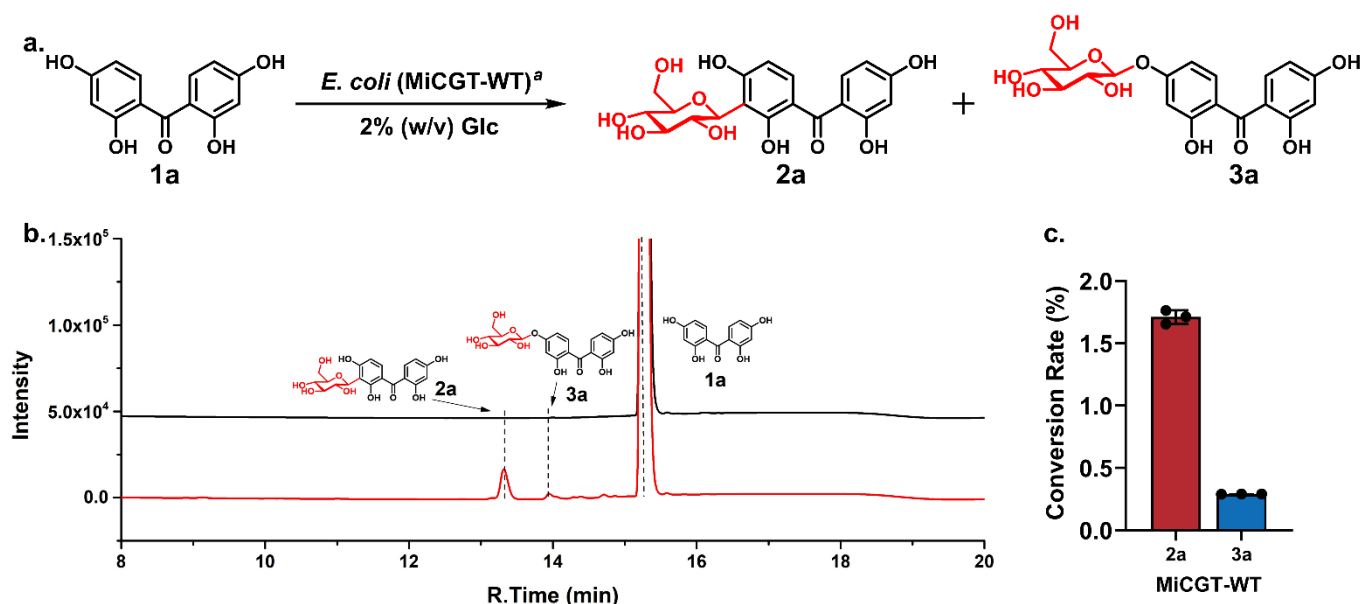
c. Ye group:



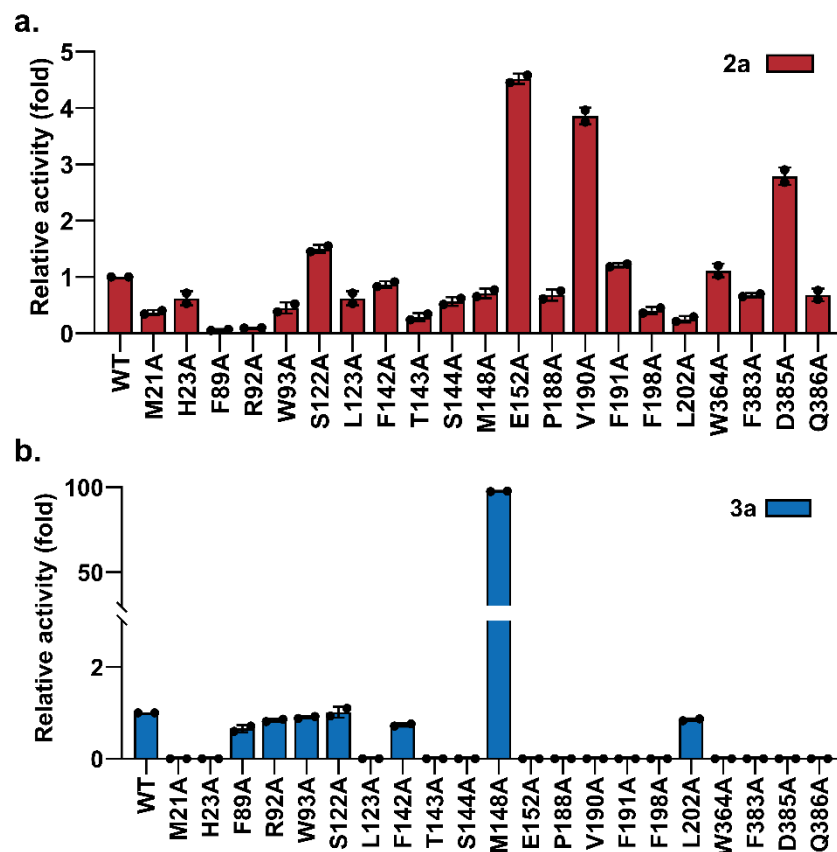
d. Teze group:



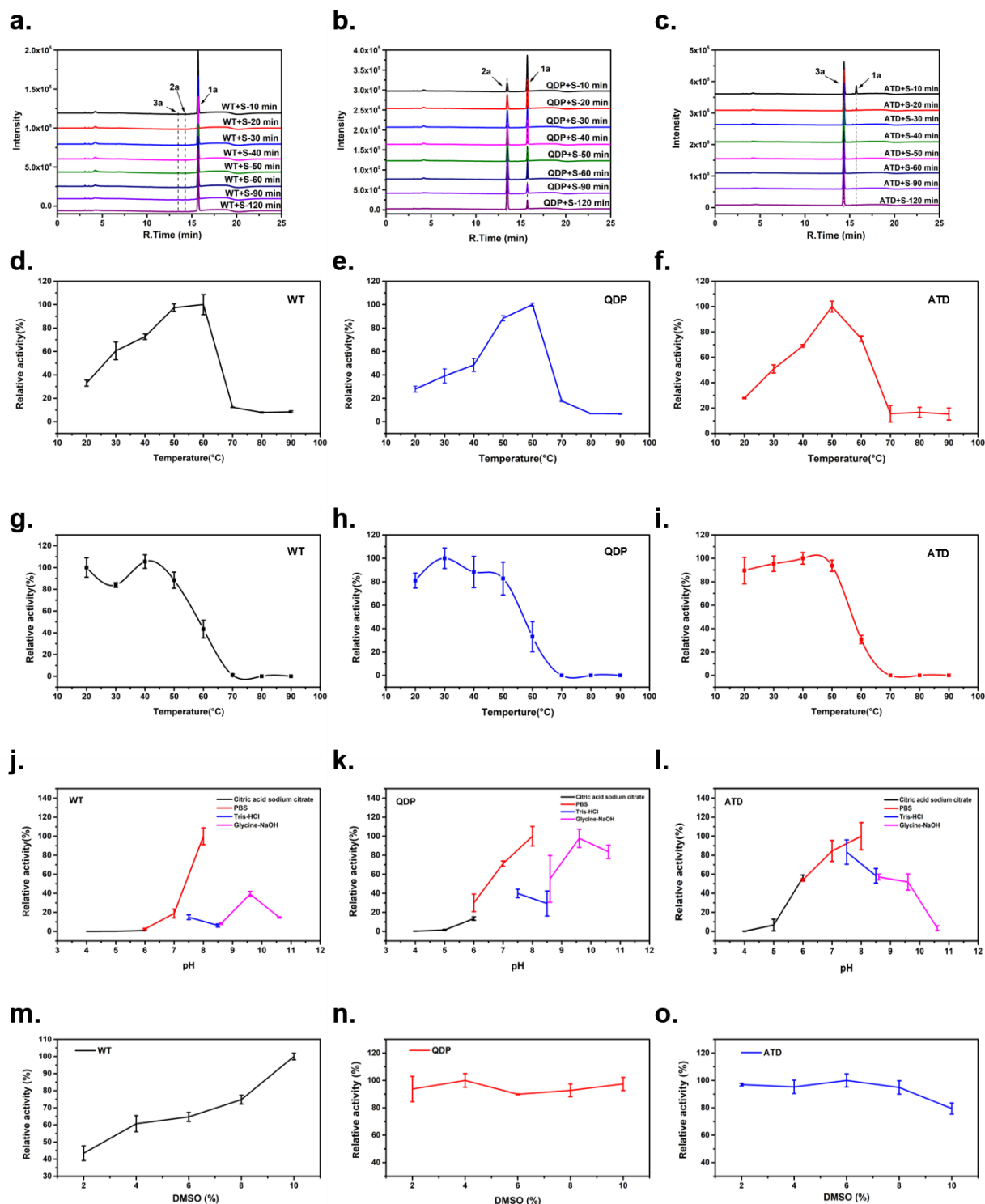
Supplementary Fig. 2 | Previous work on engineering the chemoselectivity of UGTs²⁻⁵.



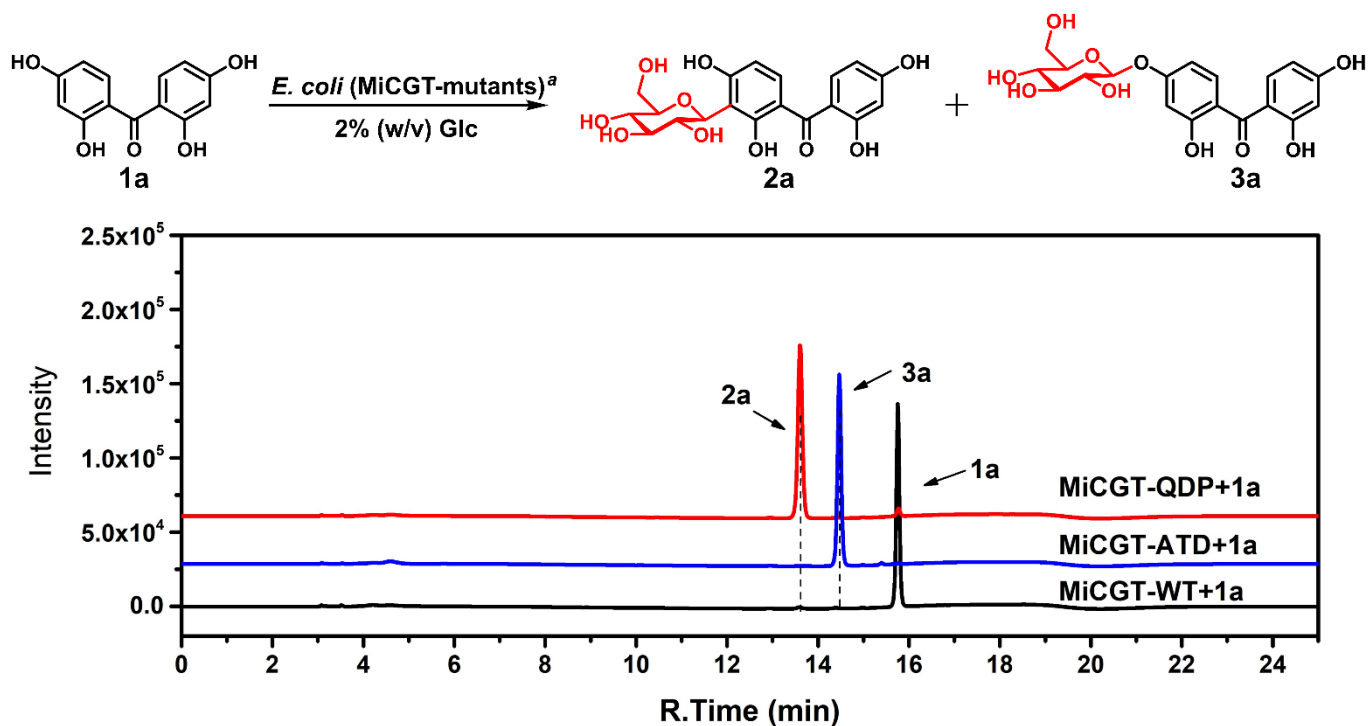
Supplementary Fig. 3 | Glycosylation of 1a catalyzed by the whole cell harboring MiCGT-WT. (a) The reaction catalyzed by the *E. coli* (MiCGT-WT). (b) HPLC chromatograms of **1a** and the enzyme product **2a** and **3a**. (c) Conversion rates of **1a**. ^a whole cells were resuspended by 100 mM Kpi buffer (pH 8.0) containing 2% (w/v) *D*-glucose and 6% (v/v) DMSO, and the cell density was adjusted to an OD_{600 nm} of 40. **1a** were added to the reaction mixture at the final concentration of 3 mM in a total volume of 2 mL, and the bioconversion reactions were kept shaking at 30 °C for 24 h. Conversion rate represent mean $\pm$ SD of three independent replicates (n = 3). WT: wild type. Source data are provided as a Source Data file.



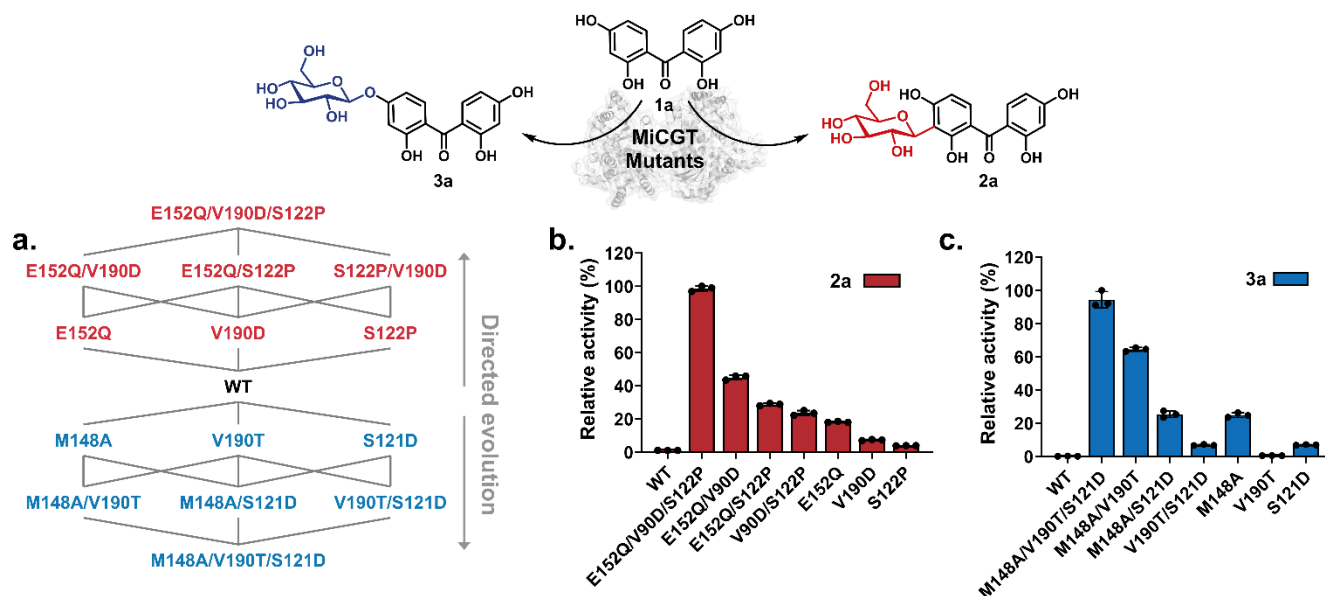
Supplementary Fig. 4 | The activity of alanine mutants catalyzing 1a to 2a or 3a relative to MiCGT-WT. Whole cells harboring MiCGT variants were resuspended by pH 9.0 Tris-HCl buffer containing 2% (w/v) *D*-glucose and 2% (v/v) DMSO, and the cell density was adjusted to an $OD_{600\text{ nm}} = 40$. **1a** were added to the reaction mixture at the final concentration of 0.4 mM in a total volume of 2 mL, and the bioconversion reactions were kept shaking at 30 °C, 1200 rpm for 24 h. Relative activity represent mean $\pm$ SD of three independent replicates ($n = 2$). WT: wild type. Source data are provided as a Source Data file.



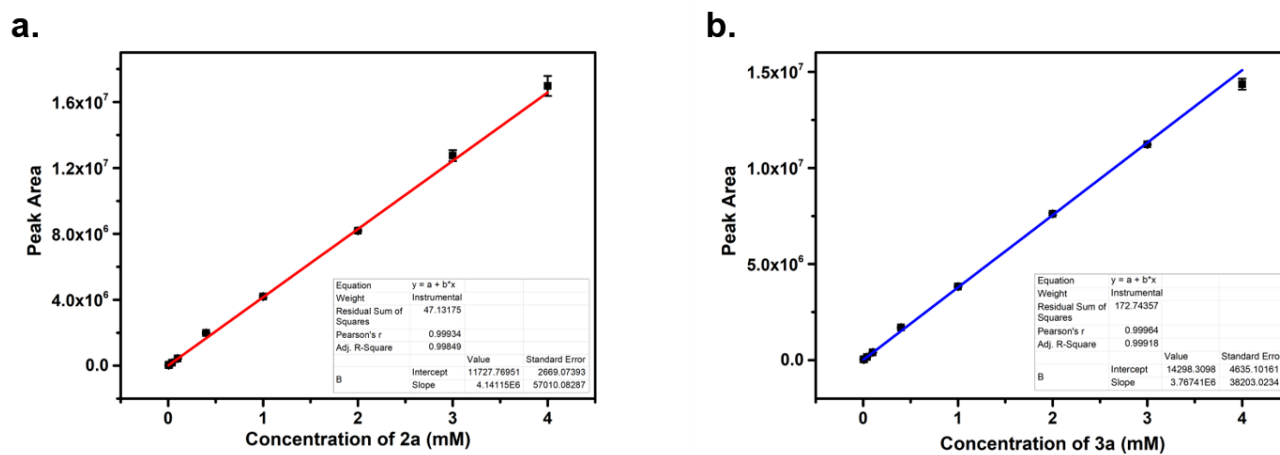
Supplementary Fig. 5 | Optimization of reaction conditions. Time-scale monitoring reactions (a-c), optimization of reaction temperature (d-f), thermostability test (g-i), optimization of reaction pH (j-l), and optimization of cosolvent concentrations (m-o). **1a** was used as the acceptor and UDP-Glc was used as the sugar donor. An optimized reaction time of 120 min was used. MiCGT exhibited its maximum activity at pH 8.0 kpi buffer containing 6% (v/v) DMSO. When the reaction activity is higher at 60 °C, the stability is significantly reduced. Therefore, all subsequent reactions were conducted at 30 °C. Relative activity represent mean $\pm$ SD of three independent replicates (n = 3). WT: wild type. Source data are provided as a Source Data file.



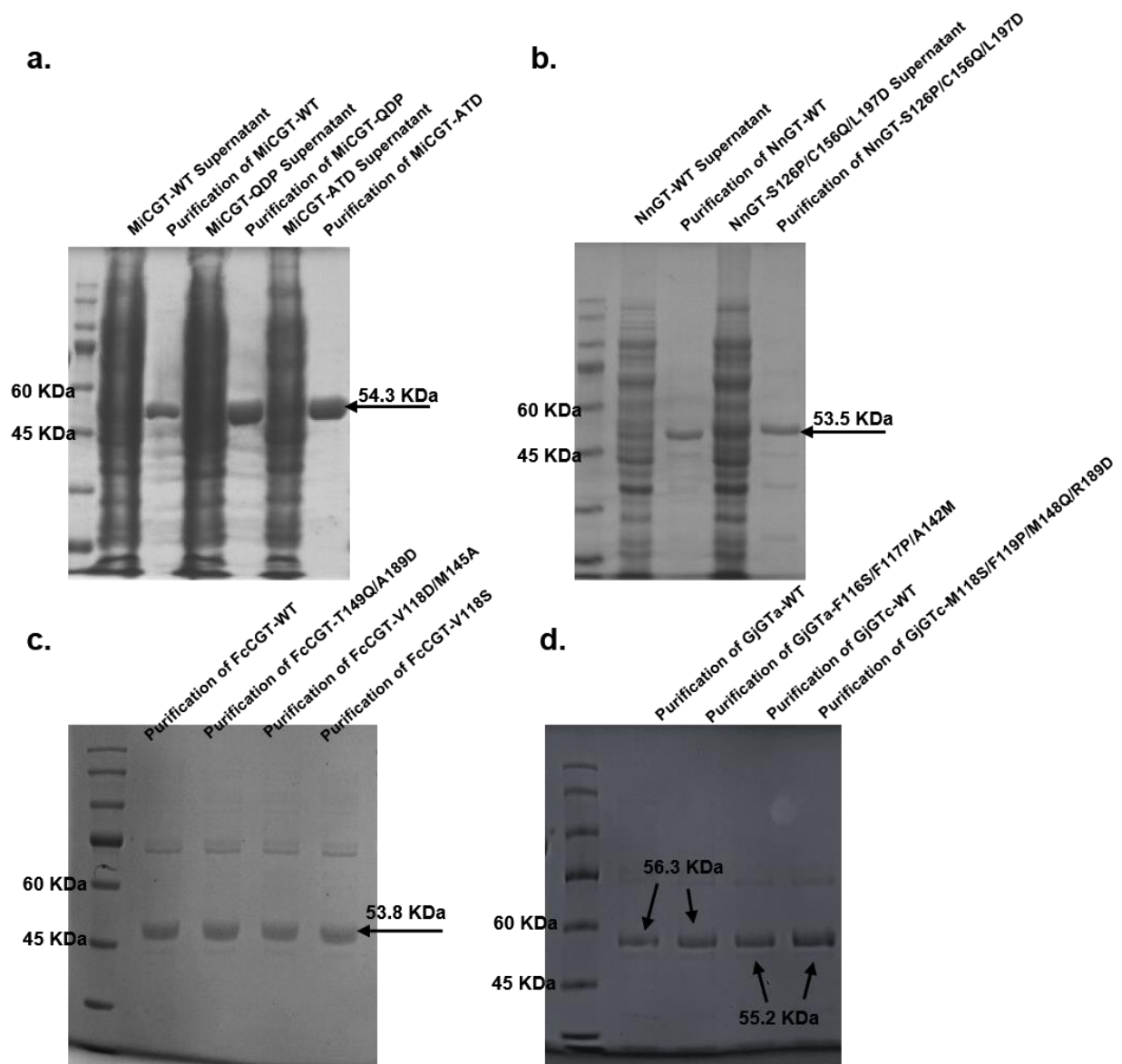
Supplementary Fig. 6 | Glycosylation activity of model substrate **1a was tested by MiCGT-mutants under optimal conditions.** ^aWhole cells were resuspended by 100 mM Kpi buffer (pH 8.0) containing 2% (w/v) *D*-glucose and 6% (v/v) DMSO, and the cell density was adjusted to an $OD_{600\text{ nm}} = 40$. **1a** were added to the reaction mixture at the final concentration of 3 mM in a total volume of 2 mL, and the bioconversion reactions were kept shaking at 30 °C, 1200 rpm for 2 h.



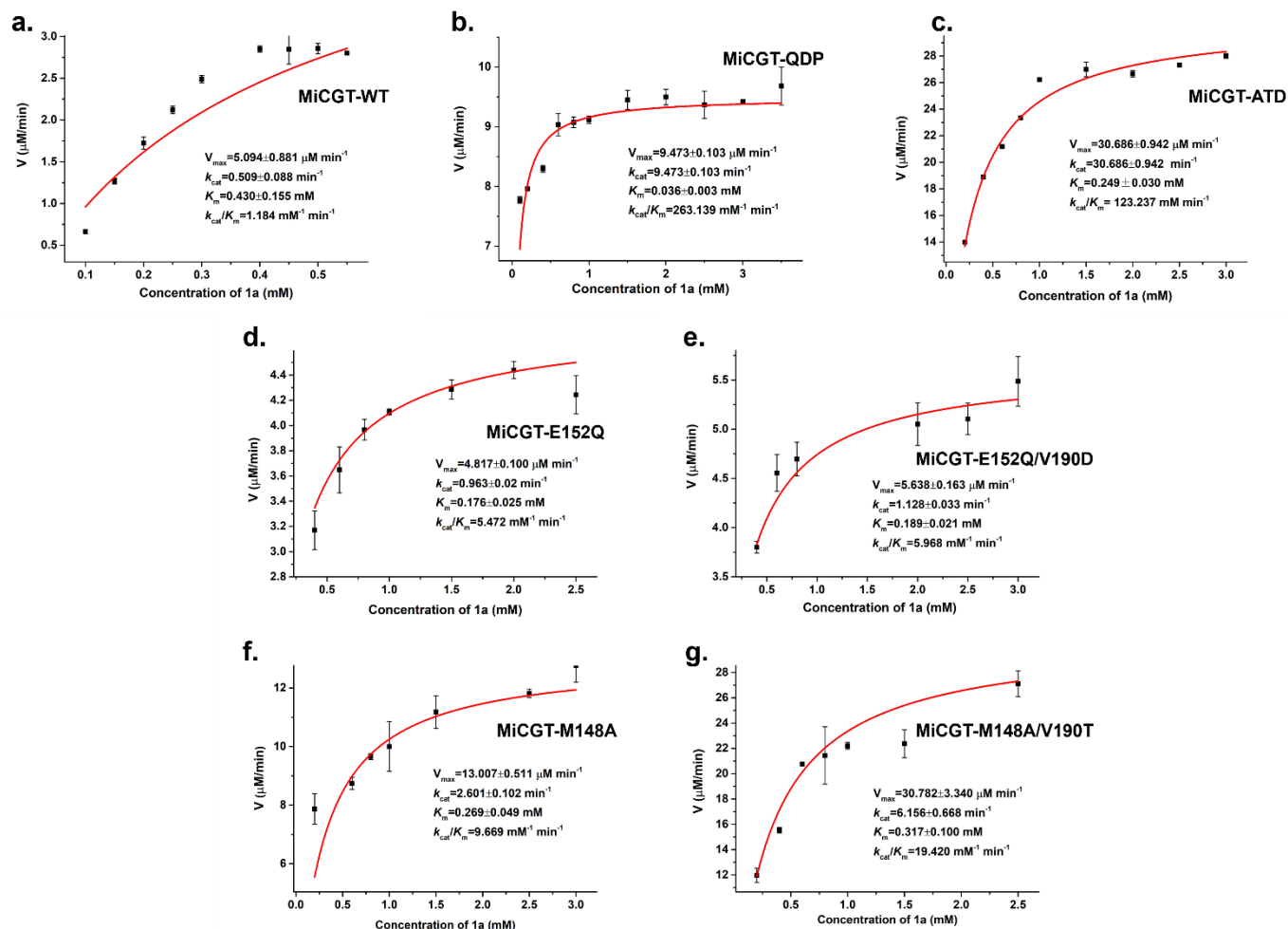
Supplementary Fig. 7 | Directed evolution campaign for chemoselective glycosylation mutants. (a) The 6 possible evolutionary trajectories between parental MiCGT-WT and “triple” mutant; (b) Relative C-glycosylation activity of deconvolutants MiCGT-QDP; (c) Relative O-glycosylation activity of deconvolutants MiCGT-ATD. Whole cells were resuspended by 100 mM Kpi buffer (pH 8.0) containing 2% (w/v) *D*-glucose and 6% (v/v) DMSO, and the cell density was adjusted to an $OD_{600\text{ nm}} = 40$. **1a** were added to the reaction mixture at the final concentration of 3 mM in a total volume of 2 mL, and the bioconversion reactions were kept shaking at 30 °C, 1200 rpm for 12 h. Relative activity represent mean $\pm$ SD of three independent replicates (n = 3). Source data are provided as a Source Data file.



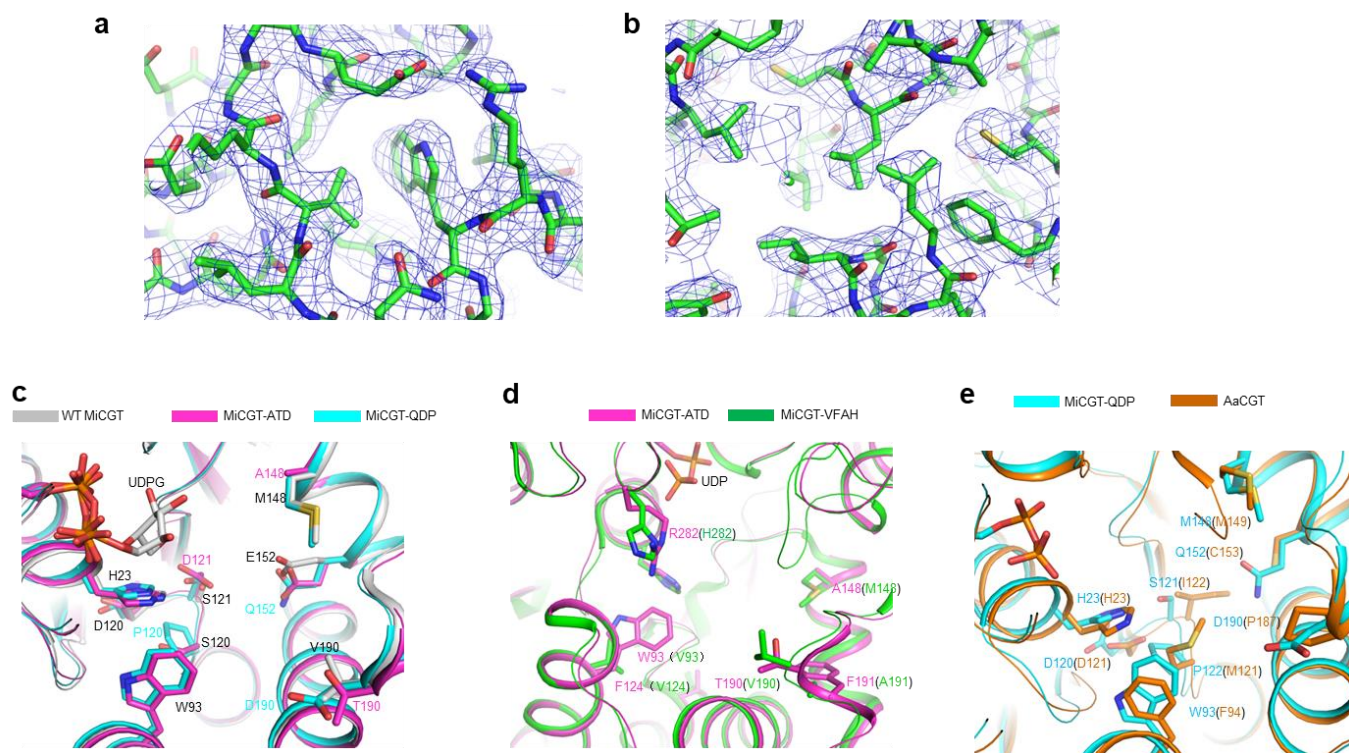
Supplementary Fig. 8 | Establishment of standard curves for glycoside products 2a and 3a.



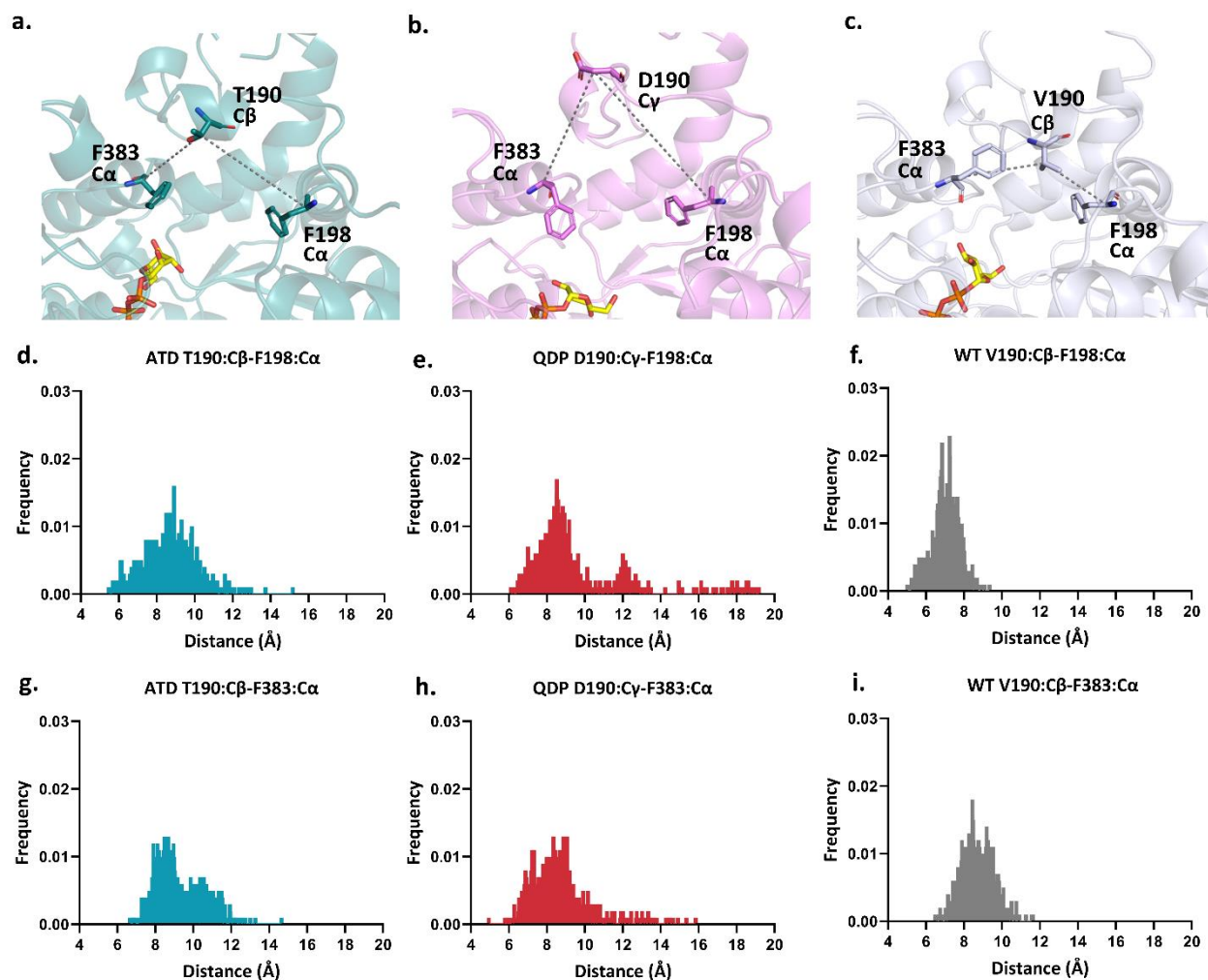
Supplementary Fig. 9 | Purification of GTs. (a) SDS-PAGE of recombinant 6His-MiCGT and its mutants purified by affinity chromatography. (b) SDS-PAGE of recombinant 6His-NnGT and its mutants purified by affinity chromatography. (c) SDS-PAGE of recombinant 6His-FcCGT and its mutants purified by affinity chromatography. (d) SDS-PAGE of recombinant 6His-GjGTa/c and its mutants purified by affinity chromatography.



Supplementary Fig. 10 | Determination of kinetic parameters for MiCGT mutants. The kinetic parameters were detected with 1a as an acceptor and UDP-Glc as a donor at 30 °C, 1200 rpm and pH 8.0.

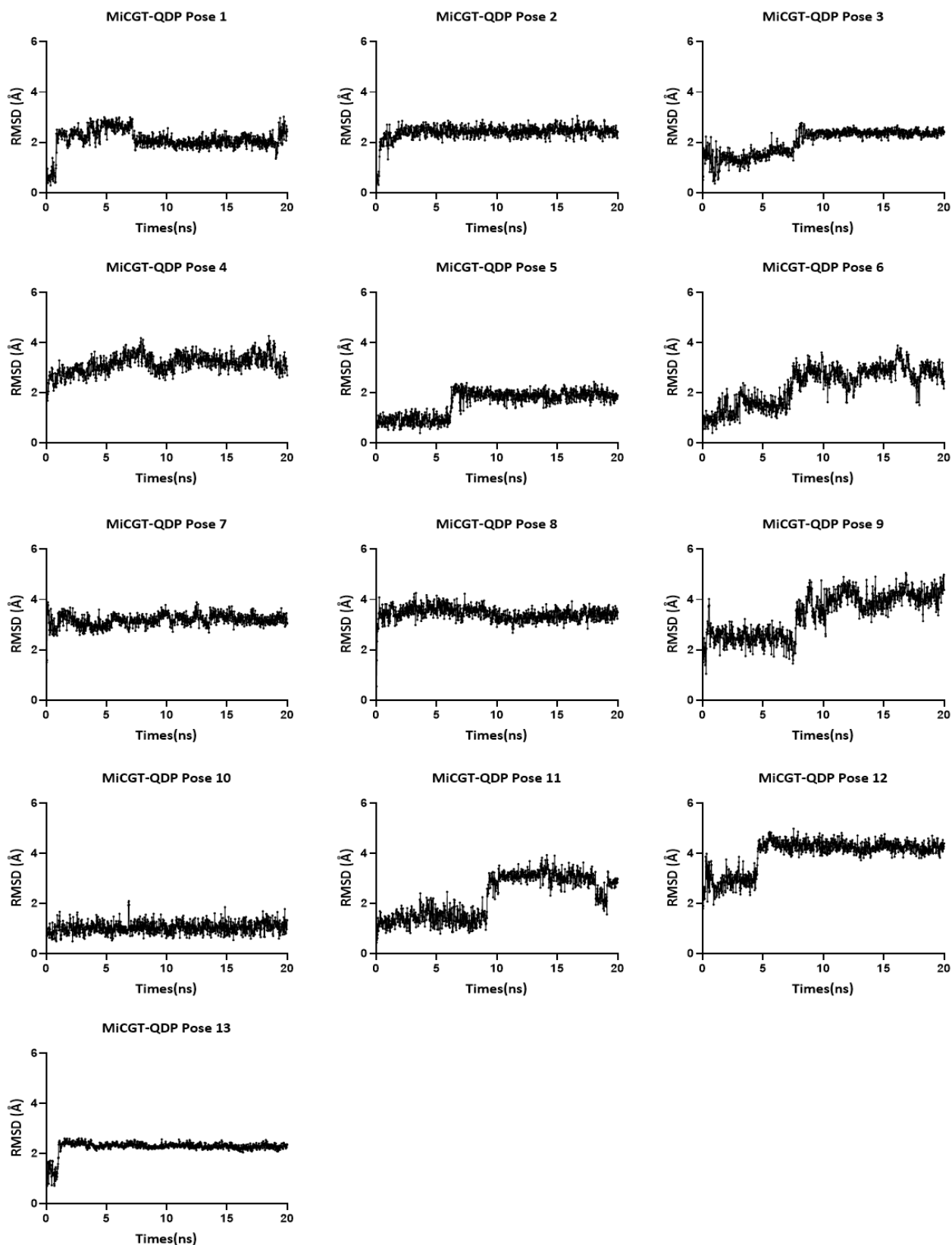


Supplementary Fig. 11 | Active site comparison of MiCGT-WT with mutants and AaCGT. (a-b) 2Fo-Fc omit maps of the structures of MiCGT-ATD (a) and MiCGT-QDP (b) at a contour level of 1.0 σ . (c) MiCGT(grey) was overlapped with MiCGT-ATD (purple) and MiCGT-QDP (cyan). Residues from WT are labelled in black, mutations in MiCGT-ATD and MiCGT-QDP are labelled in purple and cyan, respectively; (d) MiCGT-ATD was overlapped with MiCGT-VFAH (green); (e) MiCGT-QDP was overlapped with AaCGT (brown). The model of AaCGT was generated by alphafold2.

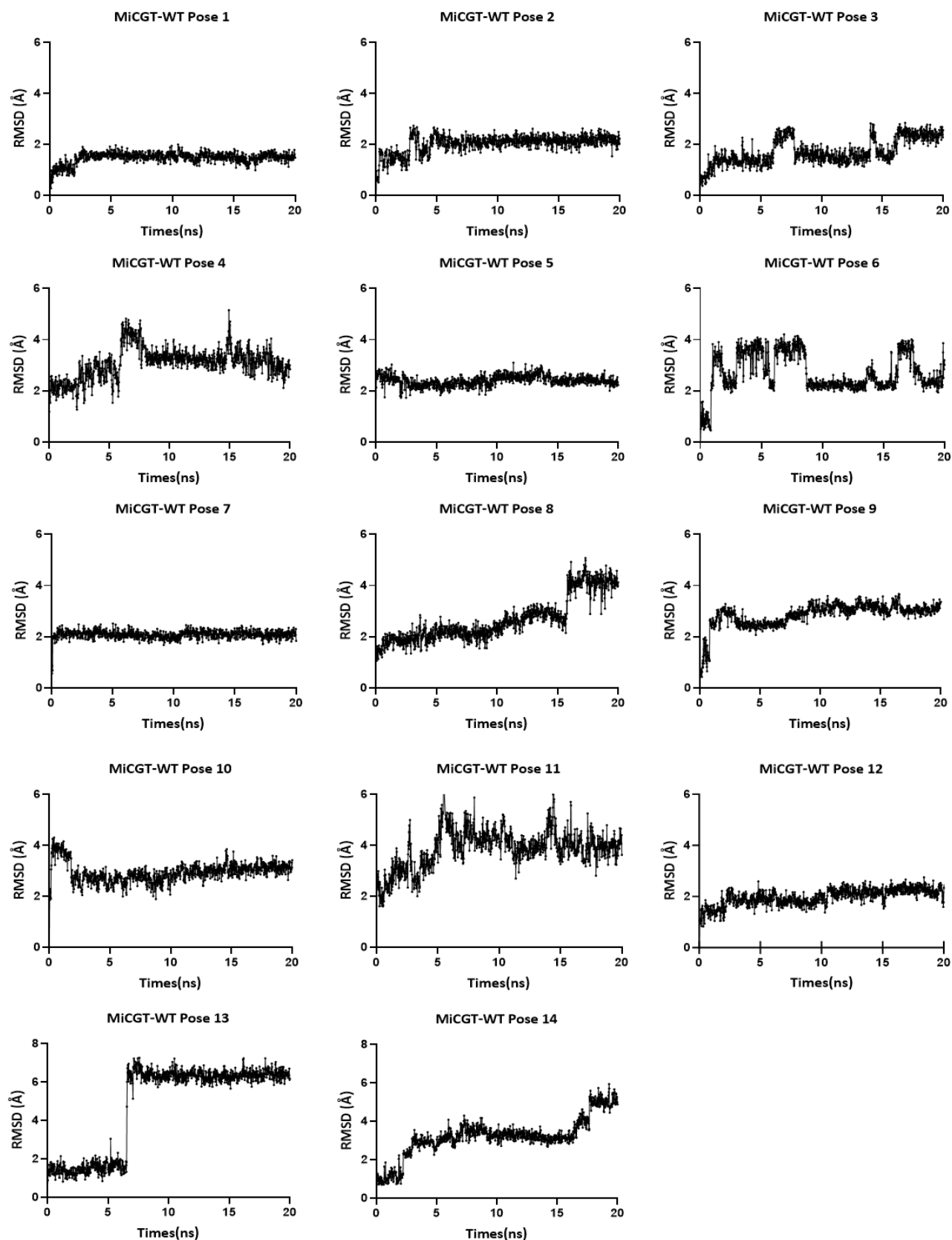


Supplementary Fig. 12 | REST2 simulations for MiCGT-QDP, MiCGT-ATD and MiCGT-WT.

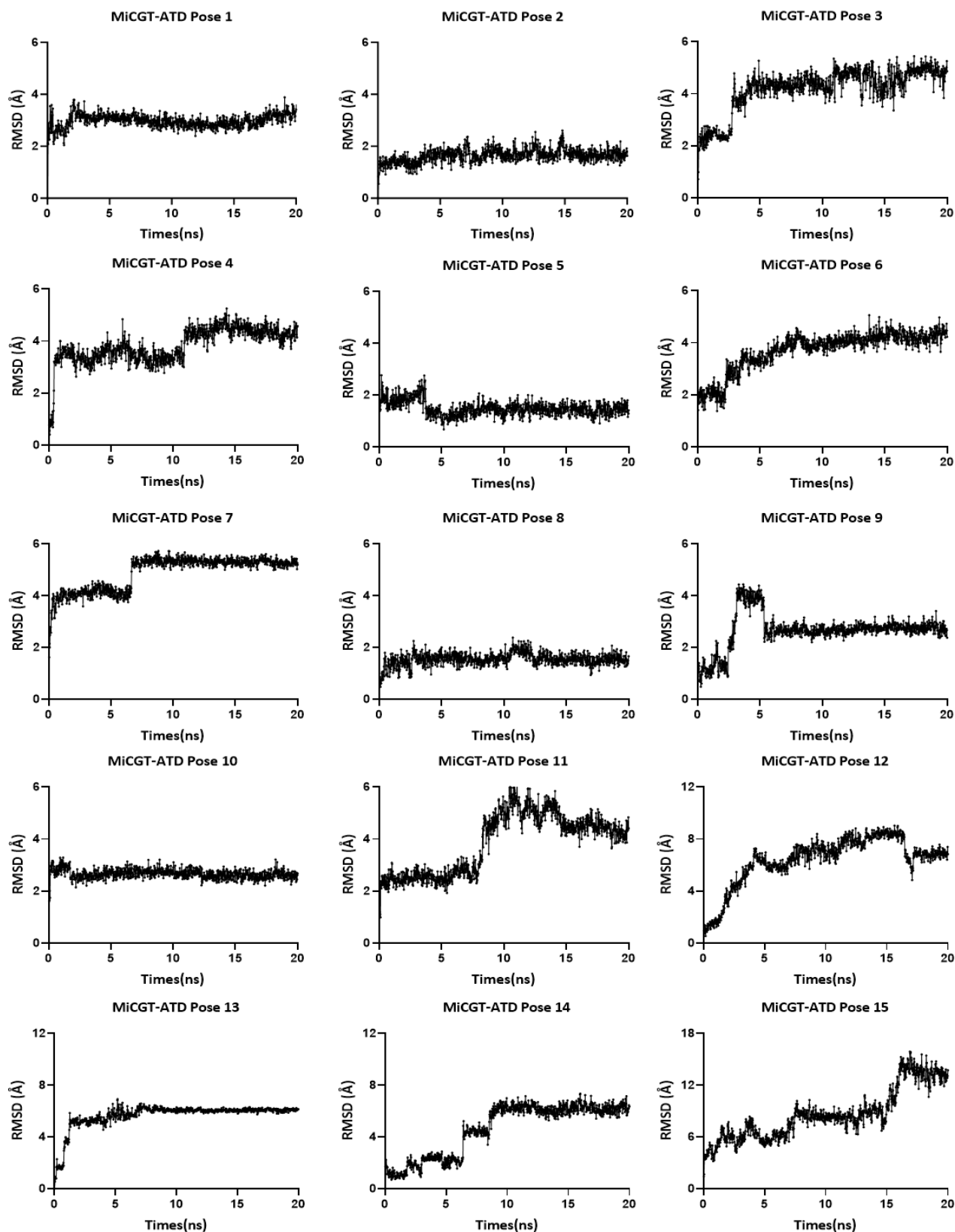
Representative structures for MiCGT-QDP (**a**), MiCGT-ATD (**b**) and MiCGT-WT (**c**); Distance distribution for MiCGT-QDP (**d-f**), MiCGT-ATD (**g-h**), and MiCGT (**i-k**), including the distance between residue 190 and 198, 190 and 383.



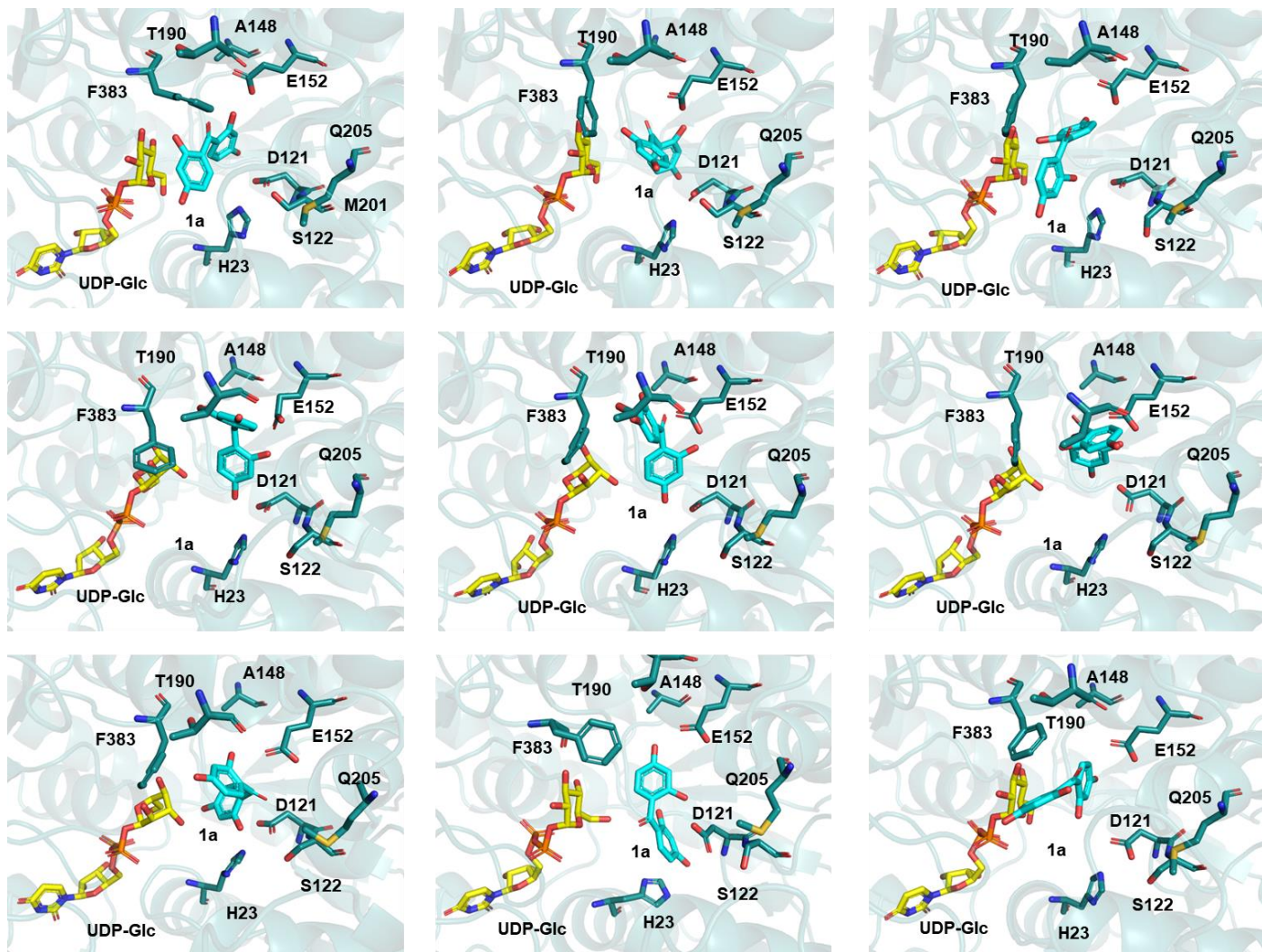
Supplementary Fig. 13 | MD simulations for substrate 1a in MiCGT-QDP starting from different binding poses. The backbone atoms of the enzymes were selected for alignment and substrate **1a** was selected for RMSD calculations.



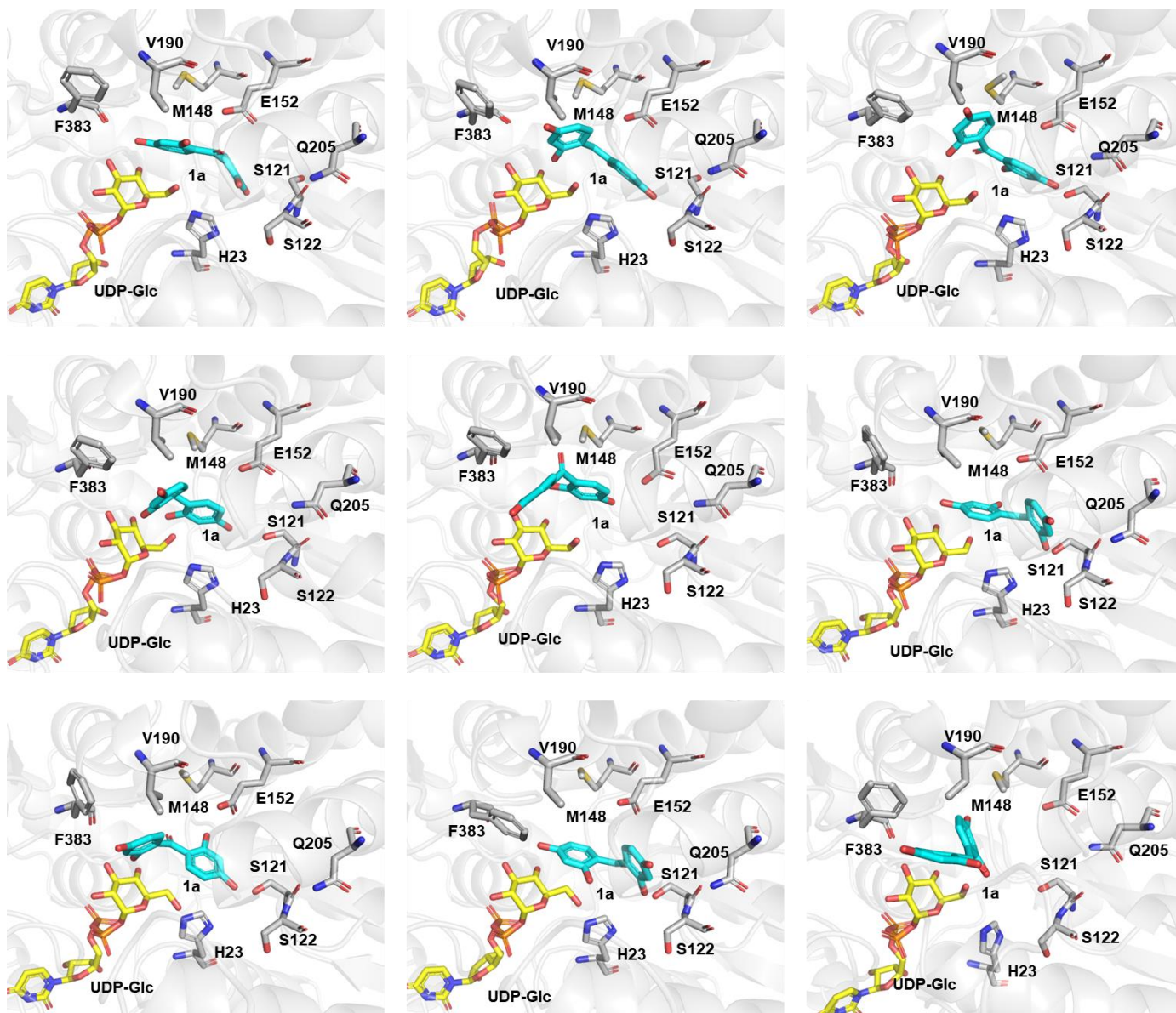
Supplementary Fig. 14 | MD simulations for substrate 1a in MiCGT-WT starting from different binding poses. The backbone atoms of the enzymes were selected for alignment and substrate **1a** was selected for RMSD calculations.



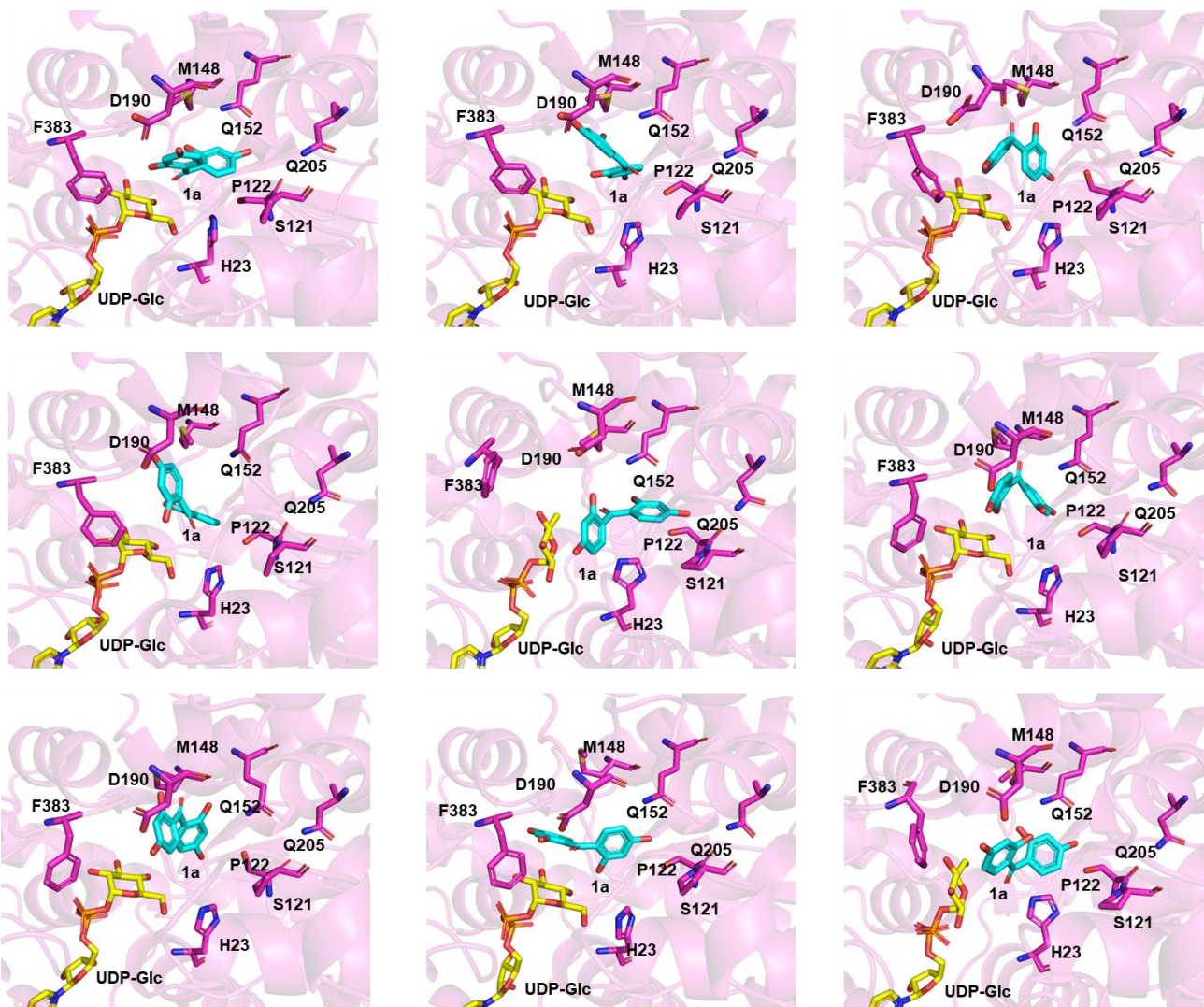
Supplementary Fig. 15 | MD simulations for substrate 1a in MiCGT-ATD starting from different binding poses. The backbone atoms of the enzymes were selected for alignment and substrate **1a** was selected for RMSD calculations.



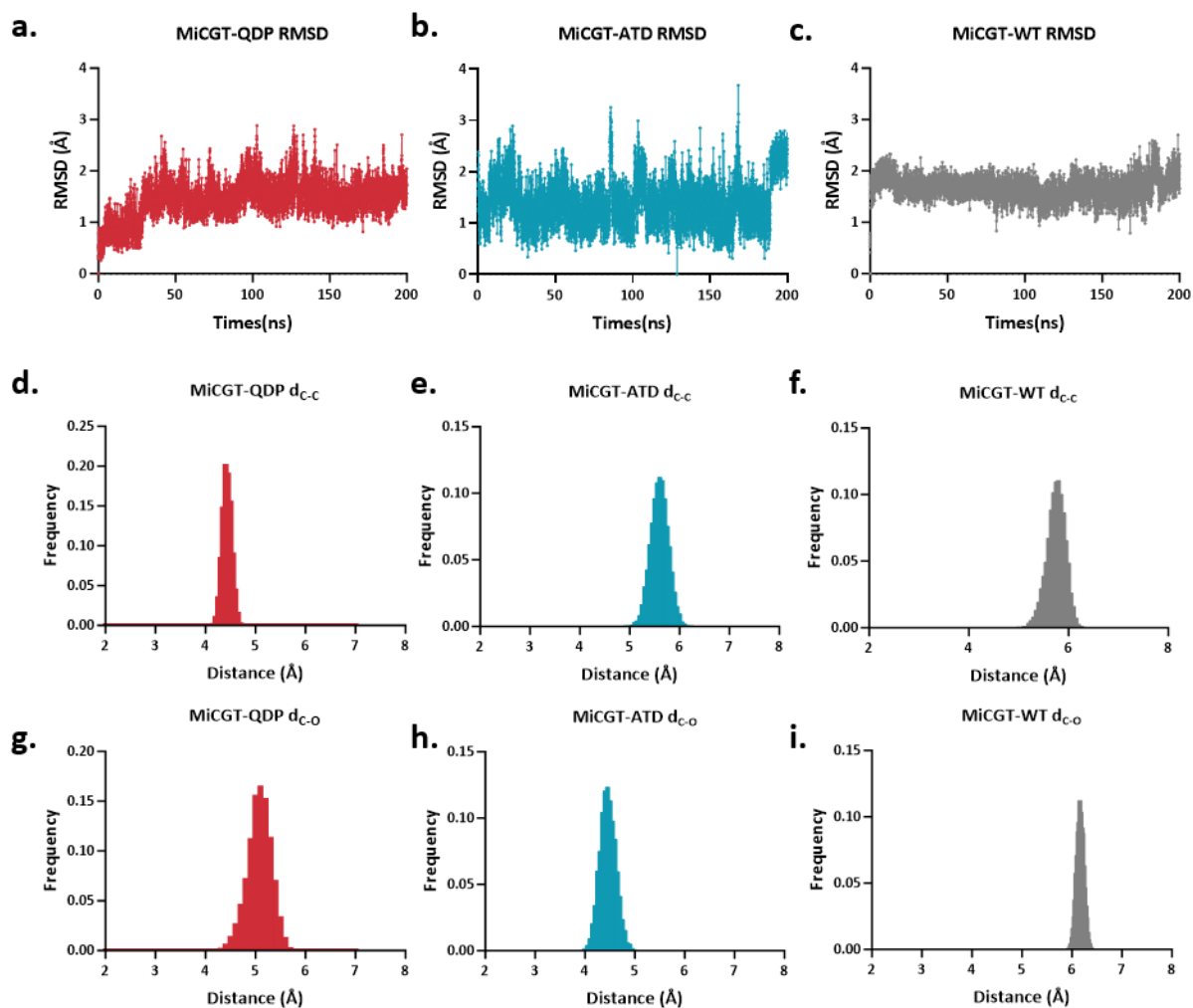
Supplementary Fig. 16 | Examples of different binding poses for substrate 1a in MiCGT-QDP.



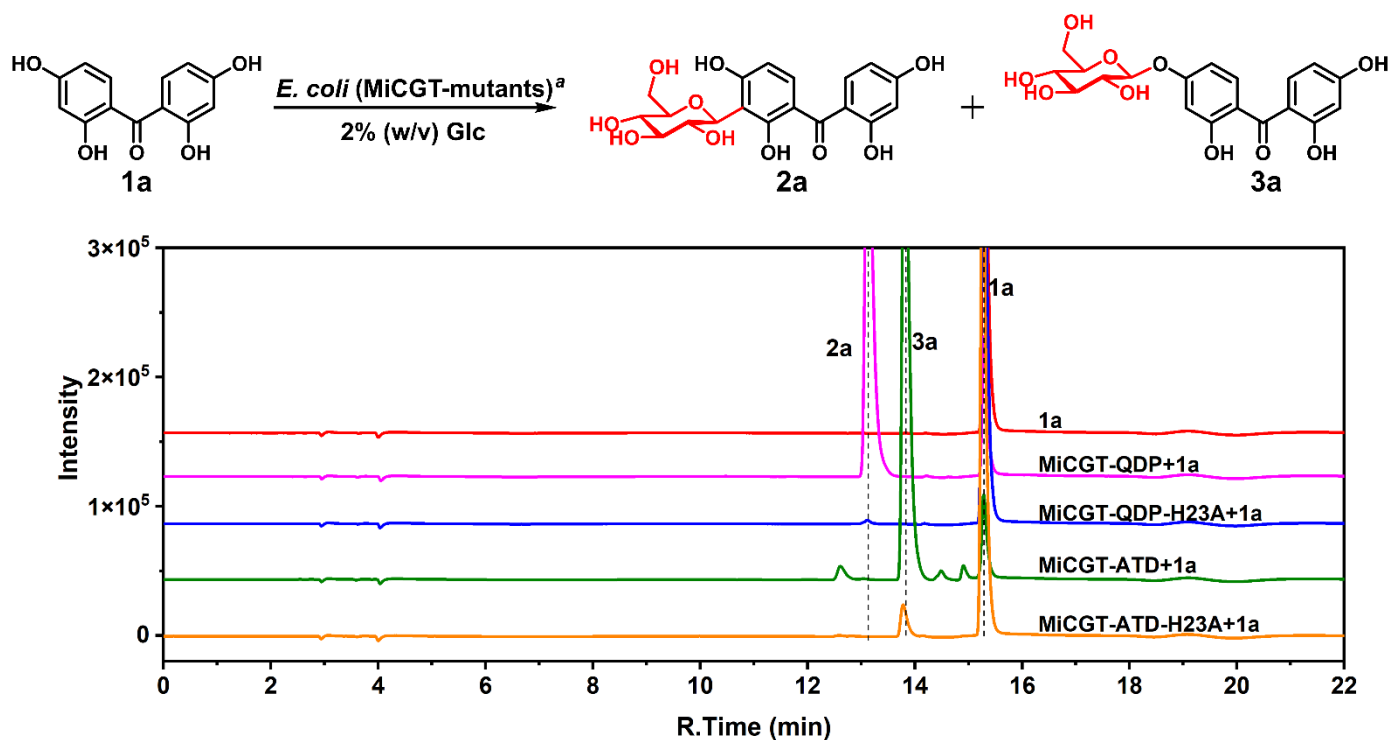
Supplementary Fig. 17 | Examples of different binding poses for substrate 1a in MiCGT-WT.



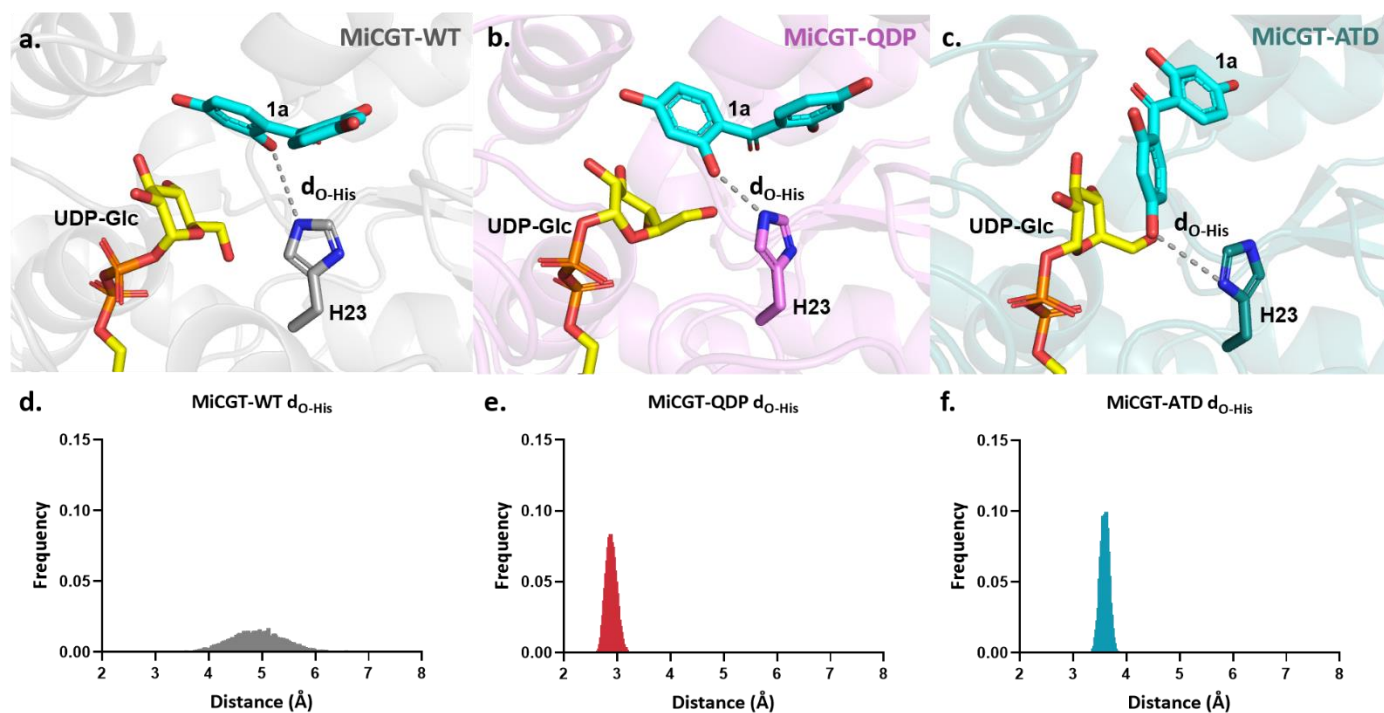
Supplementary Fig. 18 | Examples of different binding poses for substrate 1a in MiCGT-ATD.



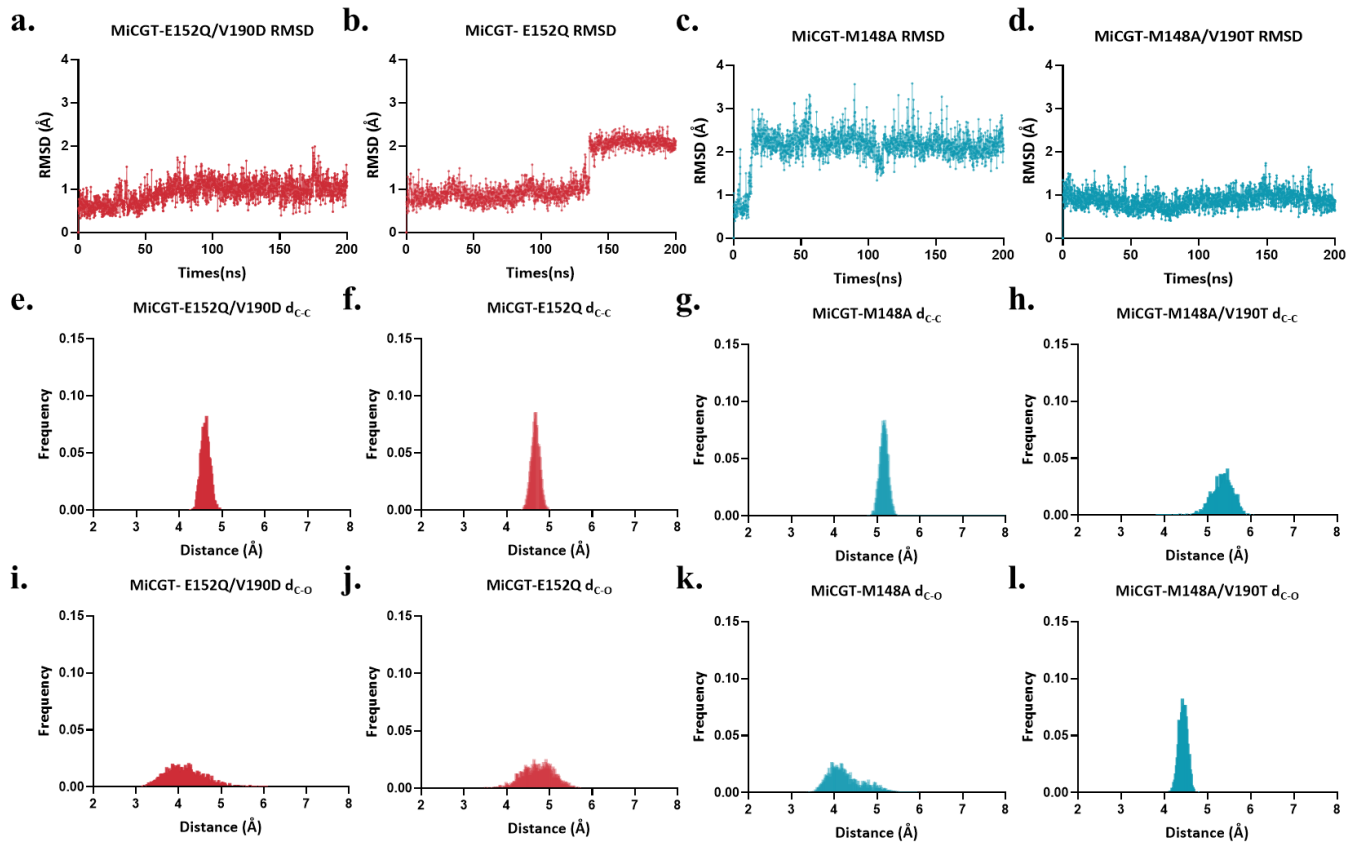
Supplementary Fig. 19 | MD simulations for MiCGT-QDP, MiCGT-ATD, and MiCGT-WT. RMSD of the substrate in MiCGT-QDP (a), MiCGT-ATD (b) and MiCGT-WT (c); Distance distribution for MiCGT-QDP (d, g), MiCGT-ATD (e, h), and MiCGT-WT (f, i), including the distance d_{C-O} and d_{C-C} .



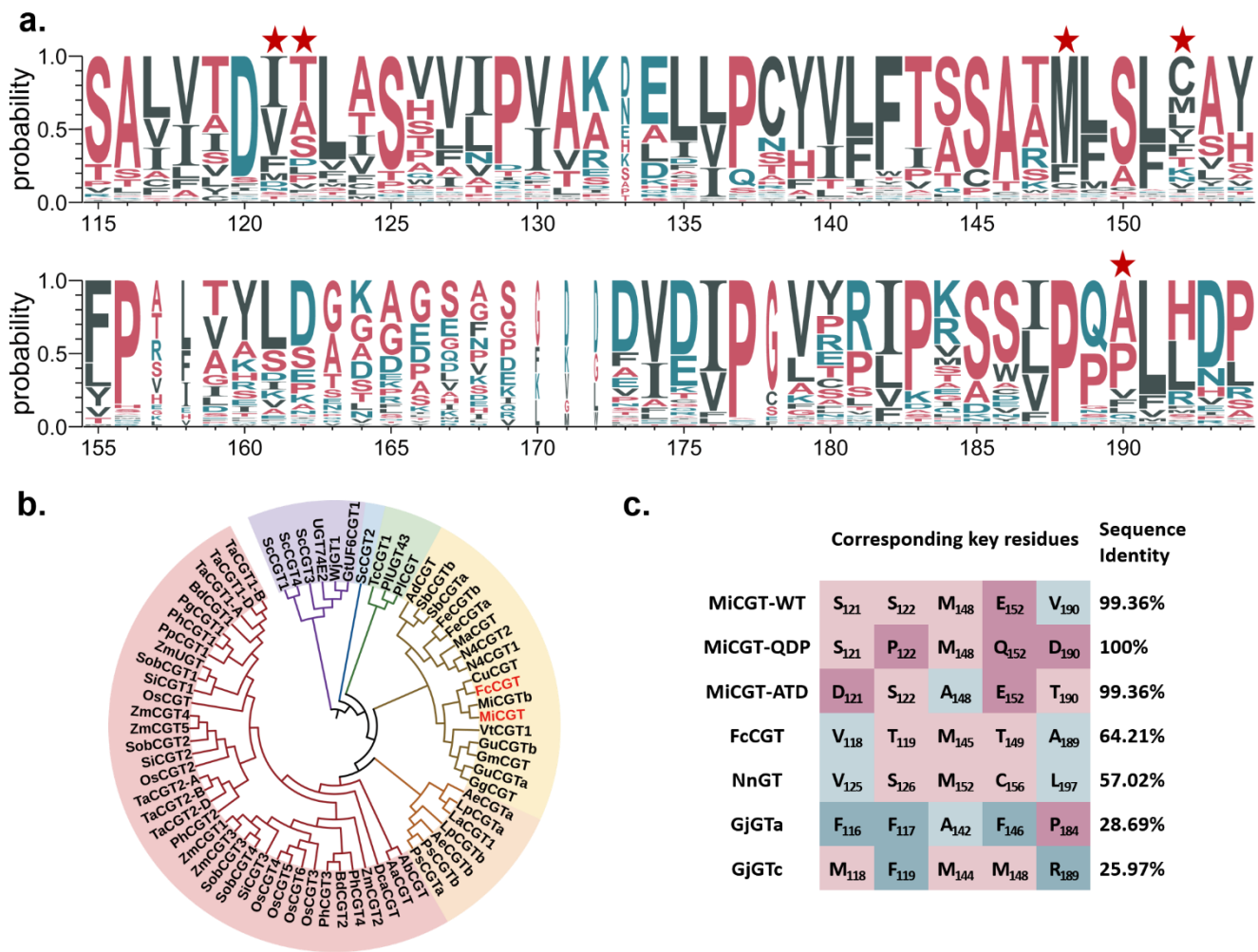
Supplementary Fig. 20 | Probing the role of His23 in the mechanism of glycosylation. ^aWhole cells were resuspended by 100 mM Kpi buffer (pH 8.0) containing 2% (w/v) *D*-glucose and 6% (v/v) DMSO, and the cell density was adjusted to an OD_{600 nm} = 40. 1a were added to the reaction mixture at the final concentration of 3 mM in a total volume of 0.5 mL, and the bioconversion reactions were kept shaking at 30 °C, 1200 rpm for 24 h.



Supplementary Fig. 21 | MD simulations for d_{O-His} in MiCGT-WT, MiCGT-QDP and MiCGT-ATD. Representative structures for MiCGT-WT (a), MiCGT-QDP (b) and MiCGT-ATD (c); Distance distribution for MiCGT-WT (d), MiCGT-QDP (e), and MiCGT-ATD (f), including the distance d_{O-His}



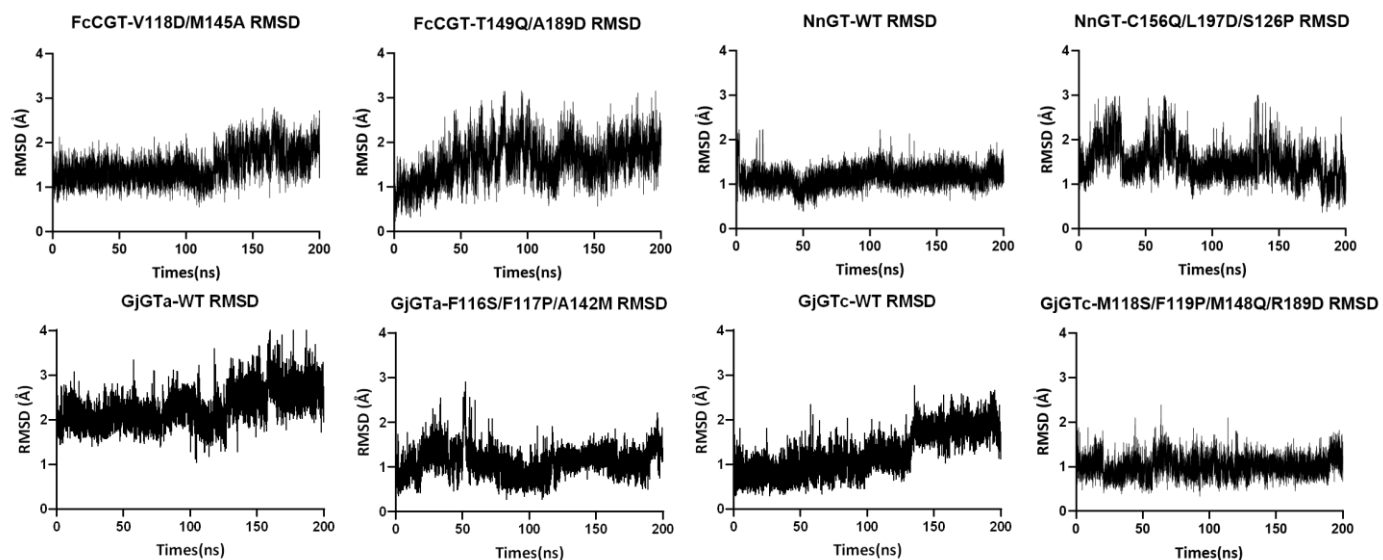
Supplementary Fig. 22 | MD simulations for MiCGT-E152Q, MiCGT-E152Q/V190D, MiCGT-M148A and MiCGT-M148A/V190T. RMSD of the substrate in MiCGT-E152Q/V190D (**a**), MiCGT-E152Q (**b**) MiCGT-M148A (**c**), and MiCGT-M148A/V190T (**d**); Distance distribution for MiCGT-E152Q/V190D (**e**, **i**), MiCGT-E152Q (**f**, **j**), MiCGT-M148A (**g**, **k**), and MiCGT-M148A/V190T (**h**, **l**), including the distance d_{C-O} and d_{C-C} .



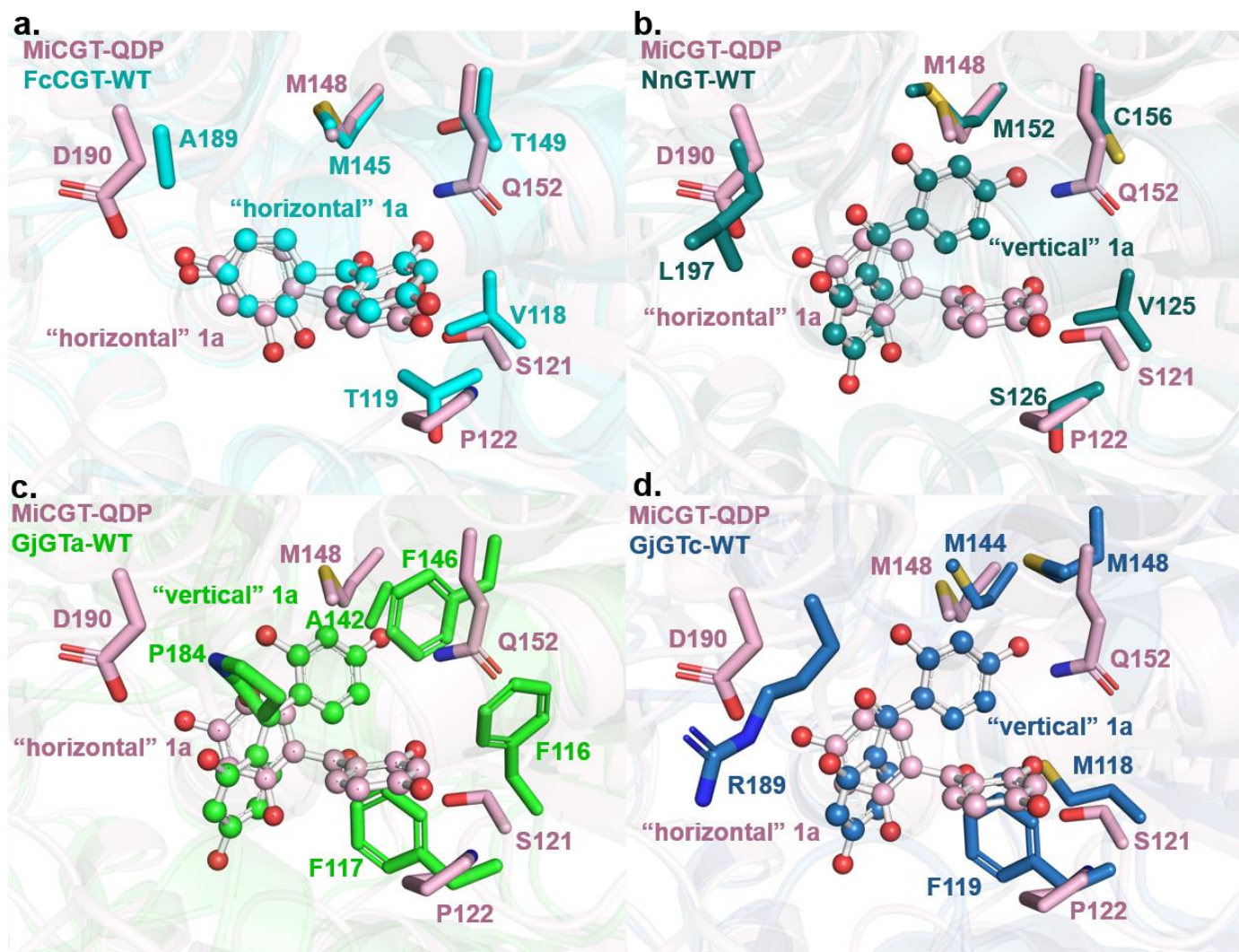
Supplementary Fig. 23 | Sequence conservation and evolutionary tree analysis of the related C-Glycosyltransferases. (a) Residue conservative analysis of 70 UGTs with C-glycosylation activity; **(b)** Phylogenetic analyses of MiCGT with other CGTs; **(c)** Alignment of the amino acid sequences of MiCGT with other plant UGTs.

MiCGT	..MSASDALNSCP.HVALLSSGMGHLTPCLRFAATLVQHH...CRVTIITNYPTVSVAESRAISLLLS	64
FcCGT	..MSDSGGFDSHP.HVALIPsAGMGHLTPFLRLAASLVQHH...CRVTLITTYPTVSLAETQHVSHFLSA	64
NnGT	MEMSSSDQLILPPHVALLPSSGMGHLVPFVRLAAALARN...CLVTFITHTPTVSLSESRLVSRLLSA	67
GjGTa	MENTLVLYPAPGIGHMISMLELAKLILRHYSNKFSRIHLINTGFRDMKSTYLDHISST	59
GjGTc	MKNQELVFIPSAVMSHLVSTVELAKLLIDRN...EHLsITVLIMKILPYEKNVGNITYPQT	57
Consensus	h a	
MiCGT	FPQITEKQFHLLPFDPSANTTDPFFLRWEAIRRSAHLLNPLSSIS...PPLSALVIDSSLVSSFPVPA	131
FcCGT	YPQVTEKRfHLLPFDPSANATDPFFLRWEAIRRSAHLLAPLIS...PPLSALITDVTLISAVLPVT	128
NnGT	FPHVTFLEFHLLPLDHSSTANSKDPFFLQFEAIRSAHLLSPLSSCS...DPPLSALITDVSLASAFISIT	135
GjGTa	NPSIVVHQFPFIQADLSSSLs..PPAIGFKFIRKNAPNVHHAHQEIS..KTSSIRALIIDFFCTsAMPYS	125
GjGTc	EASDSRIRfLELKKDESASQTVSPILFIYQFVEDHKNSARDVLTEISNSASSDLVGVVVDMFCsSMIDVA	127
Consensus	f d p l	
MiCGT	ANLDLPsYVLFSSSTRMCSLEETFFPAFVASK.TNFDS.IQLDD.VIEIPG.FSPVFPVSSVPPVFLNLNLH	197
FcCGT	INLHLENYVLFASARMFSLTASFPaIVASKSTSSGS.VEFDDDFIEIPG.LPFIPLSSVPPAVMDSKSL	196
NnGT	DELRLPNYILFTSSAWMLSLCLNFPTFVNTSTNLSSGSATASDDIEIPG.VPEVPKSWLPPLLLDLSNL	204
GjGTa	NNLGIPVYYFFTSgAAVALFLYFPTIHKQTSSESFKD...LVQTKFDVPG.LPFIIPATQMPPEVLDRDDP	191
GjGTc	NEEGVPSYVFYTSgAAMLGLMLHLQSLRDDFGEDVTN.YENSKVELAVPTYINFPVKVLPSRLFDMEG.	195
Consensus	p y t l p p p	
MiCGT	FTTMLIQNGQSFRKANGILINTFEALEGGILPGINDKRAADGLP..PYCSVGPLLP.....CKFEKTE	258
FcCGT	FATsFLENGNSFVKsNGVLINSFDALADTLVALNGRRVVAGLP..PVYAVGPLLP.....CEFEKRD	257
NnGT	FTTQFMANGQELIKsNGILINTFGNAEQATVAALNEGKVNGLP..PVTKIGPLAP.....CEFEKGS	265
GjGTa	AYDDILYYSVHLPKSSGIIVNTFDELEPIALKAITDGLCVPDAPTPPLYNIGPLIADADSRPAIDGDKGI	261
GjGTc	GGNMFLNLTKRFRETKGIVINSFFELESHAIQALSND...KTIP..PVYPVGPLLD.....LKESNGQ	253
Consensus	g n f e p p gp	
MiCGT	CSAP...VKWLDQPEGSVVYVSFGSRFALSSEQIKELGDLIRSGCRFLWVVKC.....KKVDQEDE	318
FcCGT	DPSTSLILKWLDQPEGSVVYVSFGSRLALSMEQTKELGNGLSSGCRFLWVVKG.....KTVDKEDE	320
NnGT	S.....VEWLDGQFAGSVLYVSFGSRTAMSREQIRELGDLVRSGCRFLWVVKD.....KKVDREDE	322
GjGTa	DLDQSDCFsWLDQPDQCVVFLCFGSRGTFSVEQIKEIAKGLERSGKRFLWVVKKPLRNNKSKQVEGSGG	331
GjGTc	NQETEMITKWLDIQDSSVVFLCFGSRGCFDGGQVKEIACALESGYRFLWSLRRP..PPKGKFESPGDY	321
Consensus	wld qp v fgsr q e l sg rflw k	
MiCGT	ESIDELLGRDVLKIKKYGFVIKNWVNQOEILDHRAVGGFVTHGGWNSsMEAVWHGVPMLVWPQFGDQKI	388
FcCGT	ESLKNVLGHMEKIKDQGLVVKNWVDQDKVLSHRAVGGFVSHGGWNSLVEAARHGVPVLVWPQFGDQKI	390
NnGT	EELGGIVGQELMERMKDNGLVVKNWVDQGVVLAPAVGGFVLSHCGWNSVTEAAWNGVPVLAWPQHGDQSI	392
GjGTa	FEIDSILPERFLEKTKIGLVVKSWIPLQVLRLHPAVGGFVTHCGWNSLVEAVVAGVPLVAVPLHAEQHVV	401
GjGTc	ENLEEALPEGFLQRTAEVGVKIG.WAPQAAILSHPAVGCFVSHCGWNSLVEsVWFGVPMATWPLYAEQQV	390
Consensus	g v w q l h avg f h gwns e gvp wp q	
MiCGT	NAEVIERSGLGMWVKRWG.....WGTQQLVKGEEIGERIKDLMGNNP...LRVRAKTLREEARKAIEV	448
FcCGT	NAEAVESAGLGMWVRSWG.....WGTQLRAKGDEIGLKIKDLMANDF...LREQAKRIEEERKAIGV	450
NnGT	NAQVMEKSGIGMWVKS WG.....WGRTEVVKGEEIGQKIGEMMGNDR...LKAQAANIREESRAVEV	452
GjGTa	NMAALVQD.MKMAIPVE.....QGDDGIVRGEEVEKRVRELMDSERGRELRKLSQKTRDIAAESGVH	462
GjGTc	NAFLLLKD.LGMaVDIKMDFKSTSFEPSTEIVAADLIEKATKHLMDPEN..EIRKKVKEKKEKSLSLSE	457
Consensus	m m	
MiCGT	GGSSSEKTLKELIENWKTSRK	469
FcCGT	GGSSERTFKELIDKWCNNNT	471
NnGT	GGGSYKELEGIIEKWKISRI.	472
GjGTa	LGSSSTALASLIHVVFNG...	480
GjGTc	GCPSASLGRFLDALIDNIP.	477
Consensus	g s	

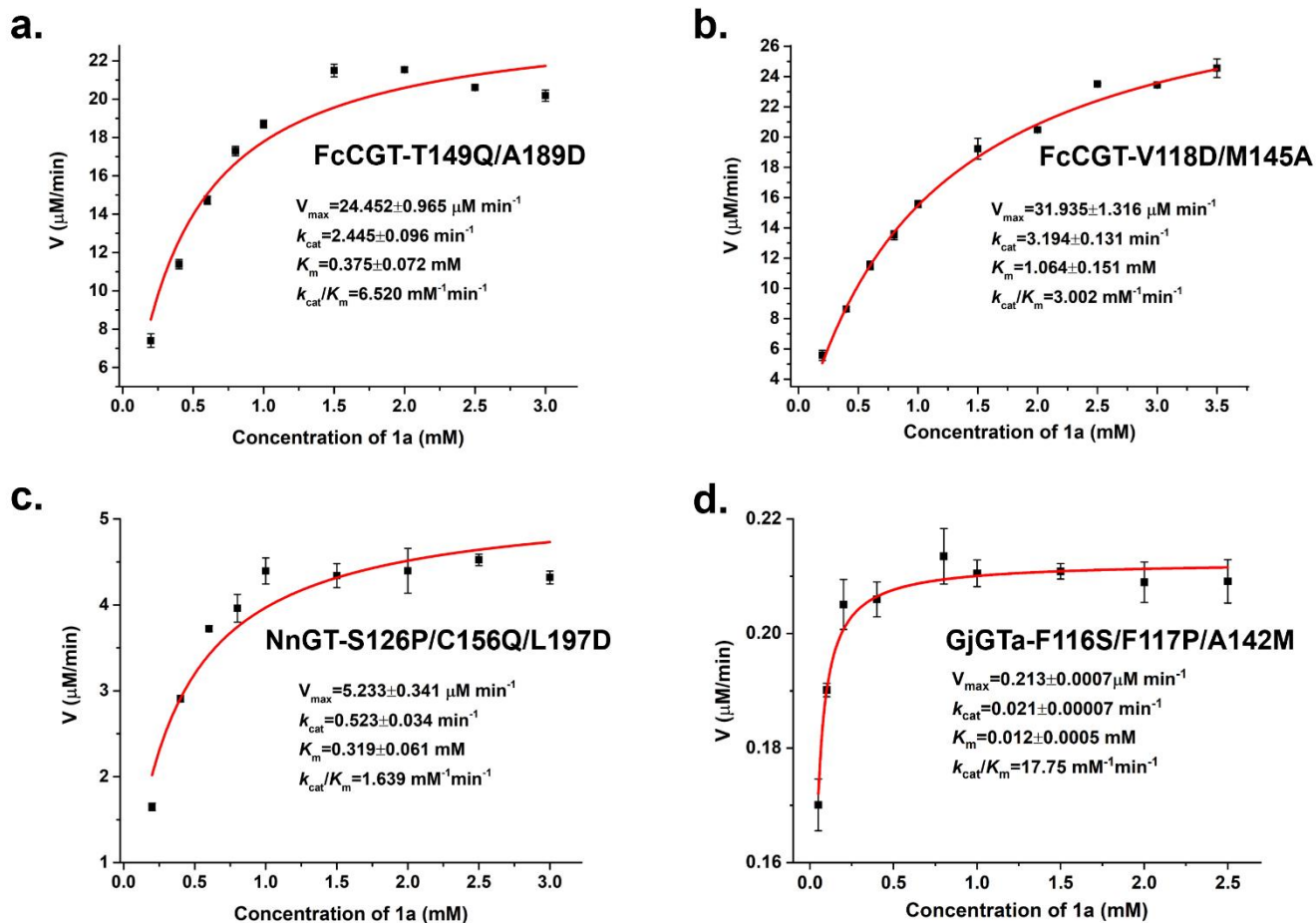
Supplementary Fig. 24 | Alignment of the amino acid sequences of MiCGT. Alignment of the amino acid sequence of MiCGT with FcCGT, NnGT, GjGTa and GjGTc. The sequences were aligned using DNAMAN 9.



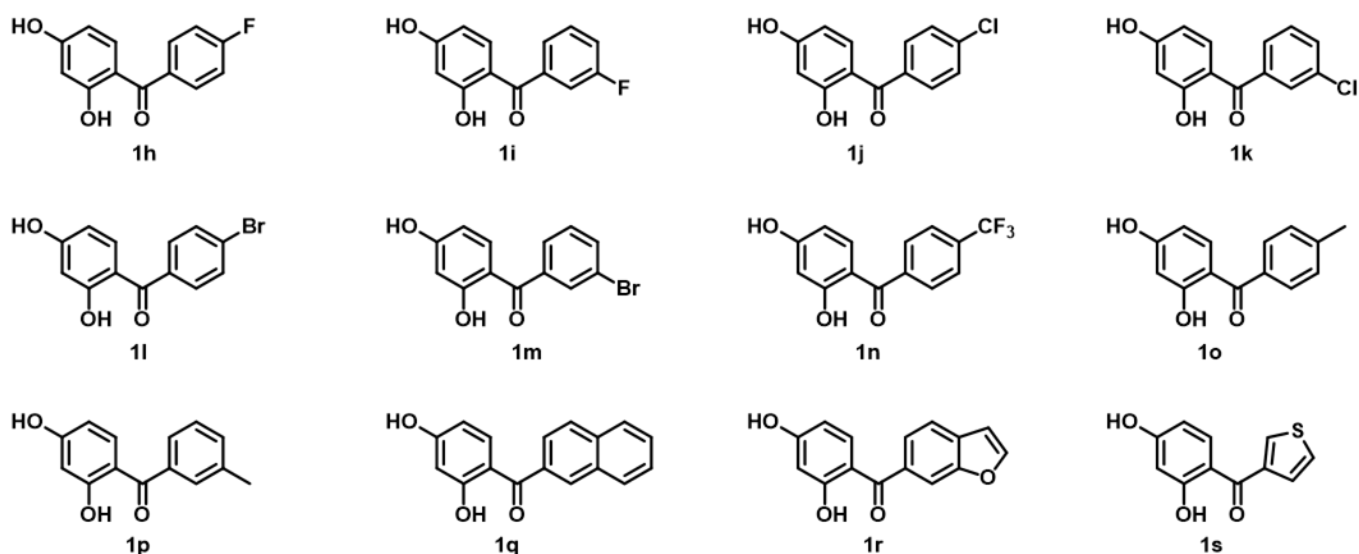
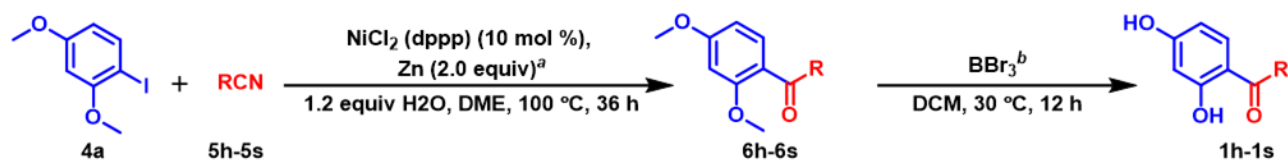
Supplementary Fig. 25 | MD simulations for new UGTs, including FcCGT, NnGT, GjGTa and GjGTc. The backbone atoms of the enzymes were selected for alignment and substrate **1a** was selected for RMSD calculations.



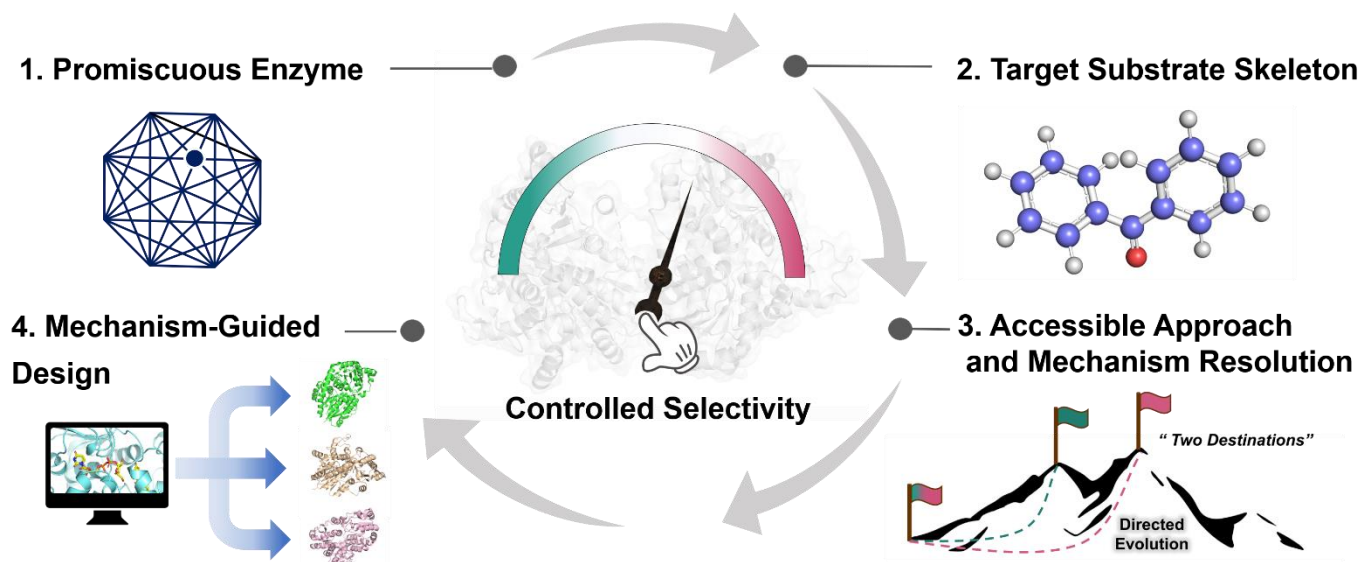
Supplementary Fig. 26 | Alignment of the structure of MiCGT with other plant GTs. Utilizing the AlphaFold2 algorithm, we predicted the respective protein structures, followed by molecular docking procedures similar to those used for MiCGT. **(a)** Comparison of key amino acid residues and glycosyl receptor substrate binding modes in the active pocket between MiCGT-QDP and FcCGT-WT. **1a** in FcCGT-WT exhibits a "horizontal" conformation. **(b)** Comparison of key amino acid residues and glycosyl receptor substrate binding modes in the active pocket between MiCGT-QDP and NnGT-WT. **1a** in NnGT-WT exhibits a "vertical" conformation. **(c)** Comparison of key amino acid residues and glycosyl receptor substrate binding modes in the active pocket between MiCGT-QDP and GjGTa-WT. **1a** in GjGTa-WT exhibits a "vertical" conformation. **(d)** Comparison of key amino acid residues and glycosyl receptor substrate binding modes in the active pocket between MiCGT-QDP and GjGTc-WT. **1a** in GjGTc-WT exhibits a "vertical" conformation.



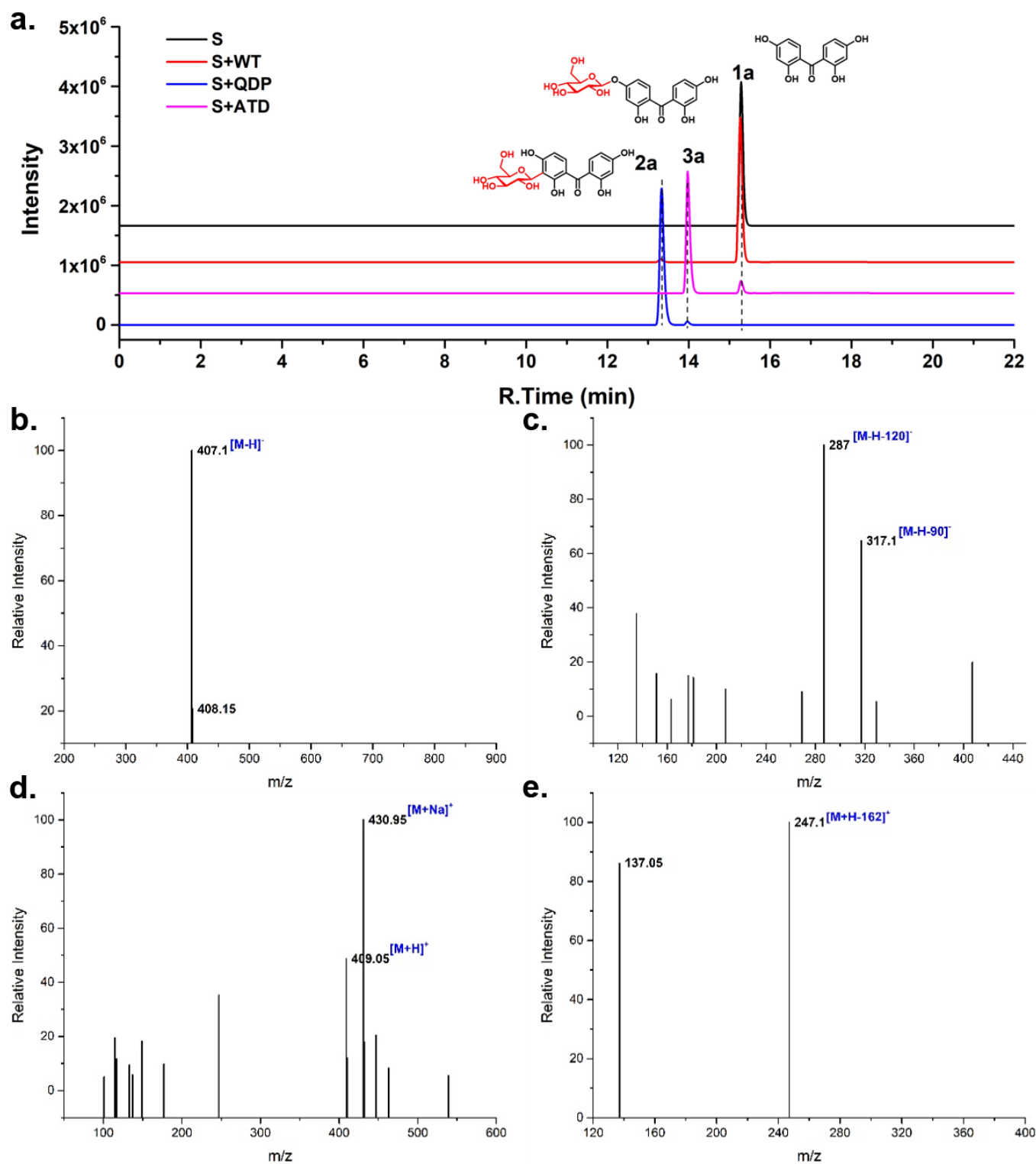
Supplementary Fig. 27 | Determination of kinetic parameters for newly designed UGTs. The kinetic parameters were detected with 1a as an acceptor and UDP-Glc as a donor at 30 °C, 1200 rpm and pH 8.0.



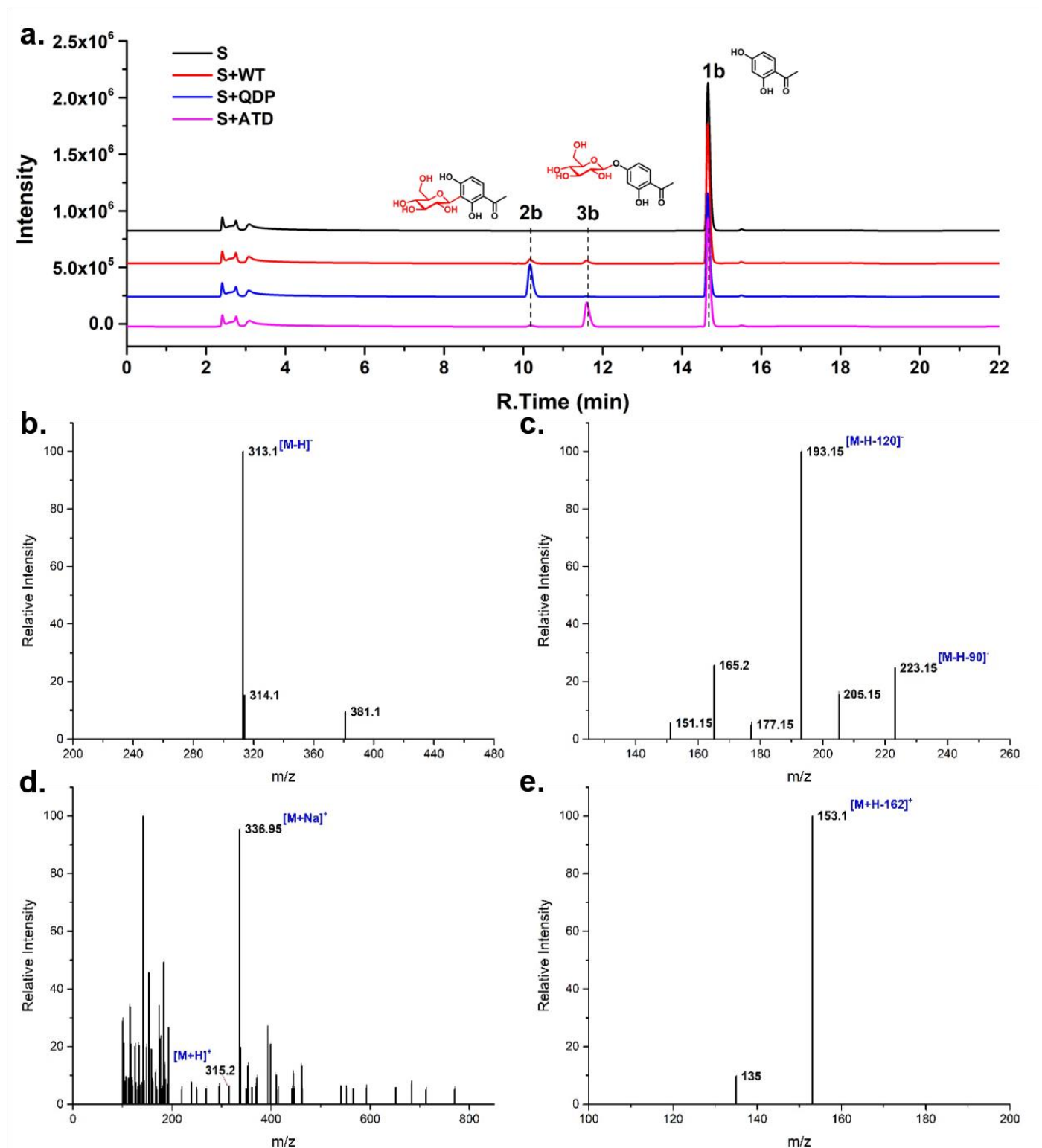
Supplementary Fig. 28 | Preparation of substrates for substrate-scope exploration. General procedure: ^aReactions were carried out using 1 mmol (1.0 equiv) aryl iodide (**4a**), 10 mol % NiCl₂(dppp), 2.0 equiv of zinc powder, and 1.2 equiv of H₂O in 0.3 mL of dimethoxyethane at 100 °C for 36 h.⁶ ^bTo a dry 5.00 mL thick walled vial equipped with a stir bar and septum was added anisole followed by the addition of dry dichloromethane (1 mL per 1 mL of BBr₃ solution). The vial was allowed to purge under nitrogen for approximately 5 min, after which BBr₃ was added slowly through the septum with stirring. The reaction was left to stir overnight before the contents were poured into ca. 1 mL of deionized H₂O.⁷



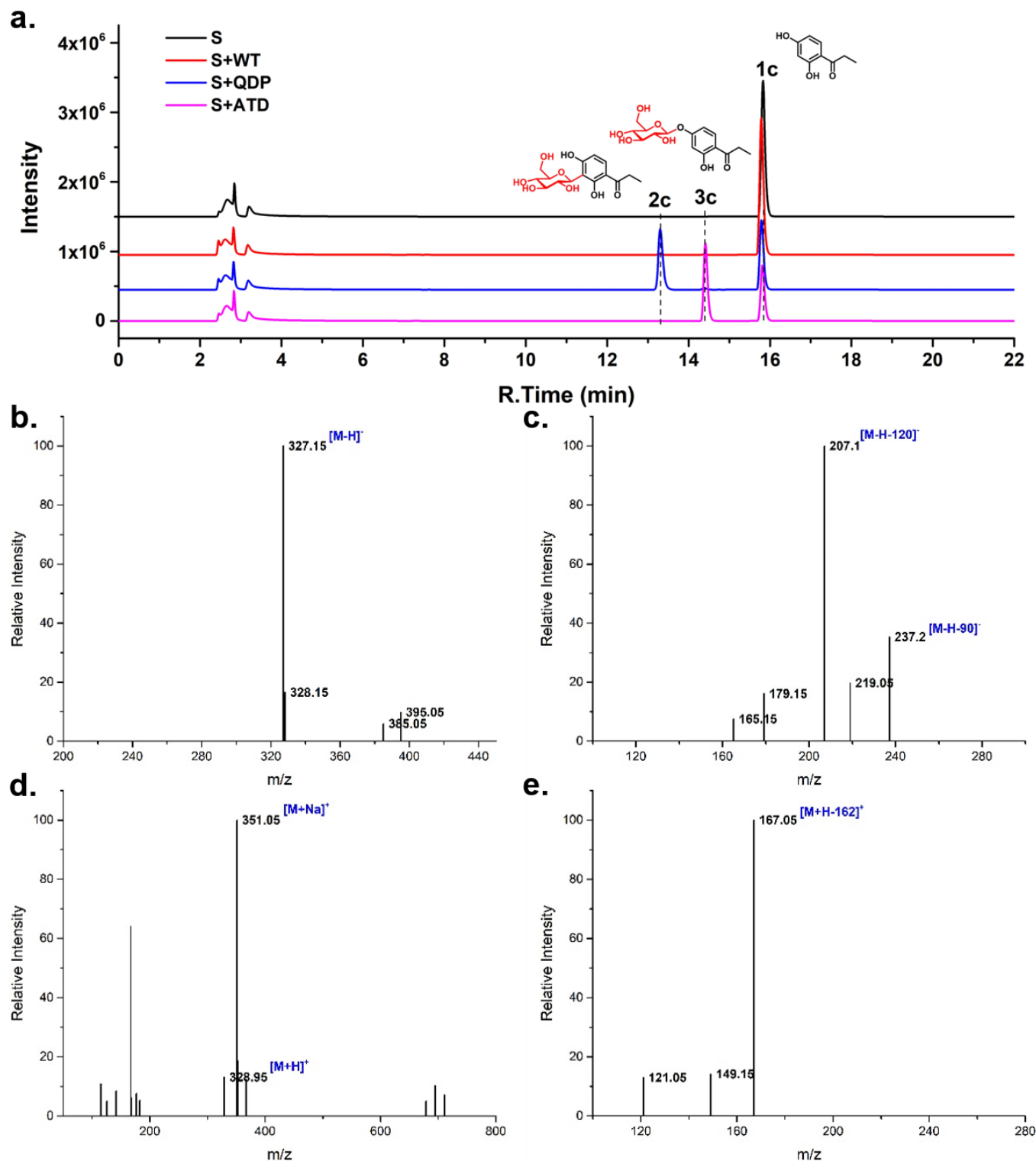
Supplementary Fig. 29 | Versatile strategy to customize UGTs by tuning selectivity.



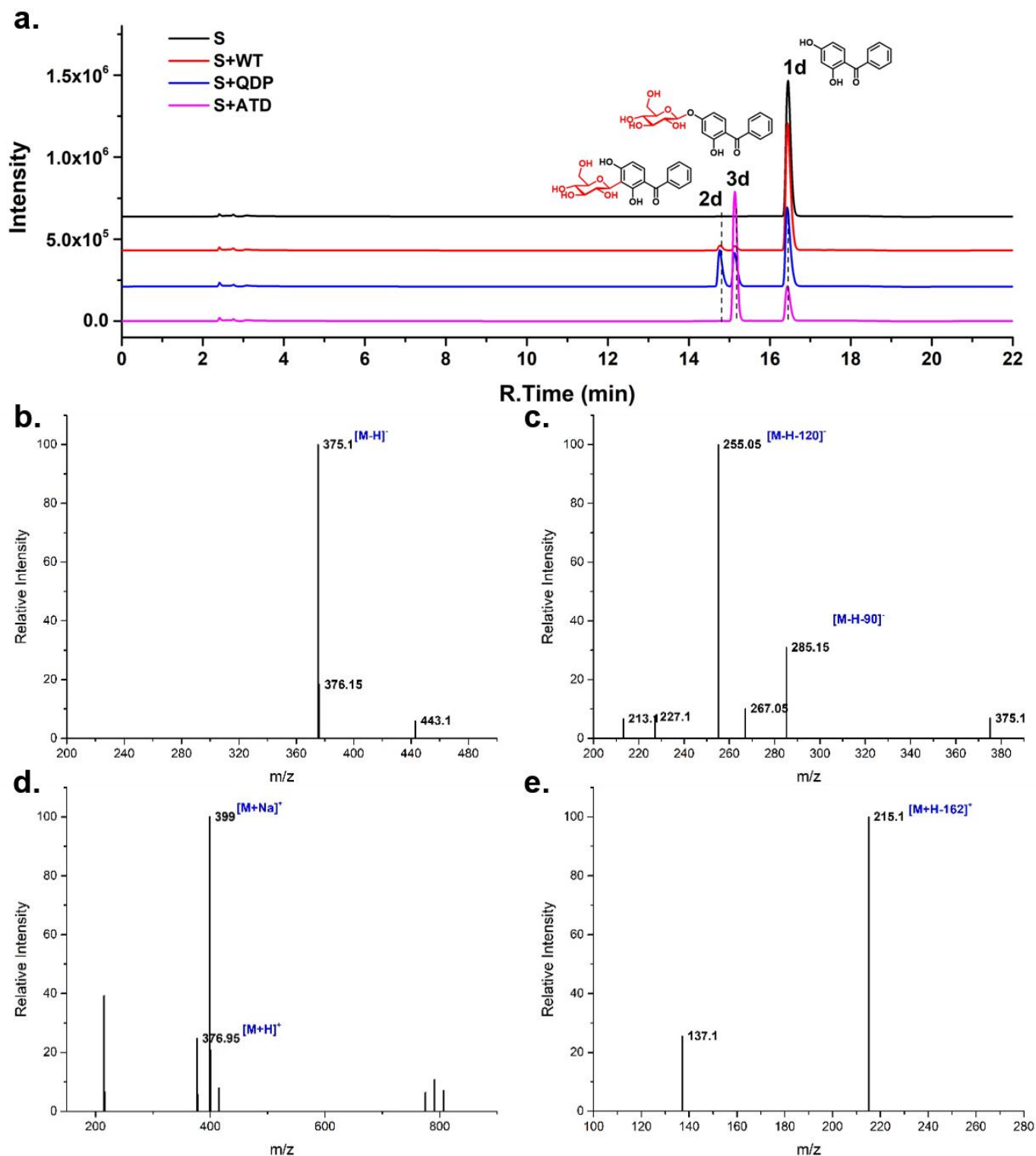
Supplementary Fig. 30 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1a by purified MiCGT variants. (a) HPLC chromatogram of **1a** and product **2a** and **3a**; (b) Typical negative ion MS spectra for **2a**; (c) Typical negative ion MS/MS for **2a**; (d) Typical positive ion MS spectra for **3a**; (e) Typical positive ion MS/MS for **3a**; **1a** MW: 246; **2a** and **3a** MW: 408.



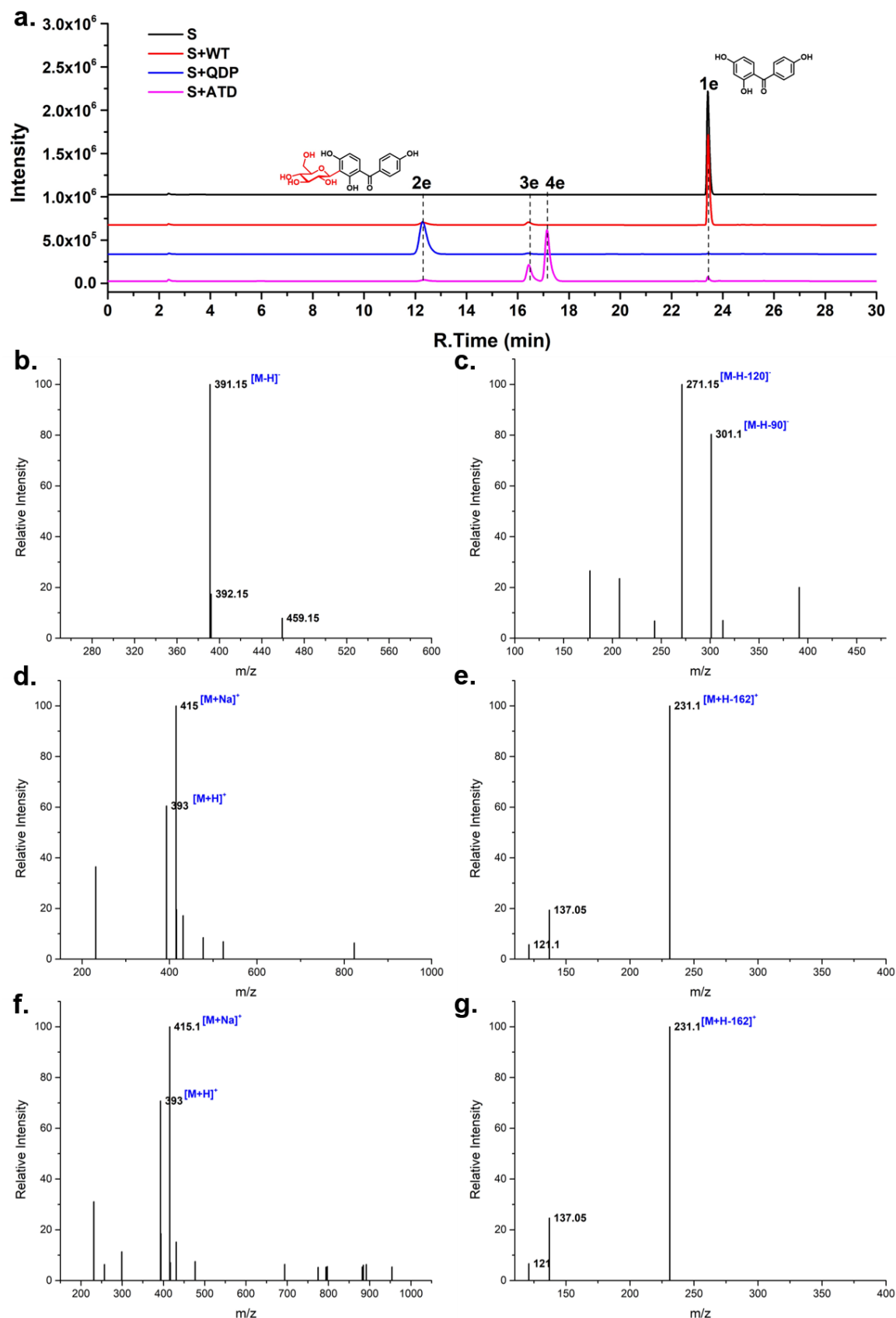
Supplementary Fig. 31 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1b by purified MiCGT variantts. (a) HPLC chromatogram of 1b and product 2b and 3b; (b) Typical negative ion MS spectra for 2b; (c) Typical negative ion MS/MS for 2b; (d) Typical positive ion MS spectra for 3b; (e) Typical positive ion MS/MS for 3b; 1b MW: 152; 2b and 3b MW: 314.



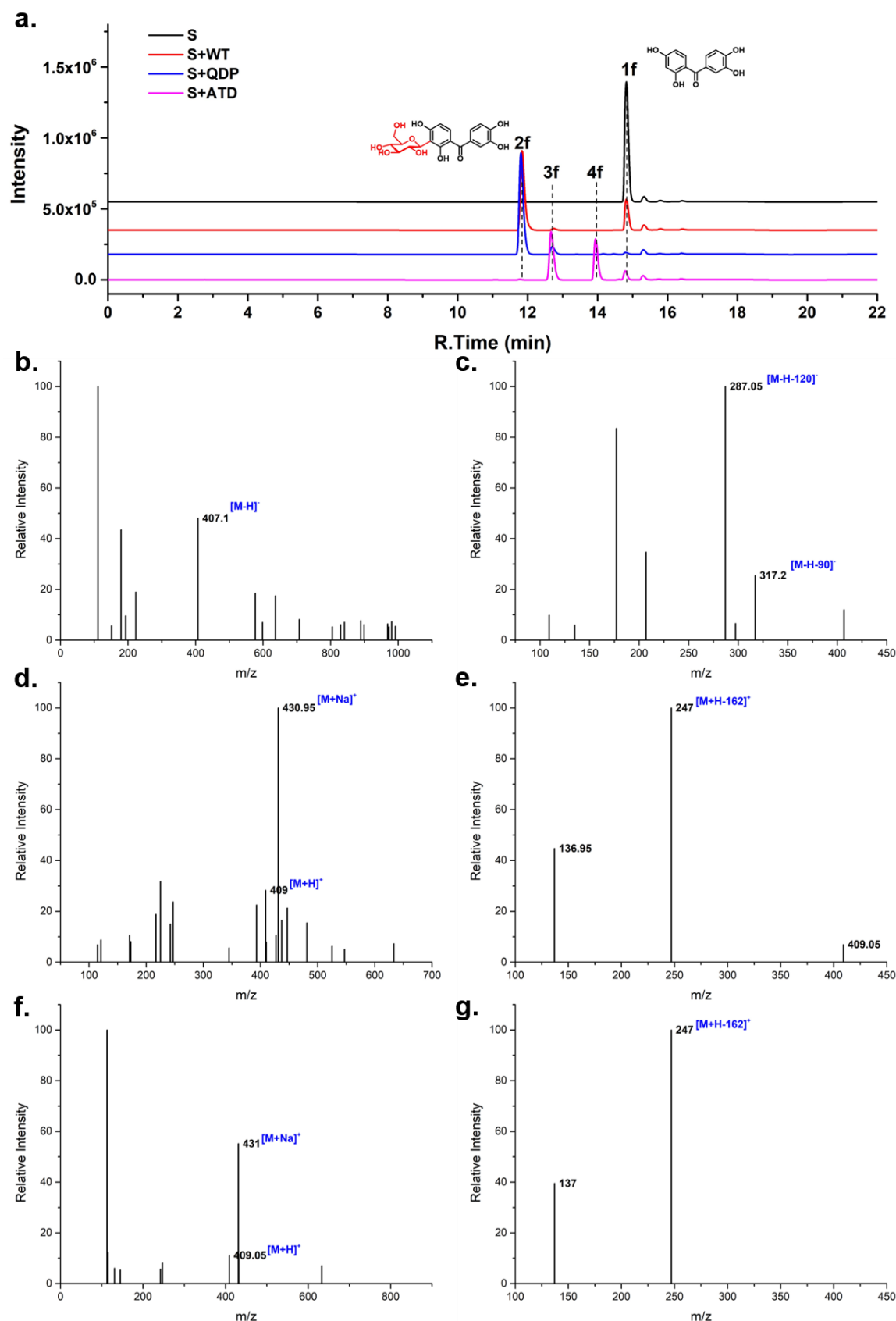
Supplementary Fig. 32 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1c by purified MiCGT variants. (a) HPLC chromatogram of **1c** and product **2c** and **3c**; (b) Typical negative ion MS spectra for **2c**; (c) Typical negative ion MS/MS for **2c**; (d) Typical positive ion MS spectra for **3c**; (e) Typical positive ion MS/MS for **3c**; **1c** MW: 166; **2c** and **3c** MW: 328.



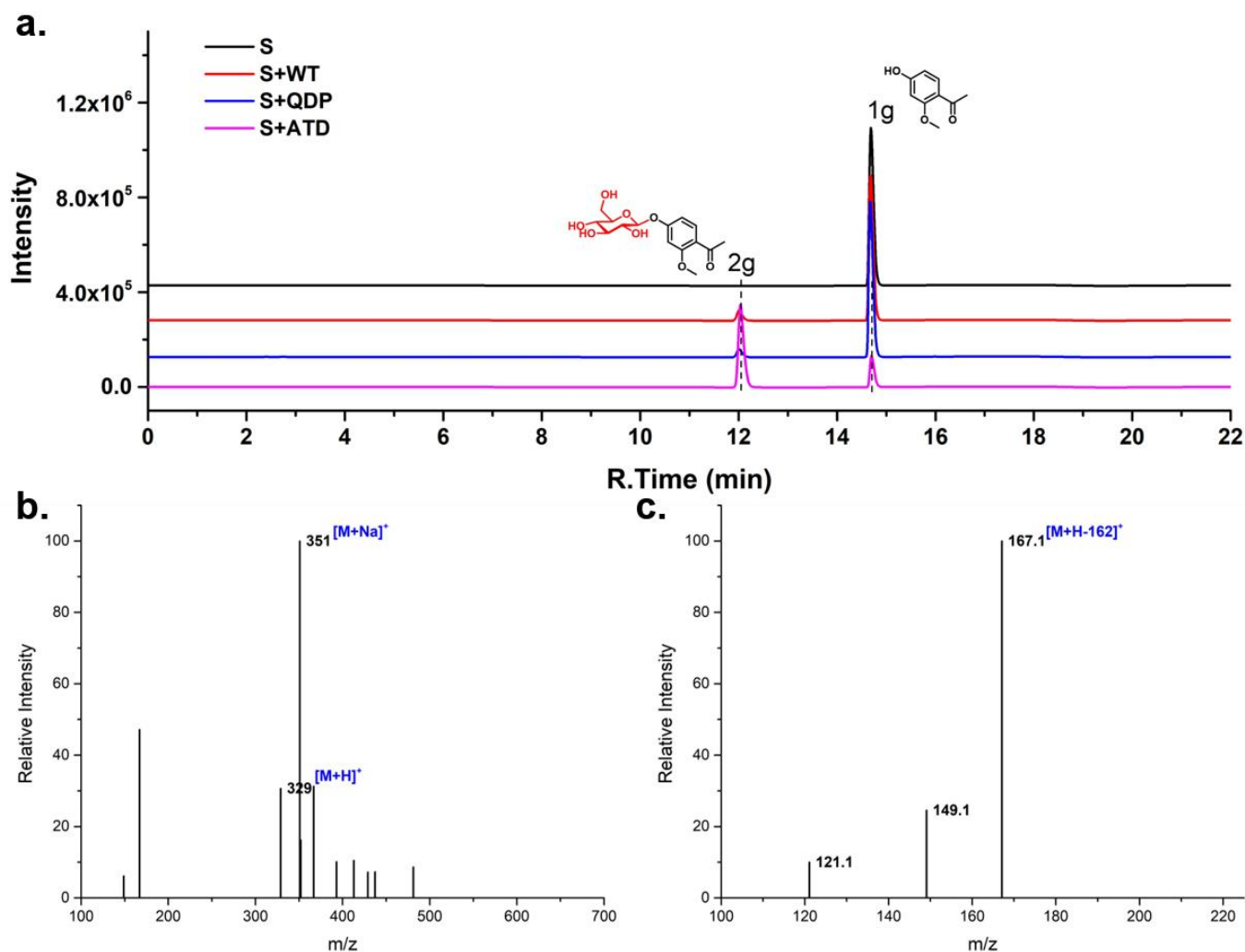
Supplementary Fig. 33 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1d by purified MiCGT variants. (a) HPLC chromatogram of **1d** and product **2d** and **3d**; (b) Typical negative ion MS spectra for **2d**; (c) Typical negative ion MS/MS for **2d**; (d) Typical positive ion MS spectra for **3d**; (e) Typical positive ion MS/MS for **3d**; **1d** MW: 214; **2d** and **3d** MW: 376.



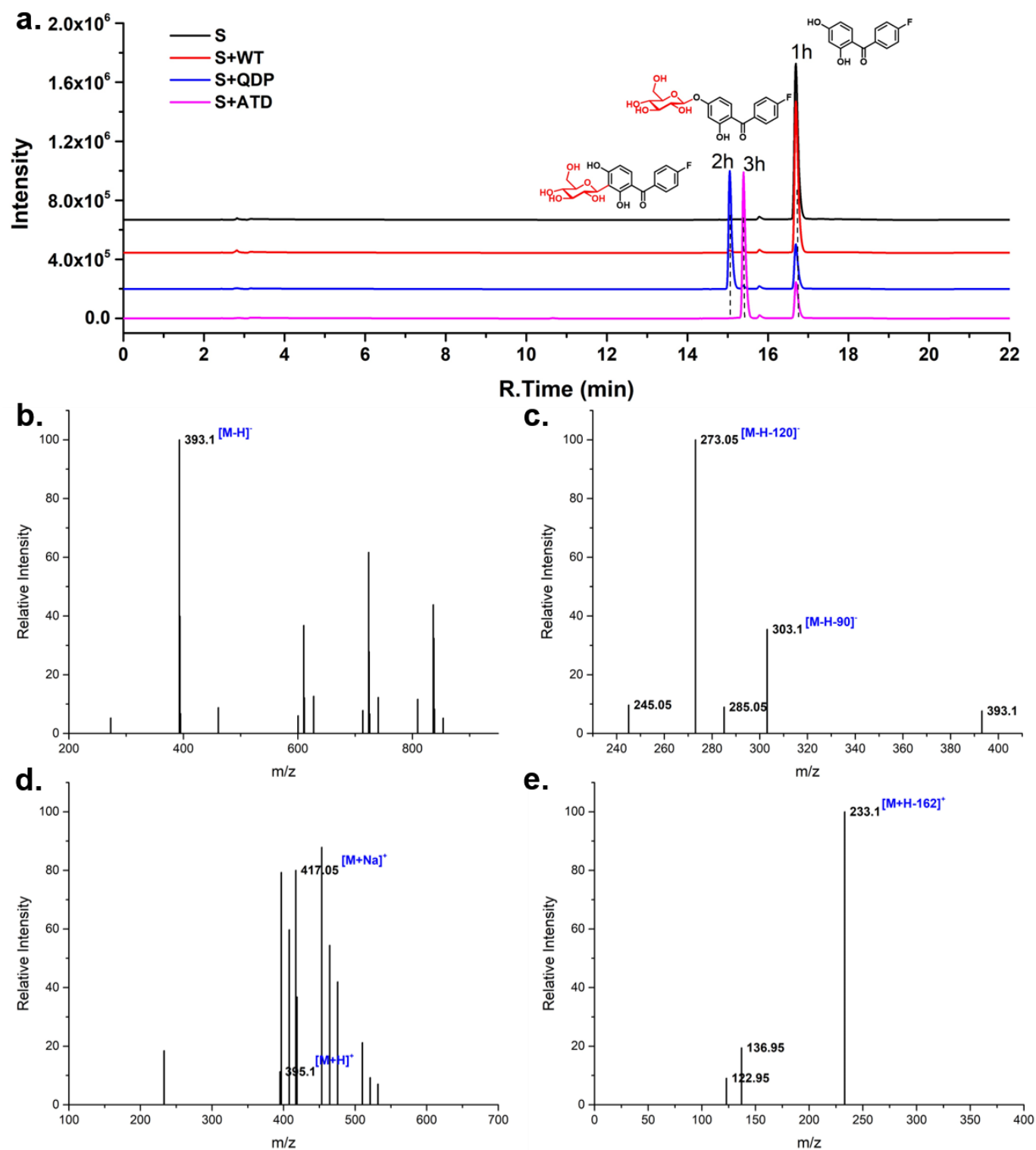
Supplementary Fig. 34 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1e by purified MiCGT variants. (a) HPLC chromatogram of **1e** and product **2e**, **3e** and **4e**; (b) Typical negative ion MS spectra for **2e**; (c) Typical negative ion MS/MS for **2e**; (d) Typical positive ion MS spectra for **3e**; (e) Typical positive ion MS/MS for **3e**. (d) Typical positive ion MS spectra for **4e**; (e) Typical positive ion MS/MS for **4e**; **1e** MW: 230; **2e**, **3e** and **4e** MW: 392.



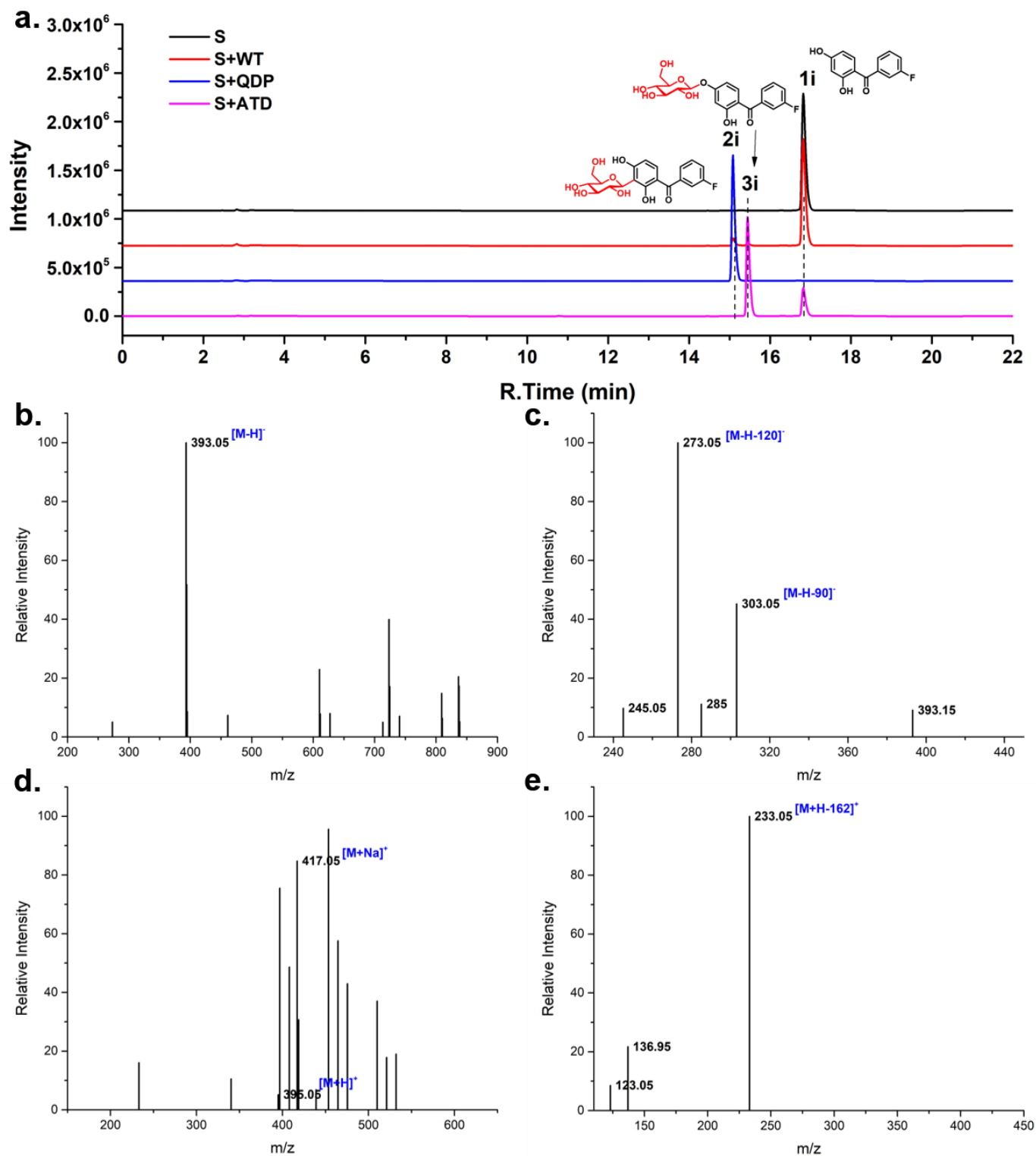
Supplementary Fig. 35 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1f by purified MiCGT variants. (a) HPLC chromatogram of 1f and product 2f, 3f and 4f; (b) Typical negative ion MS spectra for 2f; (c) Typical negative ion MS/MS for 2f; (d) Typical positive ion MS spectra for 3f; (e) Typical positive ion MS/MS for 3f; (f) Typical positive ion MS spectra for 4f; (g) Typical positive ion MS/MS for 4f; 1f MW: 246; 2f, 3f and 4f MW: 408.



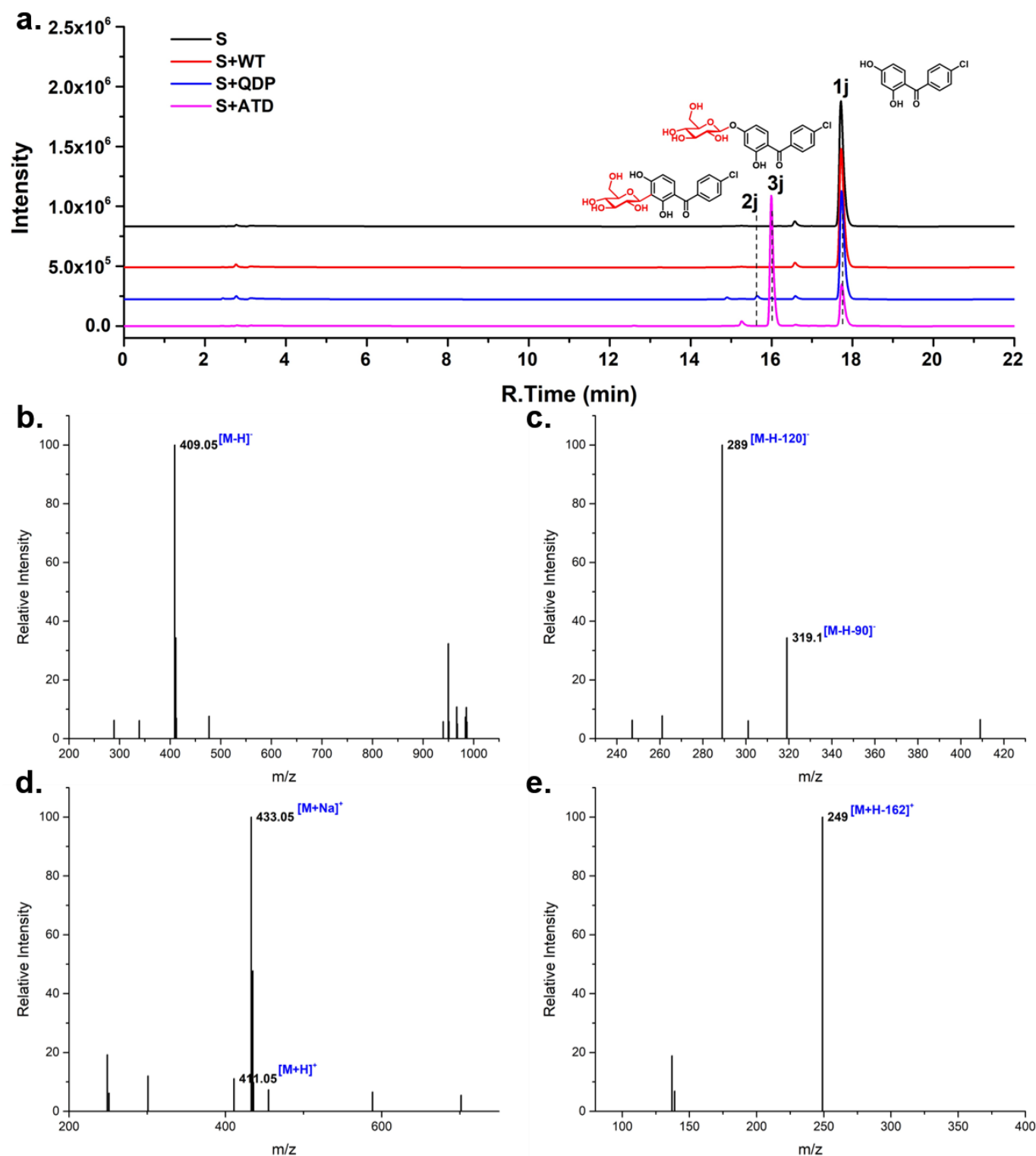
Supplementary Fig. 36 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1g by purified MiCGT variants. (a) HPLC chromatogram of 1g and enzyme product 2g; (b) Typical positive ion MS spectra for 2g; (c) Typical positive ion MS/MS for 2g; 1g MW: 166; 2g MW: 328.



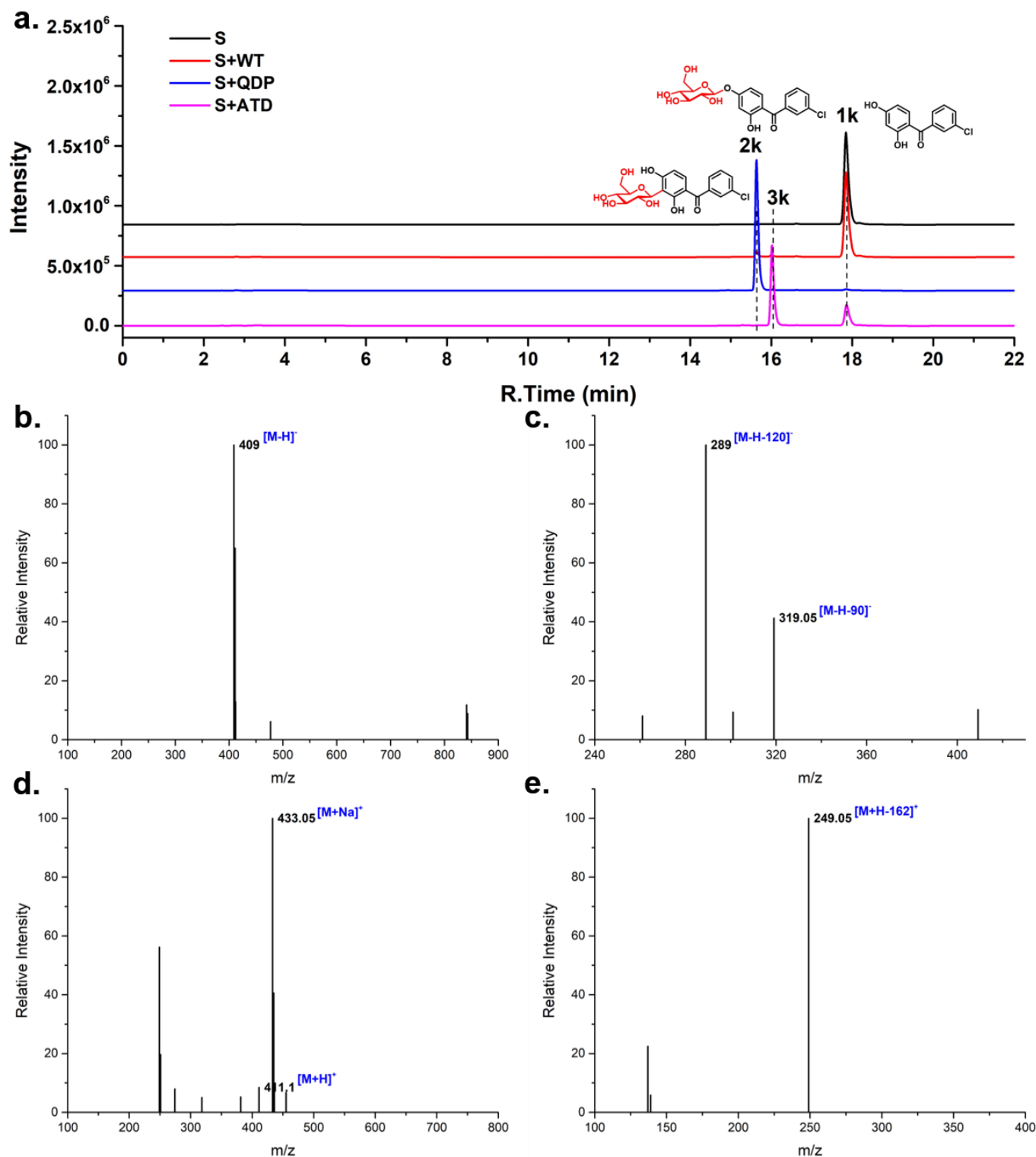
Supplementary Fig. 37 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1h by purified MiCGT variants. (a) HPLC chromatogram of **1h** and product **2h** and **3h**; (b) Typical negative ion MS spectra for **2h**; (c) Typical negative ion MS/MS for **2h**; (d) Typical positive ion MS spectra for **3h**; (e) Typical positive ion MS/MS for **3h**; **1h** MW: 232; **2h** and **3h** MW: 394.



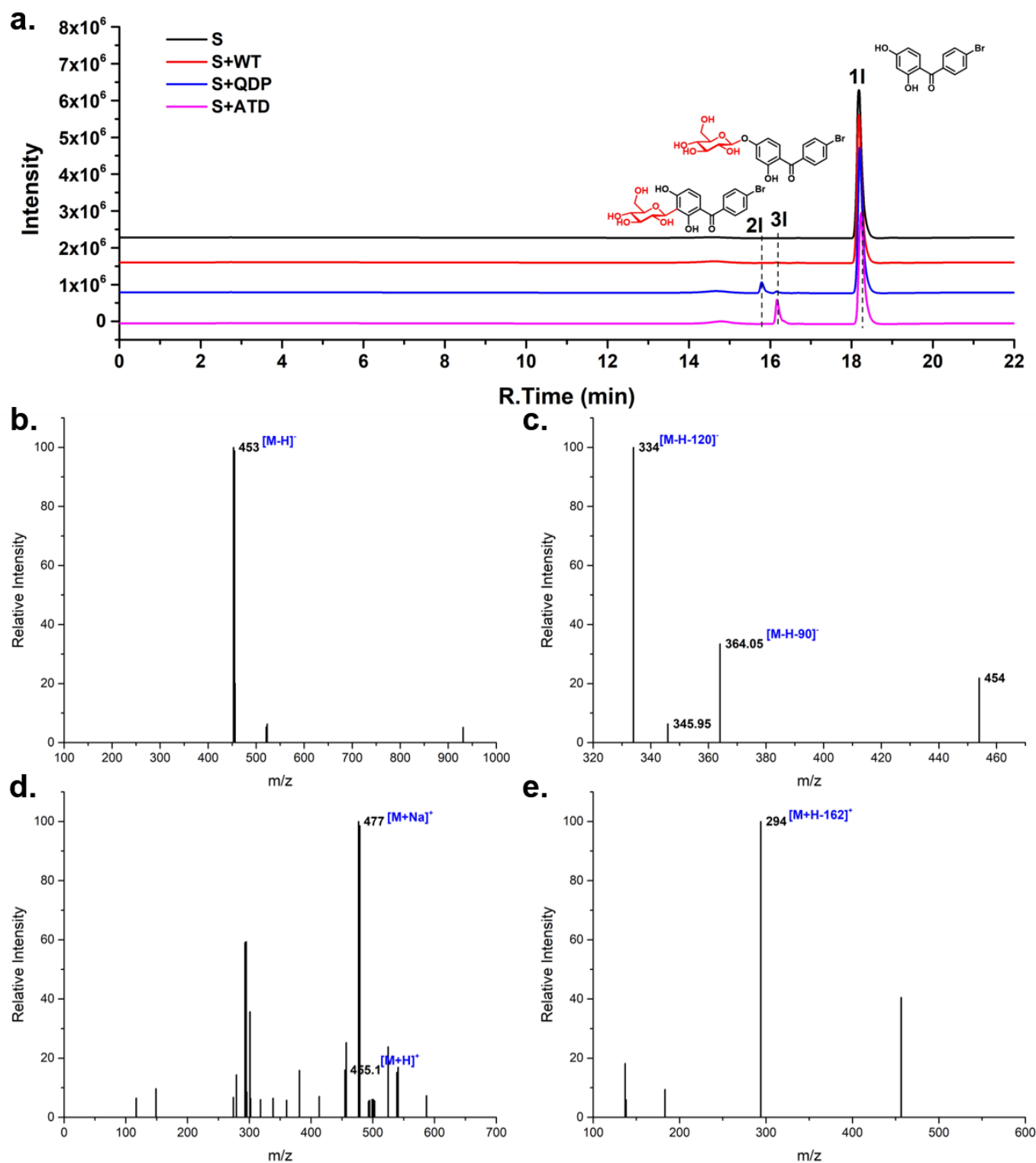
Supplementary Fig. 38 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1i by purified MiCGT variants. (a) HPLC chromatogram of 1i and product 2i and 3i; (b) Typical negative ion MS spectra for 2i; (c) Typical negative ion MS/MS for 2i; (d) Typical positive ion MS spectra for 3i; (e) Typical positive ion MS/MS for 3i; 1i MW: 232; 2i and 3i MW: 394.



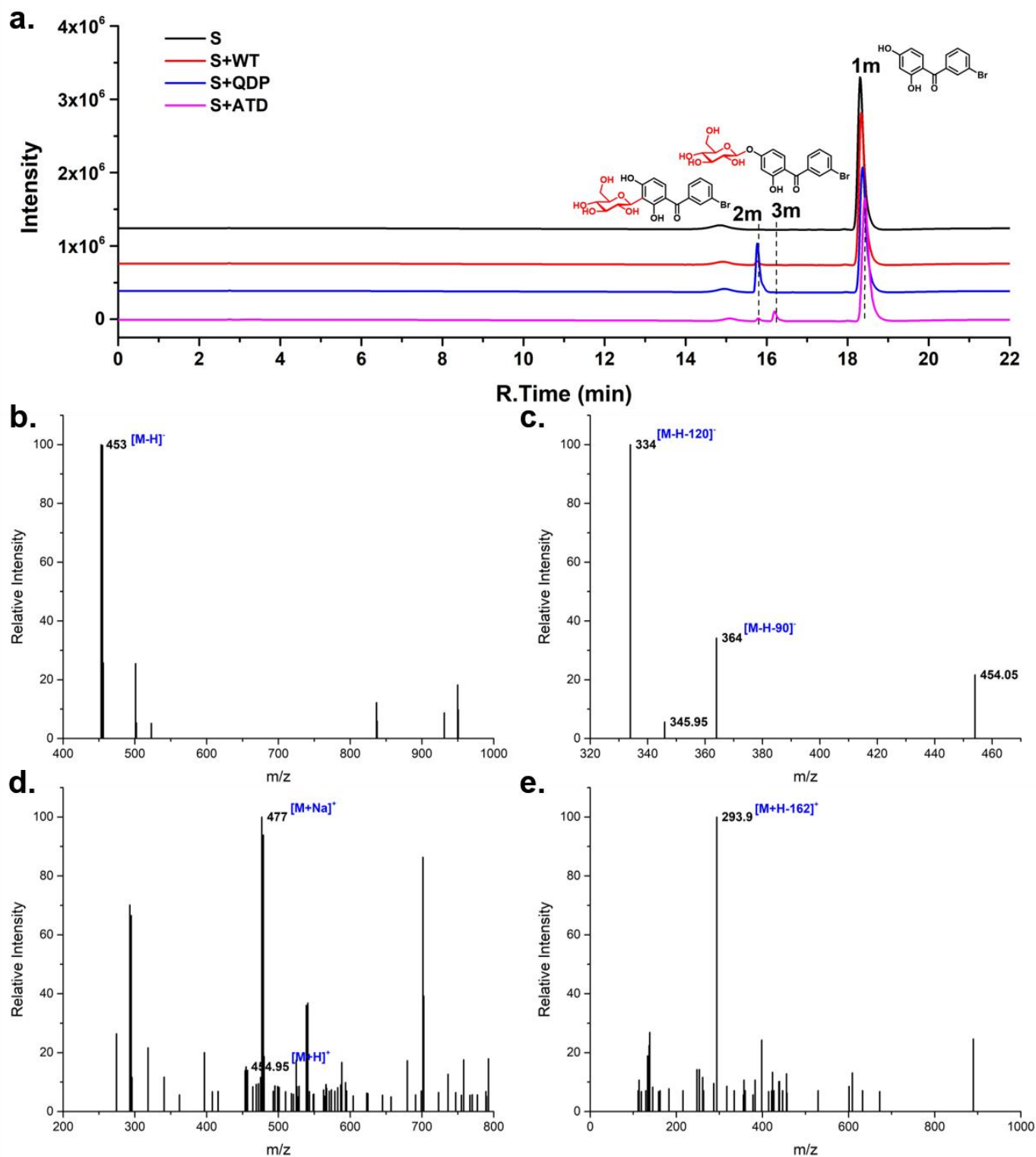
Supplementary Fig. 39 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1j by purified MiCGT variants. (a) HPLC chromatogram of 1j and product 2j and 3j; (b) Typical negative ion MS spectra for 2j; (c) Typical negative ion MS/MS for 2j; (d) Typical positive ion MS spectra for 3j; (e) Typical positive ion MS/MS for 3j; 1j MW: 248; 2j and 3j MW: 410.



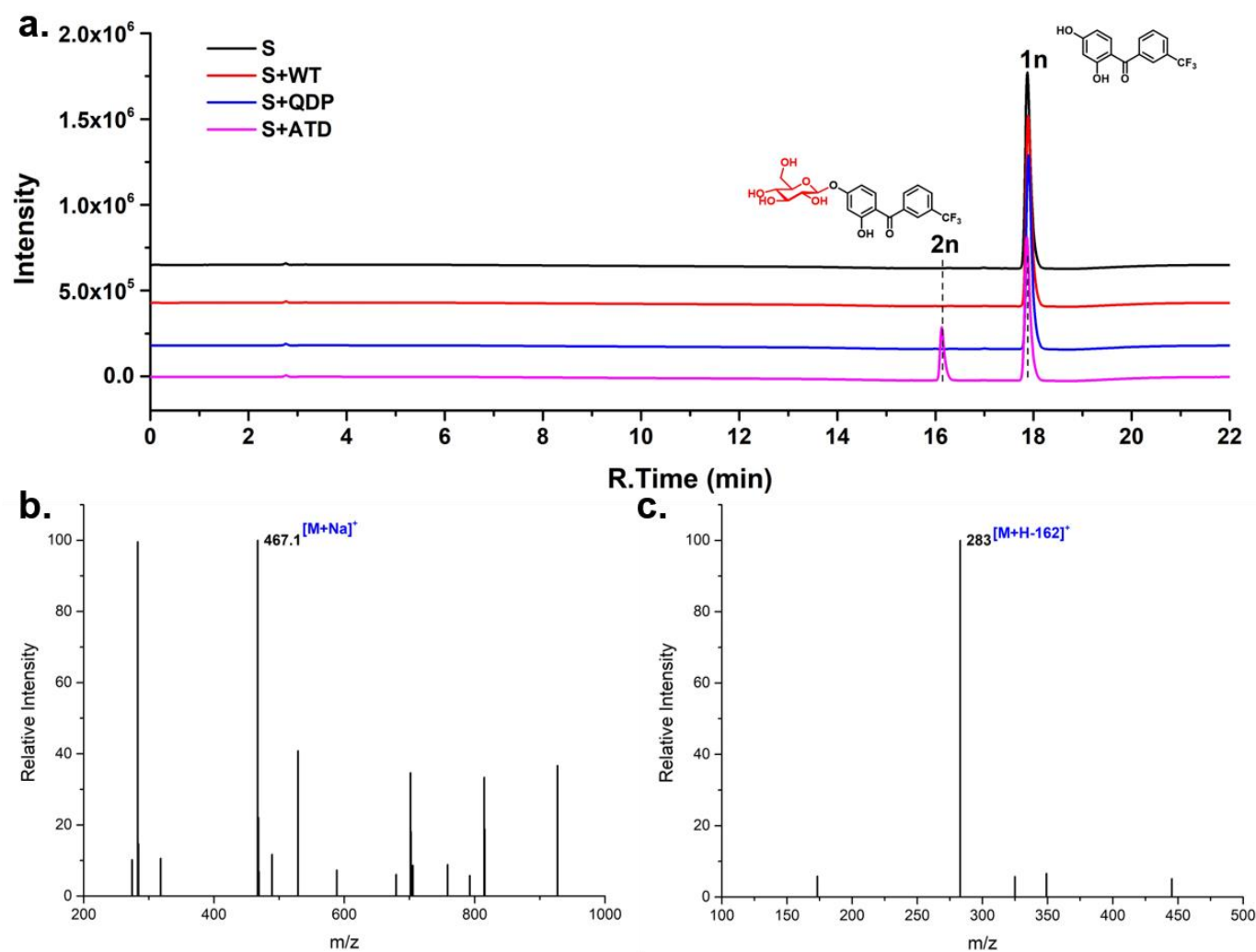
Supplementary Fig. 40 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1k by purified MiCGT variants. (a) HPLC chromatogram of **1k** and product **2k** and **3k**; (b) Typical negative ion MS spectra for **2k**; (c) Typical negative ion MS/MS for **2k**; (d) Typical positive ion MS spectra for **3k**; (e) Typical positive ion MS/MS for **3k**; **1k** MW: 248; **2k** and **3k** MW: 410.



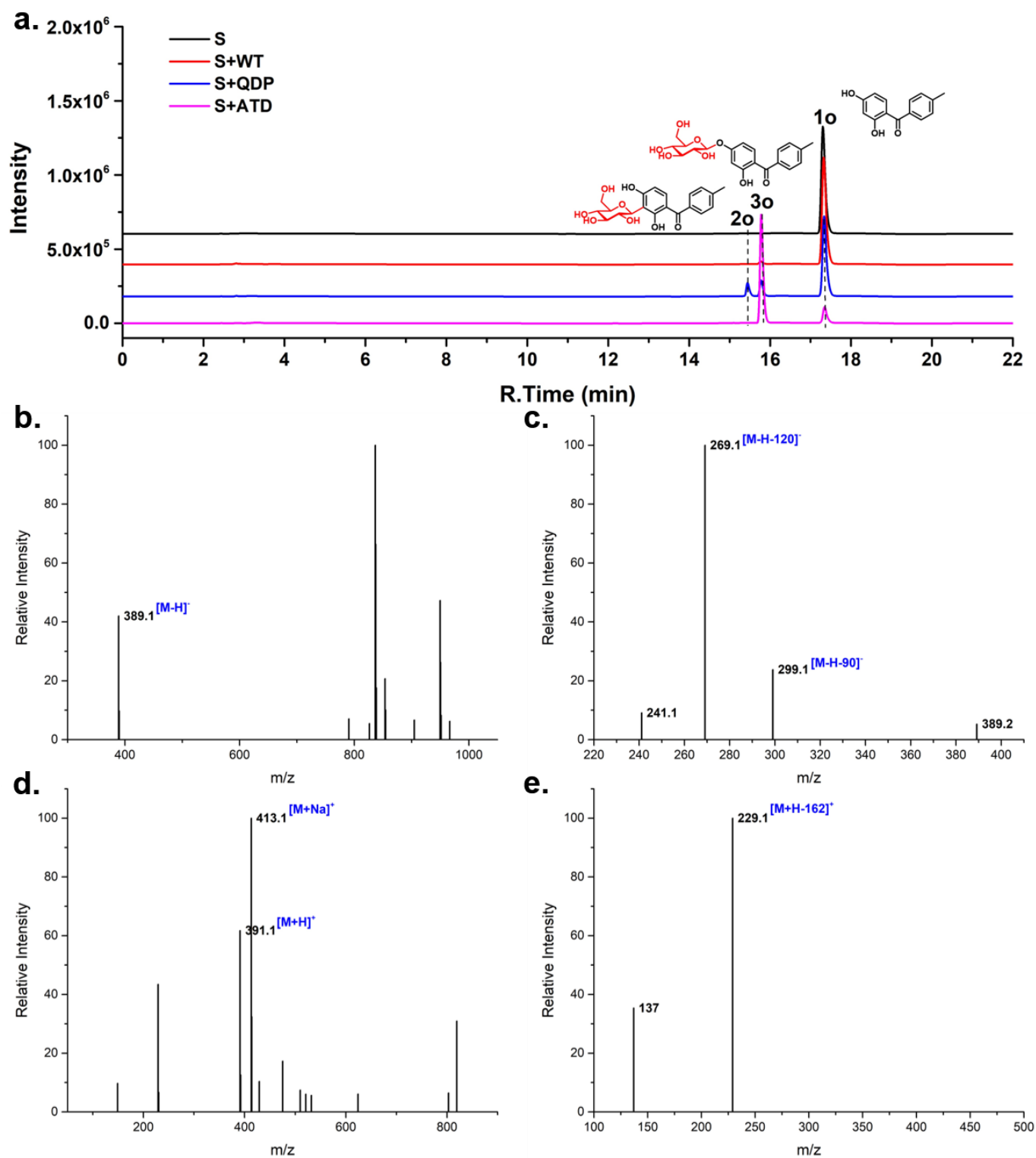
Supplementary Fig. 41 | HPLC-DAD/ESI-MS analysis of the glycosylation of 11 by purified MiCGT variants. (a) HPLC chromatogram of 11 enzyme product 21 and 31; (b) Typical negative ion MS spectra for 21; (c) Typical negative ion MS/MS for 21; (d) Typical positive ion MS spectra for 31; (e) Typical positive ion MS/MS for 31; 11 MW: 292; 21 and 31 MW: 454.



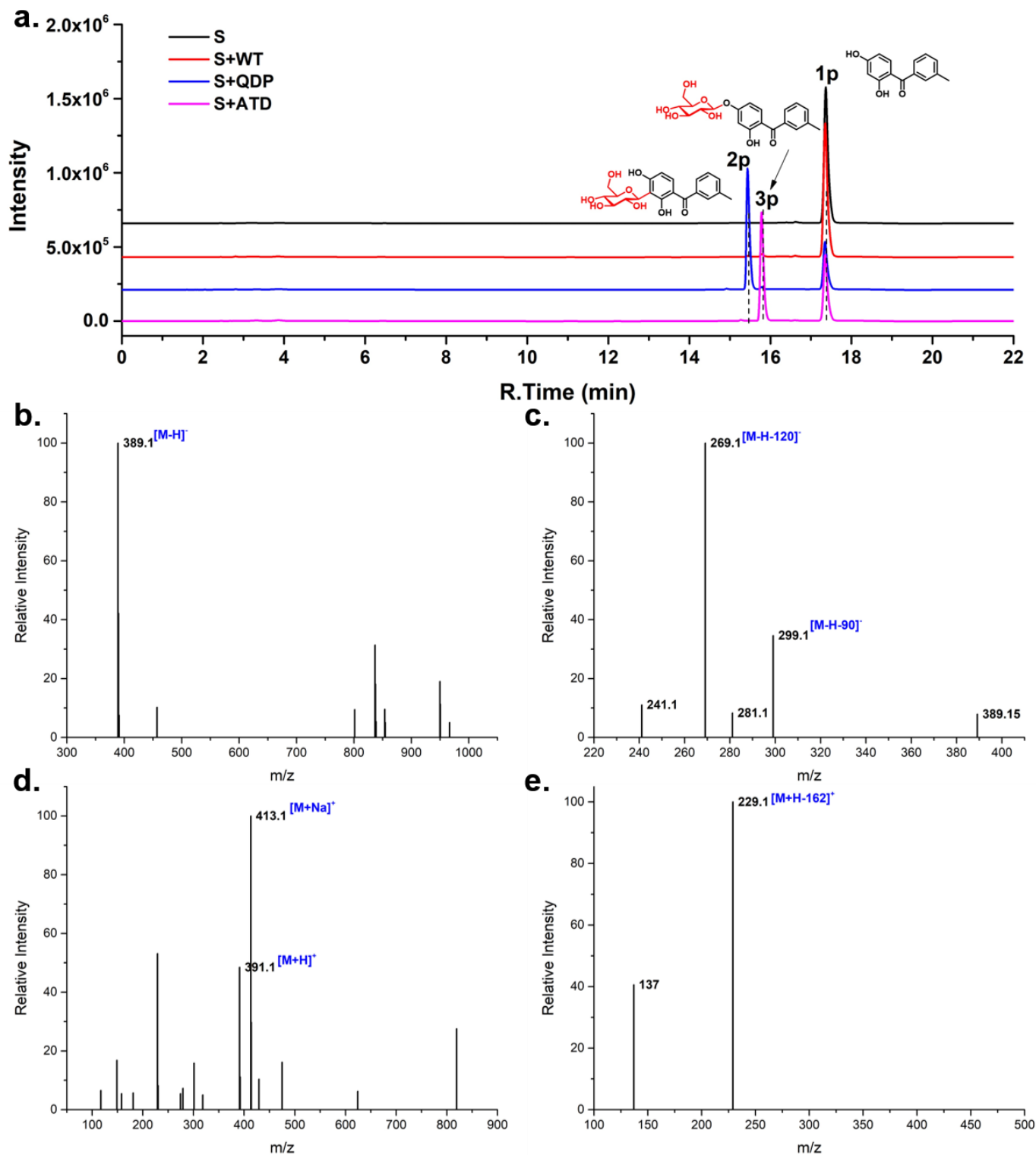
Supplementary Fig. 42 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1m by purified MiCGT variants. (a) HPLC chromatogram of **1m** and product **2m** and **3m**; (b) Typical negative ion MS spectra for **2m**; (c) Typical negative ion MS/MS for **2m**; (d) Typical positive ion MS spectra for **3m**; (e) Typical positive ion MS/MS for **3m**; **1m** MW: 292; **2m** and **3m** MW: 454.



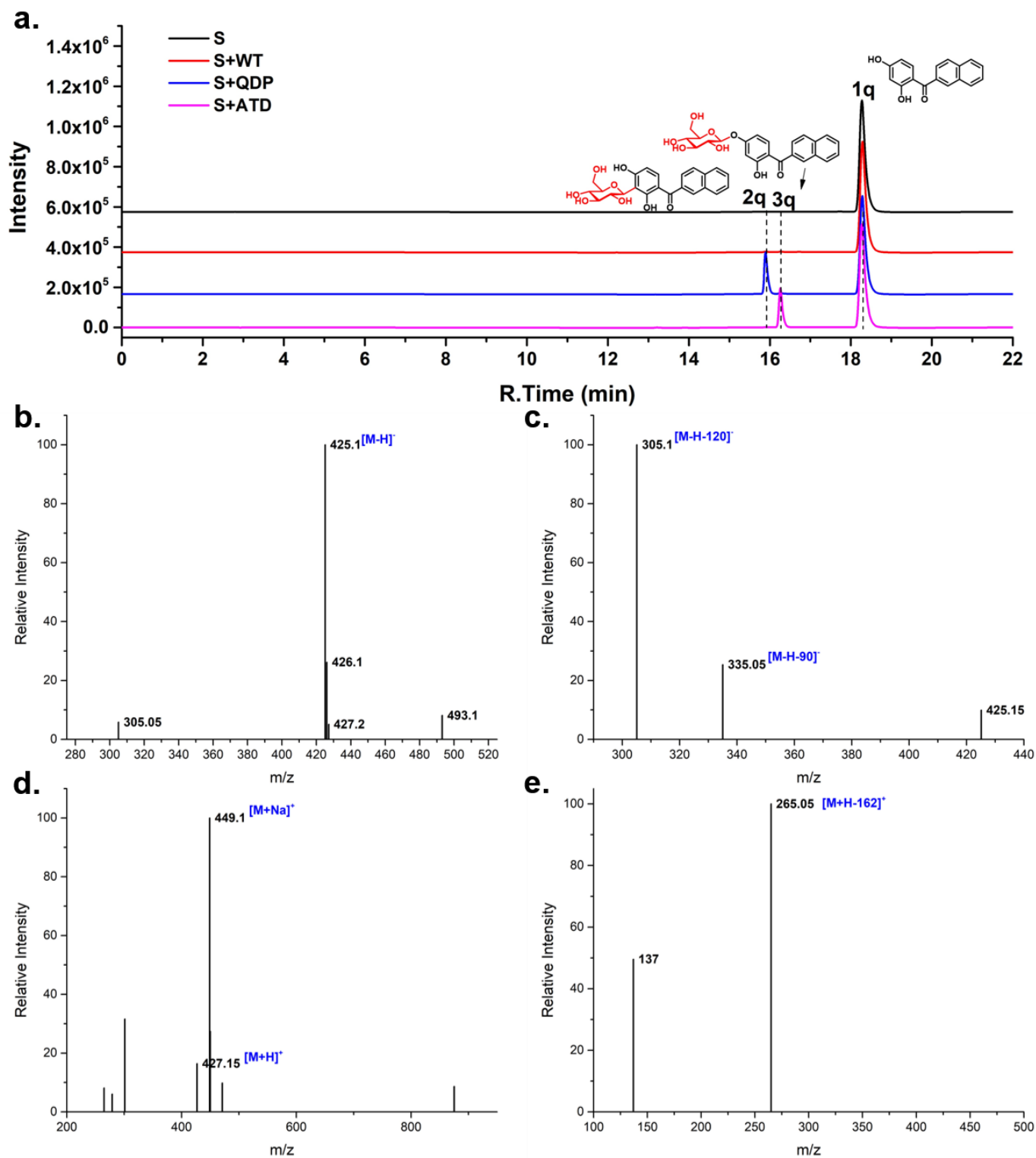
Supplementary Fig. 43 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1n by purified MiCGT variants. (a) HPLC chromatogram of **1n** and product **2n**; **(b)** Typical positive ion MS spectra for **2n**; **(c)** Typical positive ion MS/MS for **2n**; **1n** MW: 282; **2n** MW: 444.



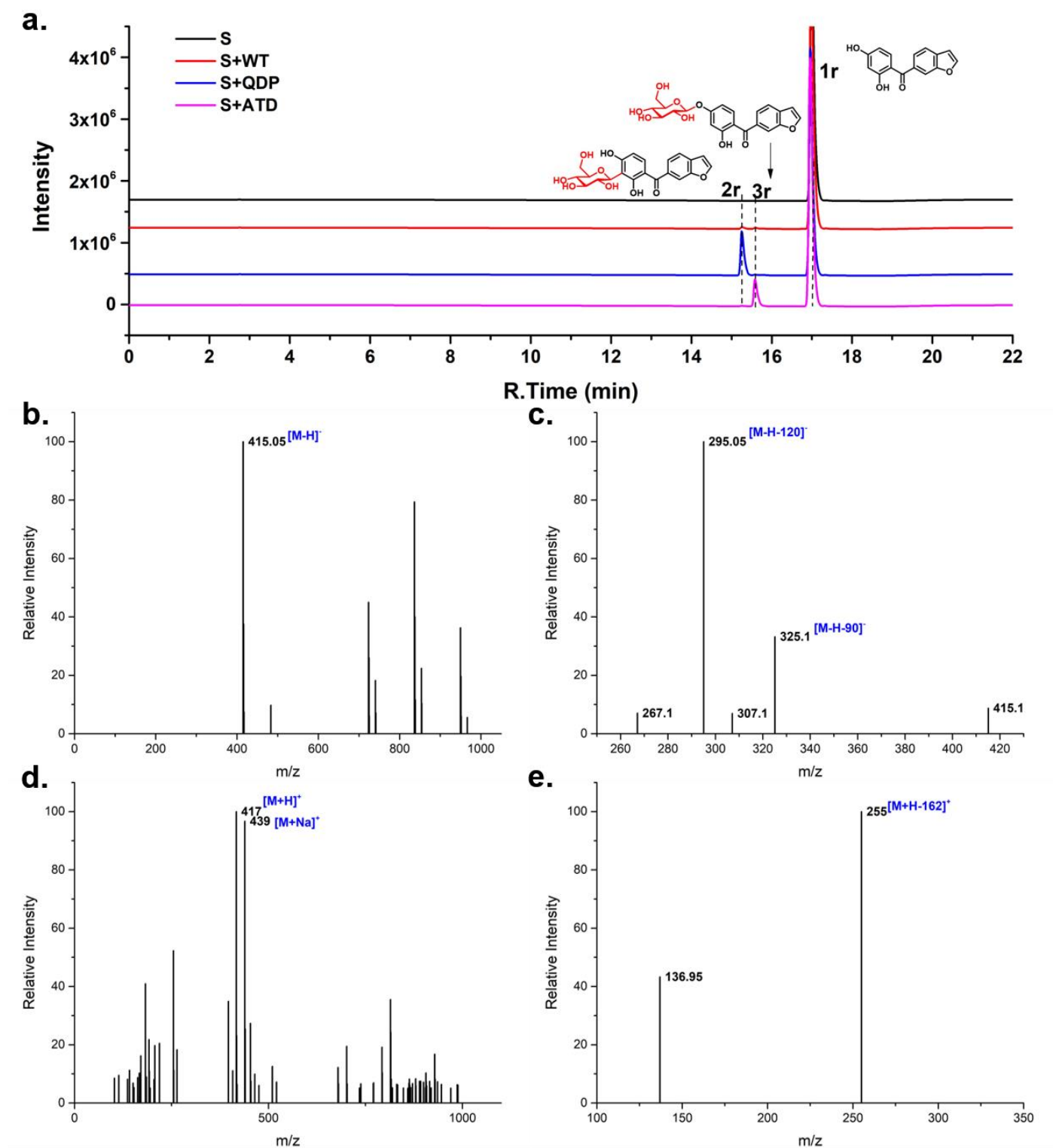
Supplementary Fig. 44 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1o by purified MiCGT variants. (a) HPLC chromatogram of 1o and product 2o and 3o; (b) Typical negative ion MS spectra for 2o; (c) Typical negative ion MS/MS for 2o; (d) Typical positive ion MS spectra for 3o; (e) Typical positive ion MS/MS for 3o; 1o MW: 228; 2o and 3o MW: 390.



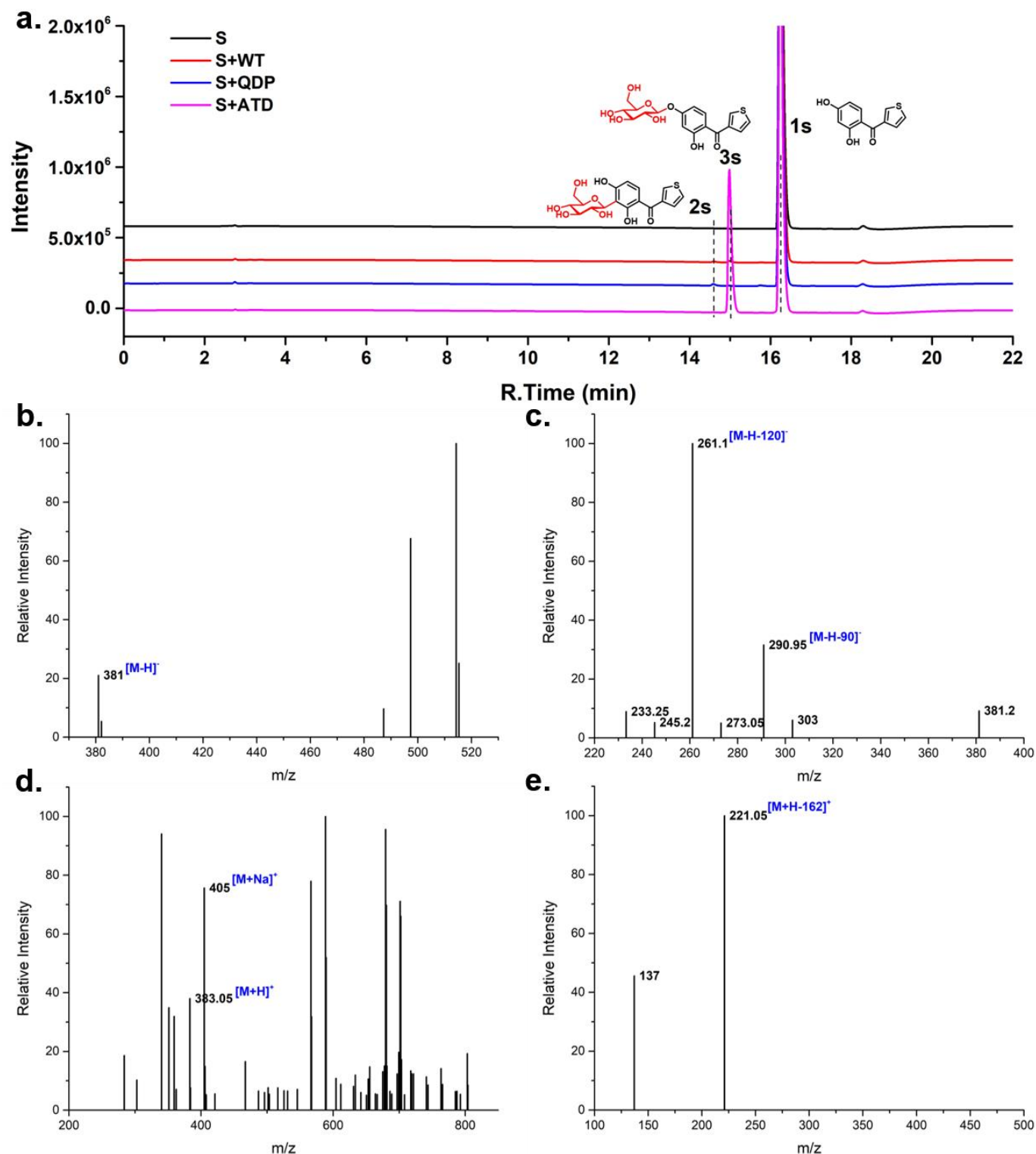
Supplementary Fig. 45 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1p by purified MiCGT variants. (a) HPLC chromatogram of **1p** and product **2p** and **3p**; (b) Typical negative ion MS spectra for **2p**; (c) Typical negative ion MS/MS for **2p**; (d) Typical positive ion MS spectra for **3p**; (e) Typical positive ion MS/MS for **3p**; **1p** MW: 228; **2p** and **3p** MW: 390.



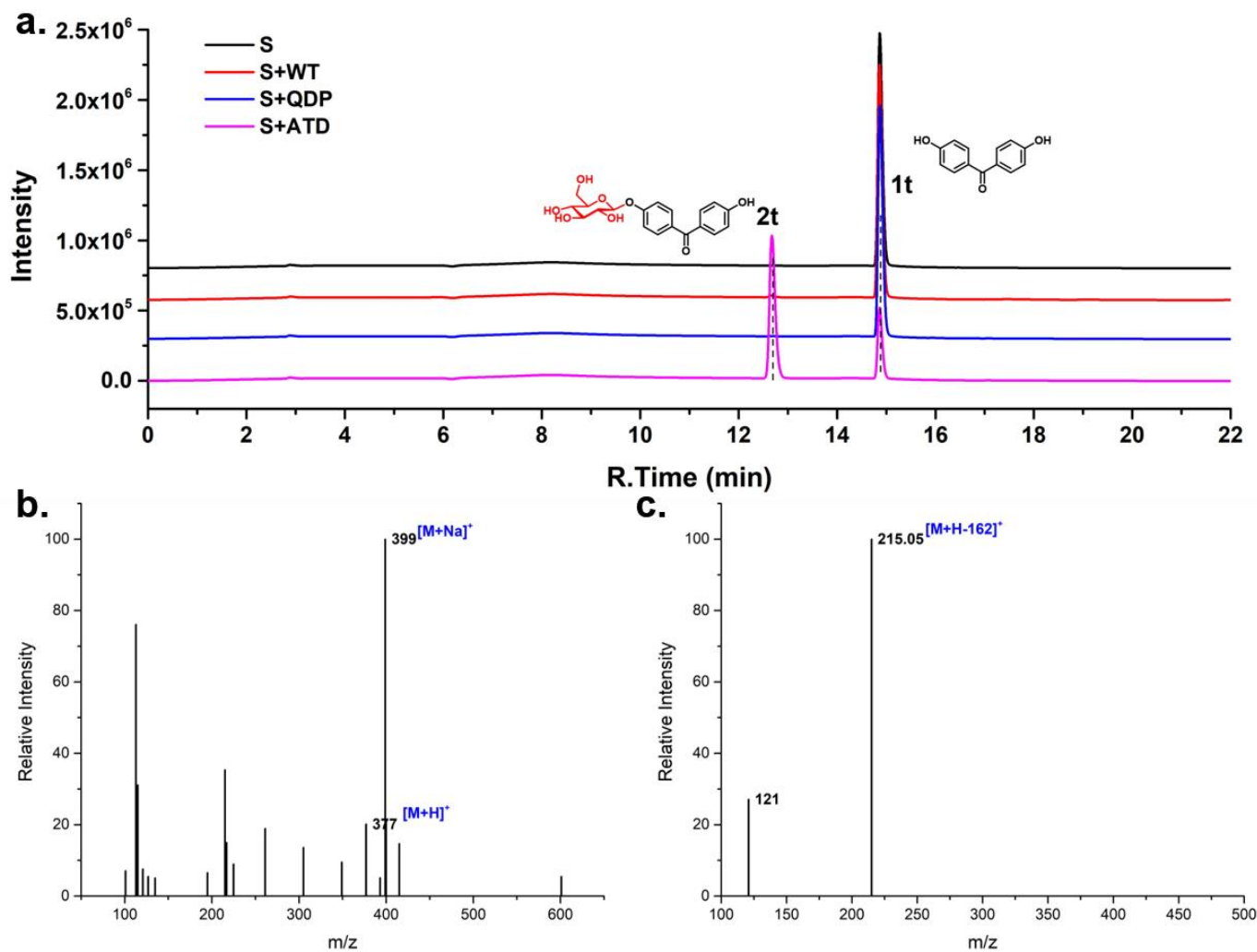
Supplementary Fig. 46 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1q by purified MiCGT variants. (a) HPLC chromatogram of **1q** and product **2q** and **3q**; (b) Typical negative ion MS spectra for **2q**; (c) Typical negative ion MS² for **2q**; (d) Typical positive ion MS spectra for **3q**; (e) Typical positive ion MS² for **3q**; **1q** MW: 264; **2q** and **3q** MW: 426.



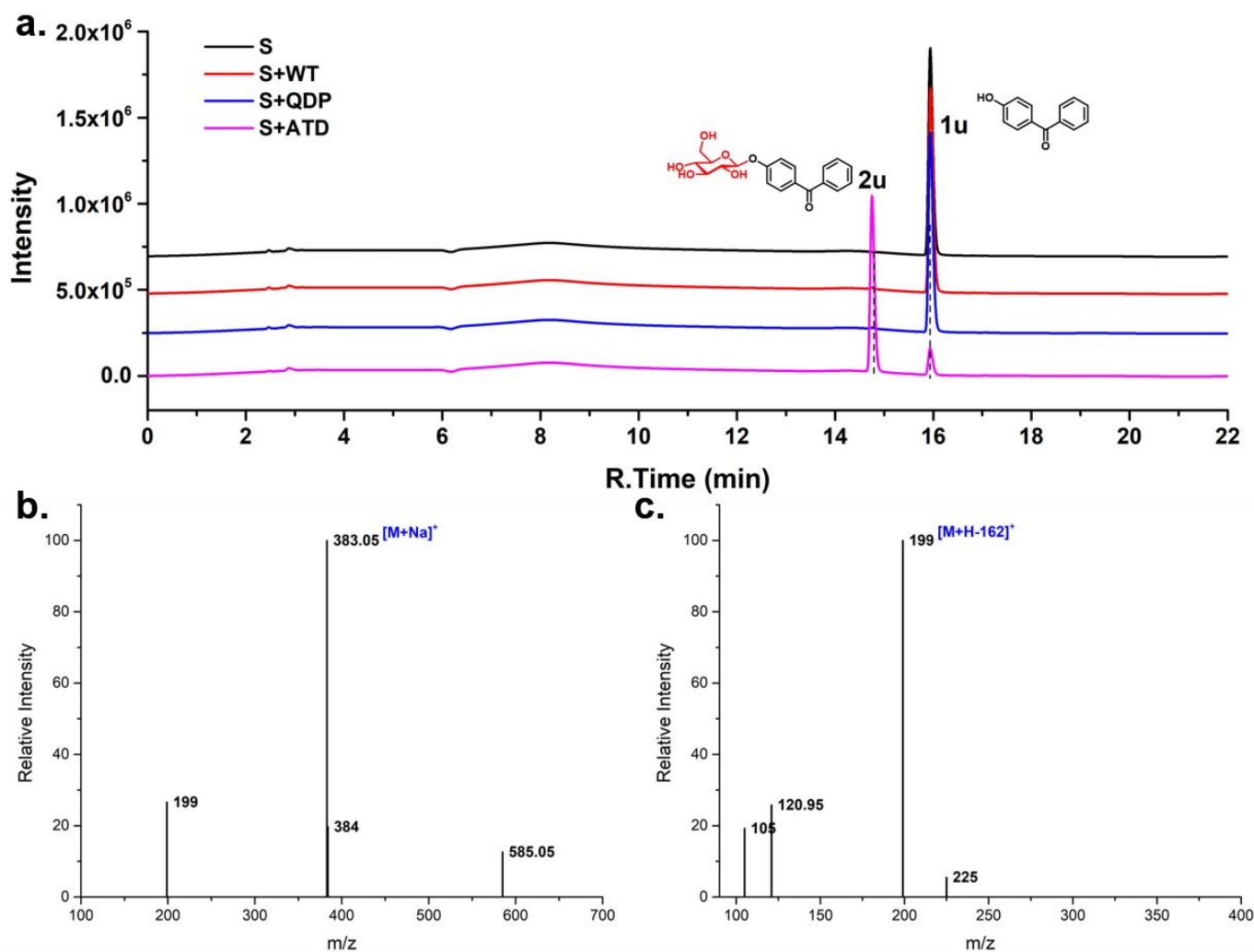
Supplementary Fig. 47 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1r by purified MiCGT variants. (a) HPLC chromatogram of **1r** enzyme product **2r** and **3r**; (b) Typical negative ion MS spectra for **2r**; (c) Typical negative ion MS/MS for **2r**; (d) Typical positive ion MS spectra for **3r**; (e) Typical positive ion MS/MS for **3r**; **1r** MW: 254; **2r** and **3r** MW: 416.



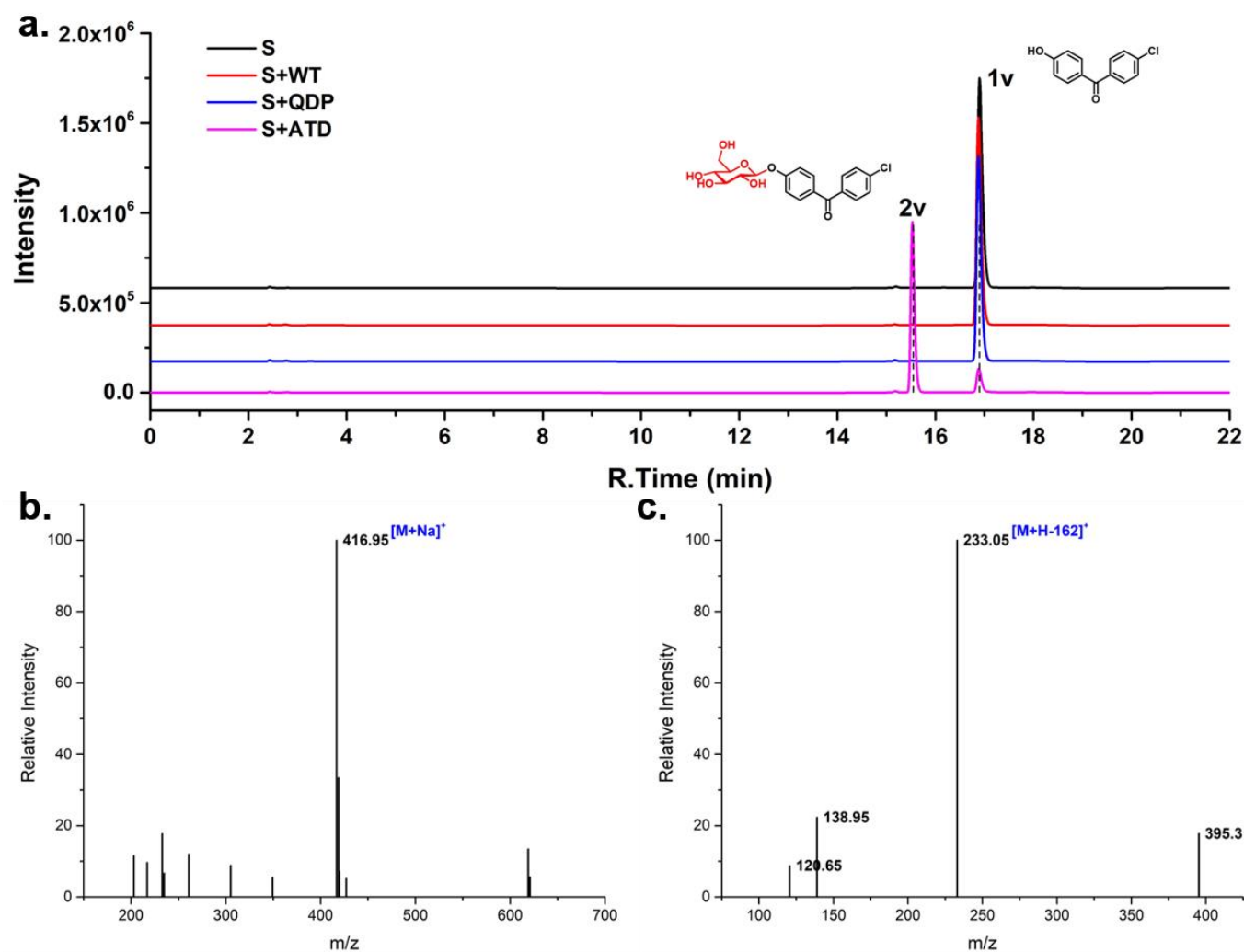
Supplementary Fig. 48 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1s by purified MiCGT variants. (a) HPLC chromatogram of 1s and product 2s and 3s; (b) Typical negative ion MS spectra for 2s; (c) Typical negative ion MS/MS for 2s; (d) Typical positive ion MS spectra for 3s; (e) Typical positive ion MS/MS for 3s; 1s MW: 220; 2s and 3s MW: 382.



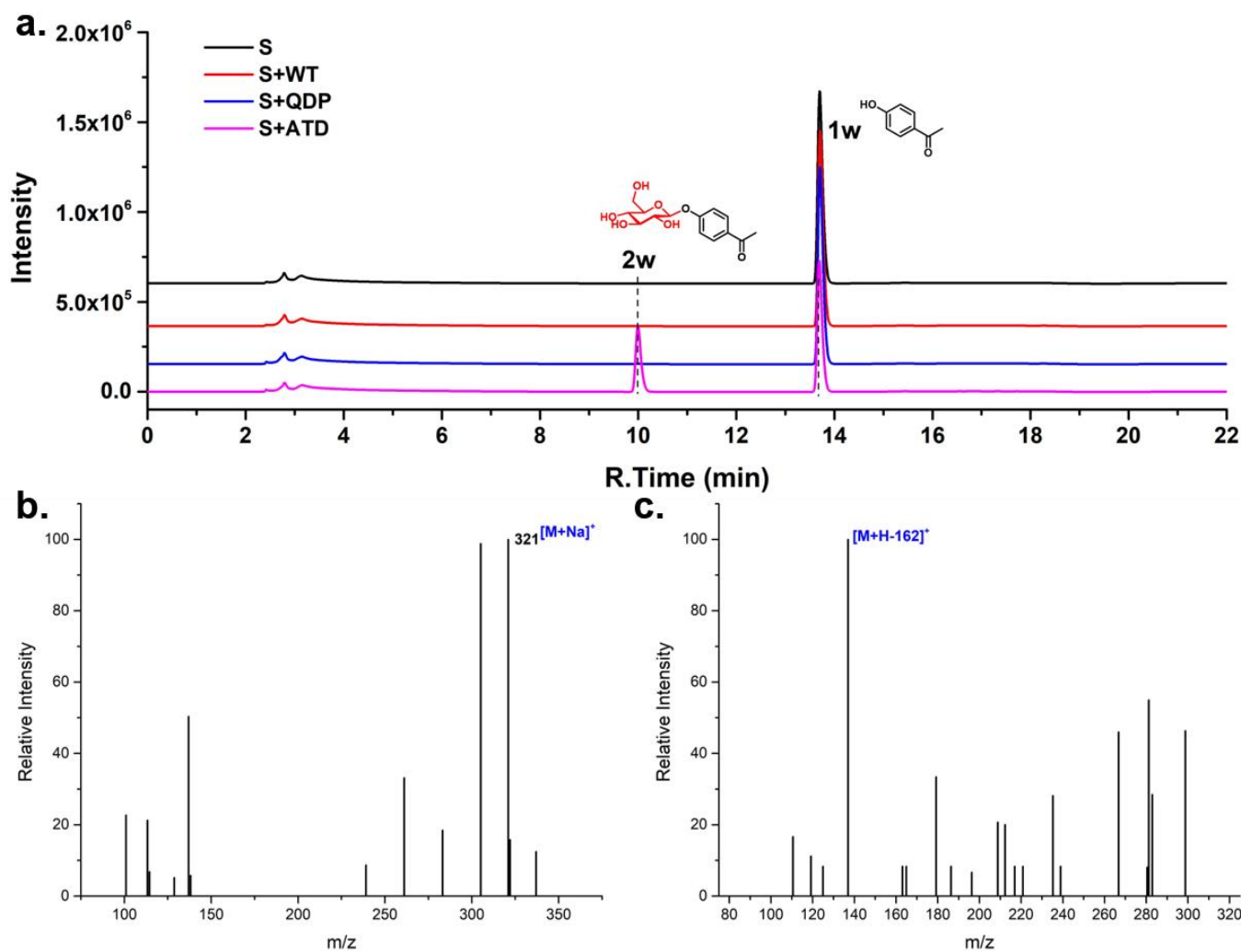
Supplementary Fig. 49 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1t by purified MiCGT variants. (a) HPLC chromatogram of 1t and product 2t; (b) Typical positive ion MS spectra for 2t; (b) Typical positive ion MS/MS for 2t; 1t MW: 214; 2t MW: 376.



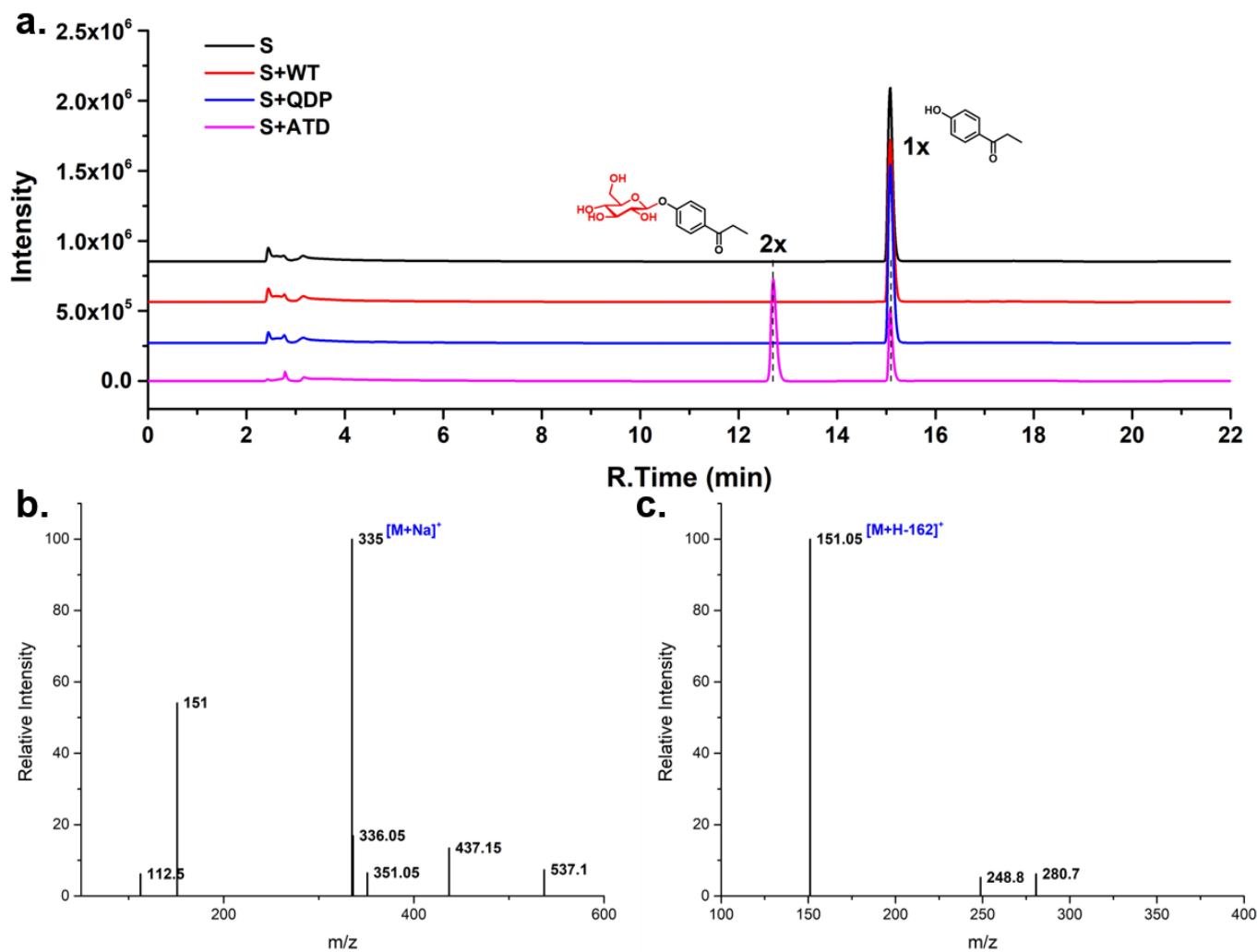
Supplementary Fig. 50 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1u by purified MiCGT variants. (a) HPLC chromatogram of 1u and product 2u; (b) Typical positive ion MS spectra for 2u; (b) Typical positive ion MS/MS for 2u; 1u MW: 198; 2u MW: 360.



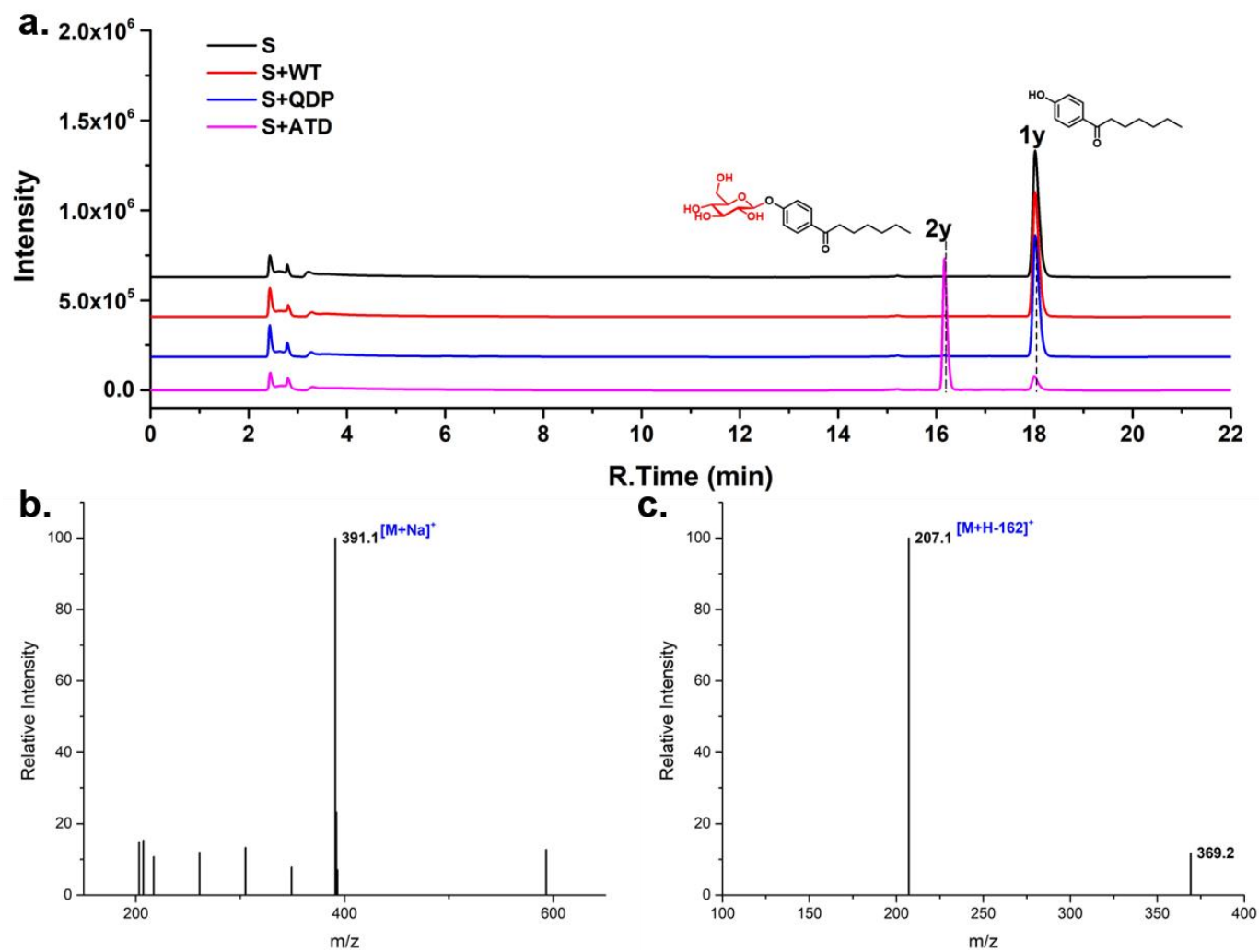
Supplementary Fig. 51 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1v by purified MiCGT variants. (a) HPLC chromatogram of 1v and product 2v; (b) Typical positive ion MS spectra for 2v; (b) Typical positive ion MS/MS for 2v; 1v MW: 232; 2v MW: 394.



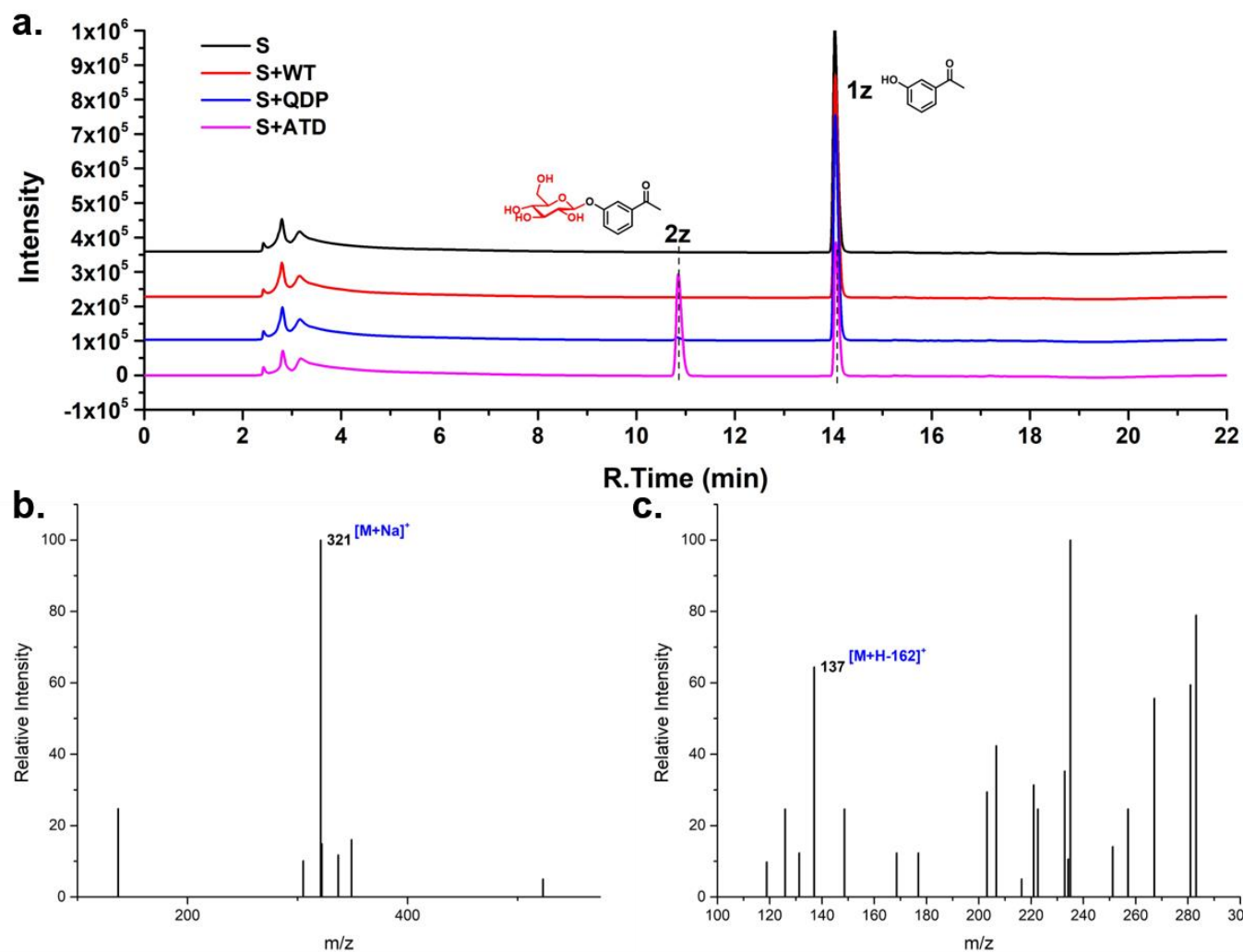
Supplementary Fig. 52 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1w by purified MiCGT variants. (a) HPLC chromatogram of 1w product 2w; (b) Typical positive ion MS spectra for 2w; (b) Typical positive ion MS/MS for 2w; 1w MW: 136; 2w MW: 298.



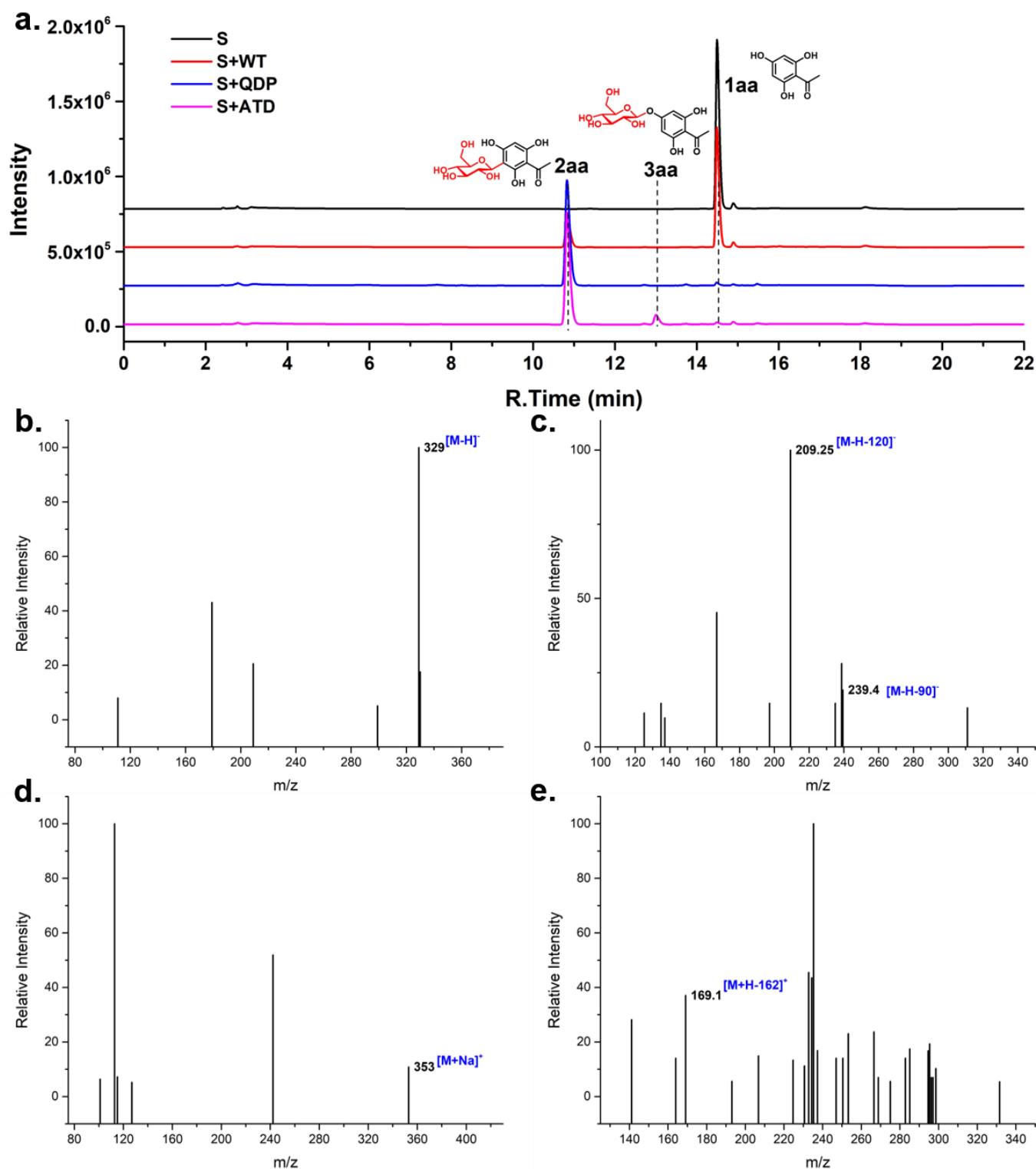
Supplementary Fig. 53 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1x by purified MiCGT variants. (a) HPLC chromatogram of 1x and product 2x; (b) Typical positive ion MS spectra for 2x; (b) Typical positive ion MS/MS for 2x; 1x MW: 150; 2x MW: 312.



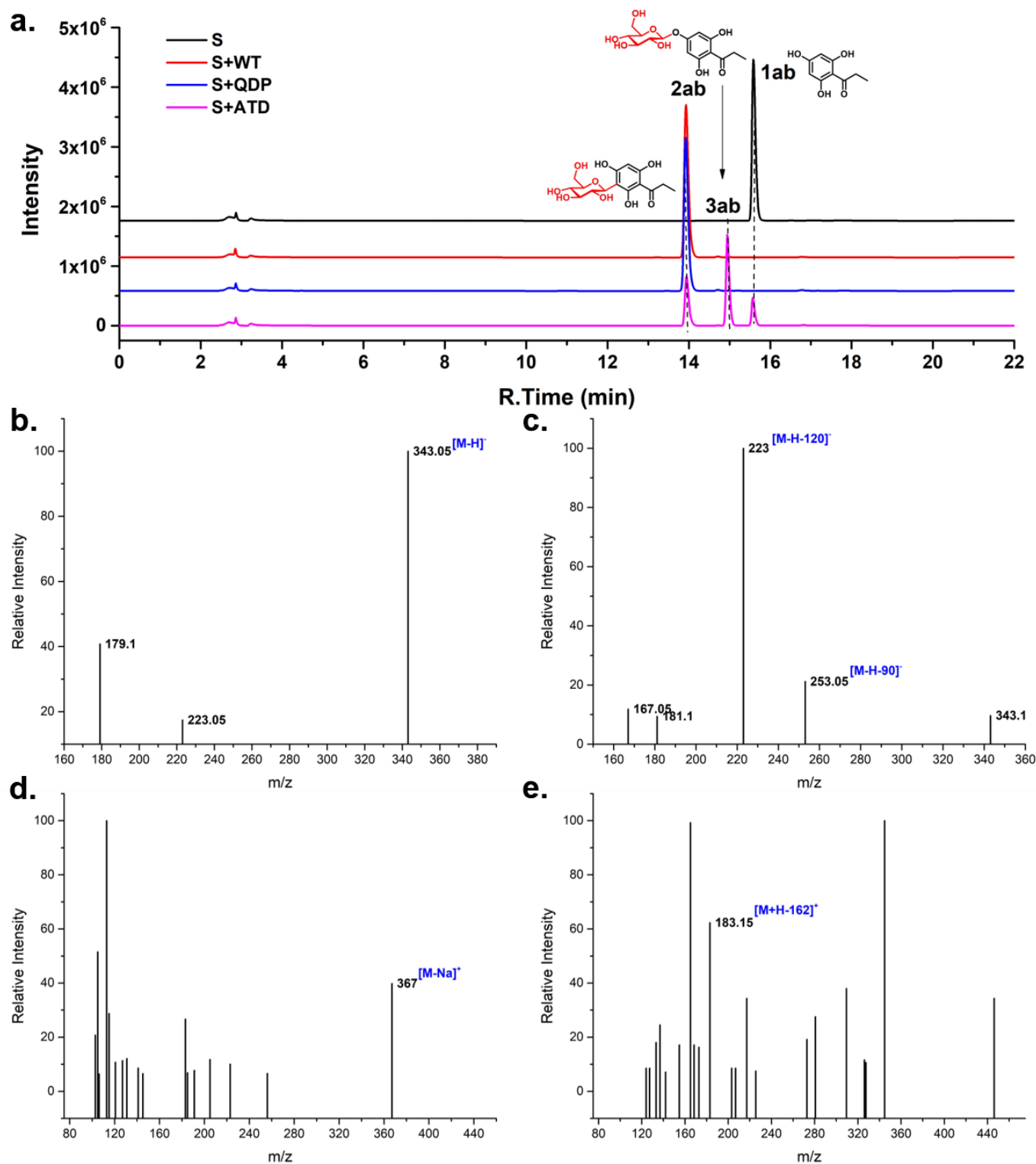
Supplementary Fig. 54 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1y by purified MiCGT variants. (a) HPLC chromatogram of **1y** and product **2y**; (b) Typical positive ion MS spectra for **2y**; (b) Typical positive ion MS/MS for **2y**; **1y** MW: 206; **2y** MW: 368.



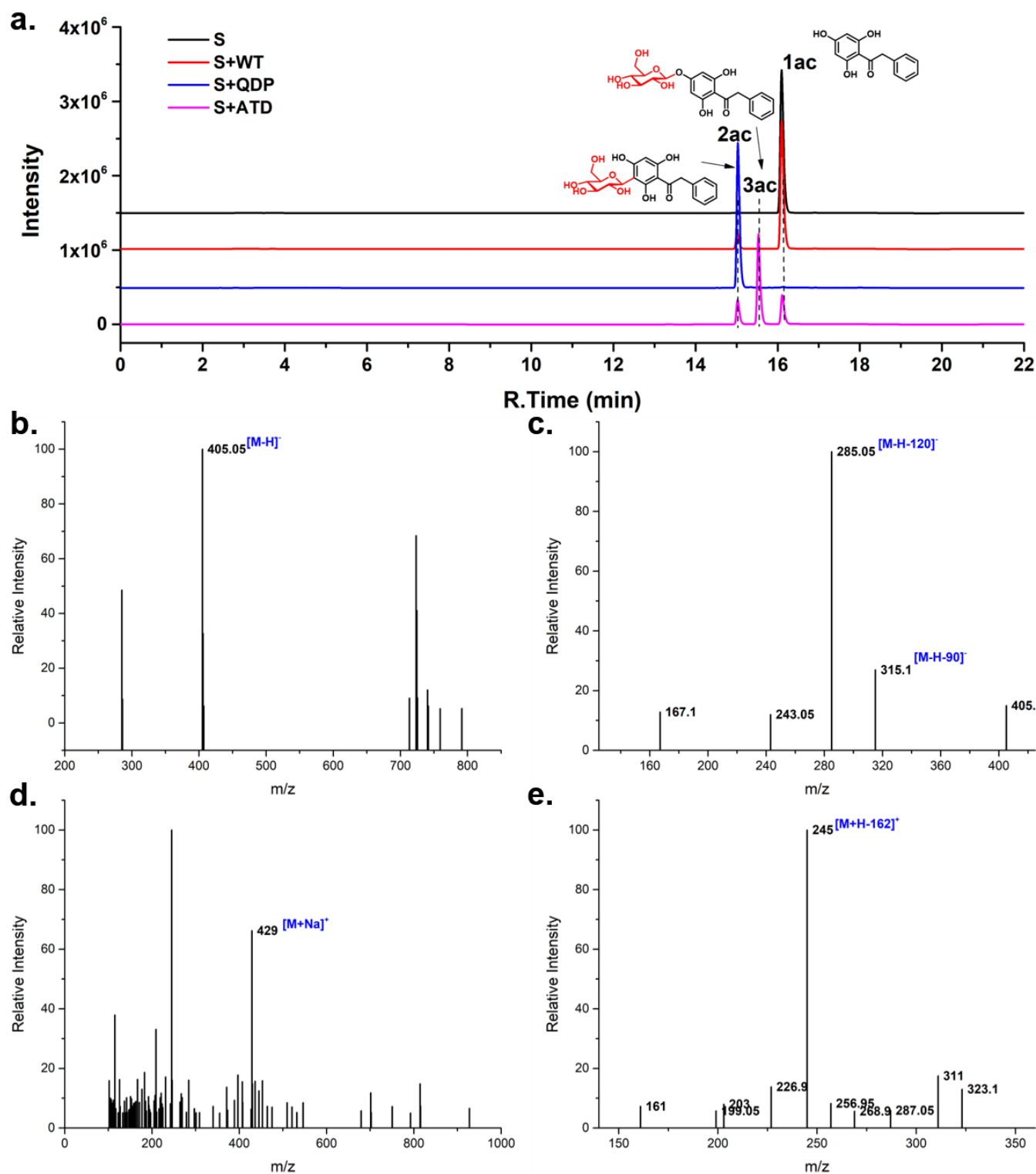
Supplementary Fig. 55 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1z by purified MiCGT variants. (a) HPLC chromatogram of **1z** and product **2z**; (b) Typical positive ion MS spectra for **2z**; (b) Typical positive ion MS/MS for **2z**; **1z** MW: 136; **2z** MW: 298.



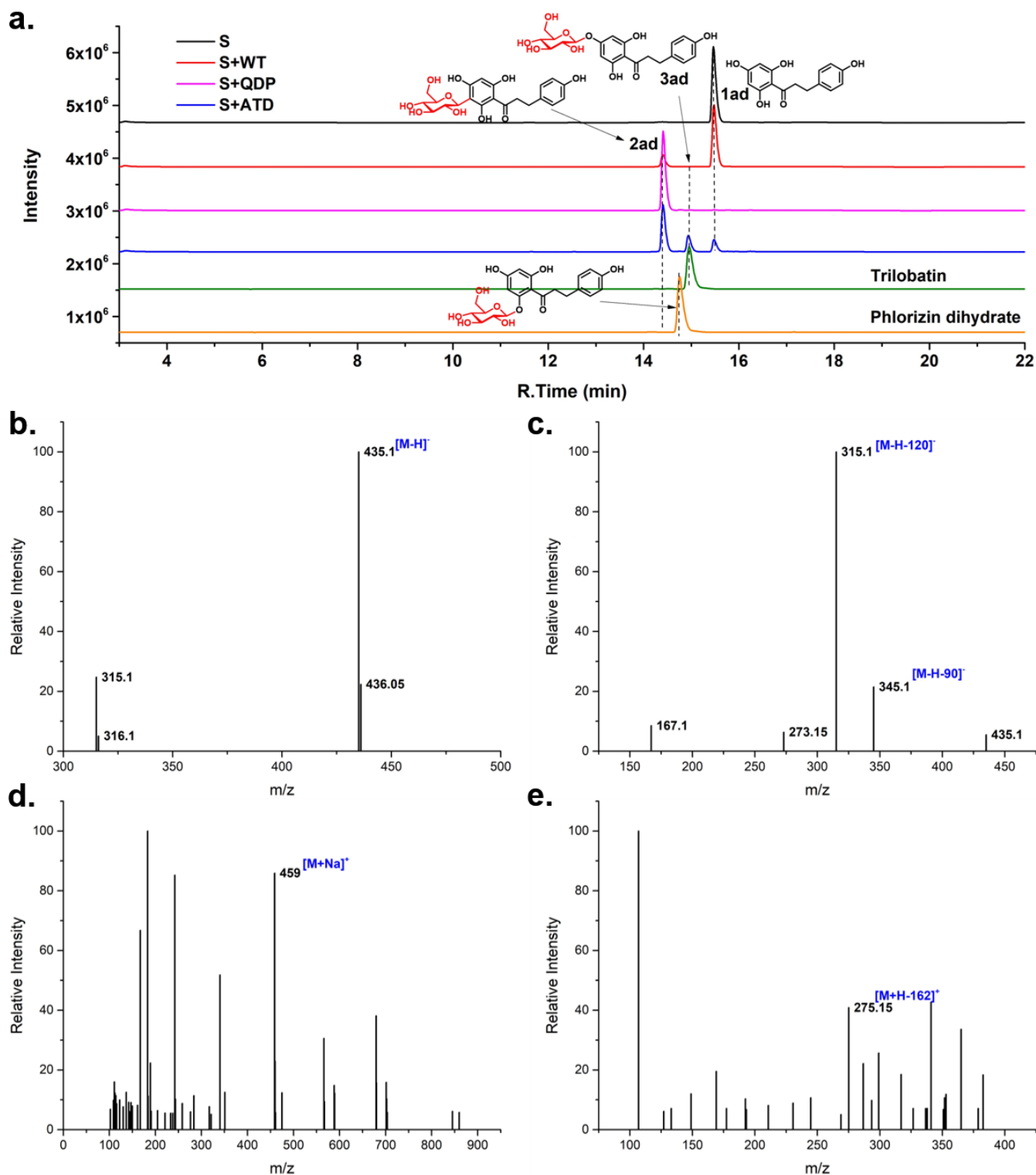
Supplementary Fig. 56 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1aa by purified MiCGT variants. (a) HPLC chromatogram of 1aa and product 2aa and 3aa; (b) Typical negative ion MS spectra for 2aa; (c) Typical negative ion MS/MS for 2aa; (d) Typical positive ion MS spectra for 3aa; (e) Typical positive ion MS/MS for 3aa; 1aa MW: 168; 2aa and 3aa MW: 330.



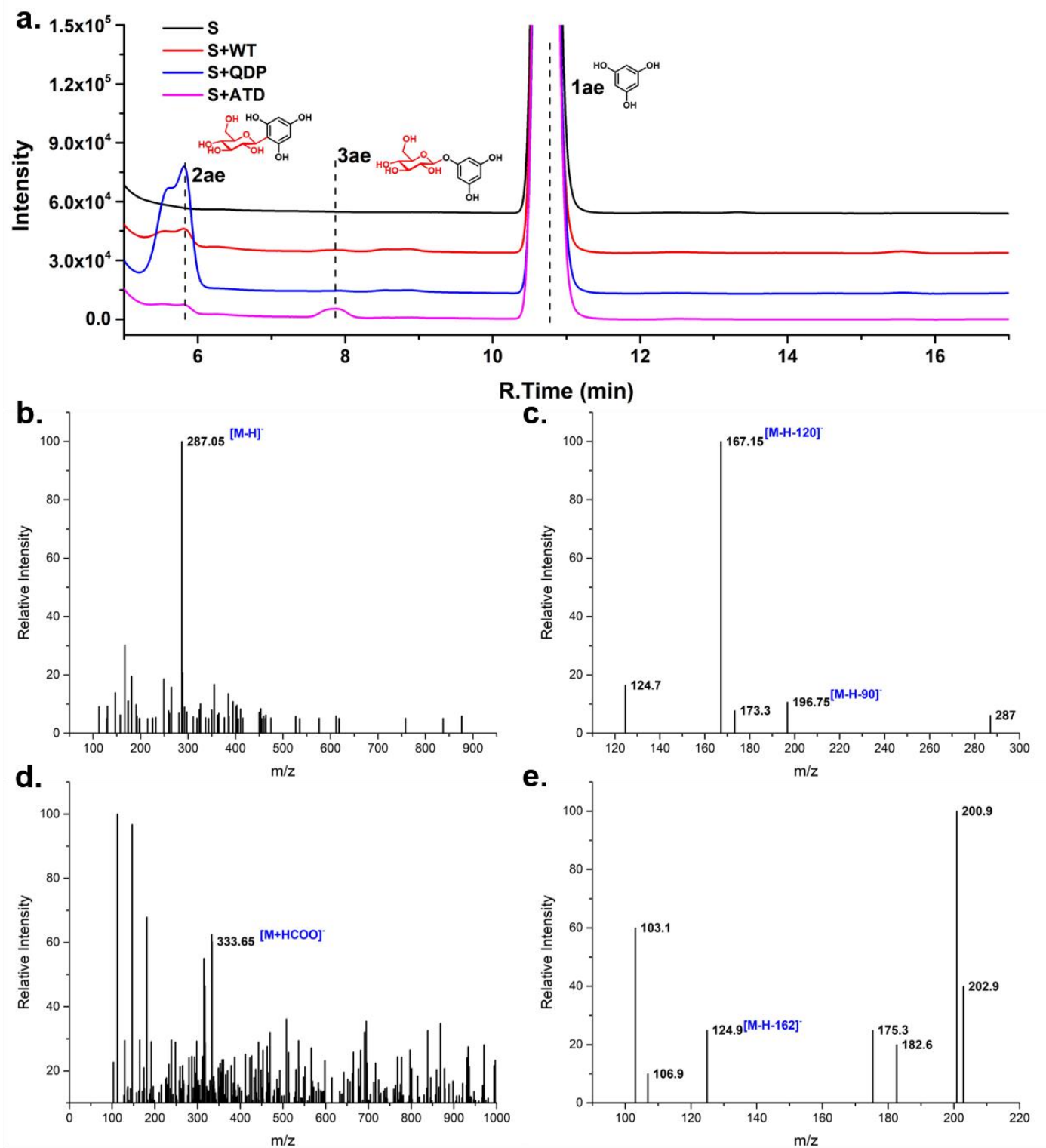
Supplementary Fig. 57 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1ab by purified MiCGT variants. (a) HPLC chromatogram of **1ab** and product **2ab** and **3ab**; (b) Typical negative ion MS spectra for **2ab**; (c) Typical negative ion MS/MS for **2ab**; (d) Typical positive ion MS spectra for **3ab**; (e) Typical positive ion MS/MS for **3ab**; **1ab** MW: 182; **2ab** and **3ab** MW: 344.



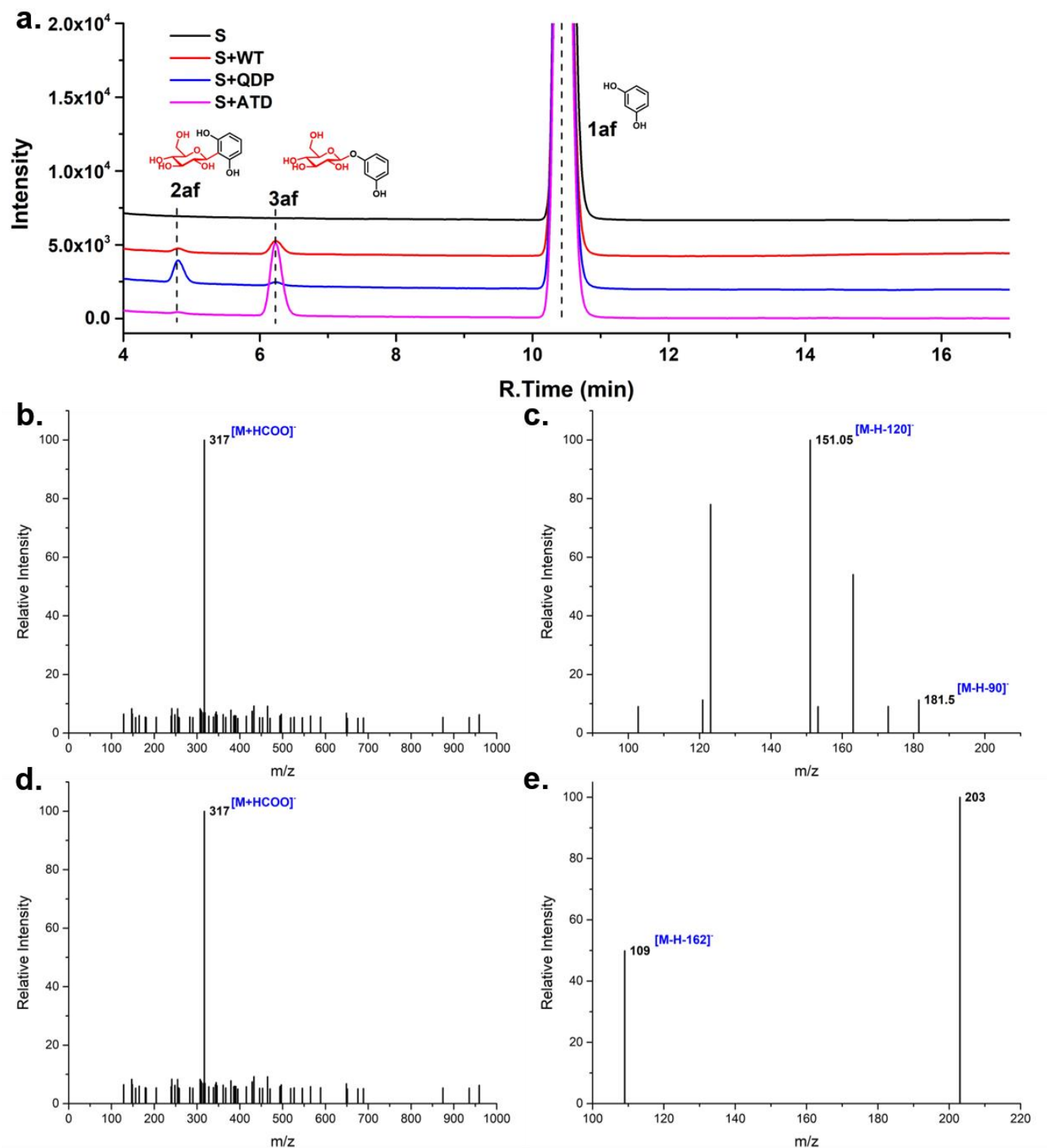
Supplementary Fig. 58 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1ac by purified MiCGT variants. (a) HPLC chromatogram of **1ac** and product **2ac** and **3ac**; (b) Typical negative ion MS spectra for **2ac**; (c) Typical negative ion MS/MS for **2ac**; (d) Typical positive ion MS spectra for **3ac**; (e) Typical positive ion MS/MS for **3ac**; **1ac** MW: 244; **2ac** and **3ac** MW: 406.



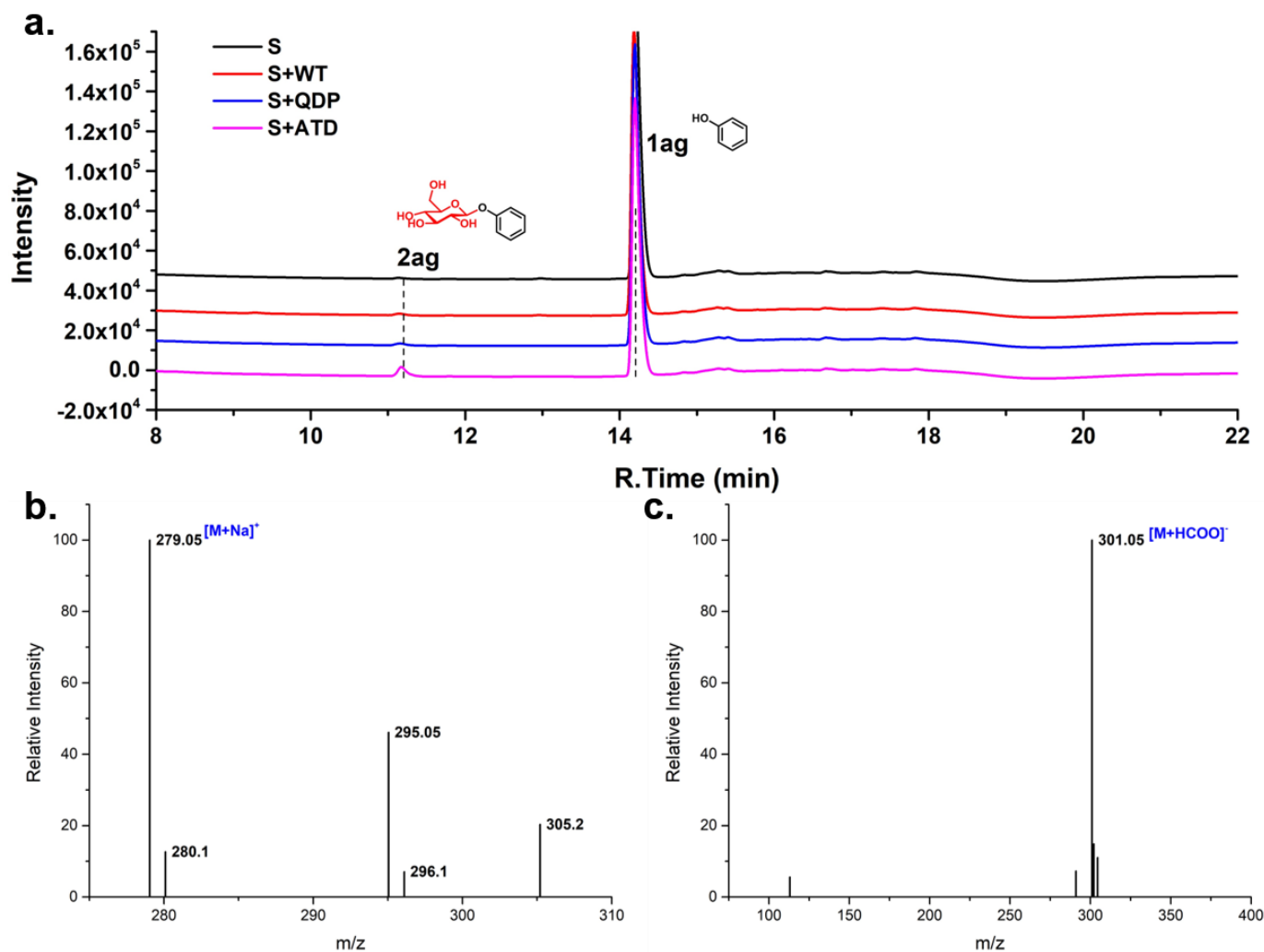
Supplementary Fig. 59 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1ad by purified MiCGT variants. (a) HPLC chromatogram of **1ad** and product **2ad** and **3ad**; (b) Typical negative ion MS spectra for **2ad**; (c) Typical negative ion MS/MS for **2ad**; (d) Typical positive ion MS spectra for **3ad**; (e) Typical positive ion MS/MS for **3ad**; **1ad** MW: 274; **2ad** and **3ad** MW: 436.



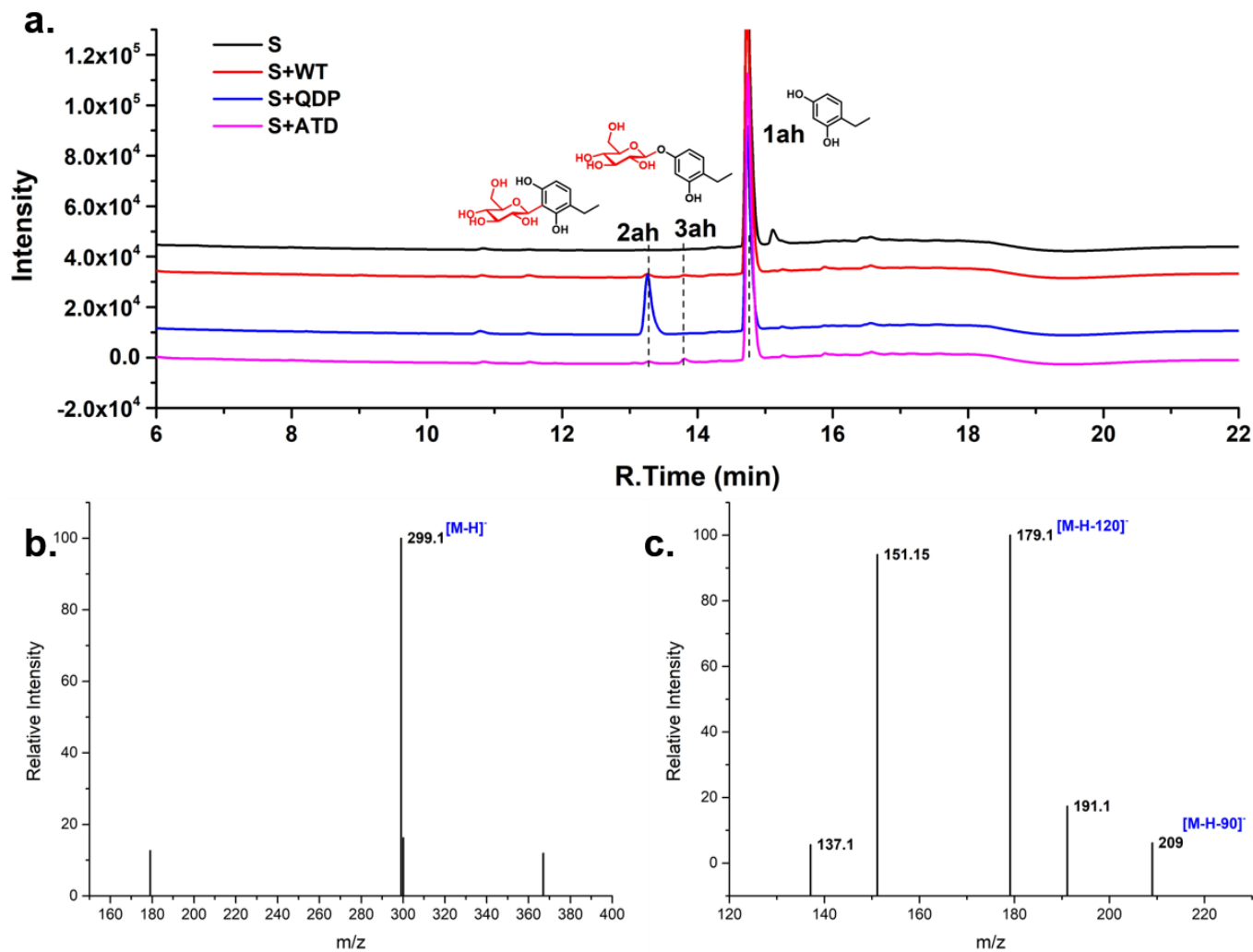
Supplementary Fig. 60 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1ae by purified MiCGT variants. (a) HPLC chromatogram of 1ae and product 2ae and 3ae; (b) Typical negative ion MS spectra for 2ae; (c) Typical negative ion MS/MS for 2ae; (d) Typical positive ion MS spectra for 3ae; (e) Typical positive ion MS/MS for 3ae; 1ae MW: 126; 2ae and 3ae MW: 288.



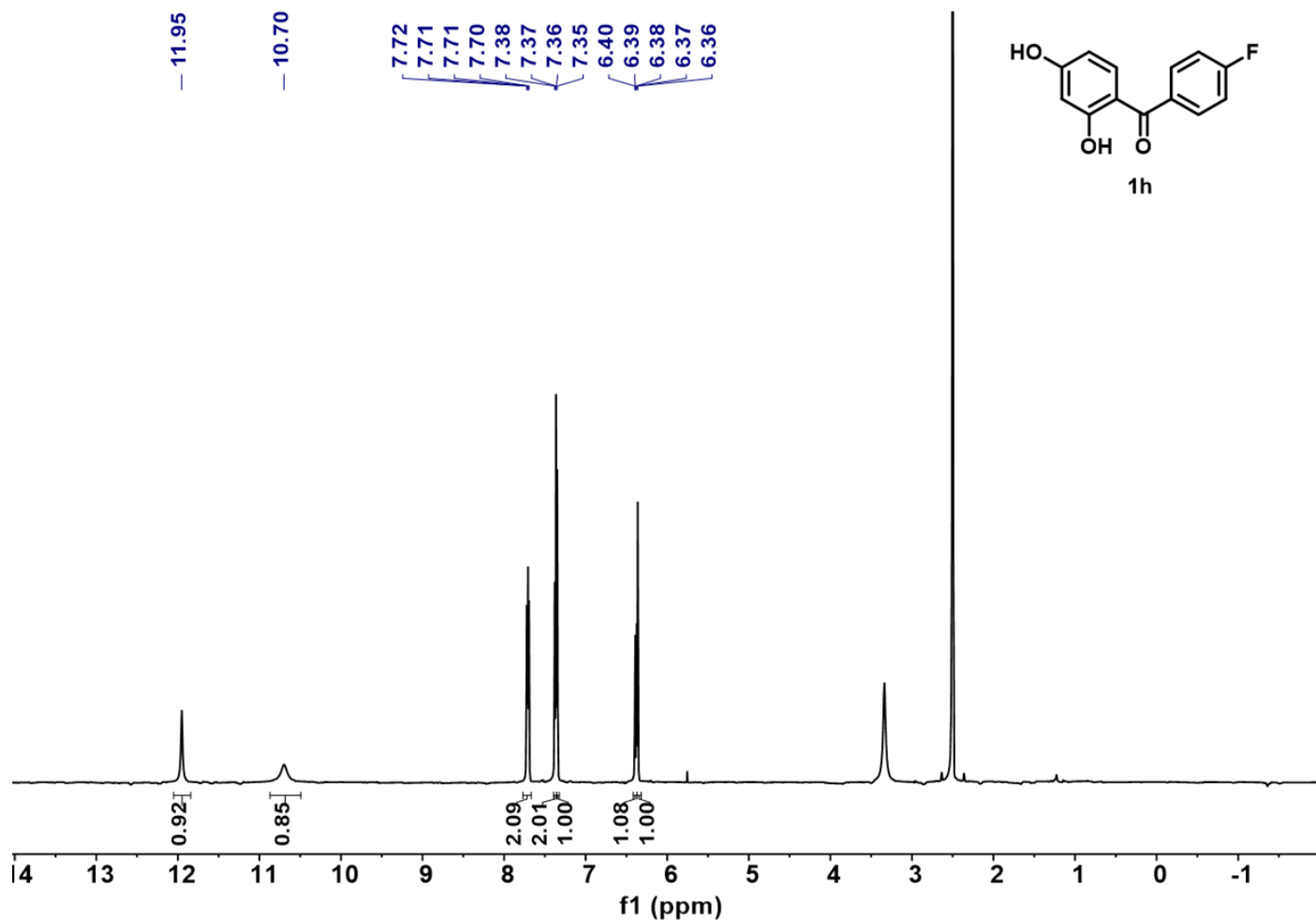
Supplementary Fig. 61 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1af by purified MiCGT variants. (a) HPLC chromatogram of **1af** and product **2af** and **3af**; (b) Typical negative ion MS spectra for **2af**; (c) Typical negative ion MS/MS for **2af**; (d) Typical positive ion MS spectra for **3af**; (e) Typical positive ion MS/MS for **3af**; **1af** MW: 110; **2af** and **3af** MW: 277.



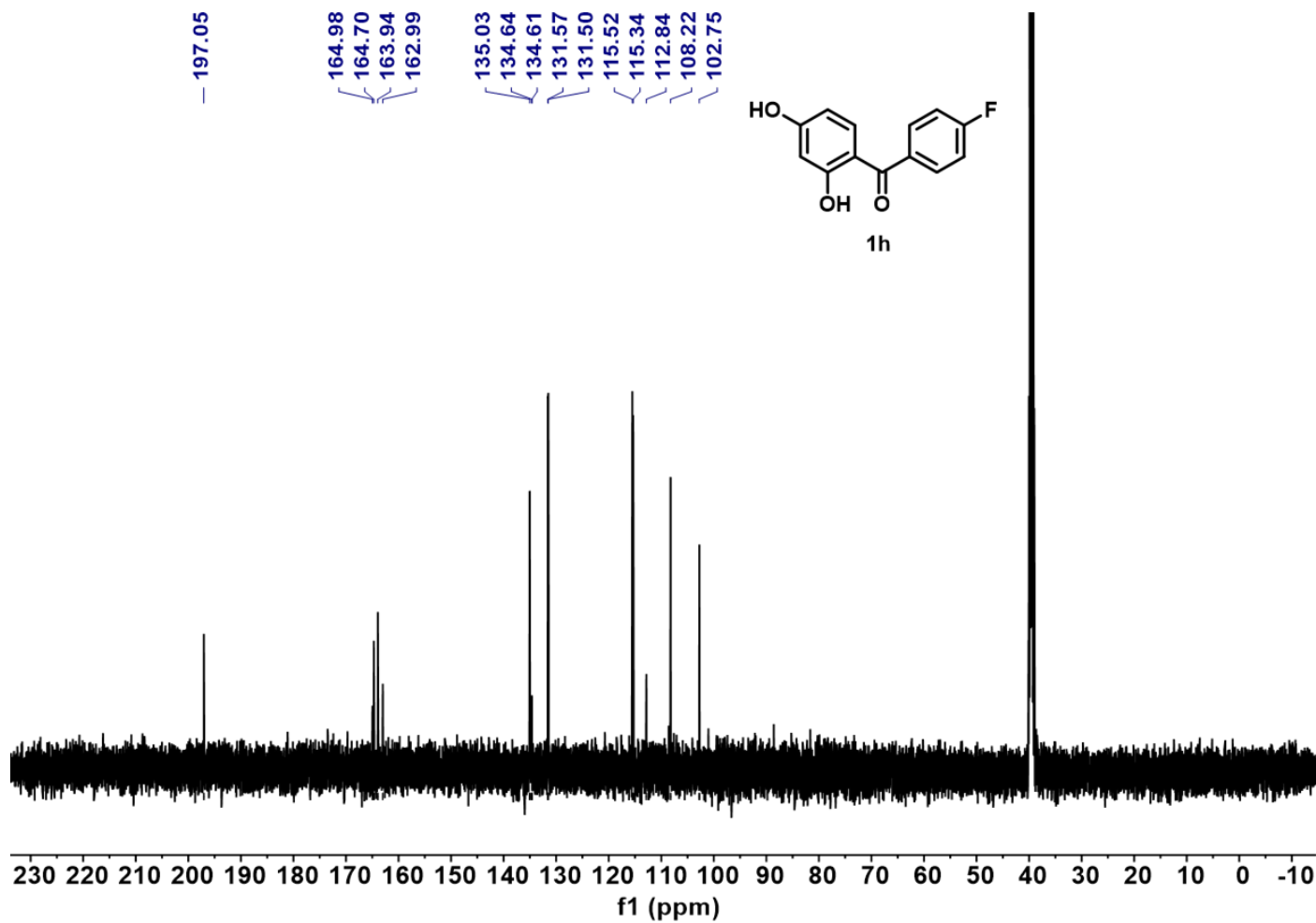
Supplementary Fig. 62 | HPLC-DAD/ESI-MS analysis of the glycosylation of 1ag by purified MiCGT variants. (a) HPLC chromatogram of **1ag** and product **2ag**; (b) Typical positive ion MS spectra for **2ag**; (c) Typical negative ion MS spectra for **2ag**; **1ag** MW: 94; **2ag** and **3ag** MW: 256.



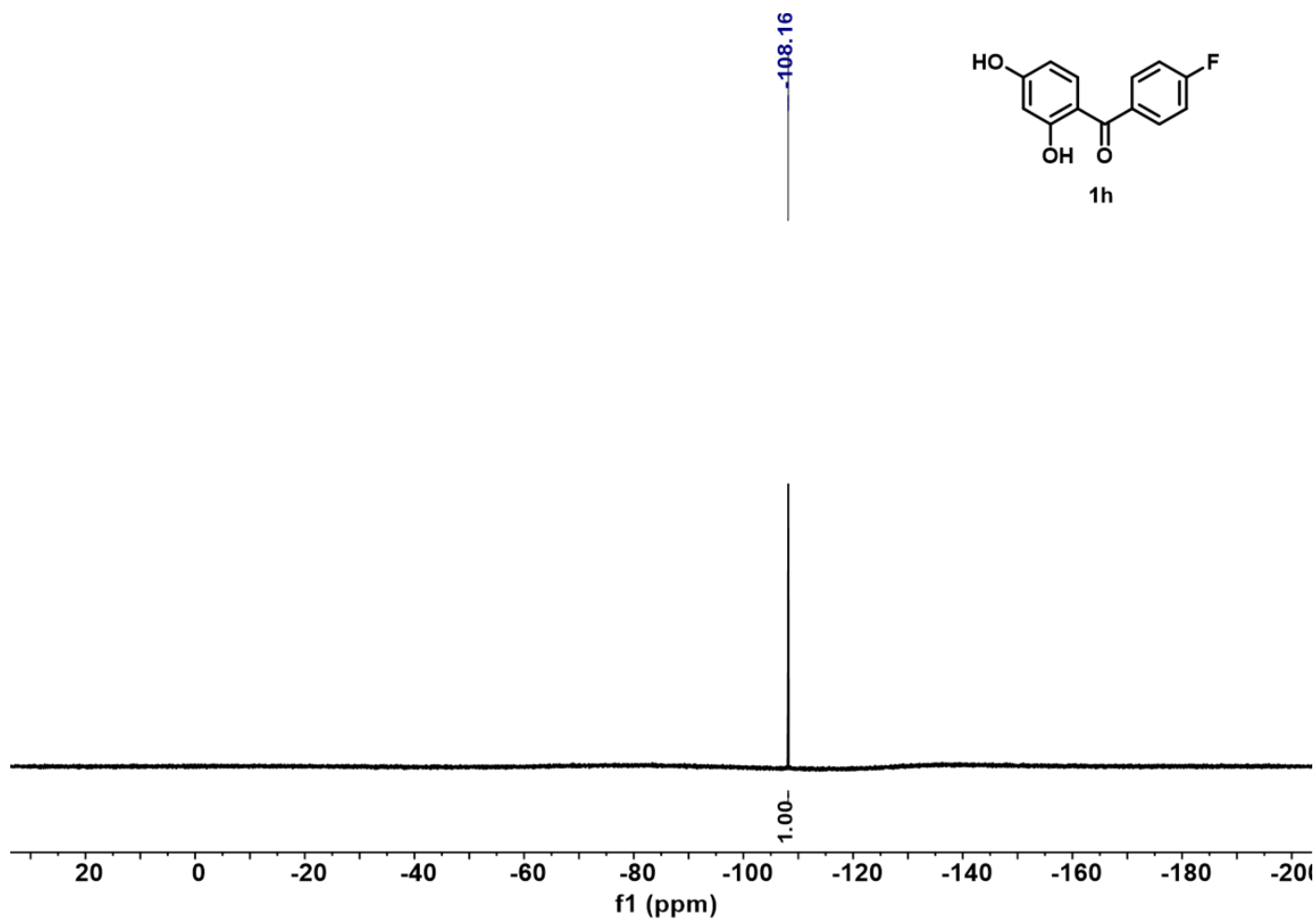
Supplementary Fig. 63 | HPLC-DAD/ESI-MS analysis of MiCGT enzymatic product using 1ah as an aglycon acceptor. (a) HPLC chromatogram of 1ah and enzyme product 2ah and 3ah; (b) Typical negative ion MS spectra for 2ah; (c) Typical negative ion MS/MS for 2ah; 1ah MW: 138; 2ah and 3ah MW: 300.



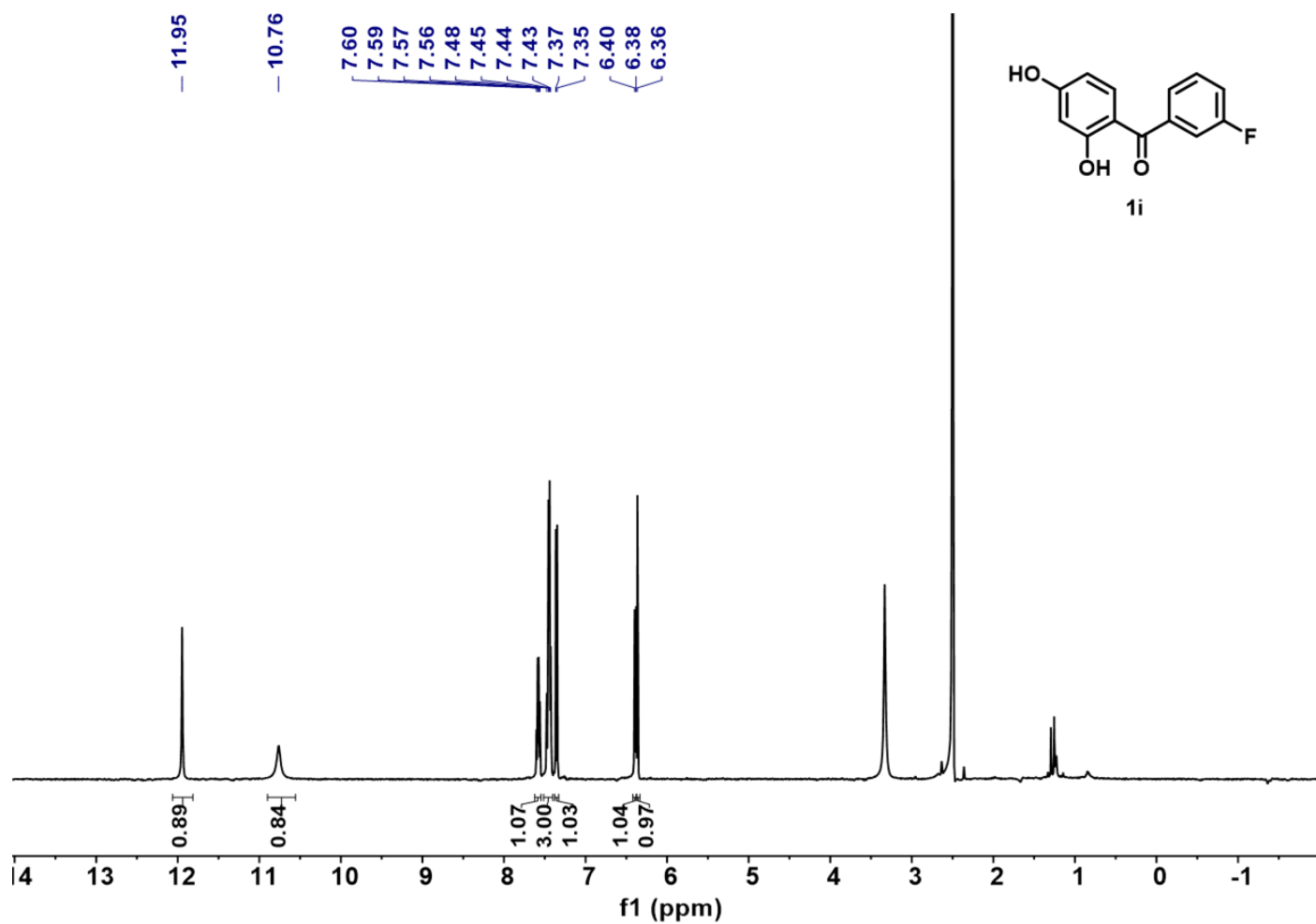
Supplementary Fig. 64 | ¹H NMR spectrum of **1h** (500 MHz, DMSO-*d*₆).



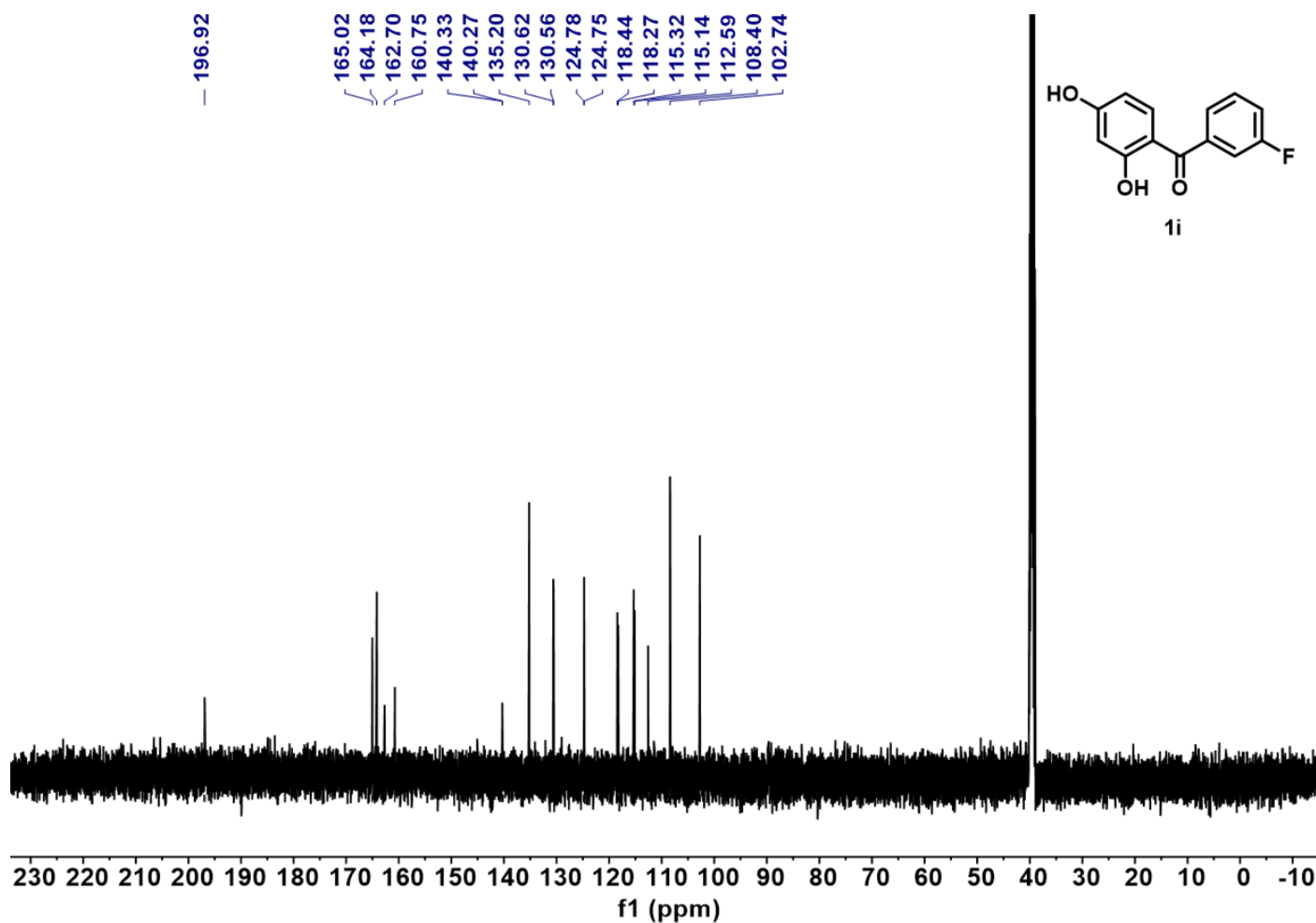
Supplementary Fig. 65 | ¹³C NMR spectrum of **1h** (126 MHz, DMSO-*d*₆).



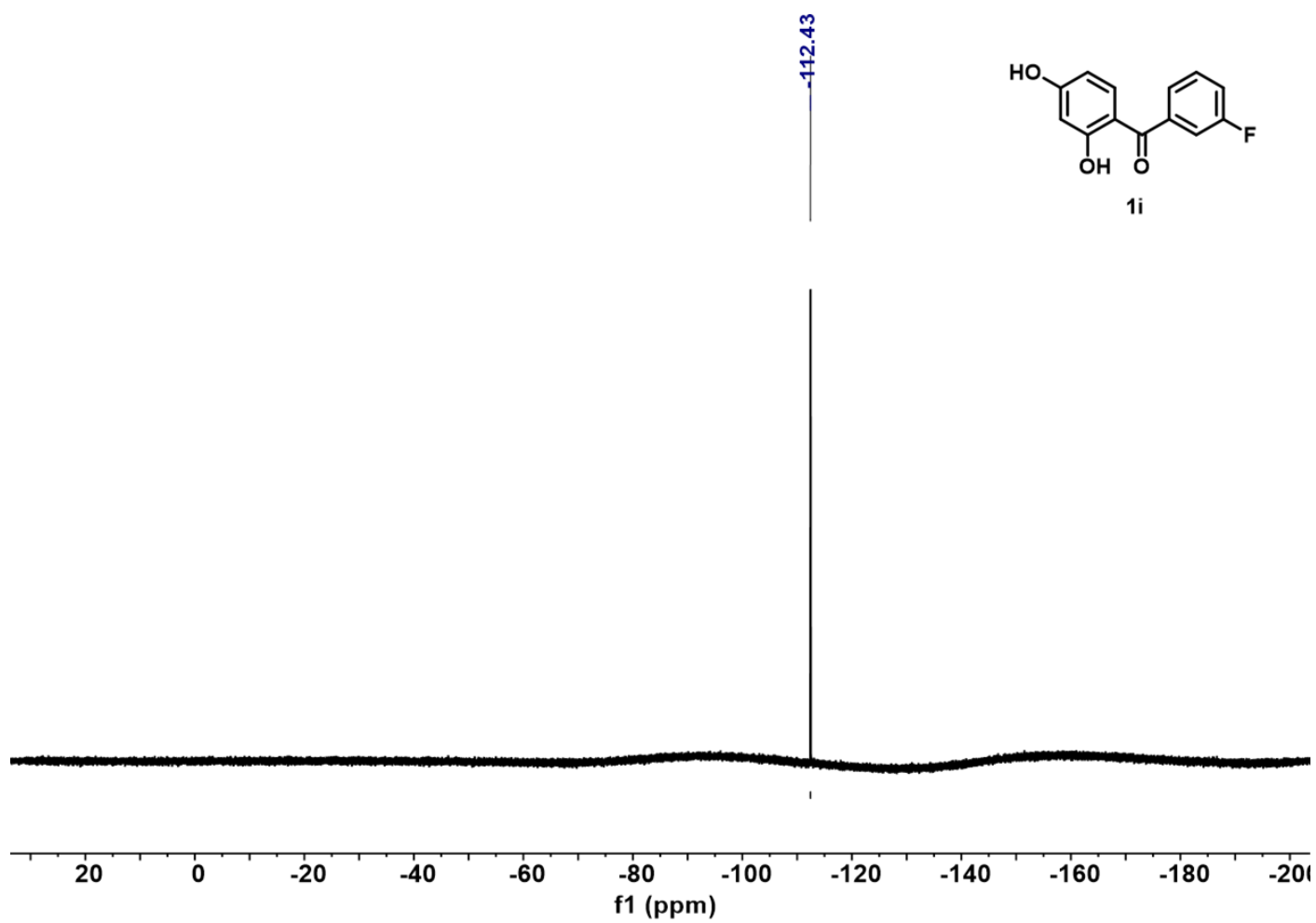
Supplementary Fig. 66 | ^{19}F NMR spectrum of **1h** (376 MHz, $\text{DMSO}-d_6$).



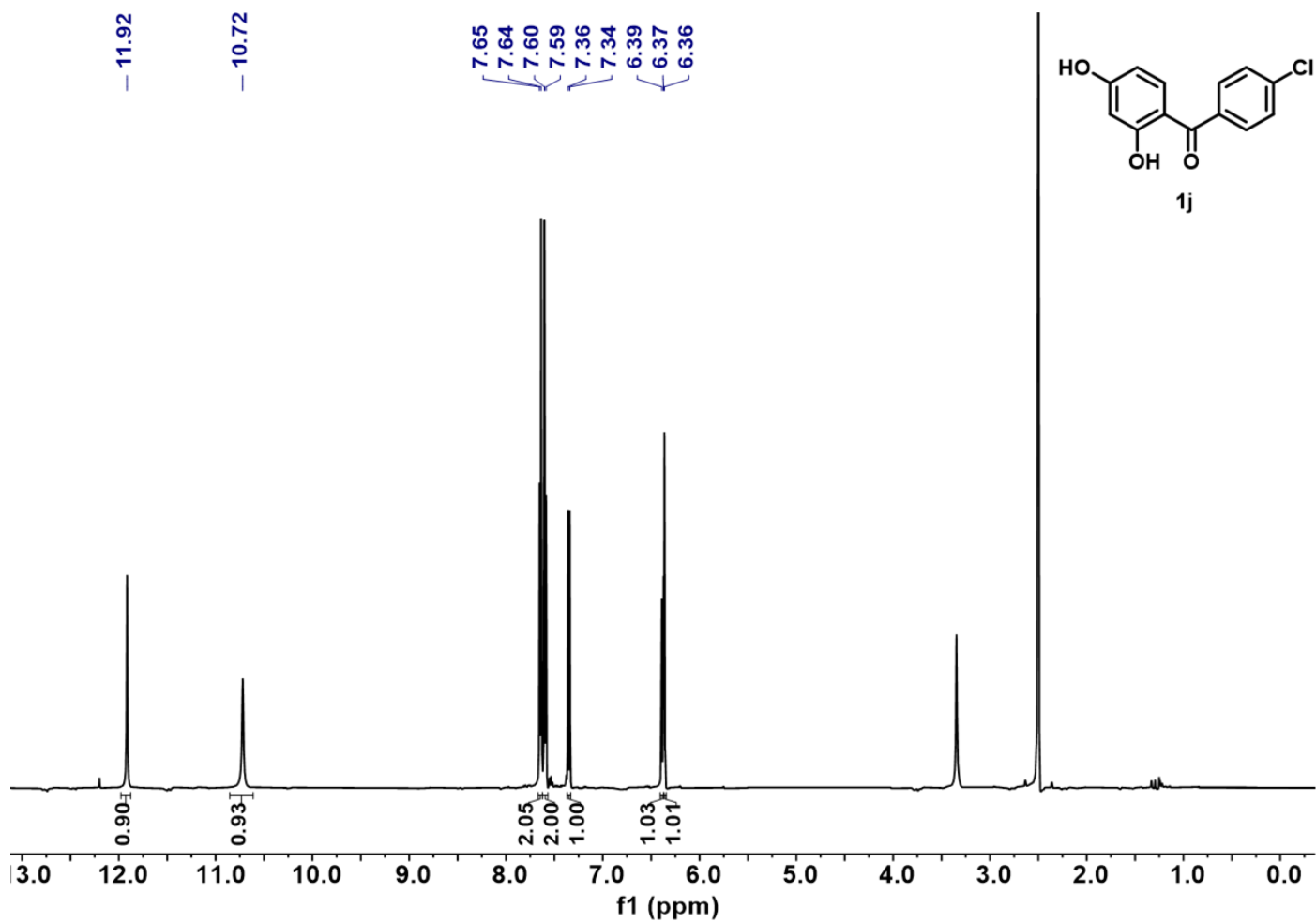
Supplementary Fig. 67 | ¹H NMR spectrum of **1i** (500 MHz, DMSO-*d*₆).



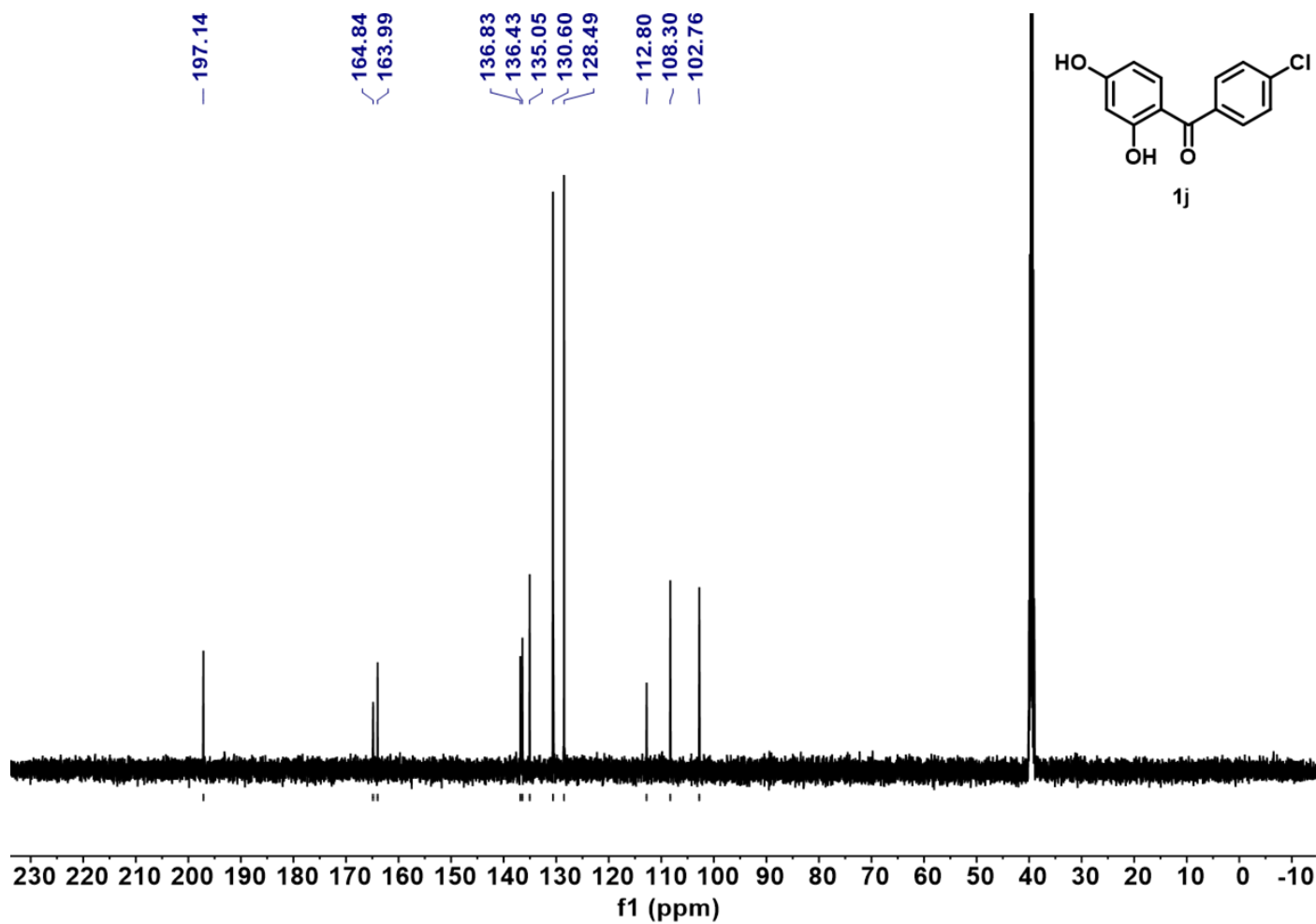
Supplementary Fig. 68 | ¹³C NMR spectrum of **1i** (126 MHz, DMSO-*d*₆).



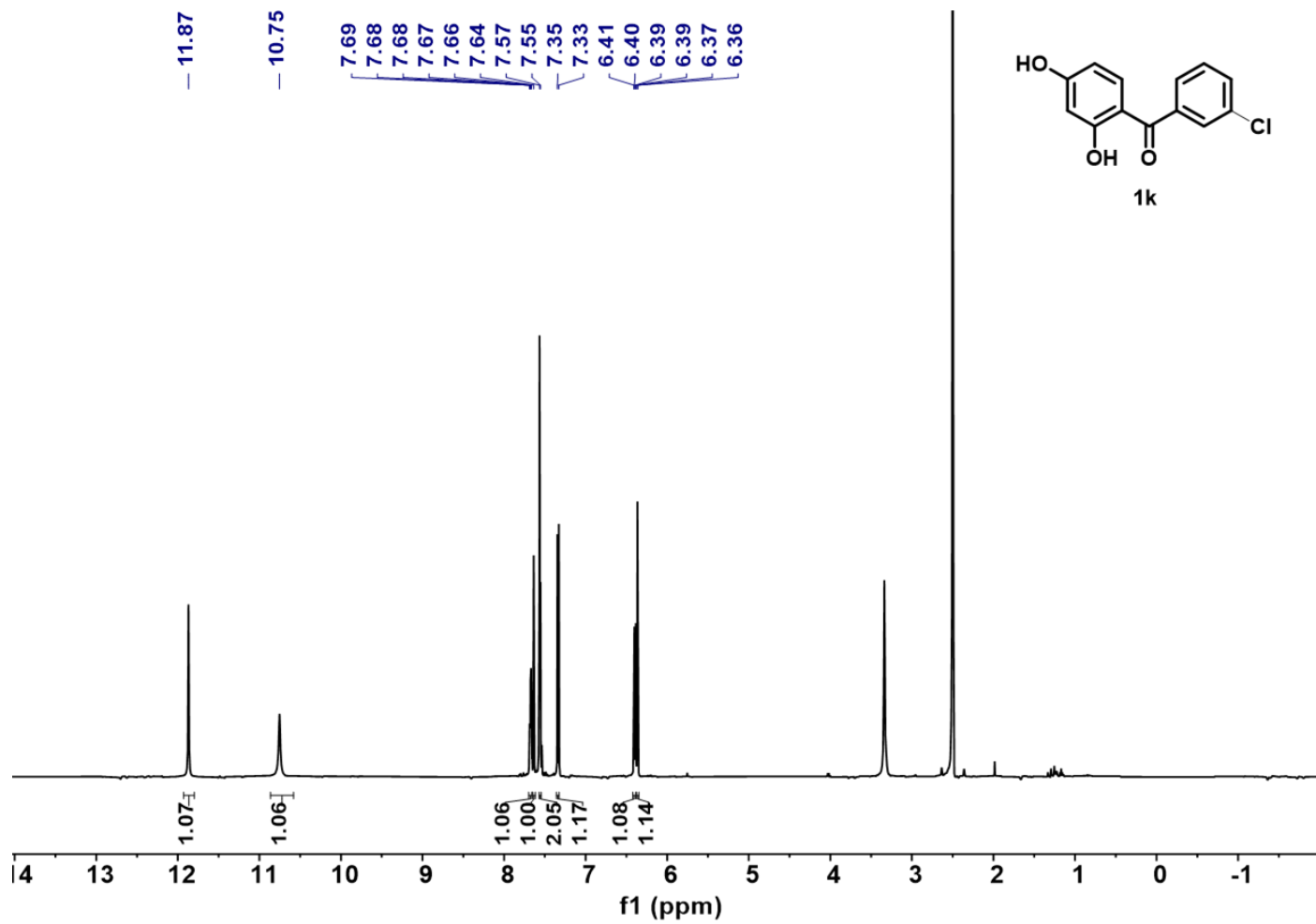
Supplementary Fig. 69 | ^{19}F NMR spectrum of **1i** (376 MHz, $\text{DMSO}-d_6$).



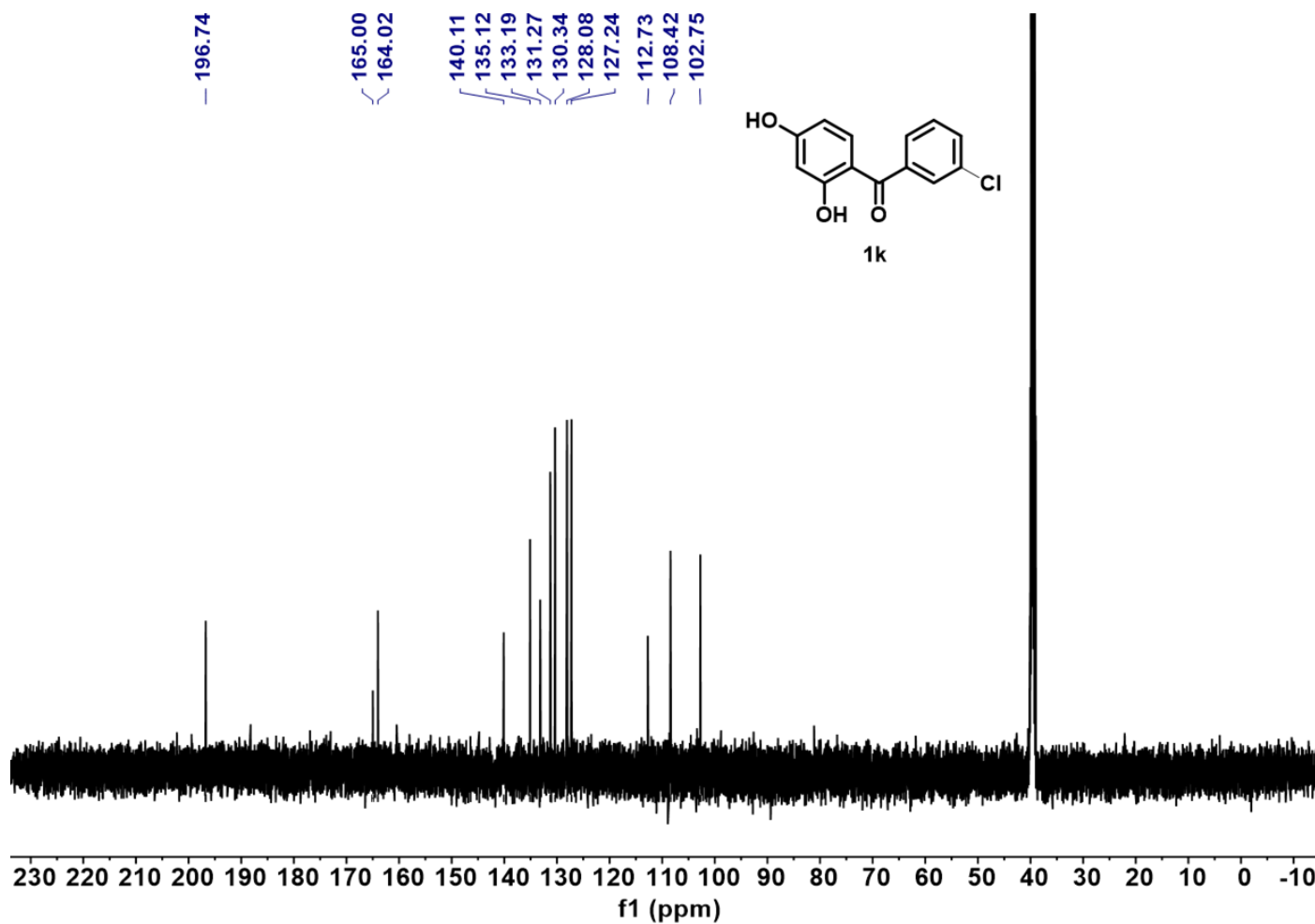
Supplementary Fig. 70 | ¹H NMR spectrum of **1j** (500 MHz, DMSO-*d*₆).



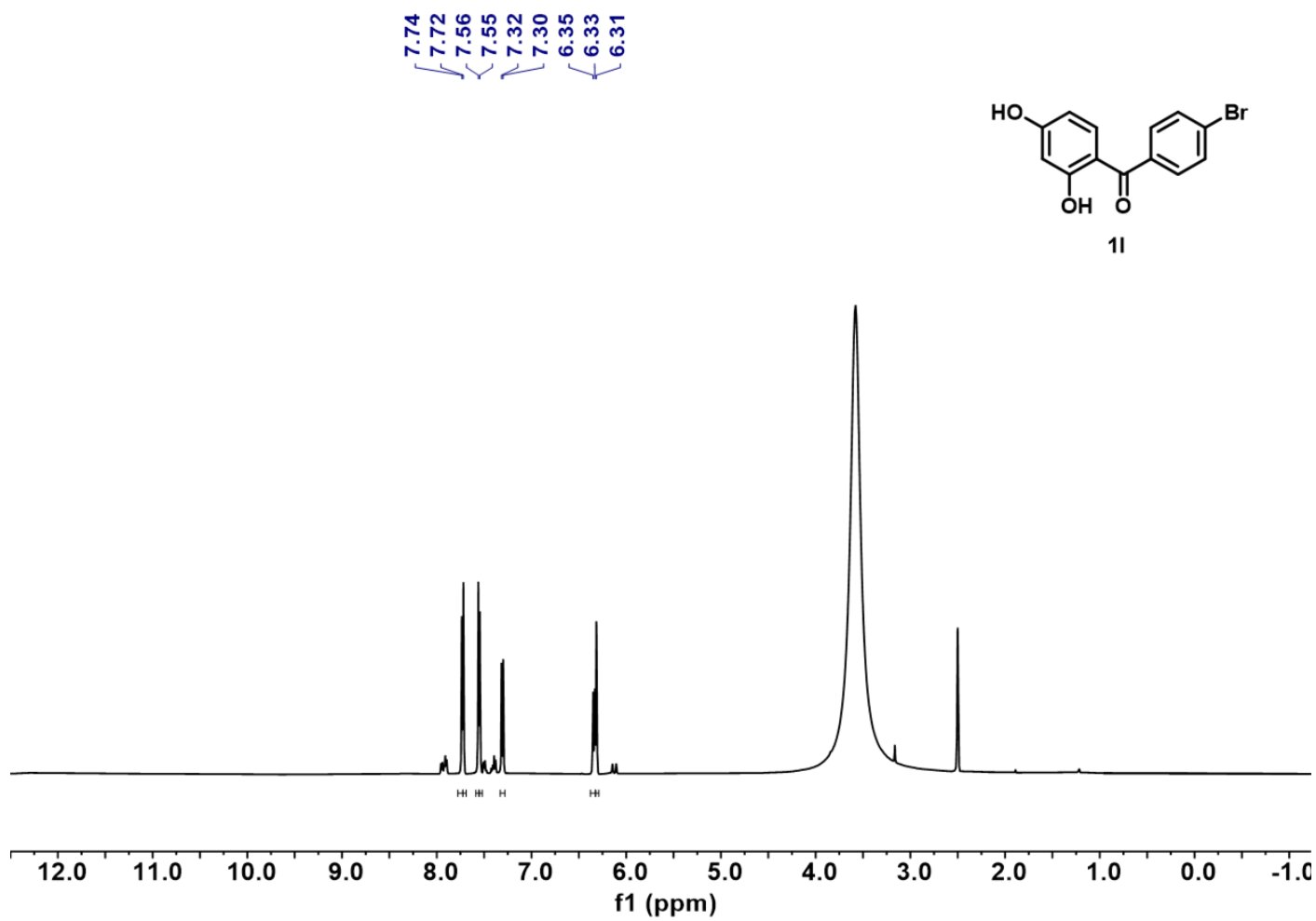
Supplementary Fig. 71 | ¹³C NMR spectrum of **1j** (126 MHz, DMSO-*d*₆).



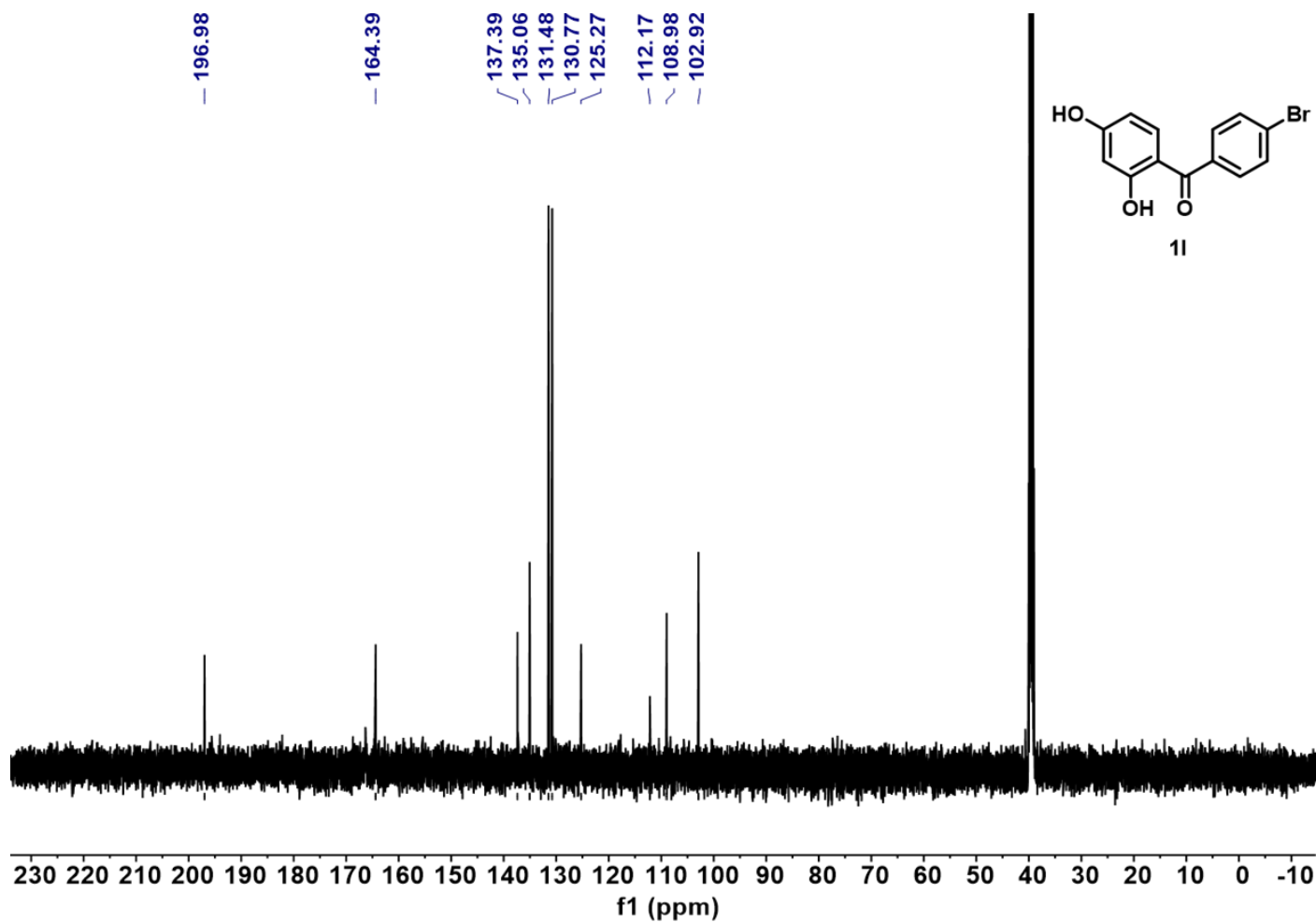
Supplementary Fig. 72 | ¹H NMR spectrum of **1k** (500 MHz, DMSO-*d*₆).



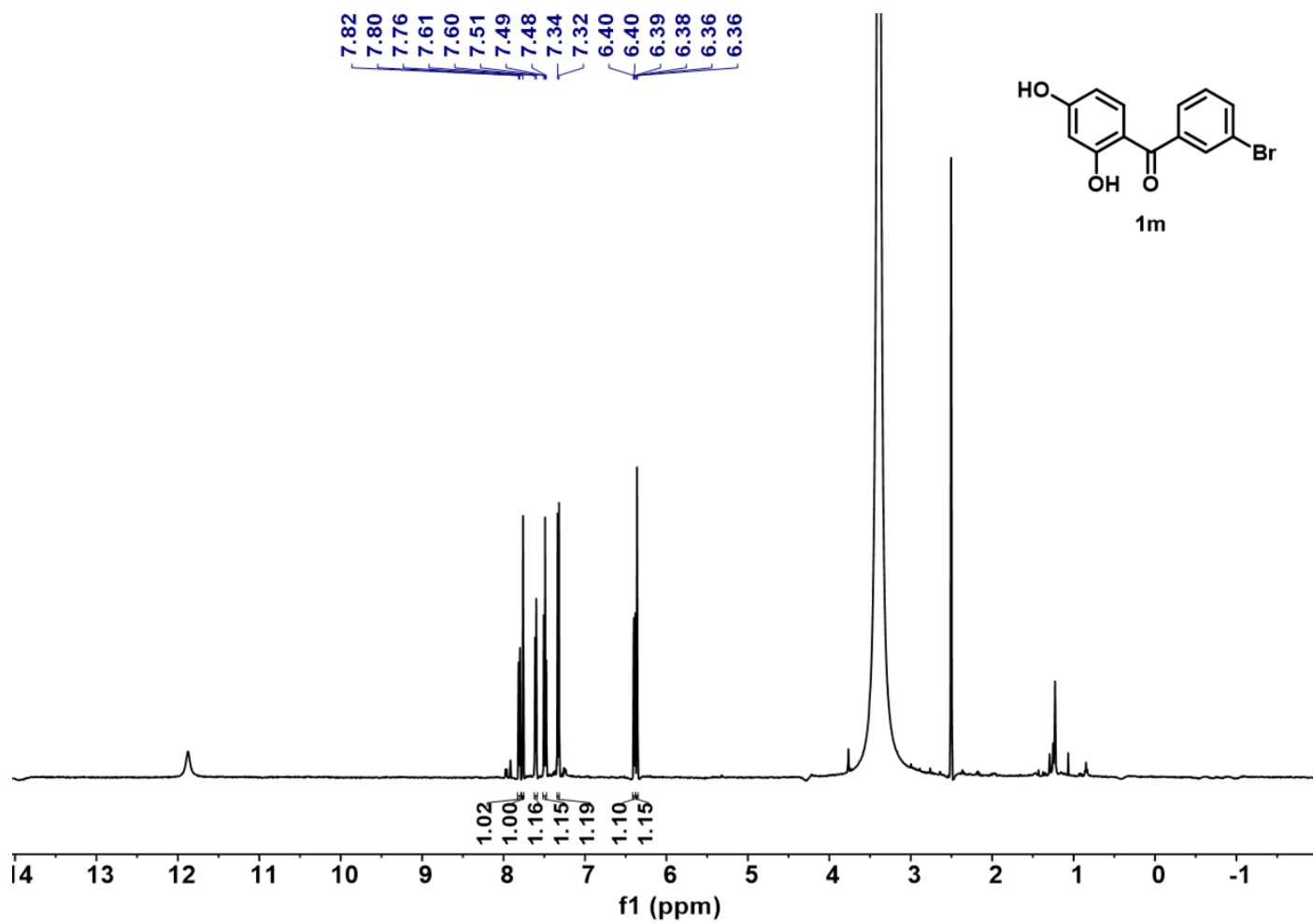
Supplementary Fig. 73 | ^{13}C NMR spectrum of **1k** (126 MHz, $\text{DMSO}-d_6$).



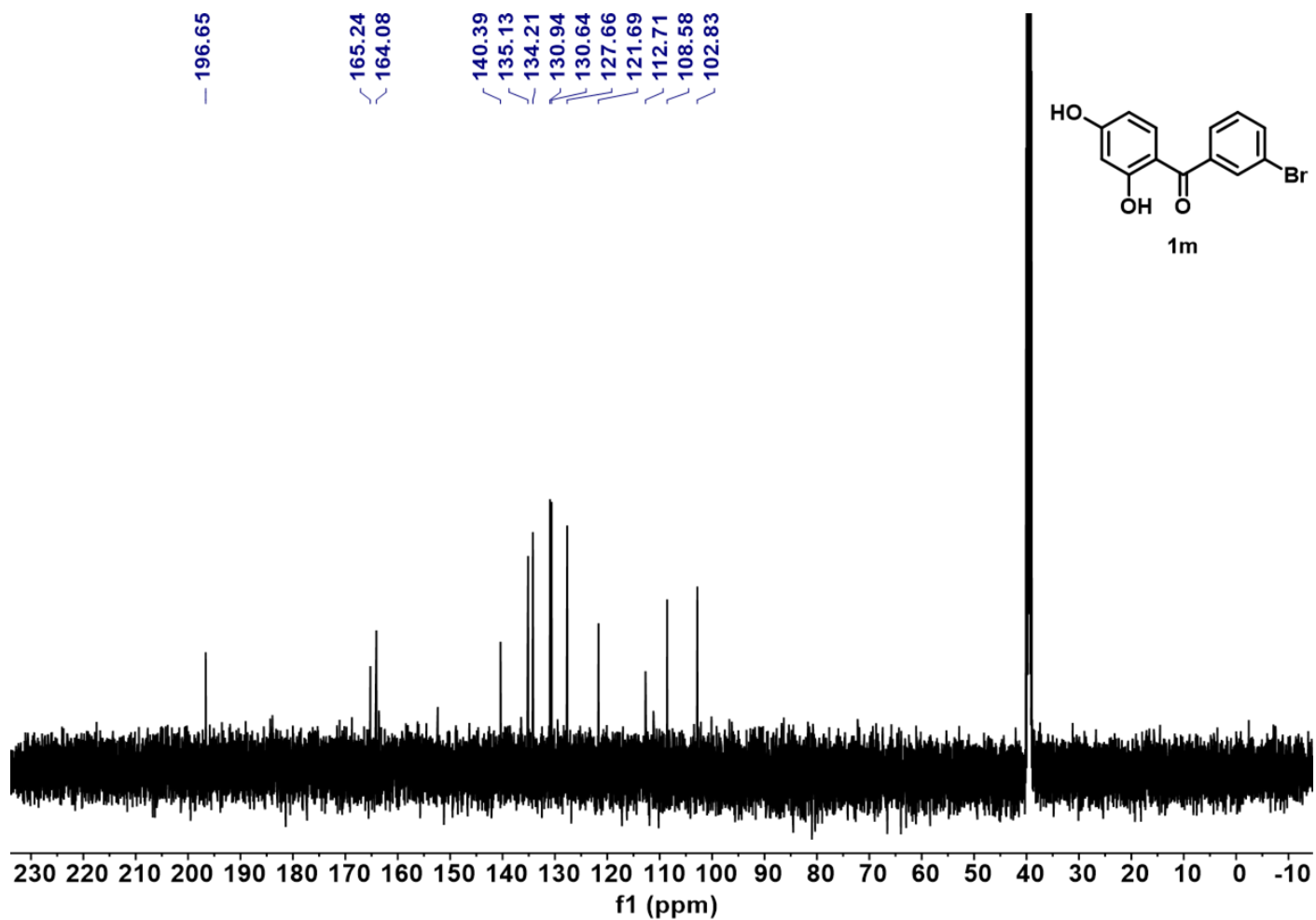
Supplementary Fig. 74 | ¹H NMR spectrum of **11** (500 MHz, DMSO-*d*₆).



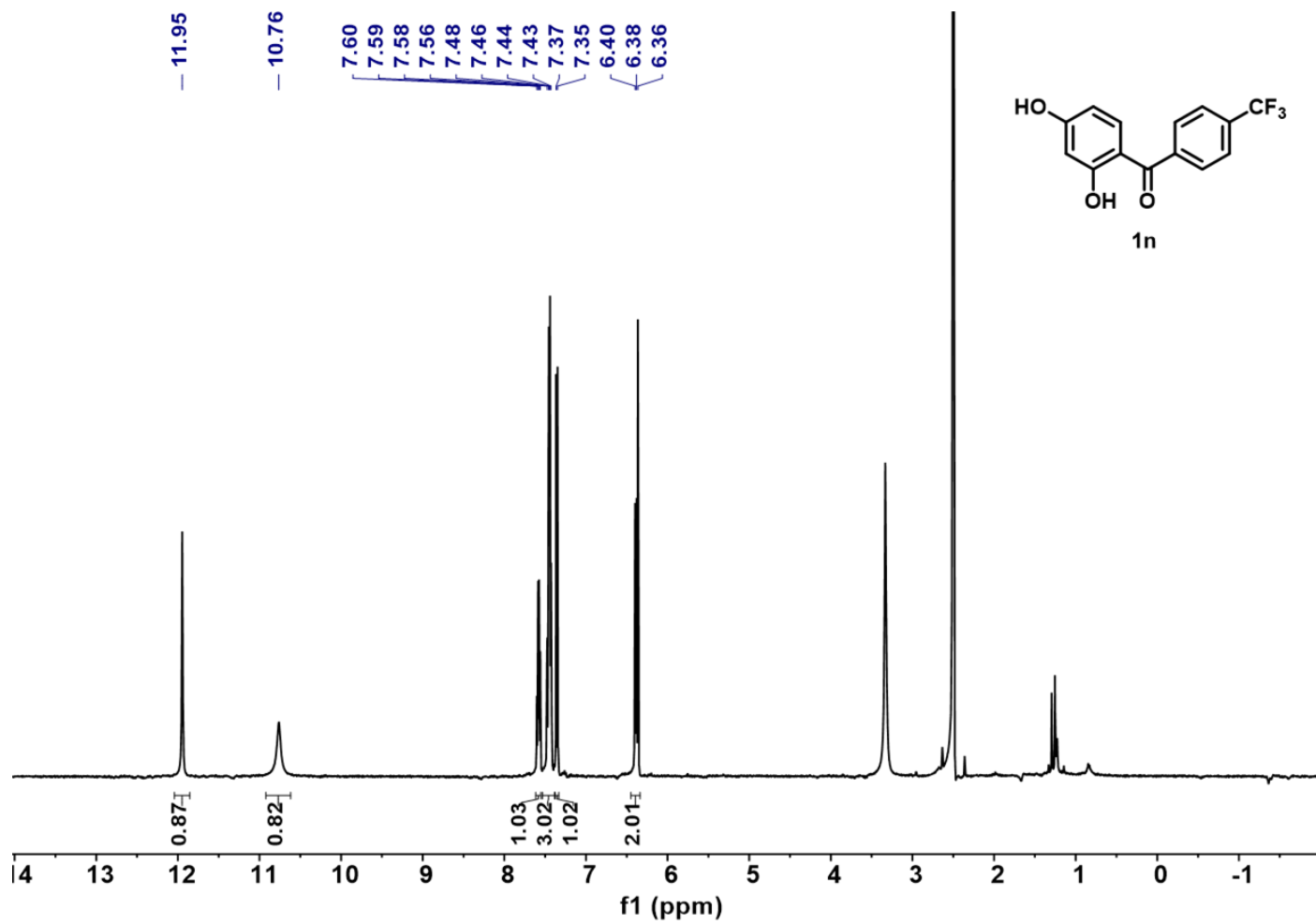
Supplementary Fig. 75 | ^{13}C NMR spectrum of **1I** (126 MHz, $\text{DMSO}-d_6$).



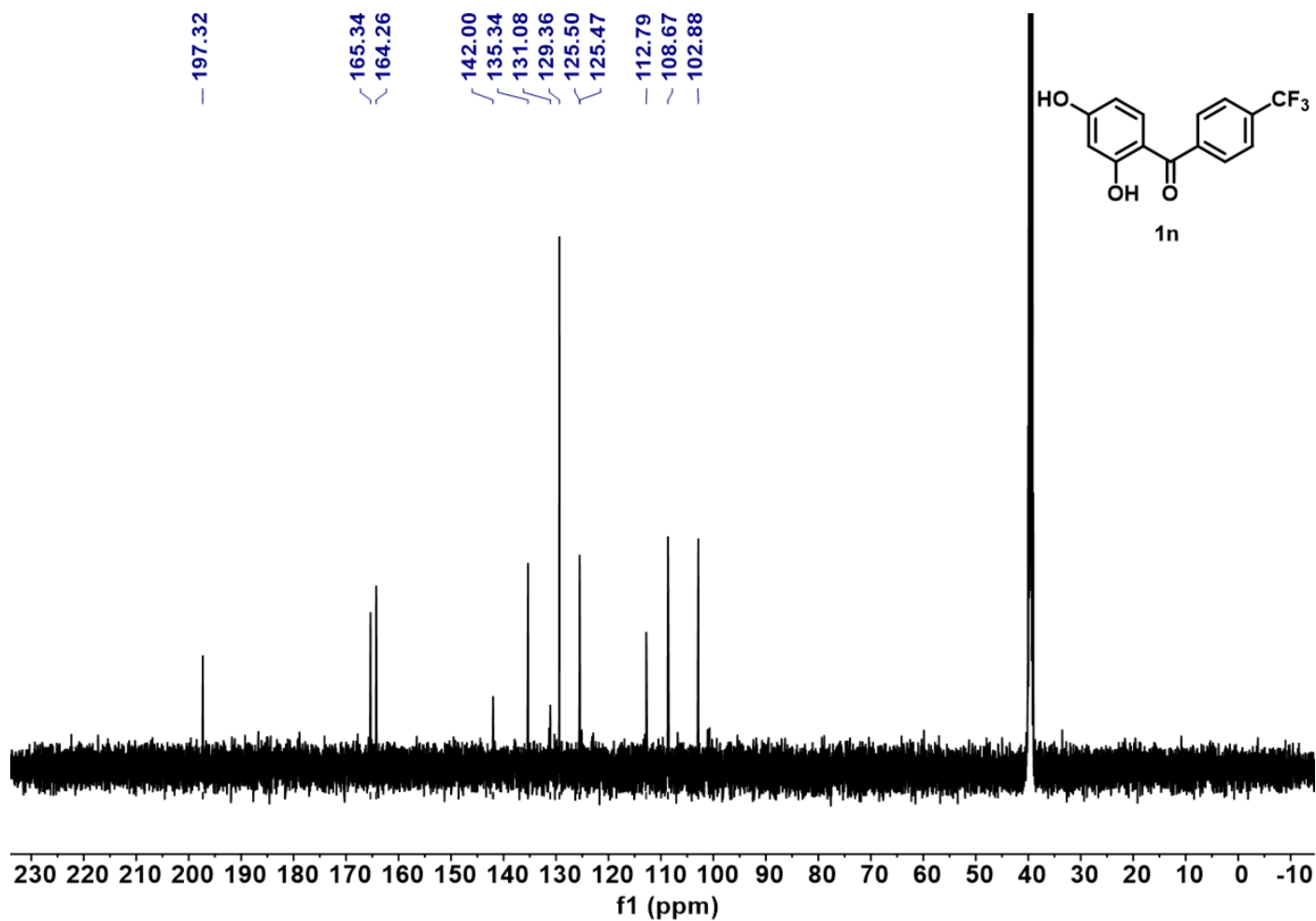
Supplementary Fig. 76 | ¹H NMR spectrum of **1m** (500 MHz, DMSO-*d*₆).



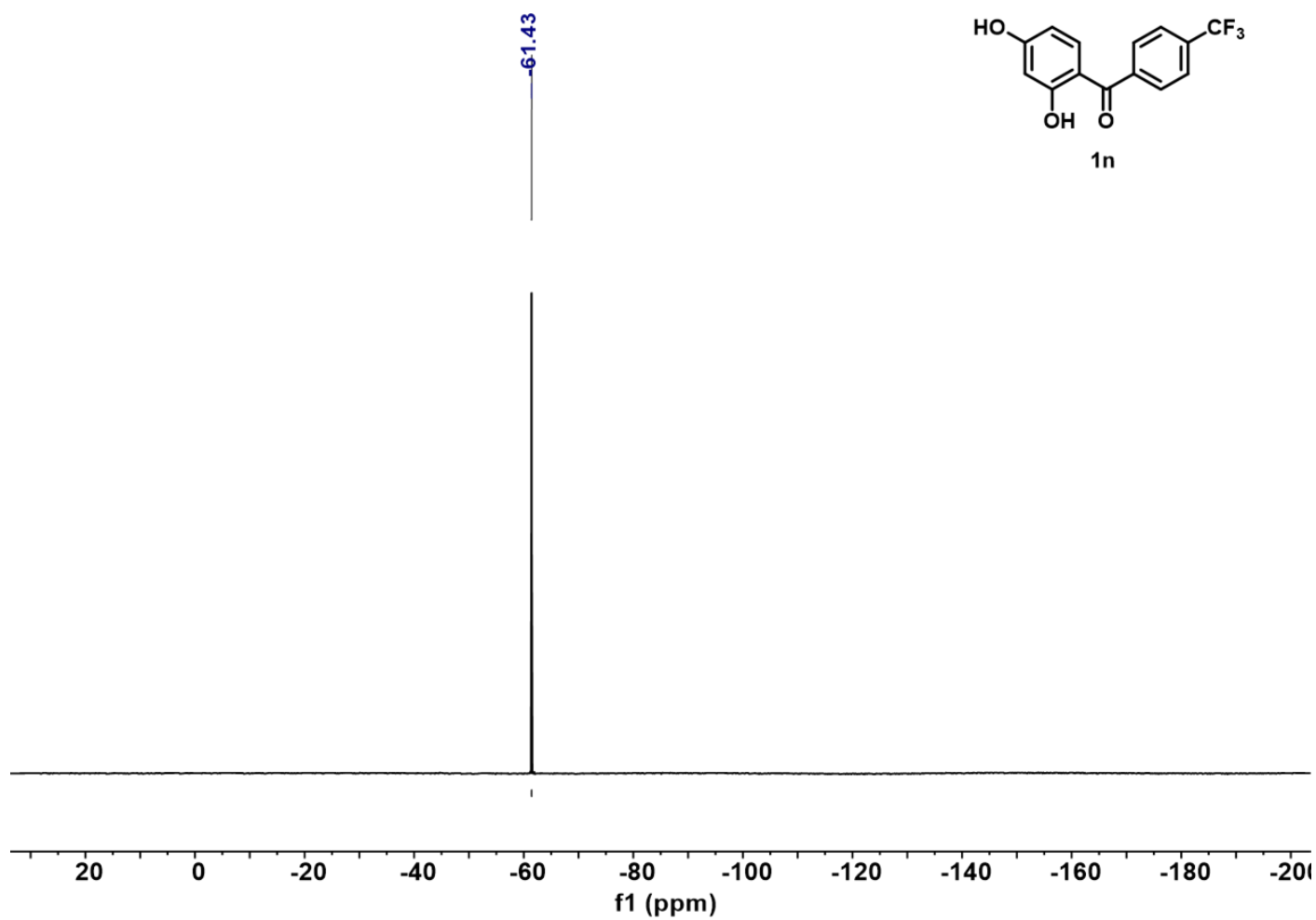
Supplementary Fig. 77 | ¹³C NMR spectrum of **1m** (126 MHz, D MSO-*d*₆).



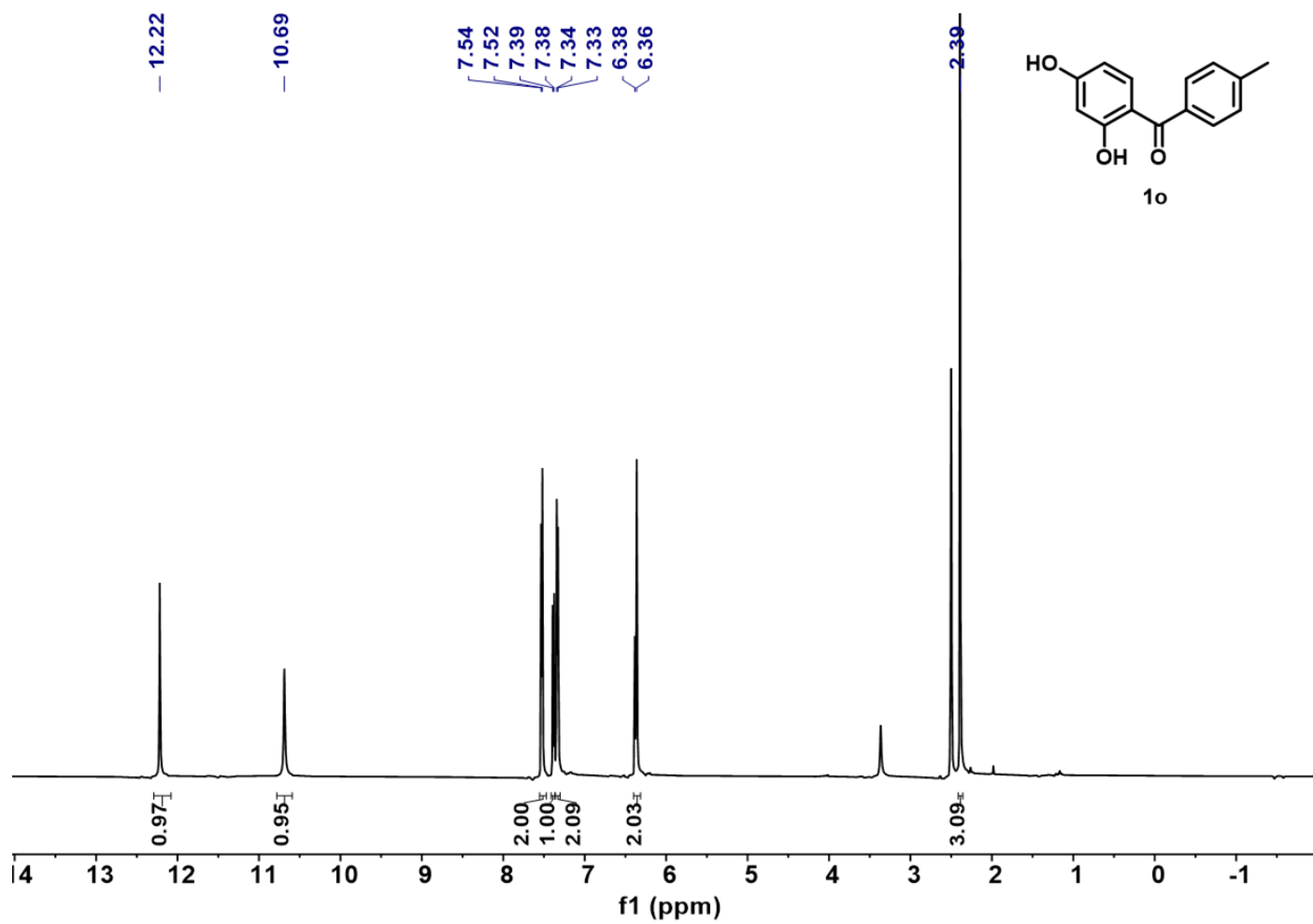
Supplementary Fig. 78 | ¹H NMR spectrum of **1n** (500 MHz, DMSO-*d*₆).



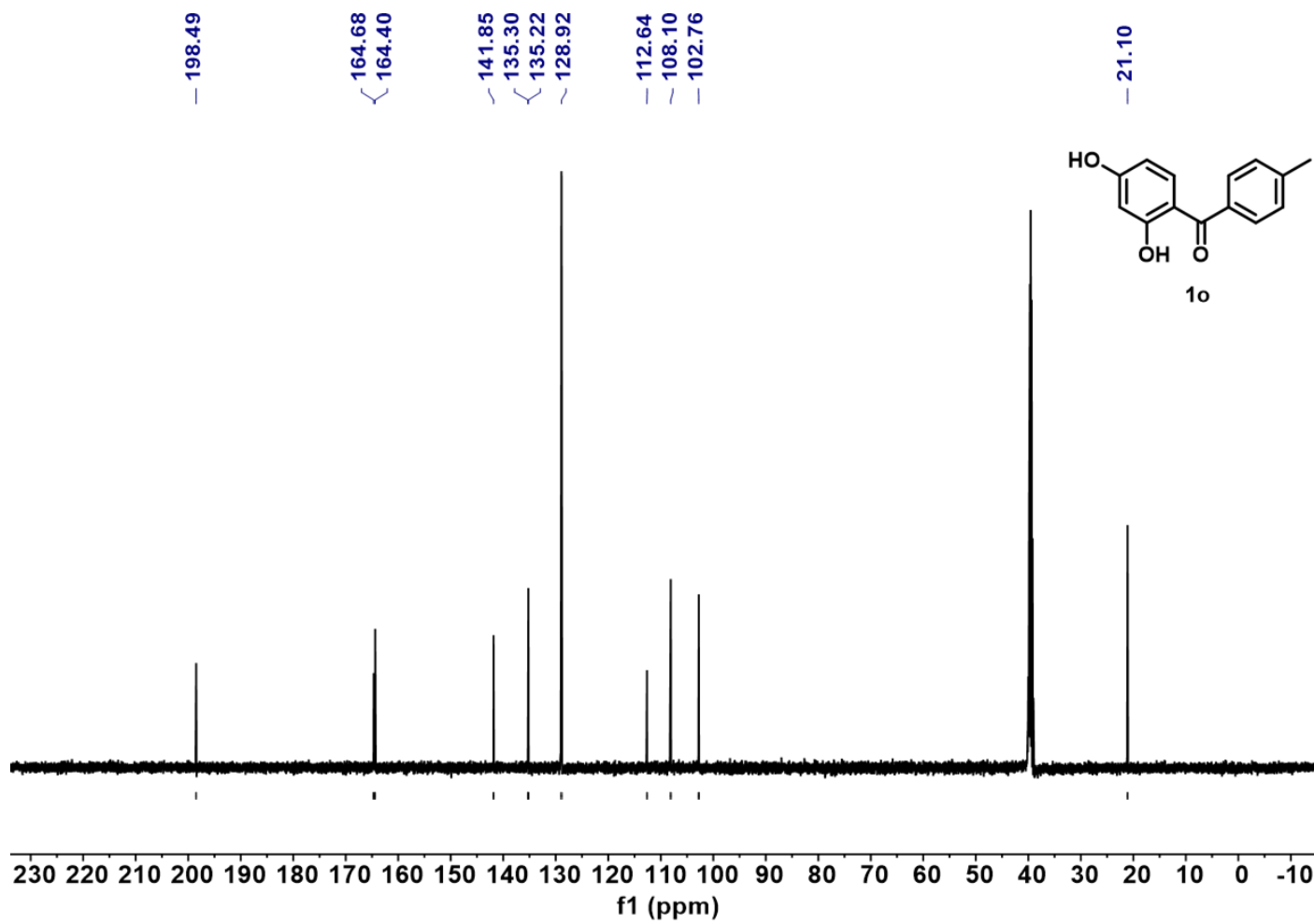
Supplementary Fig. 79 | ¹³C NMR spectrum of **1n** (126 MHz, D MSO-*d*₆).



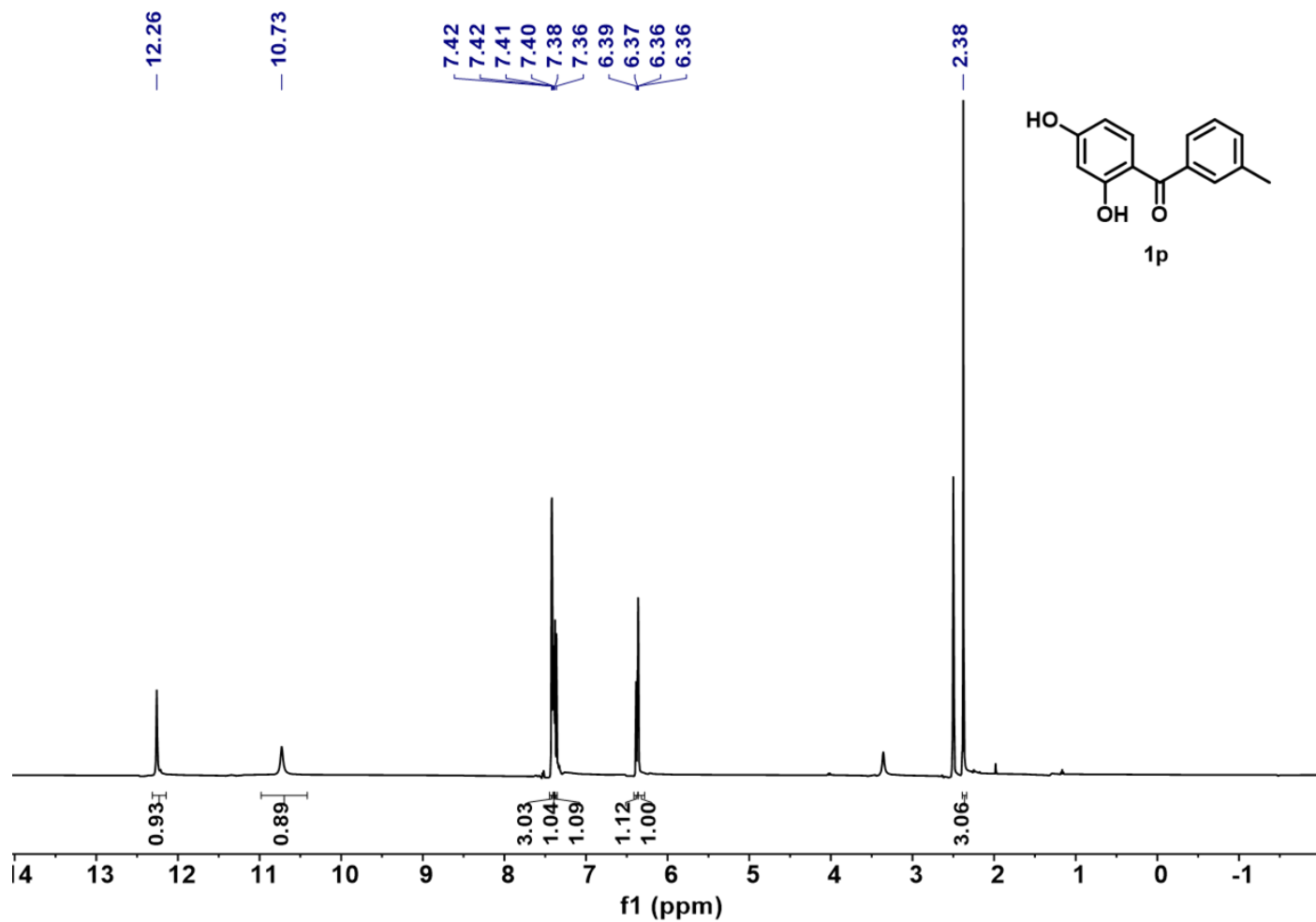
Supplementary Fig. 80 | ^{19}F NMR spectrum of **1n** (376 MHz, $\text{DMSO-}d_6$).



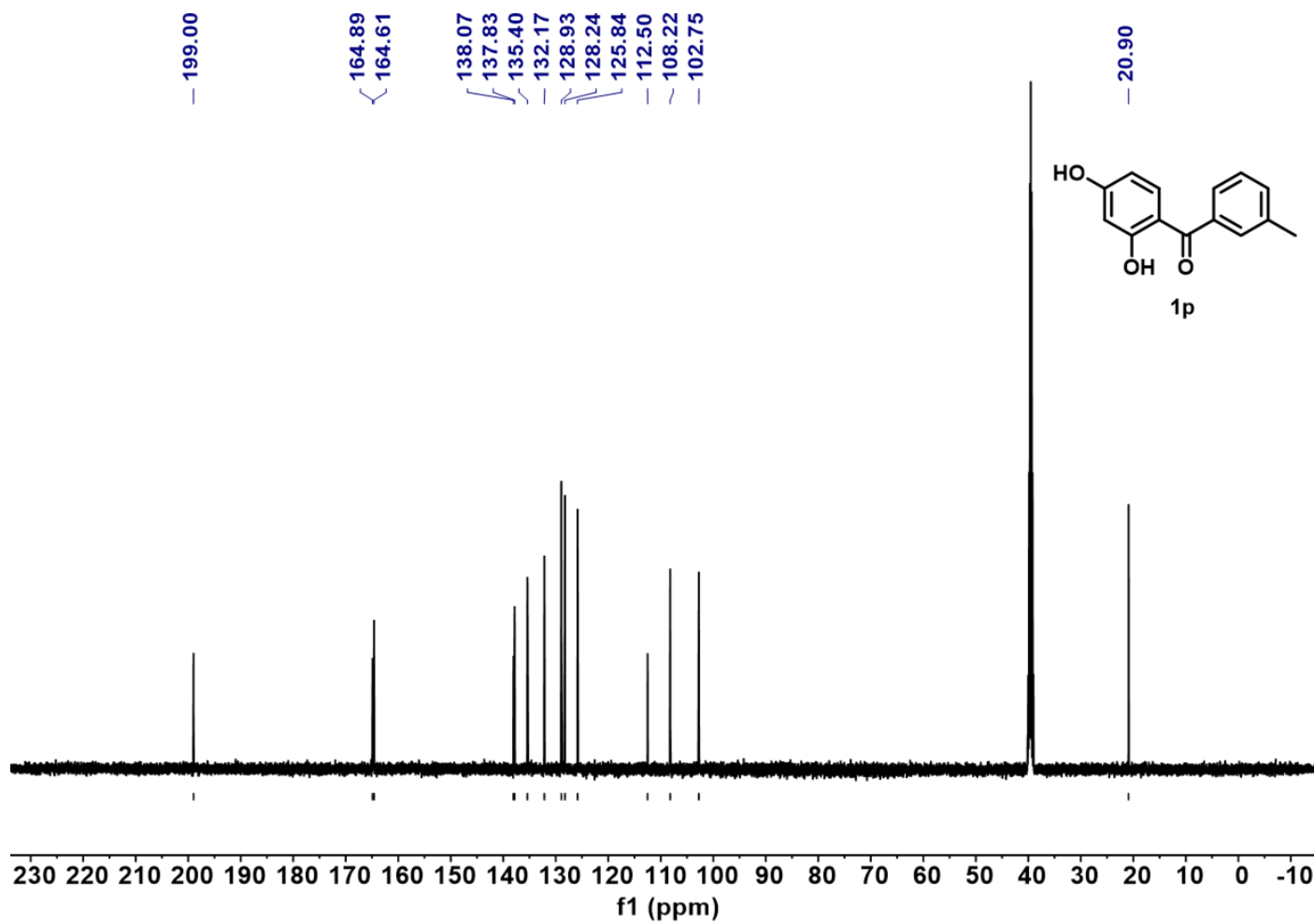
Supplementary Fig. 81 | ¹H NMR spectrum of **1o** (500 MHz, DMSO-*d*₆).



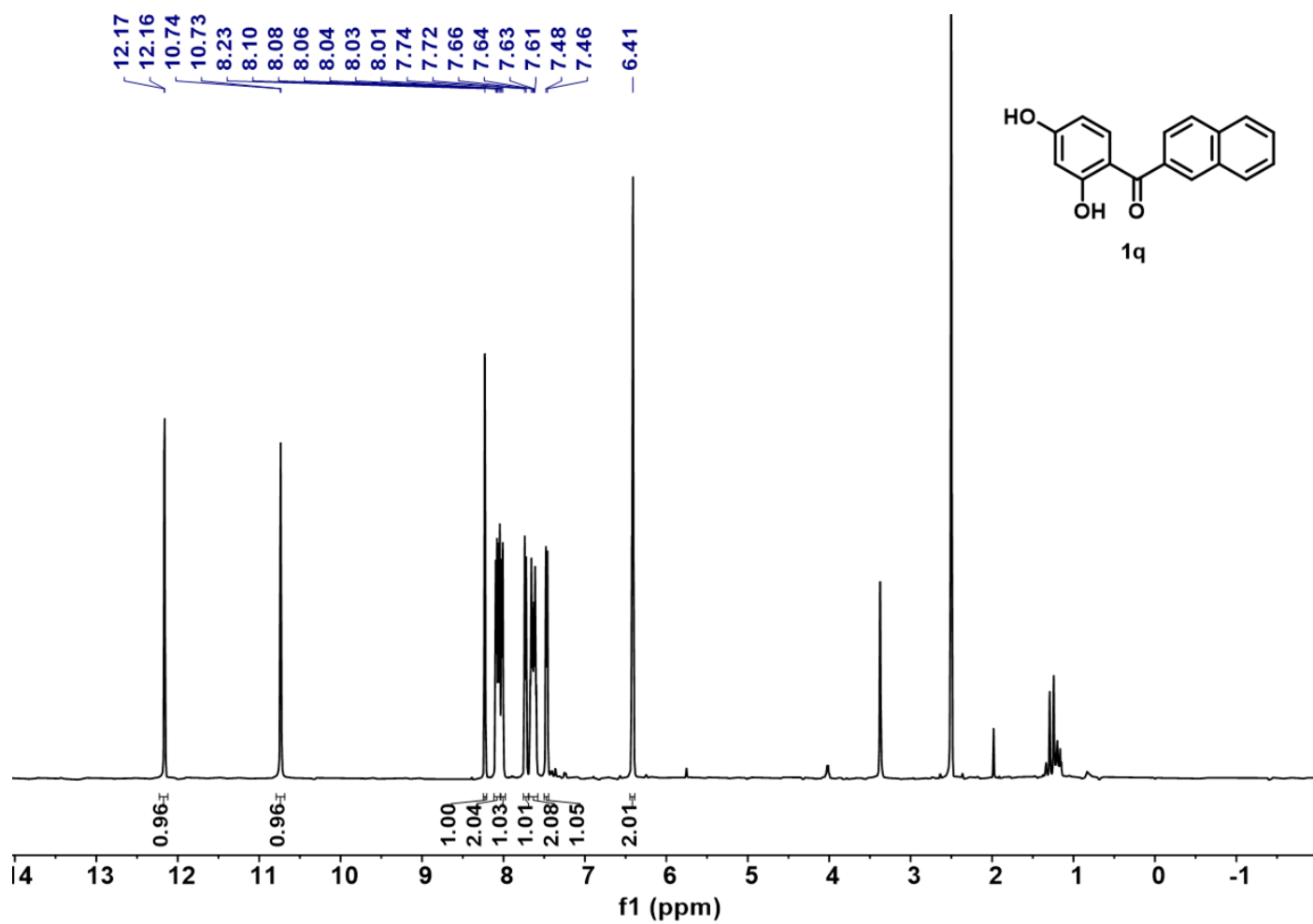
Supplementary Fig. 82 | ¹³C NMR spectrum of **1o** (126 MHz, DMSO-*d*₆).



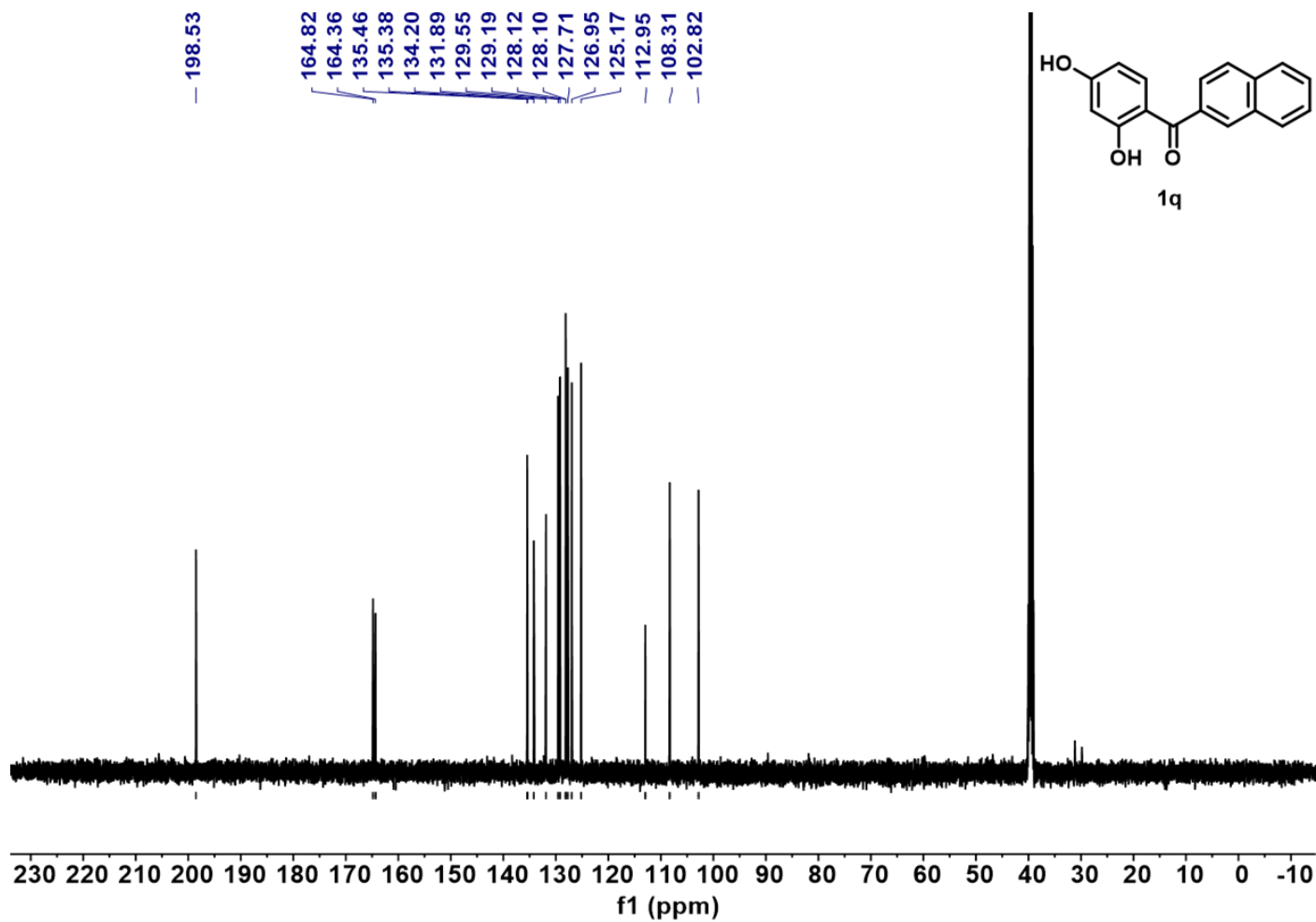
Supplementary Fig. 83 | ¹H NMR spectrum of **1p** (500 MHz, DMSO-*d*₆).



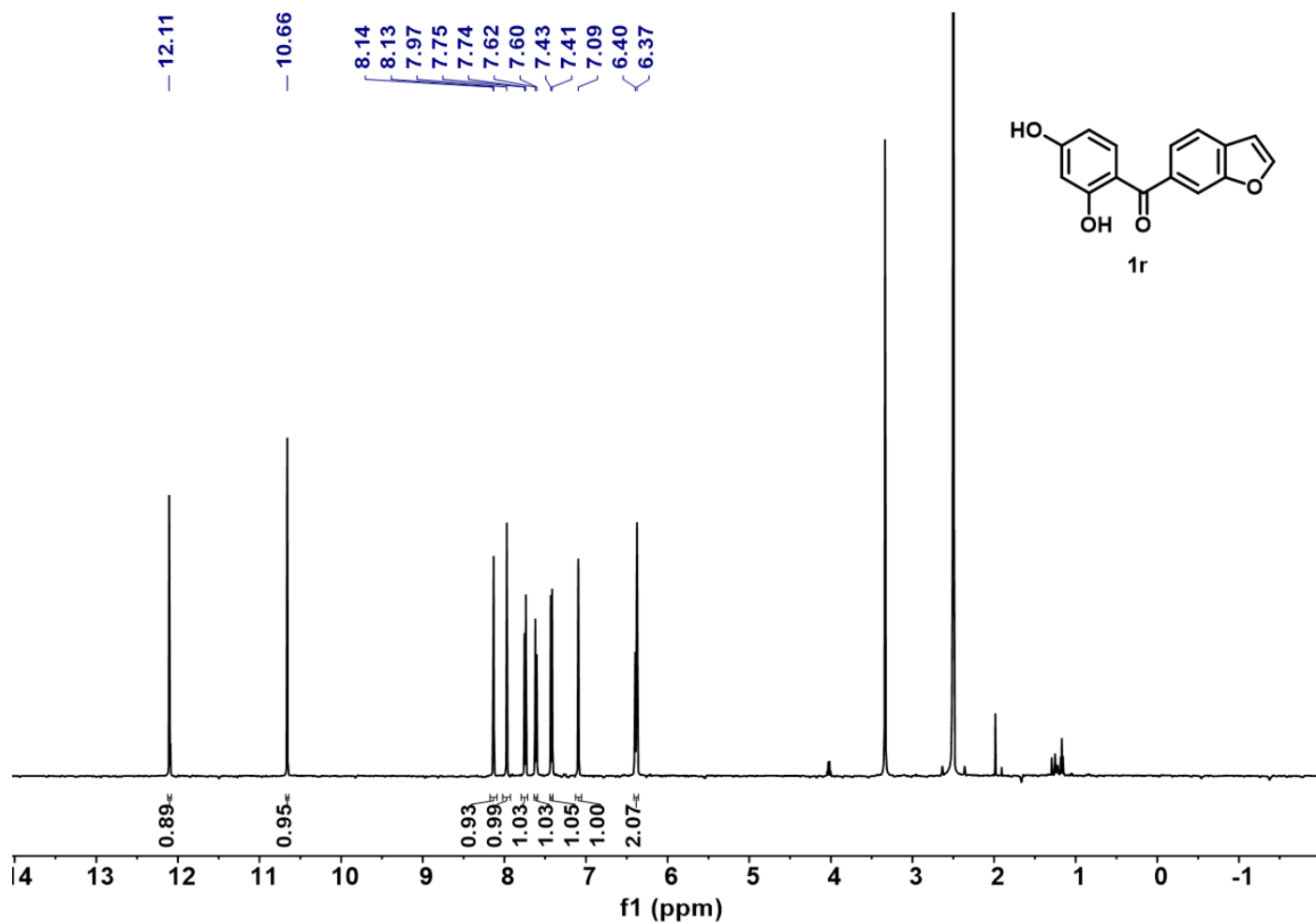
Supplementary Fig. 84 | ¹³C NMR spectrum of **1p** (126 MHz, D MSO-*d*₆).



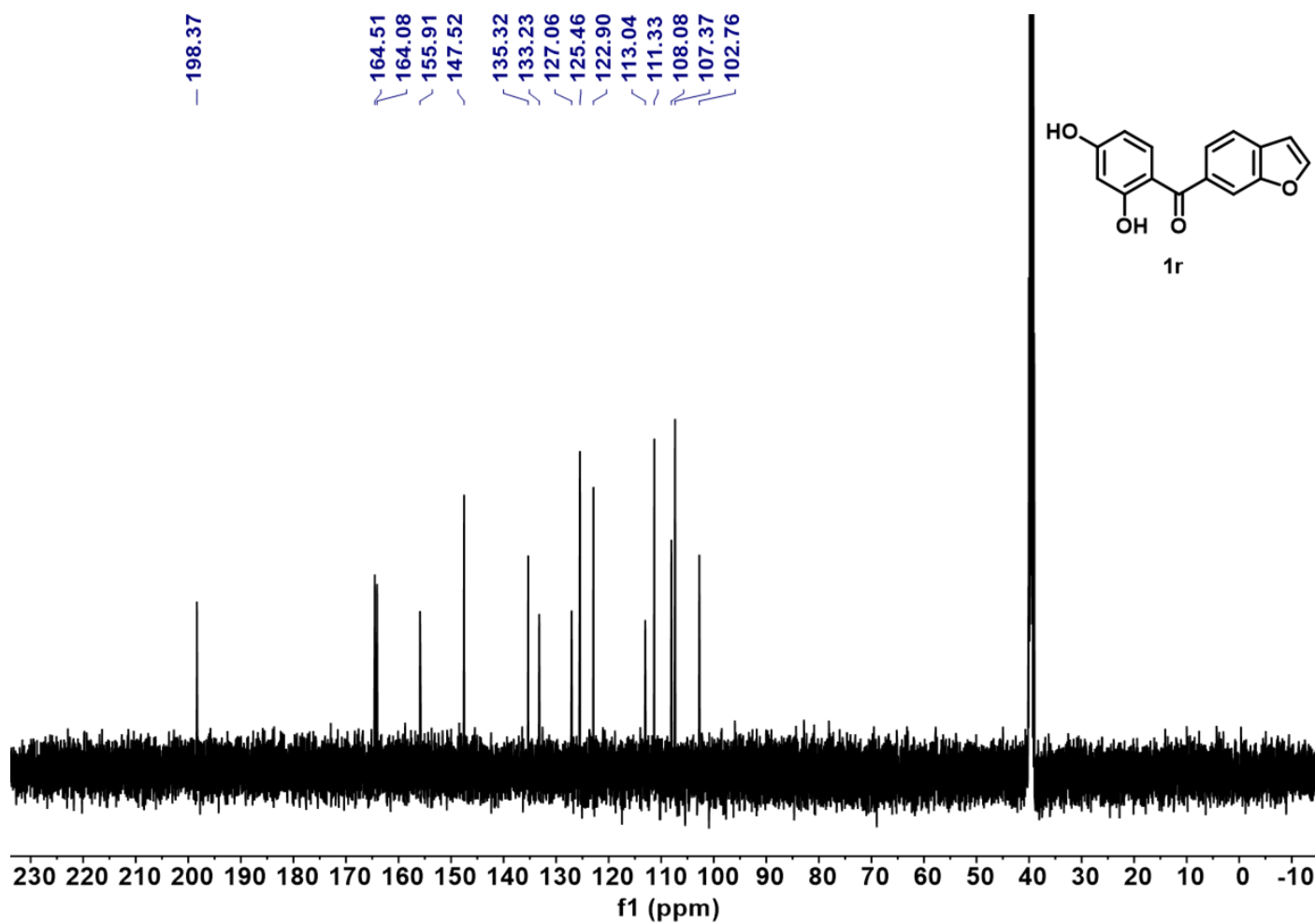
Supplementary Fig. 85 | ¹H NMR spectrum of **1q** (500 MHz, DMSO-*d*₆).



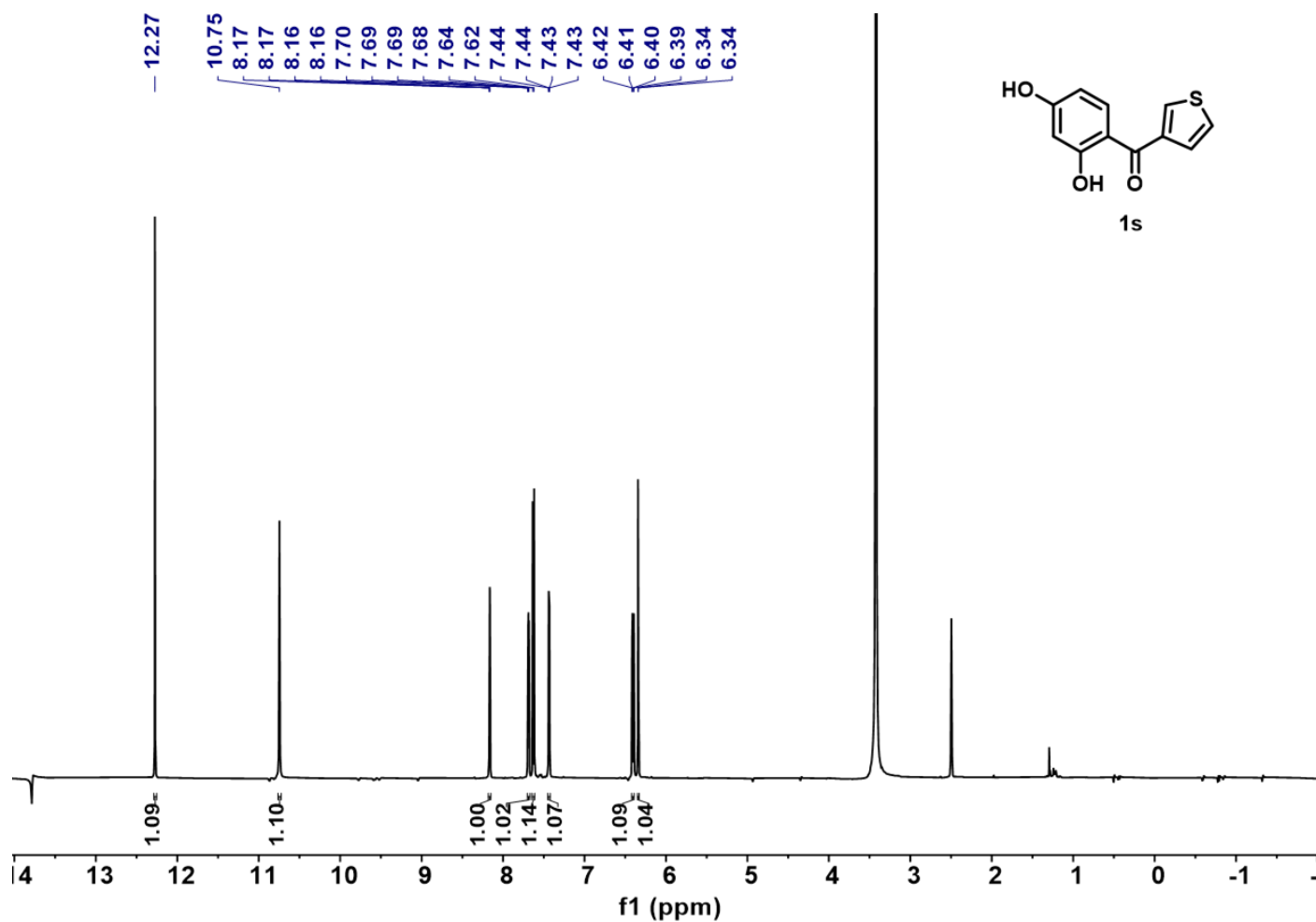
Supplementary Fig. 86 | ¹³C NMR spectrum of **1q** (126 MHz, D MSO-*d*₆).



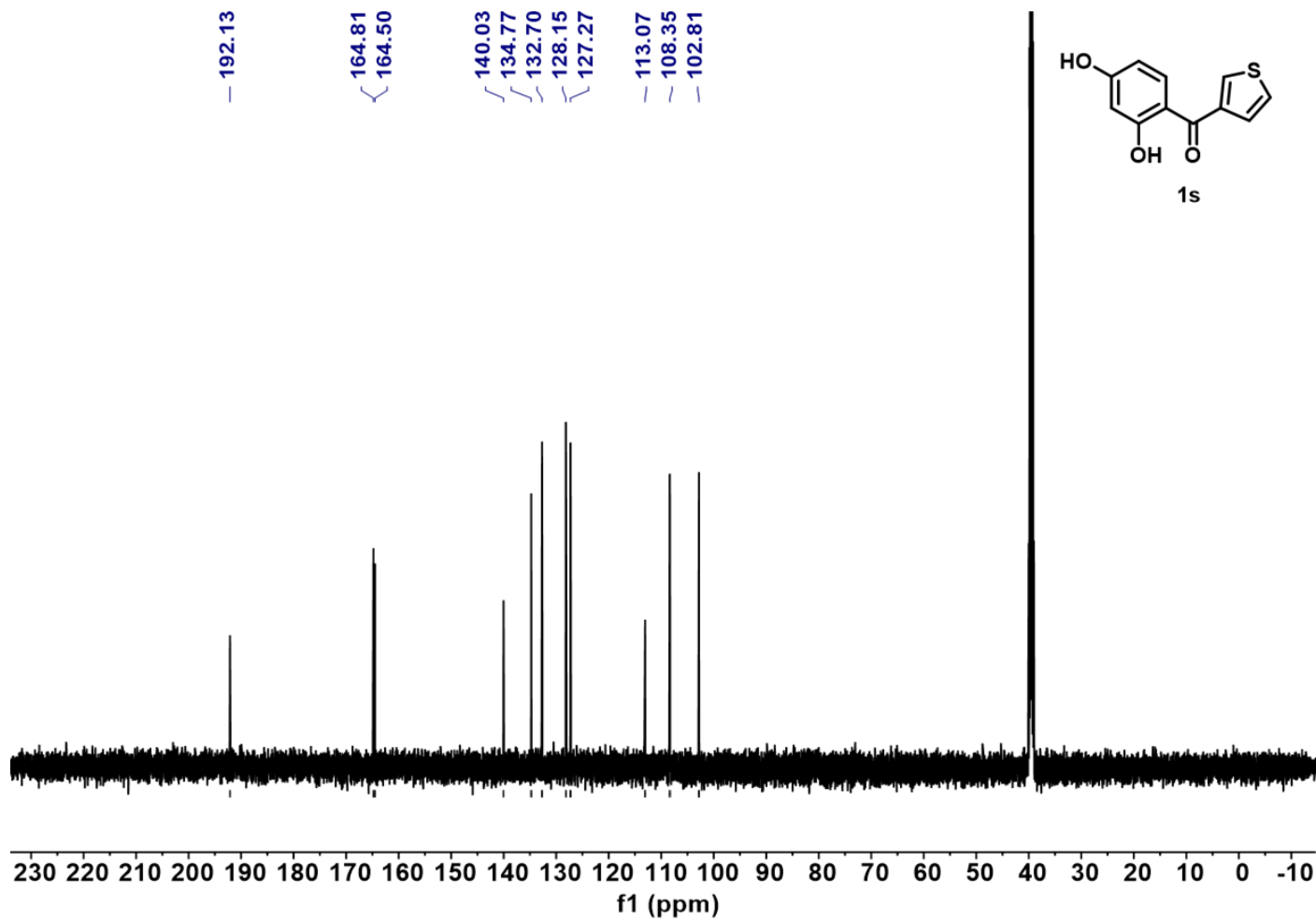
Supplementary Fig. 87 | ¹H NMR spectrum of **1r** (500 MHz, DMSO-*d*₆).



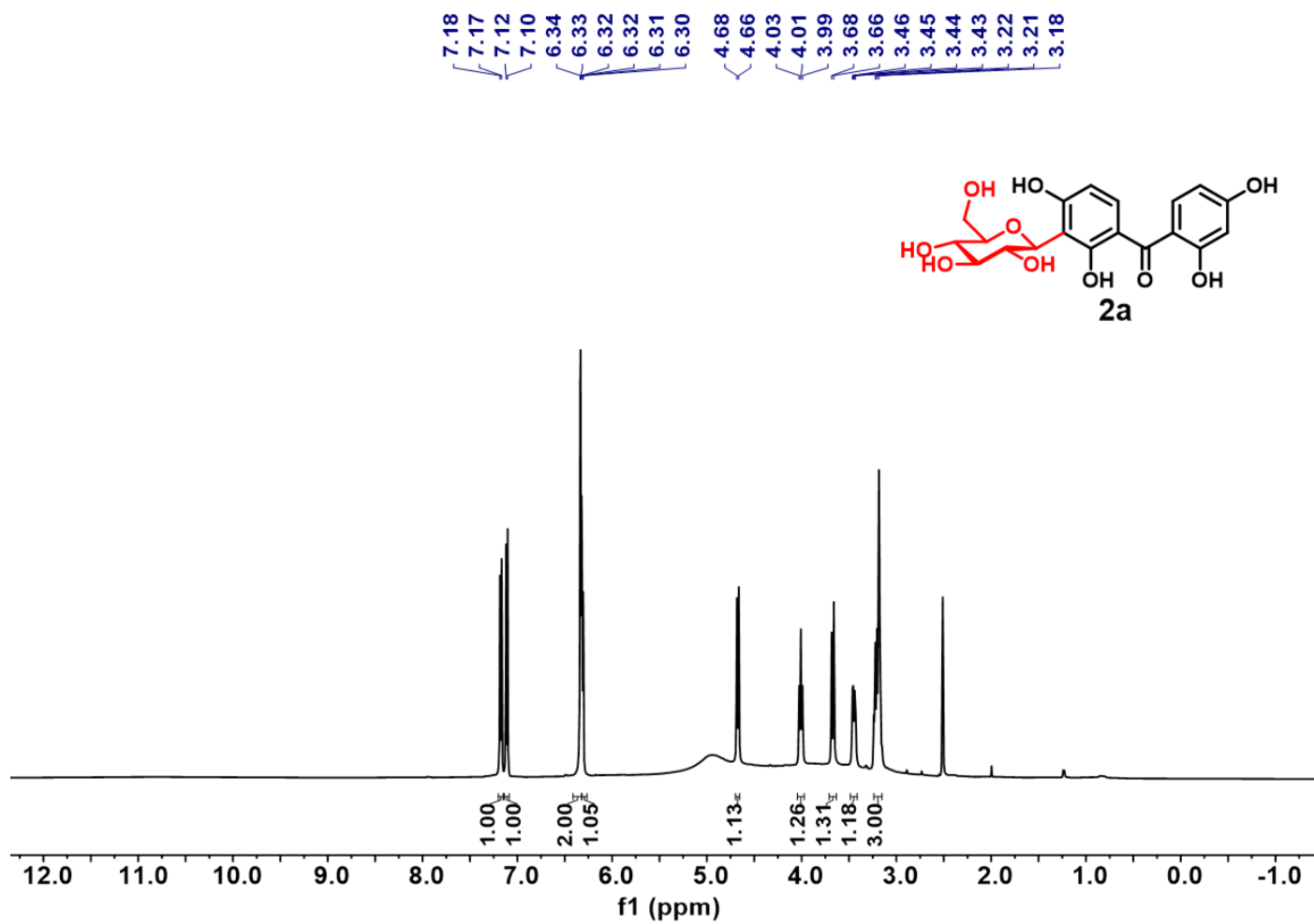
Supplementary Fig. 88 | ¹³C NMR spectrum of **1r** (126 MHz, DMSO-*d*₆).



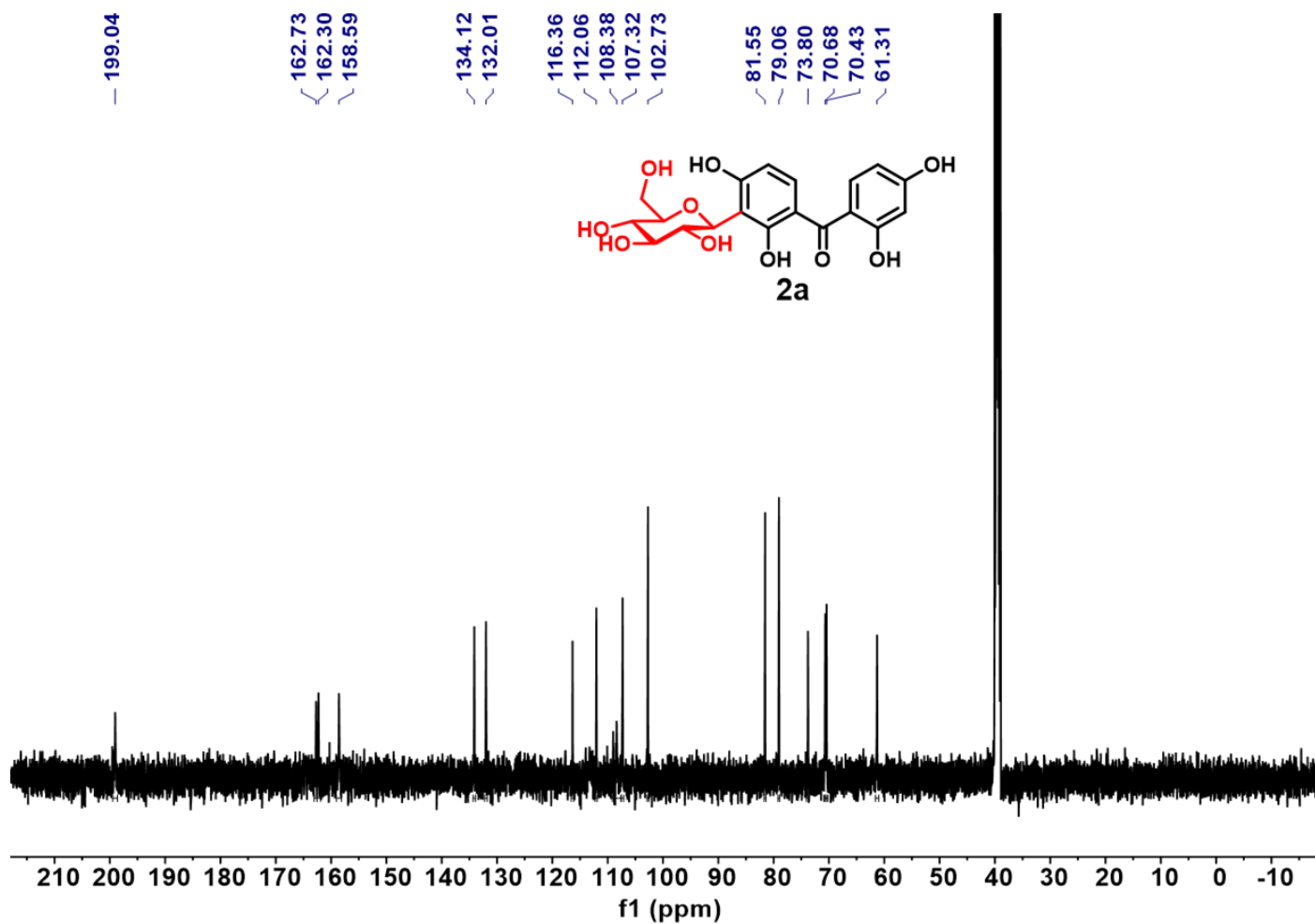
Supplementary Fig. 89 | ¹H NMR spectrum of **1s** (500 MHz, DMSO-*d*₆).



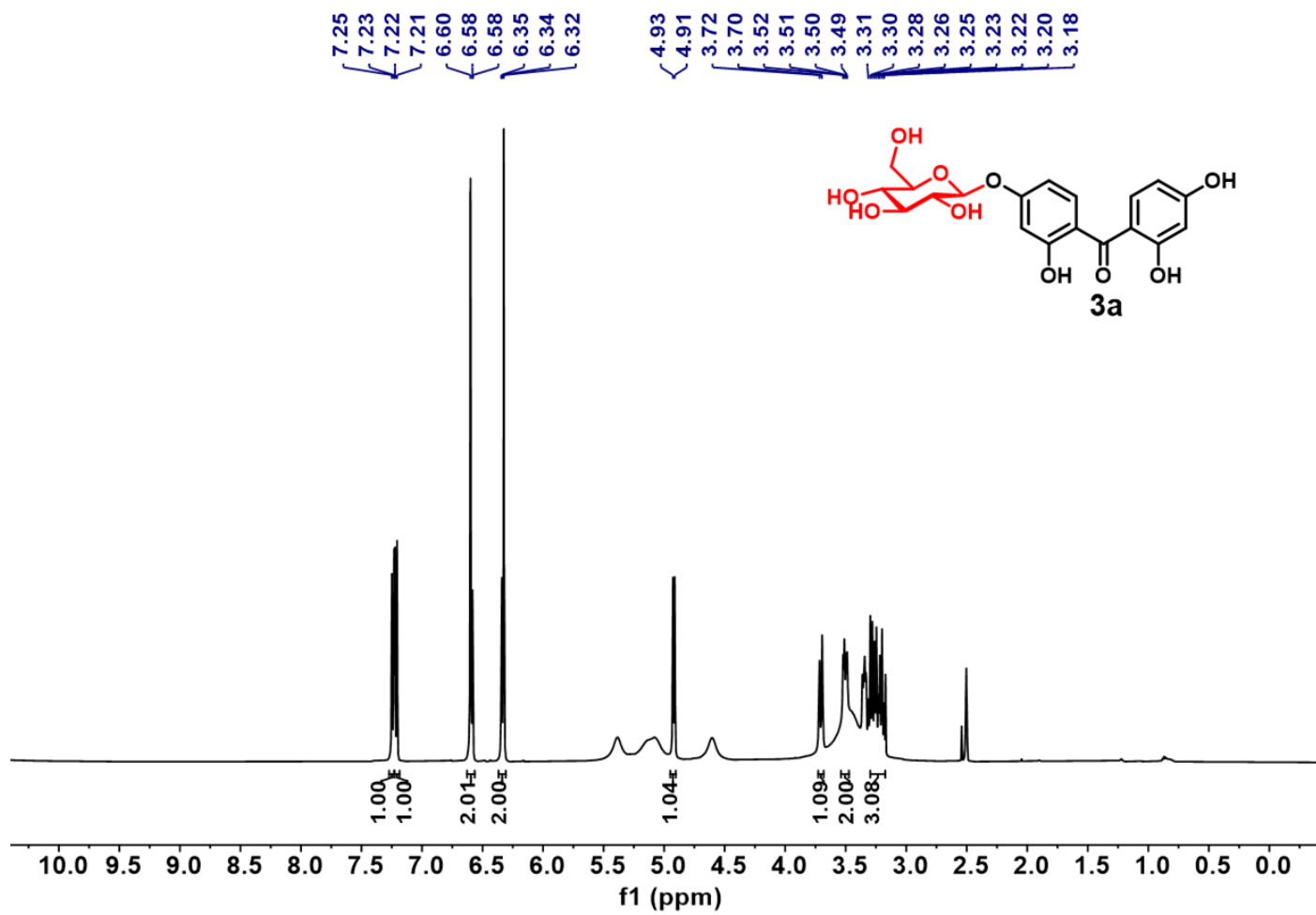
Supplementary Fig. 90 | ¹³C NMR spectrum of **1s** (126 MHz, DMSO-*d*₆).



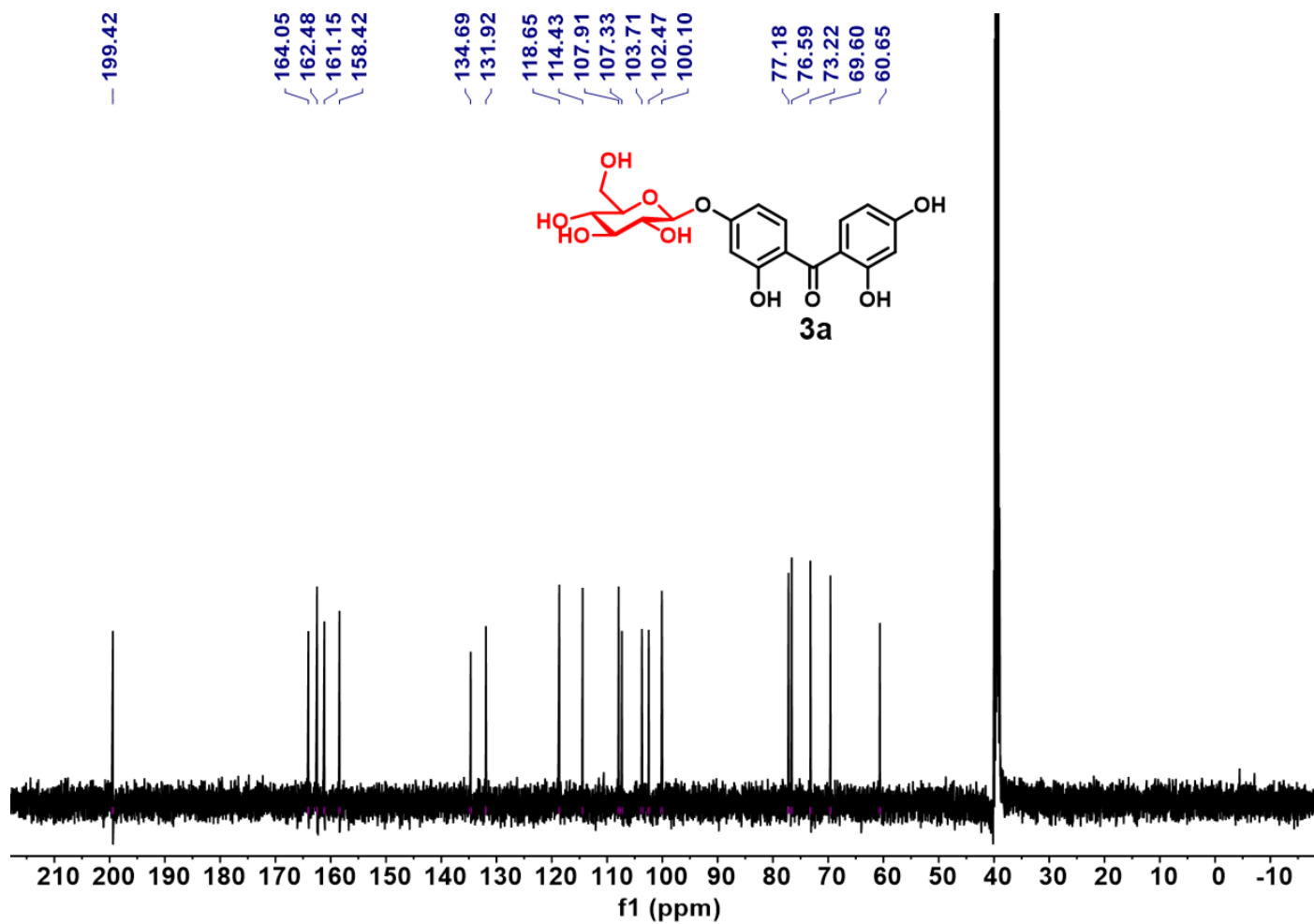
Supplementary Fig. 91 | ^1H NMR spectrum of **2a** (500 MHz, $\text{DMSO-}d_6$).



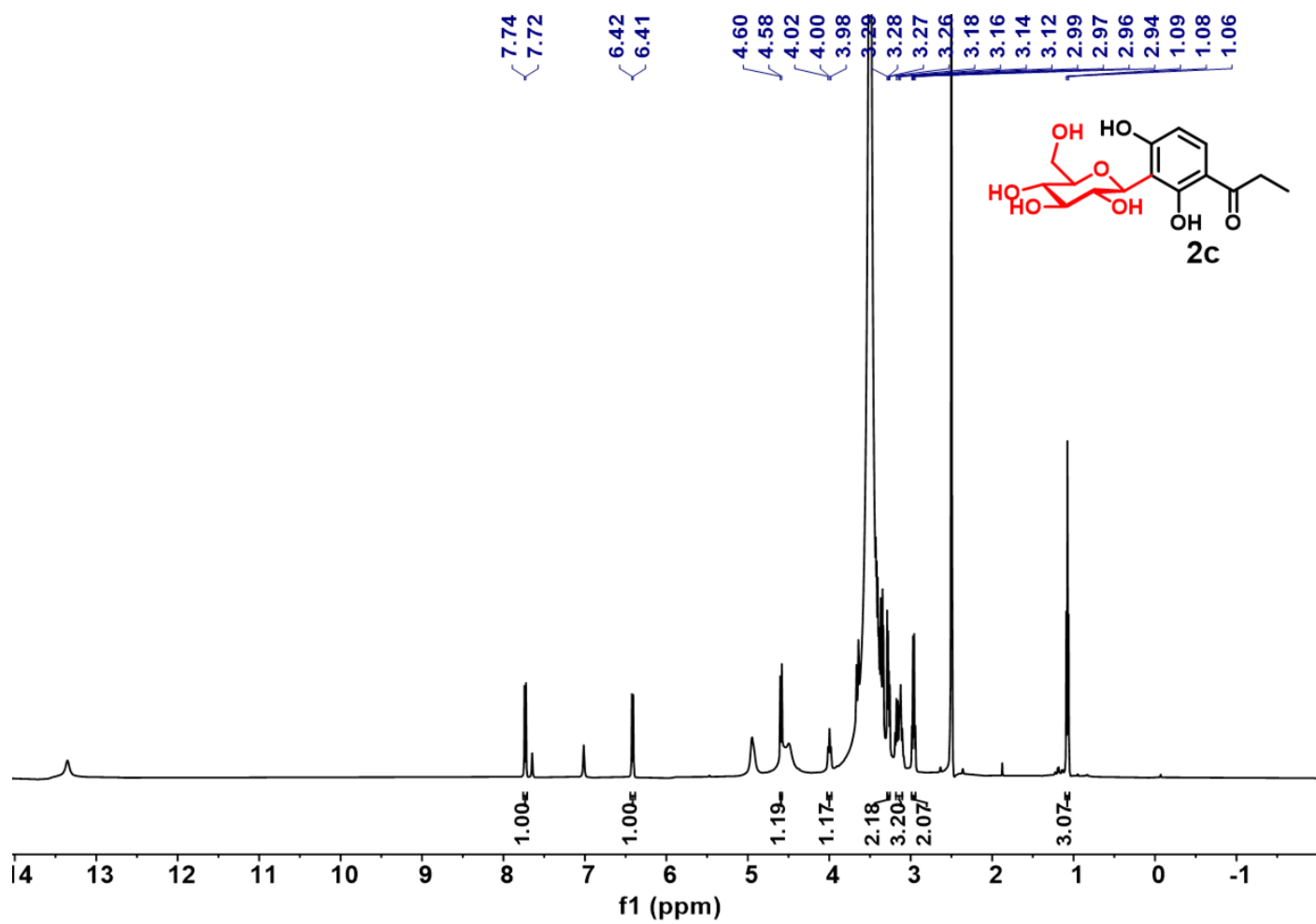
Supplementary Fig. 92 | ¹³C NMR spectrum of **2a** (126 MHz, DMSO-*d*₆).



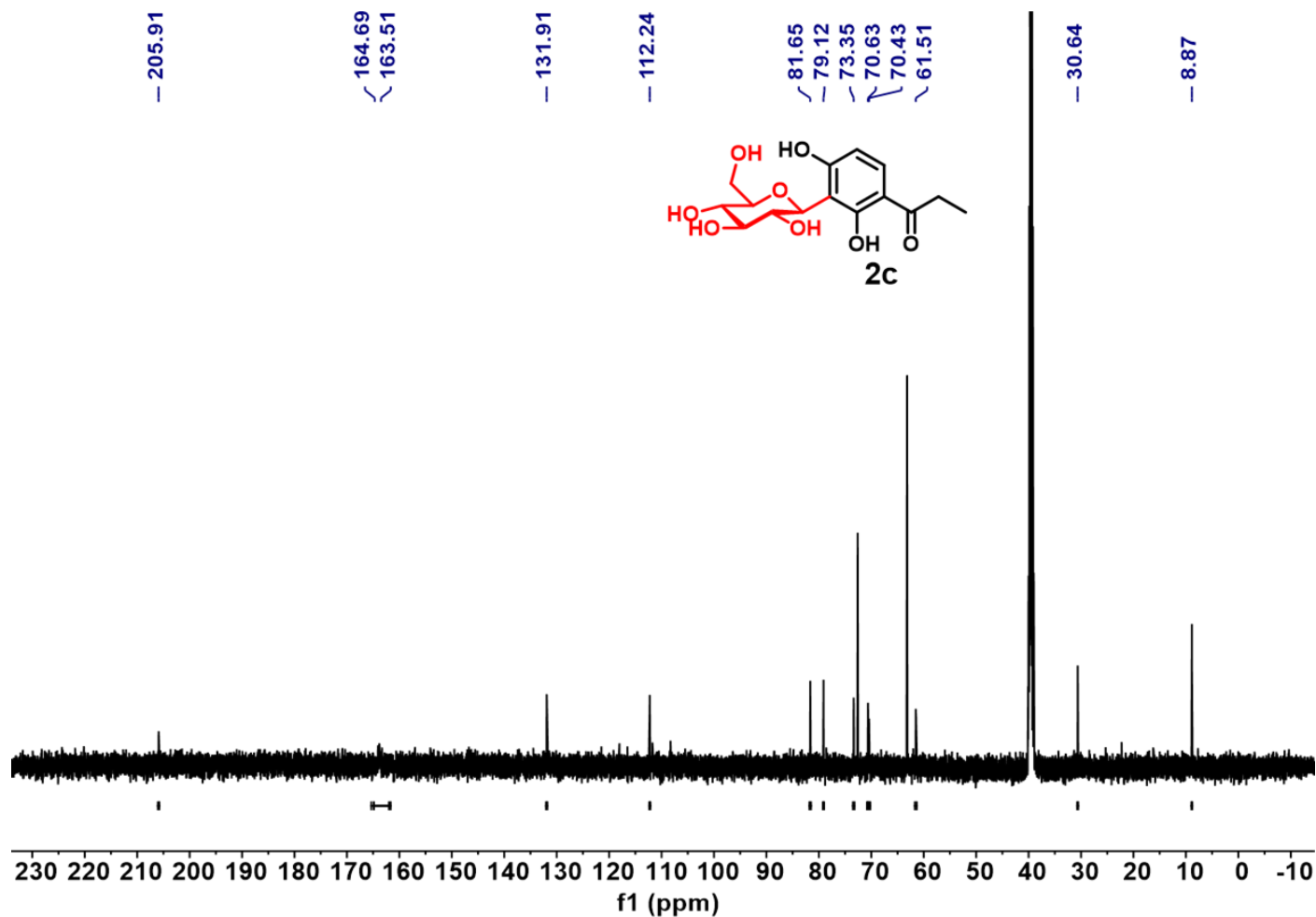
Supplementary Fig. 93 | ¹H NMR spectrum of **3a** (500 MHz, DMSO-*d*₆).



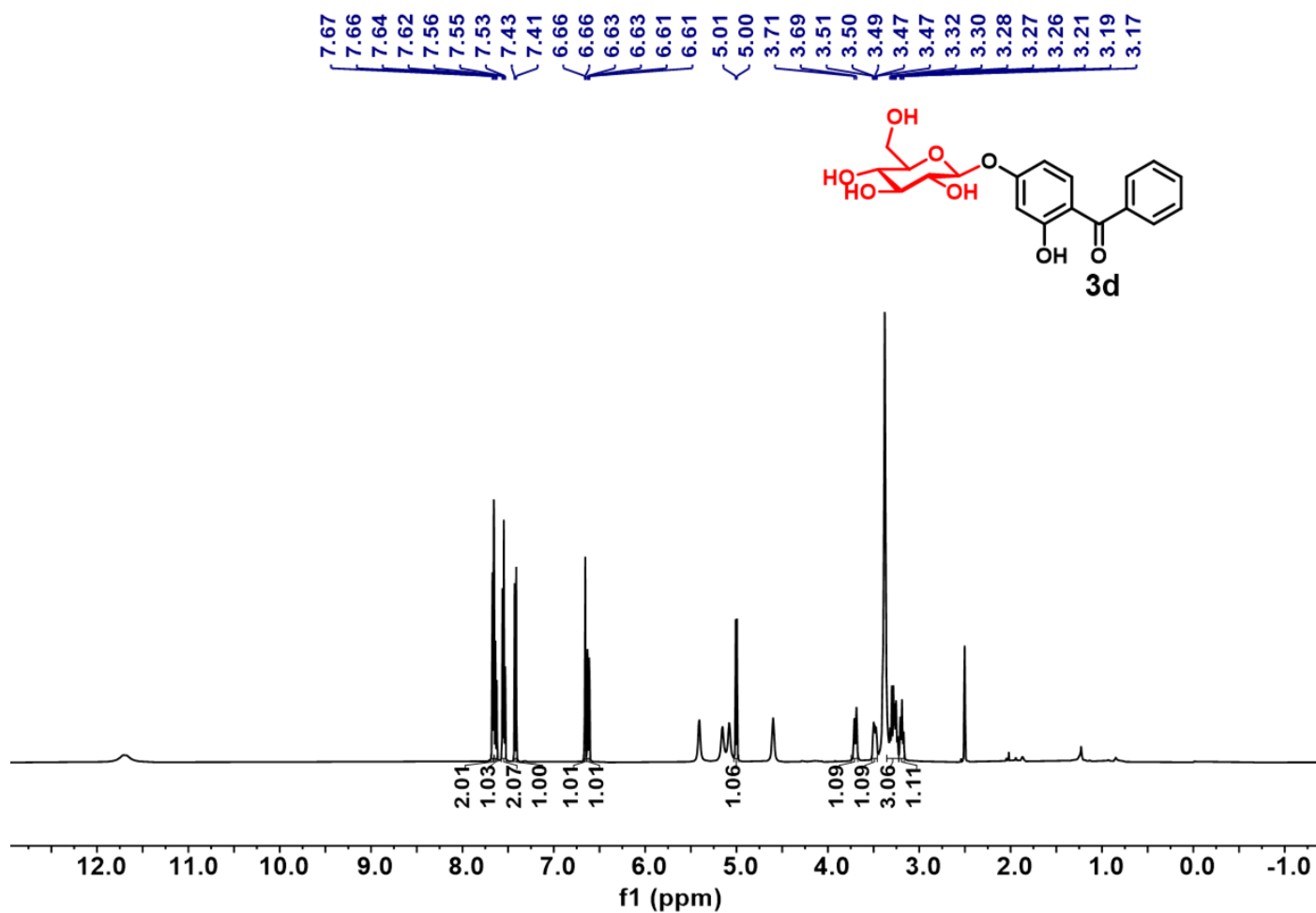
Supplementary Fig. 94 | ^{13}C NMR spectrum of **3a** (126 MHz, $\text{DMSO}-d_6$).



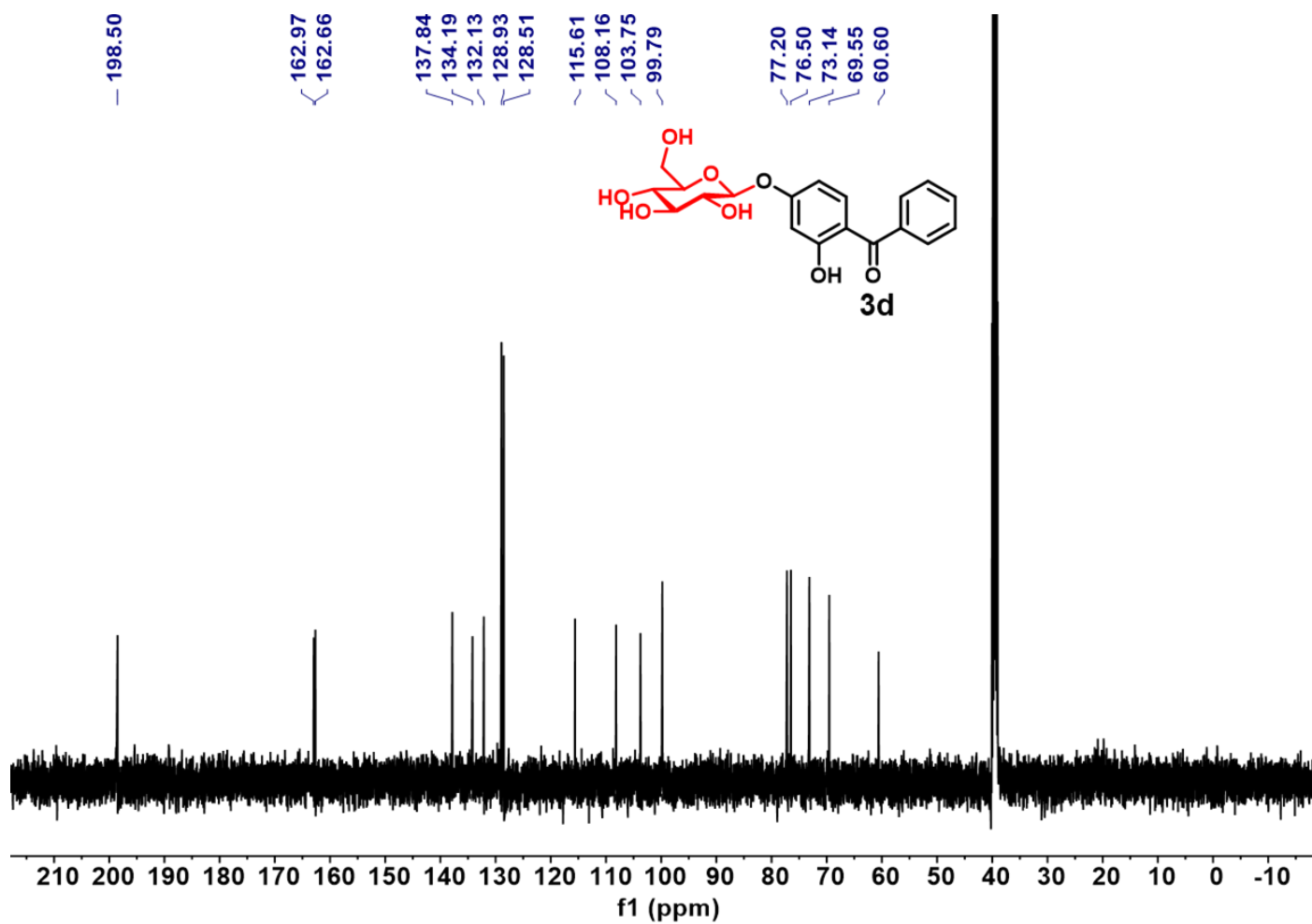
Supplementary Fig. 95 | ^1H NMR spectrum of **2c** (500 MHz, $\text{DMSO}-d_6$).



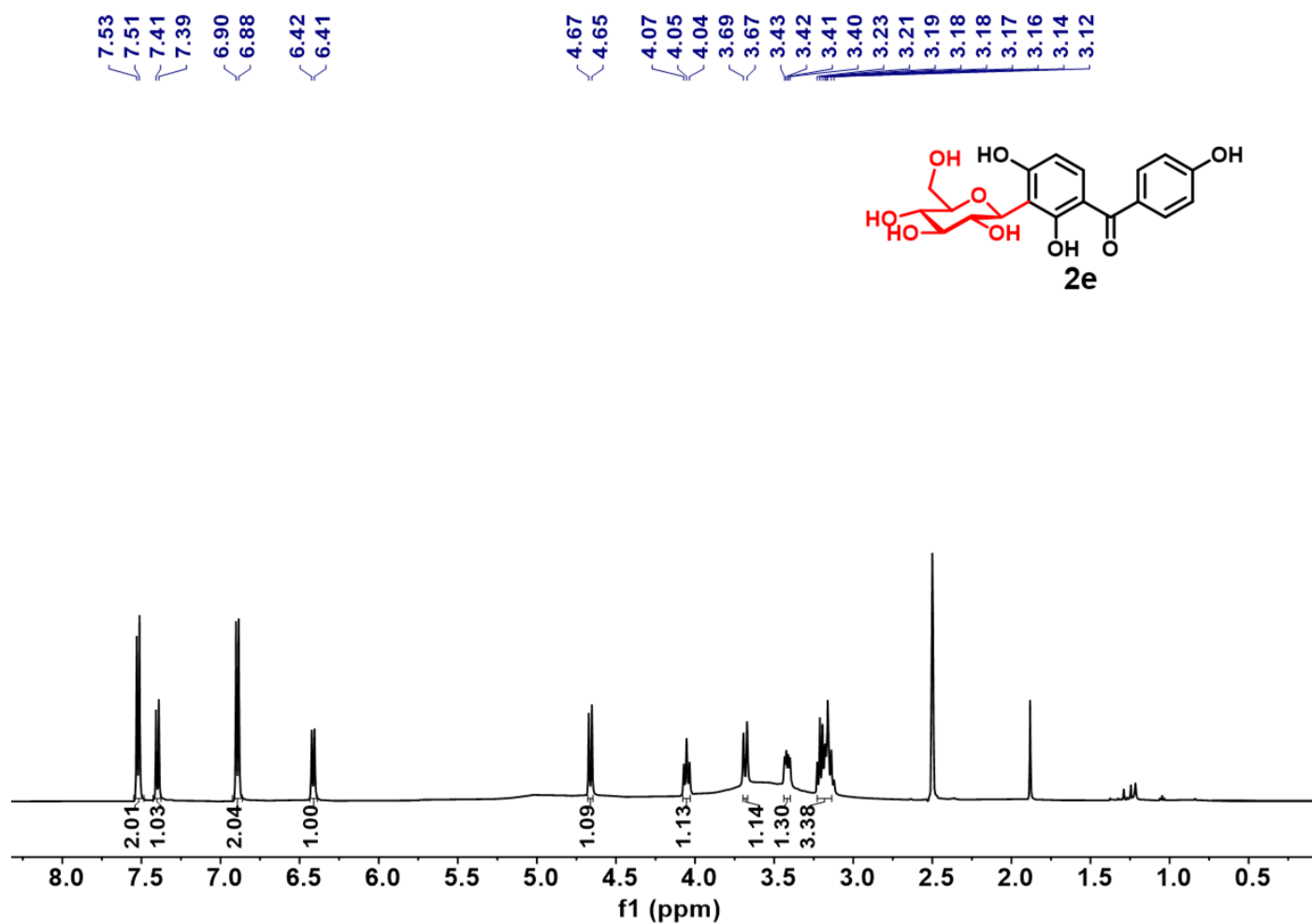
Supplementary Fig. 96 | ^{13}C NMR spectrum of **2c** (126 MHz, $\text{DMSO-}d_6$). Glycerol: ^{13}C NMR (126 MHz, $\text{DMSO-}d_6$) δ 72.5, 63.1.



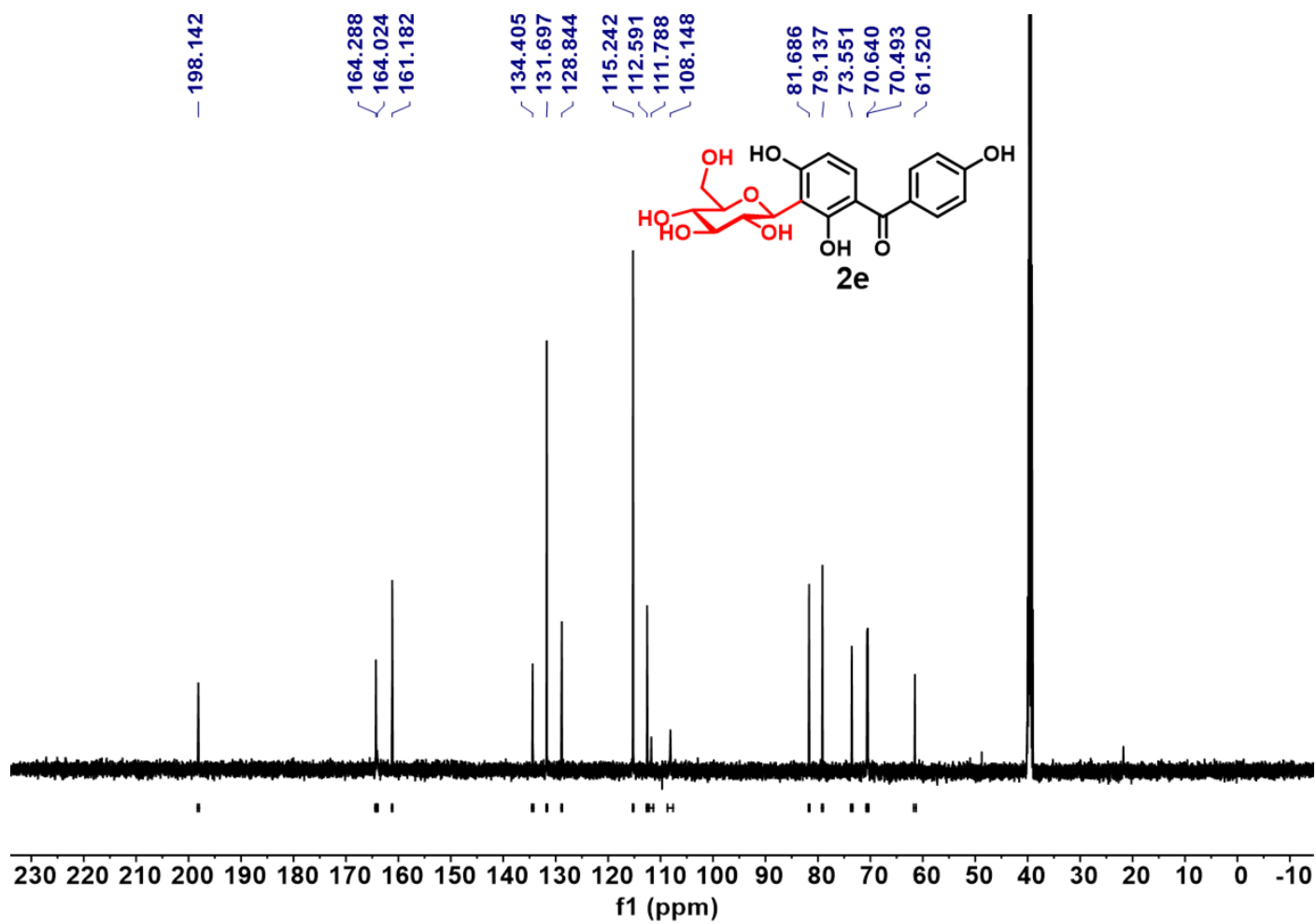
Supplementary Fig. 97 | ^1H NMR spectrum of **3d** (500 MHz, $\text{DMSO-}d_6$).



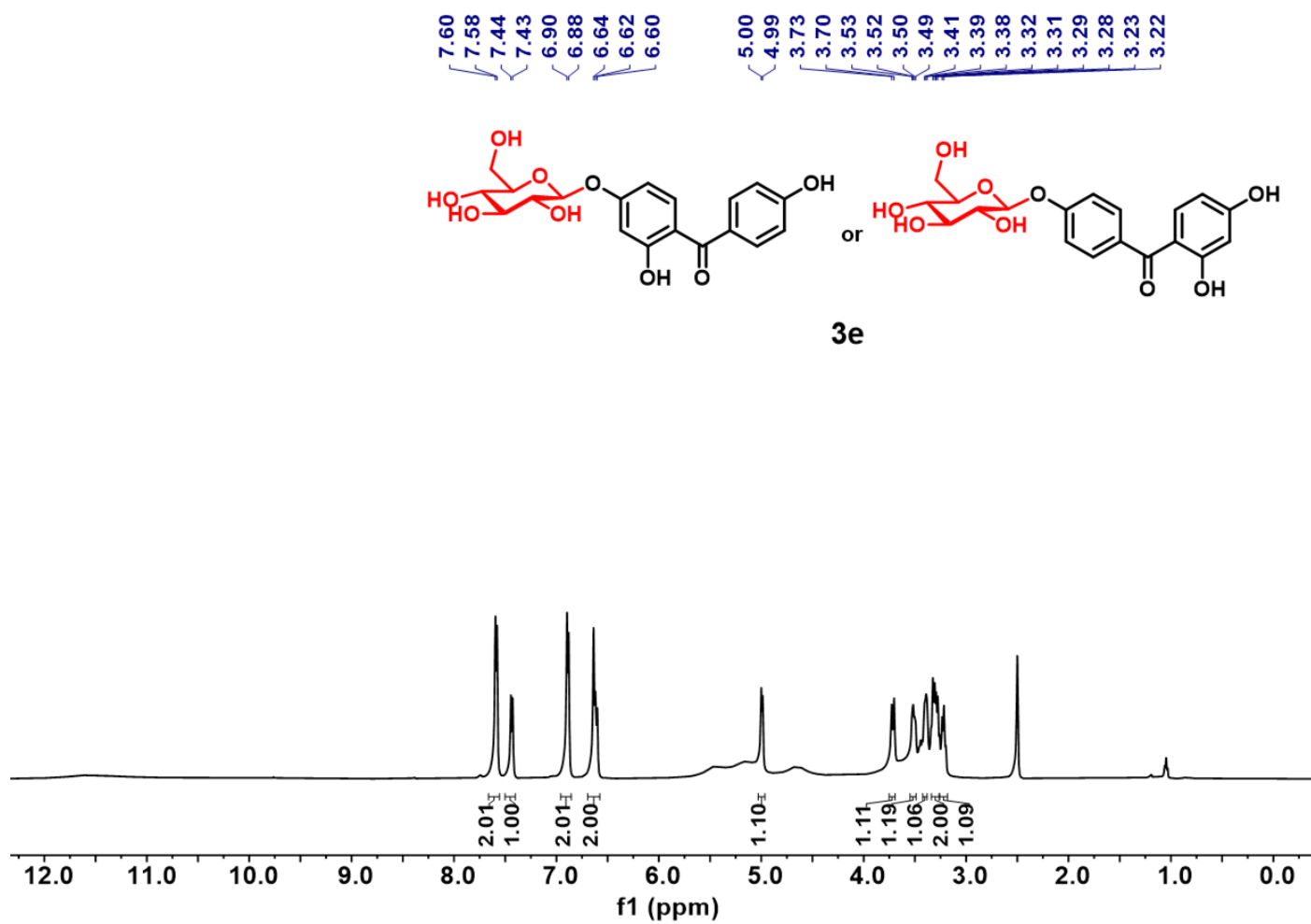
Supplementary Fig. 98 | ^{13}C NMR spectrum of **3d** (126 MHz, $\text{DMSO}-d_6$).



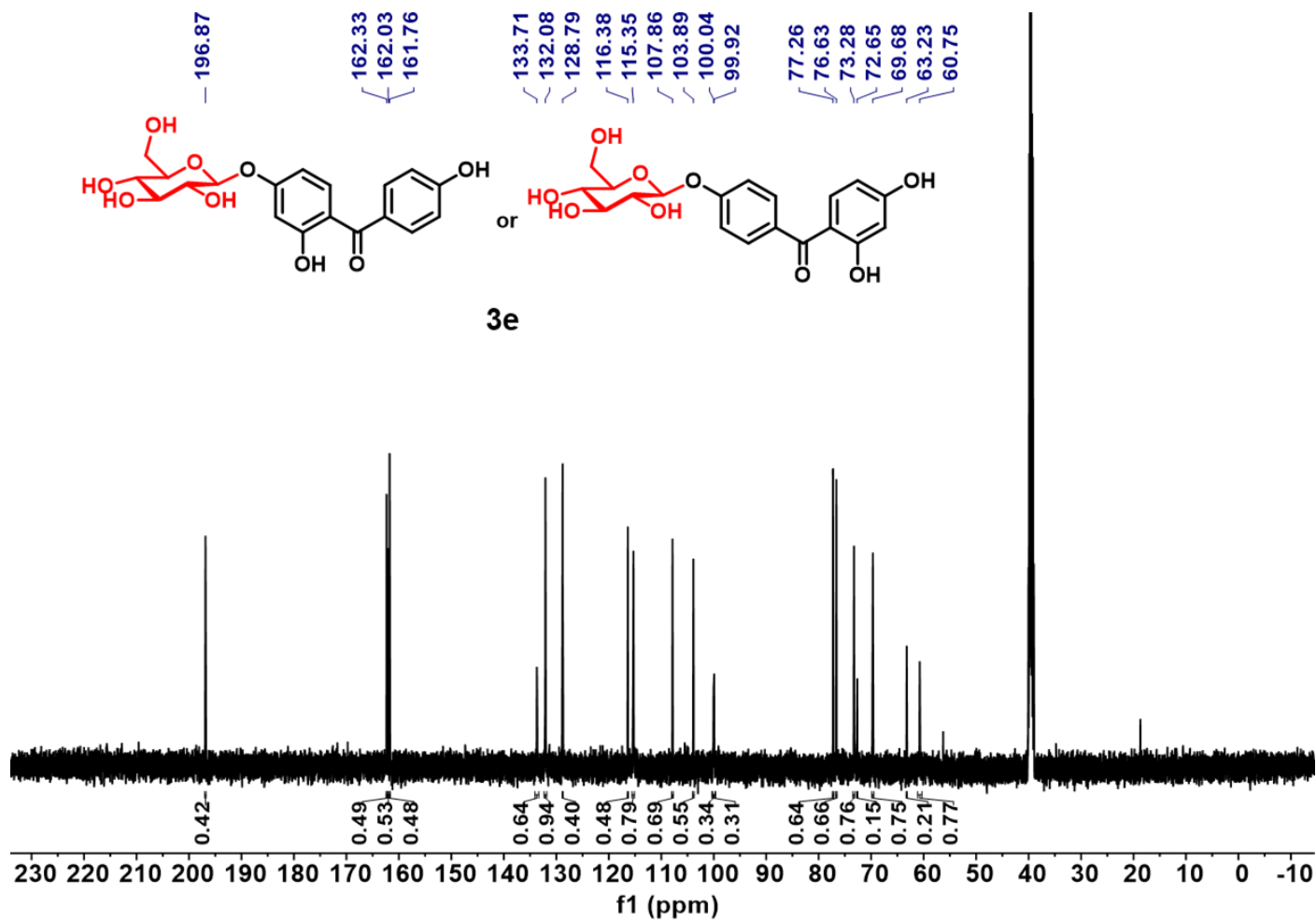
Supplementary Fig. 99 | ¹H NMR spectrum of **2e** (500 MHz, DMSO-*d*₆).



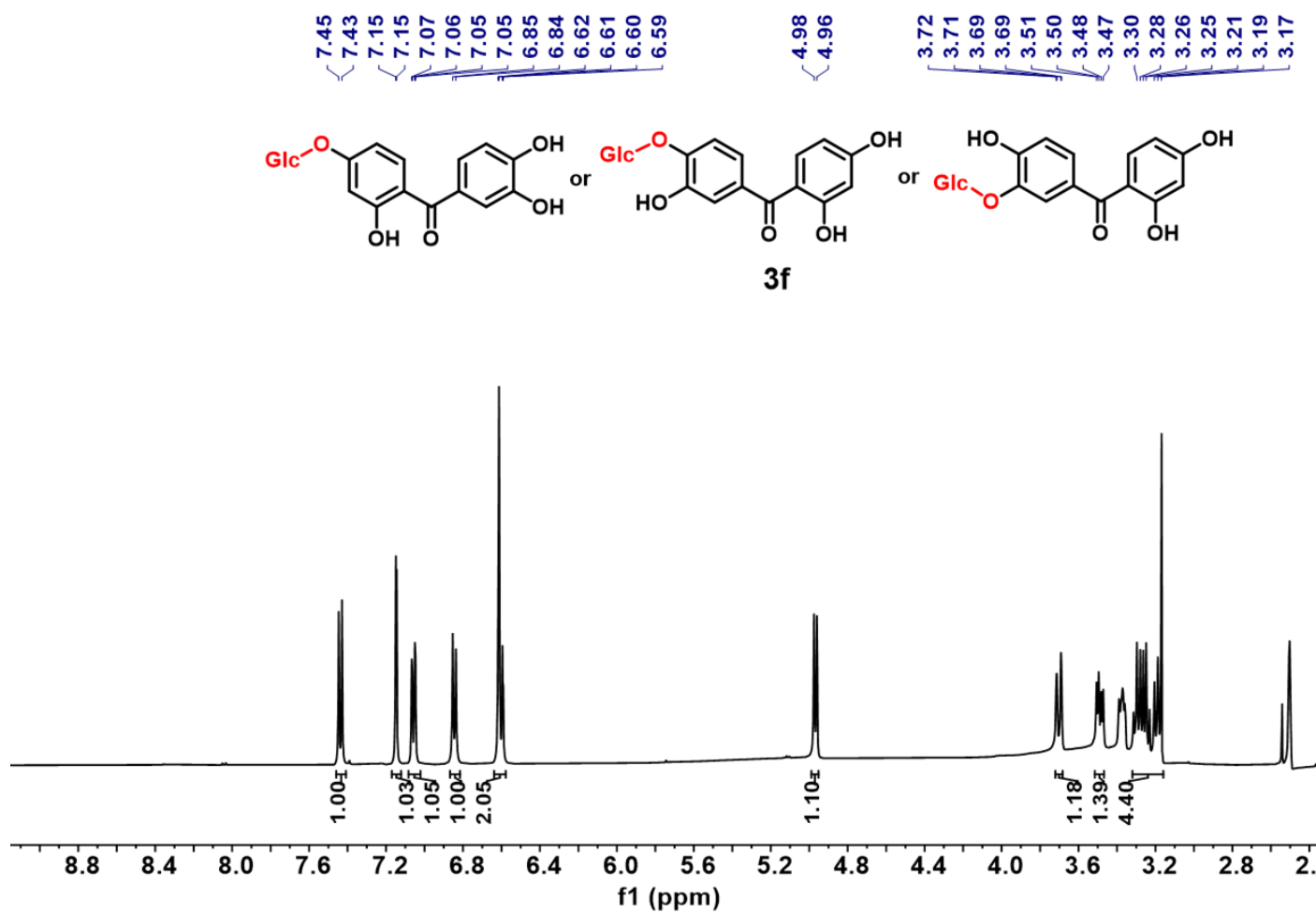
Supplementary Fig. 100 | ^{13}C NMR spectrum of **2e** (126 MHz, $\text{DMSO}-d_6$).



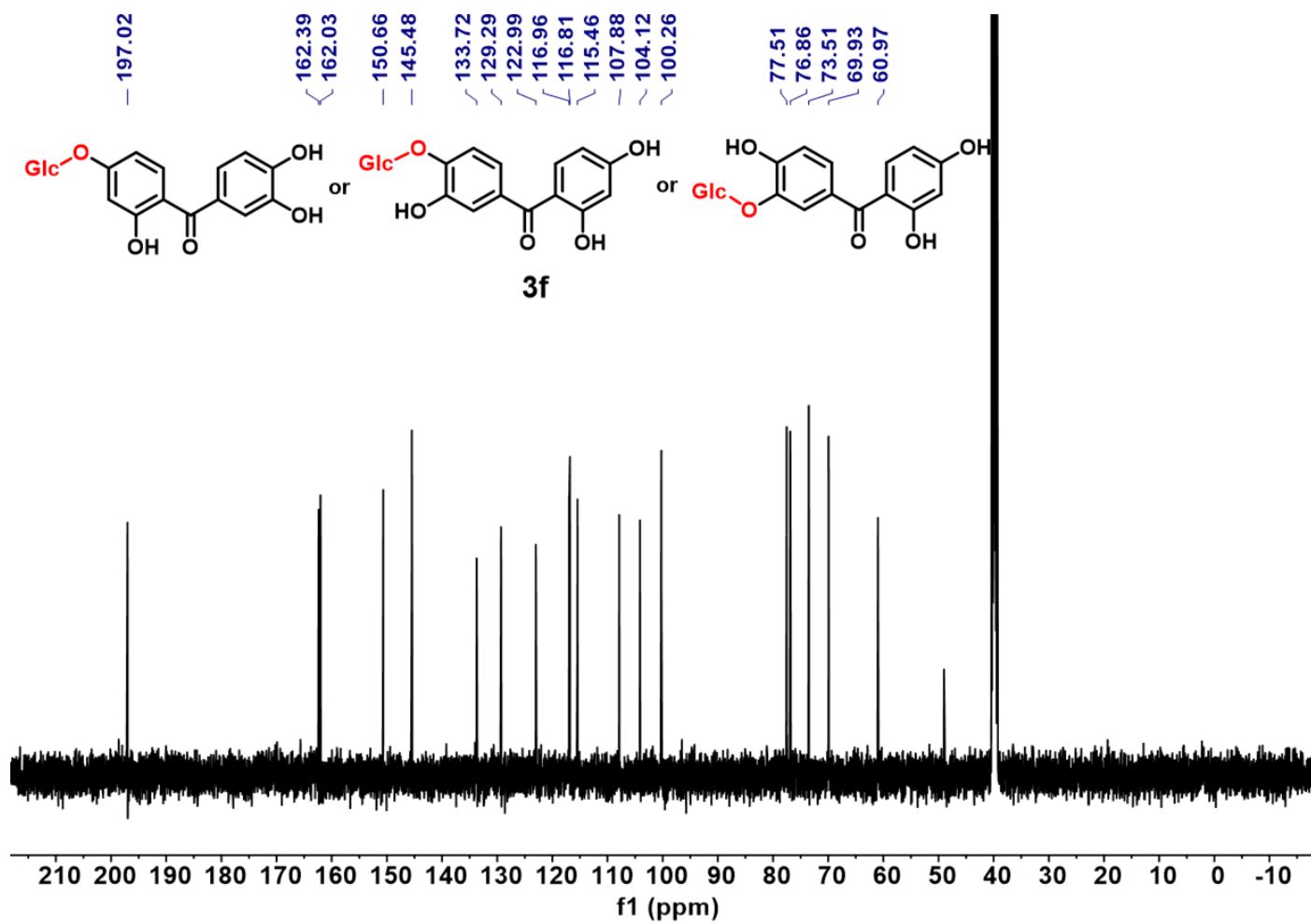
Supplementary Fig. 101 | ^1H NMR spectrum of **3e** (500 MHz, $\text{DMSO-}d_6$).



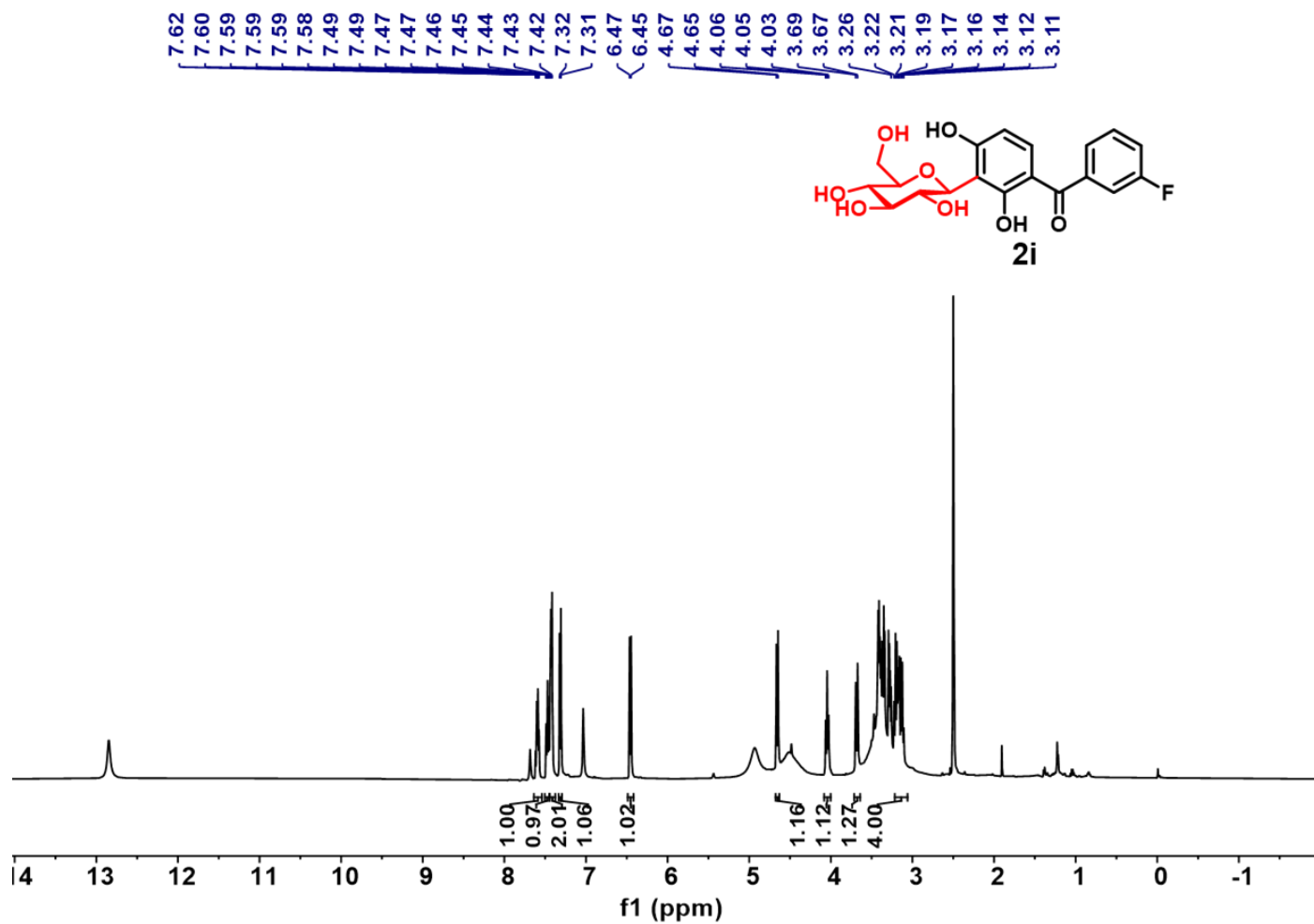
Supplementary Fig. 102 | ^{13}C NMR spectrum of **3e** (126 MHz, $\text{DMSO-}d_6$).



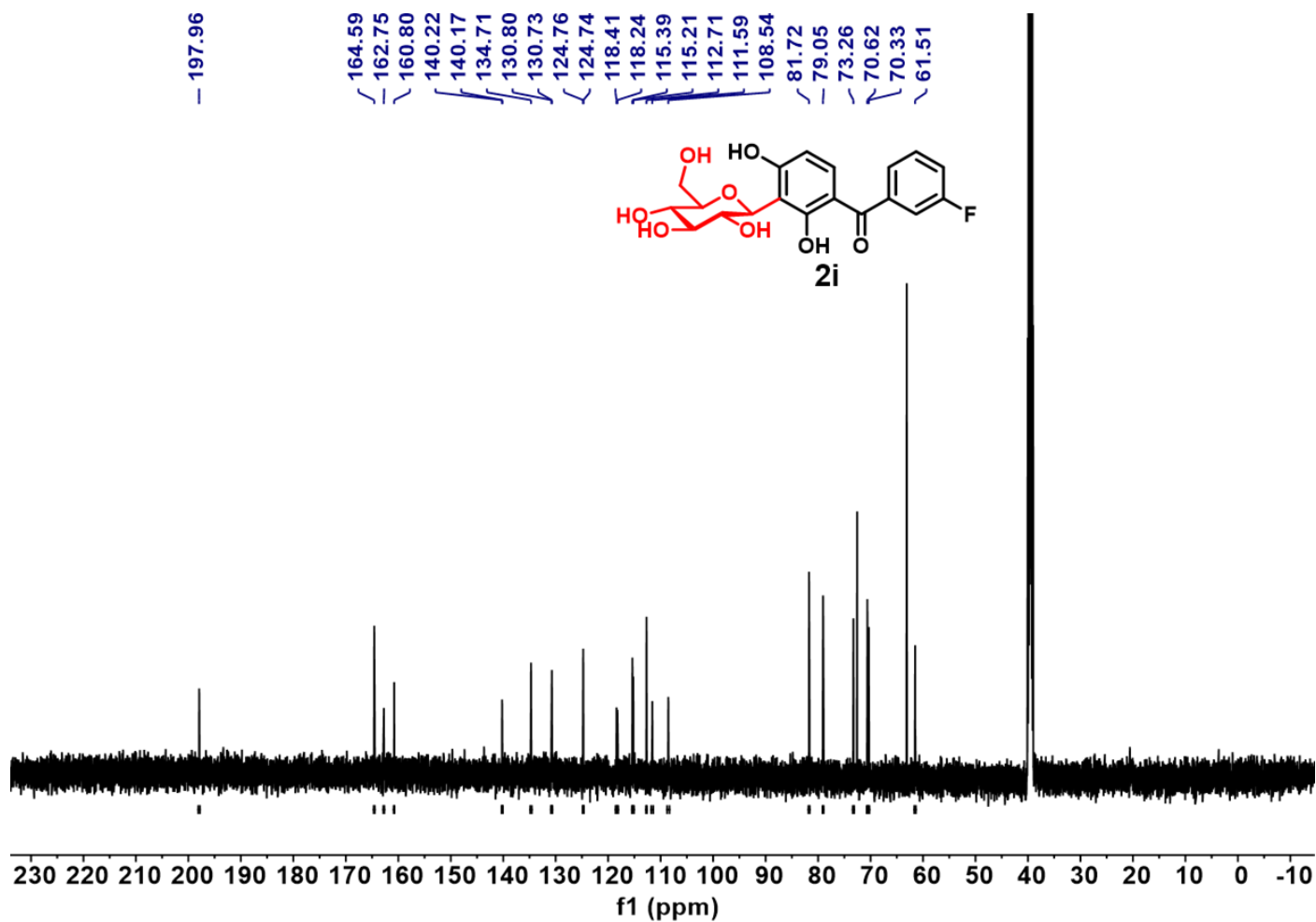
Supplementary Fig. 103 | ^1H NMR spectrum of **3f** (500 MHz, $\text{DMSO}-d_6$).



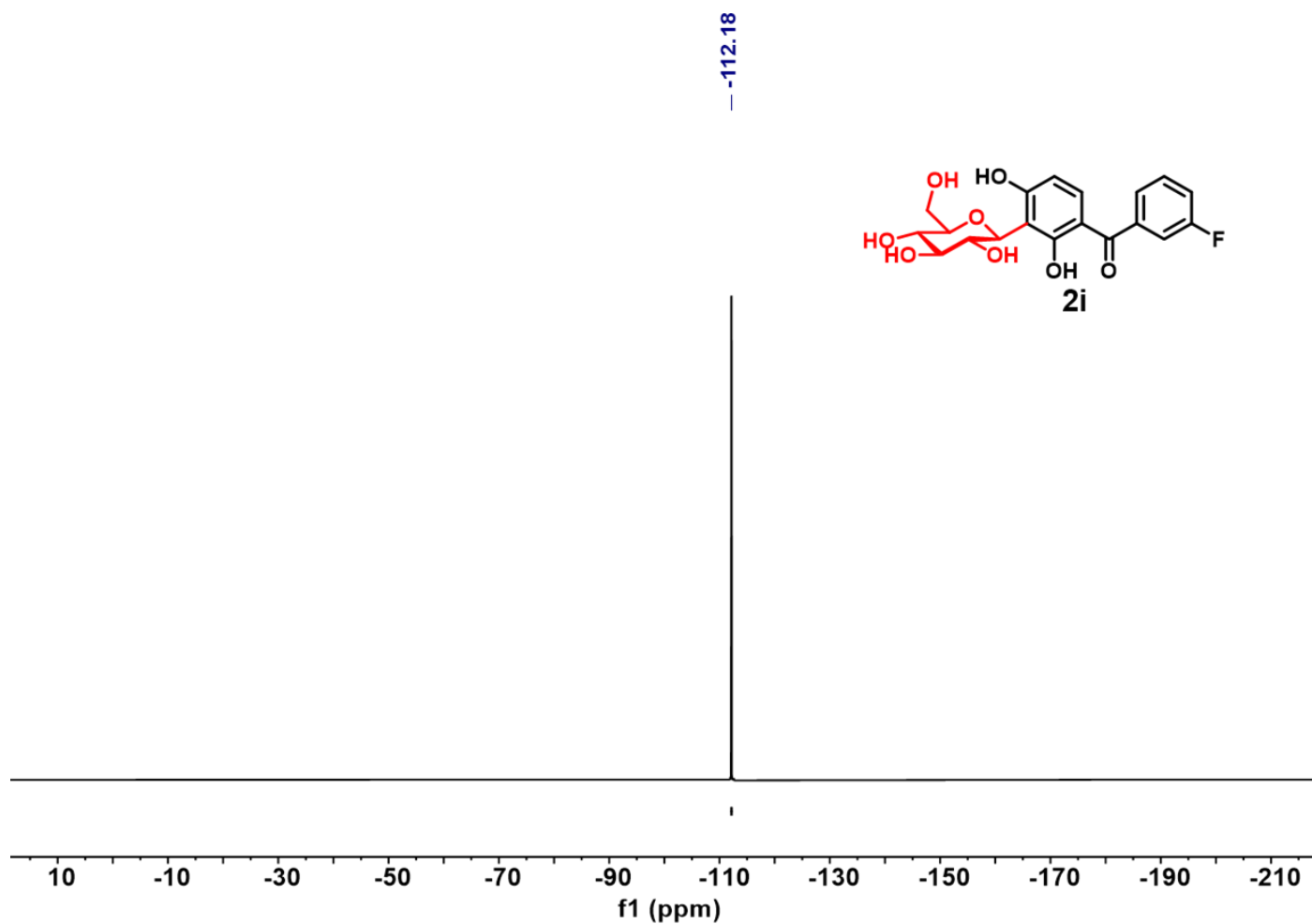
Supplementary Fig. 104 | ^{13}C NMR spectrum of **3f** (126 MHz, $\text{DMSO-}d_6$).



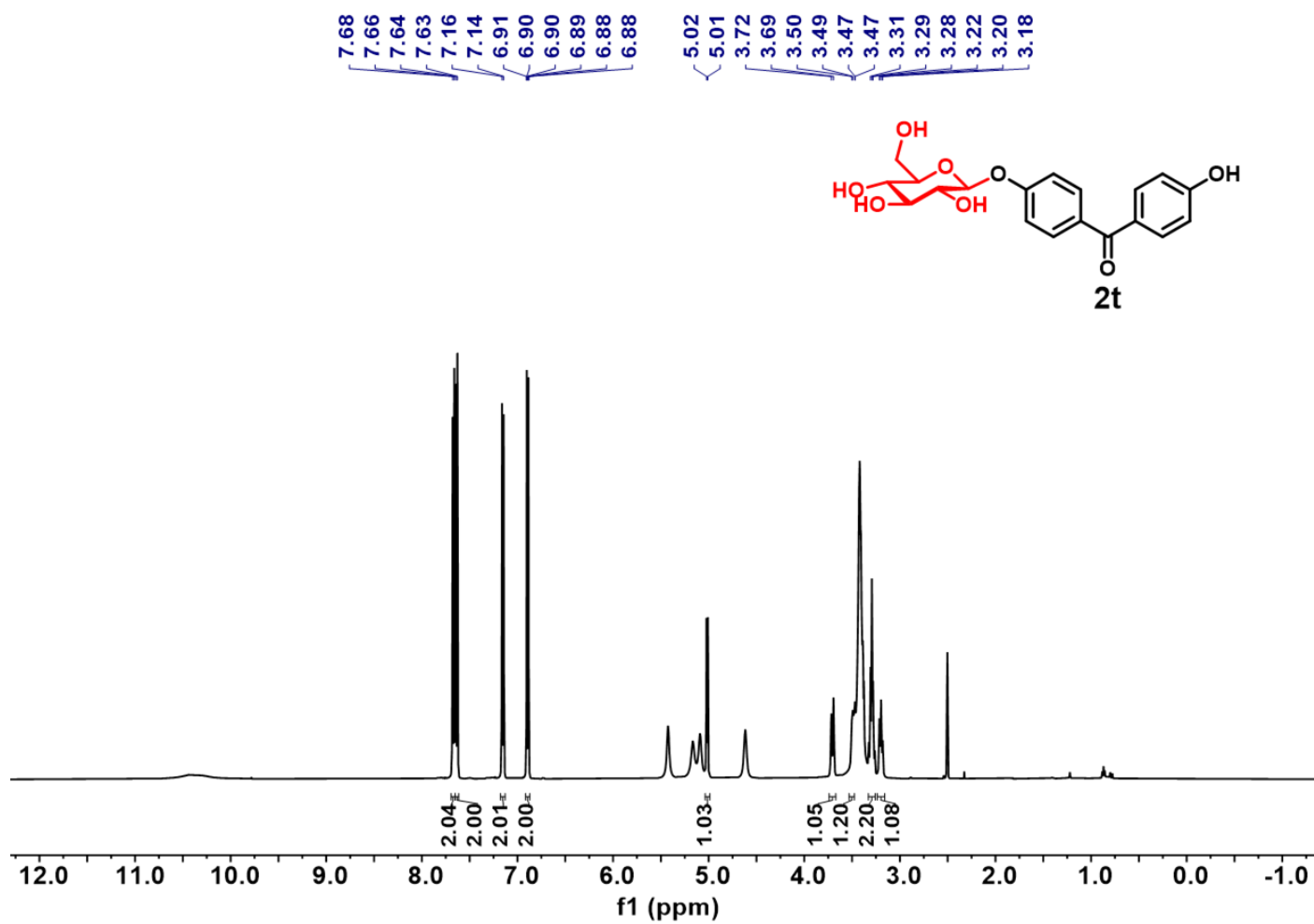
Supplementary Fig. 105 | ¹H NMR spectrum of **2i** (500 MHz, DMSO-*d*₆).



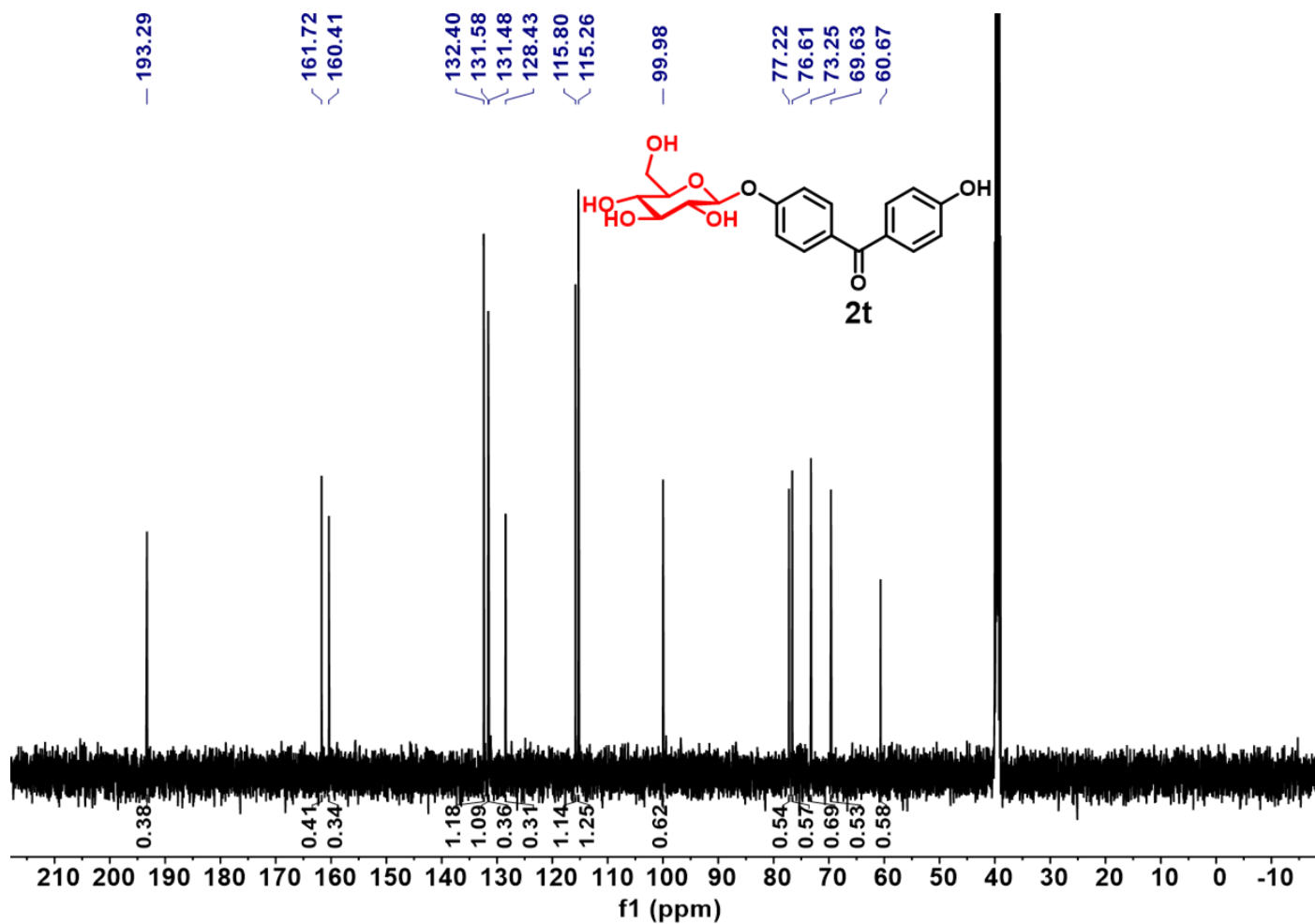
Supplementary Fig. 106 | ¹³C NMR spectrum of **2i** (126 MHz, DMSO-*d*₆). Glycerol: ¹³C NMR (126 MHz, DMSO-*d*₆) δ 72.5, 63.1.



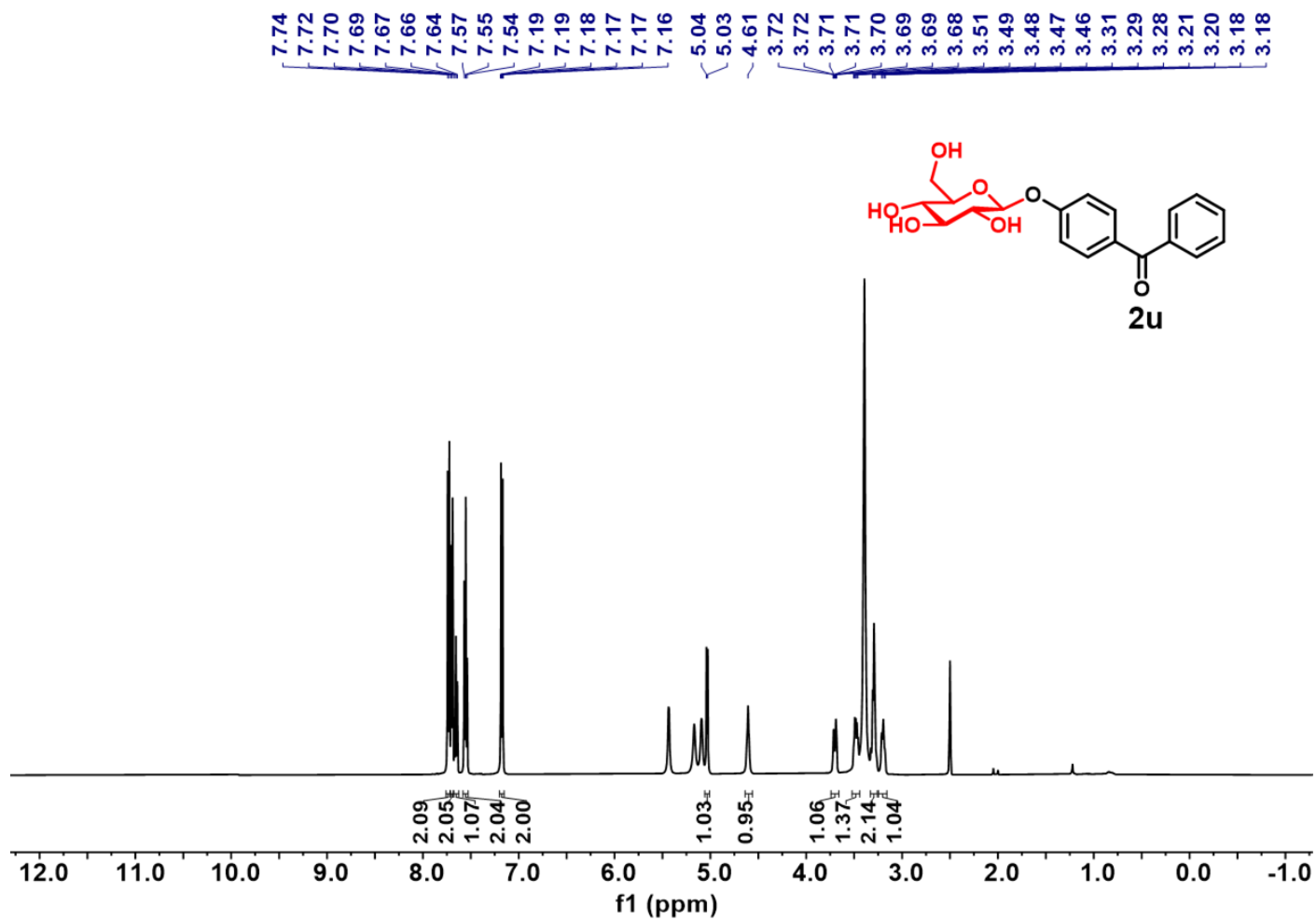
Supplementary Fig. 107 | ^{19}F NMR spectrum of **2i** (376 MHz, $\text{DMSO-}d_6$).



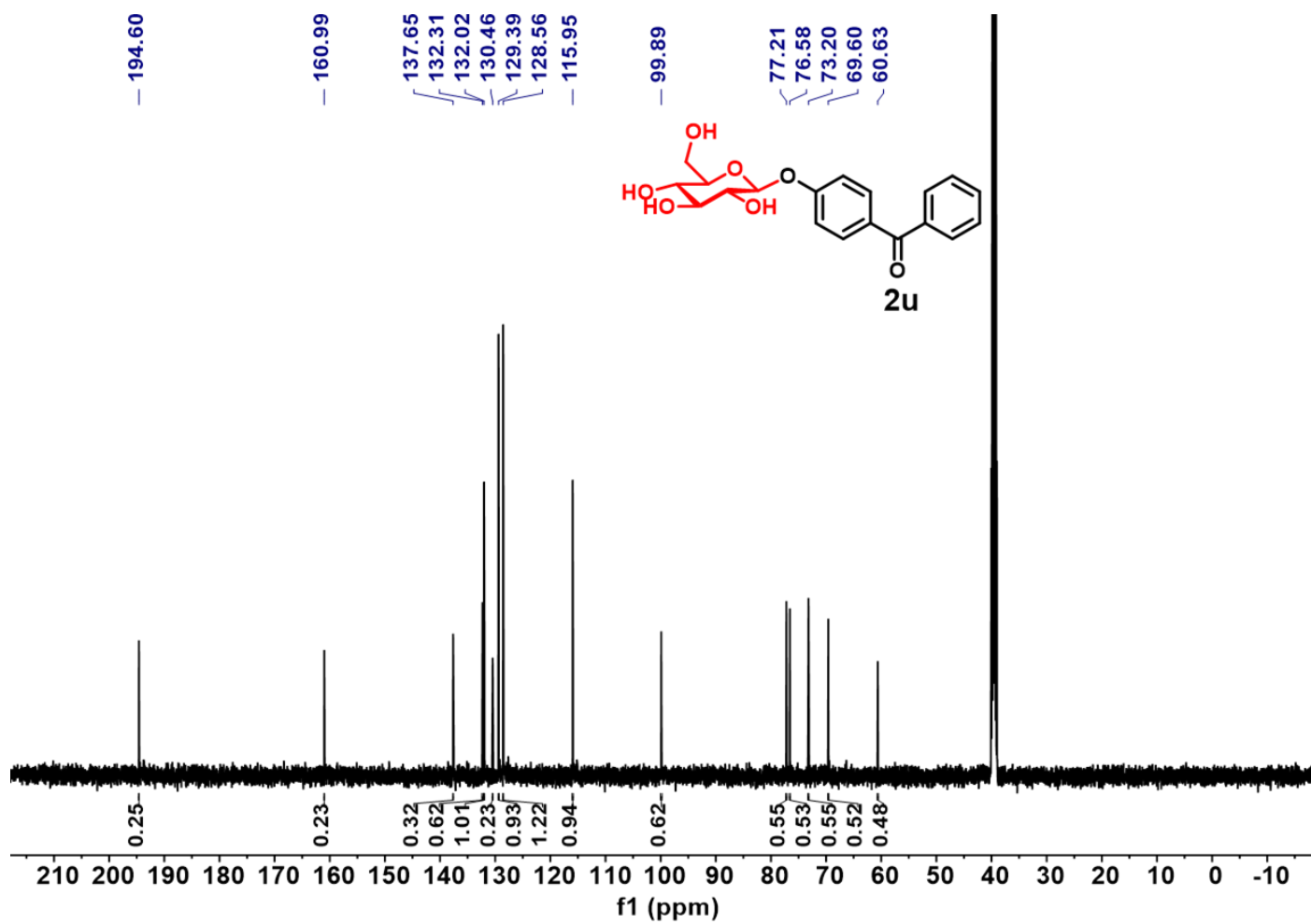
Supplementary Fig. 108 | ¹H NMR spectrum of **2t** (500 MHz, DMSO-*d*₆).



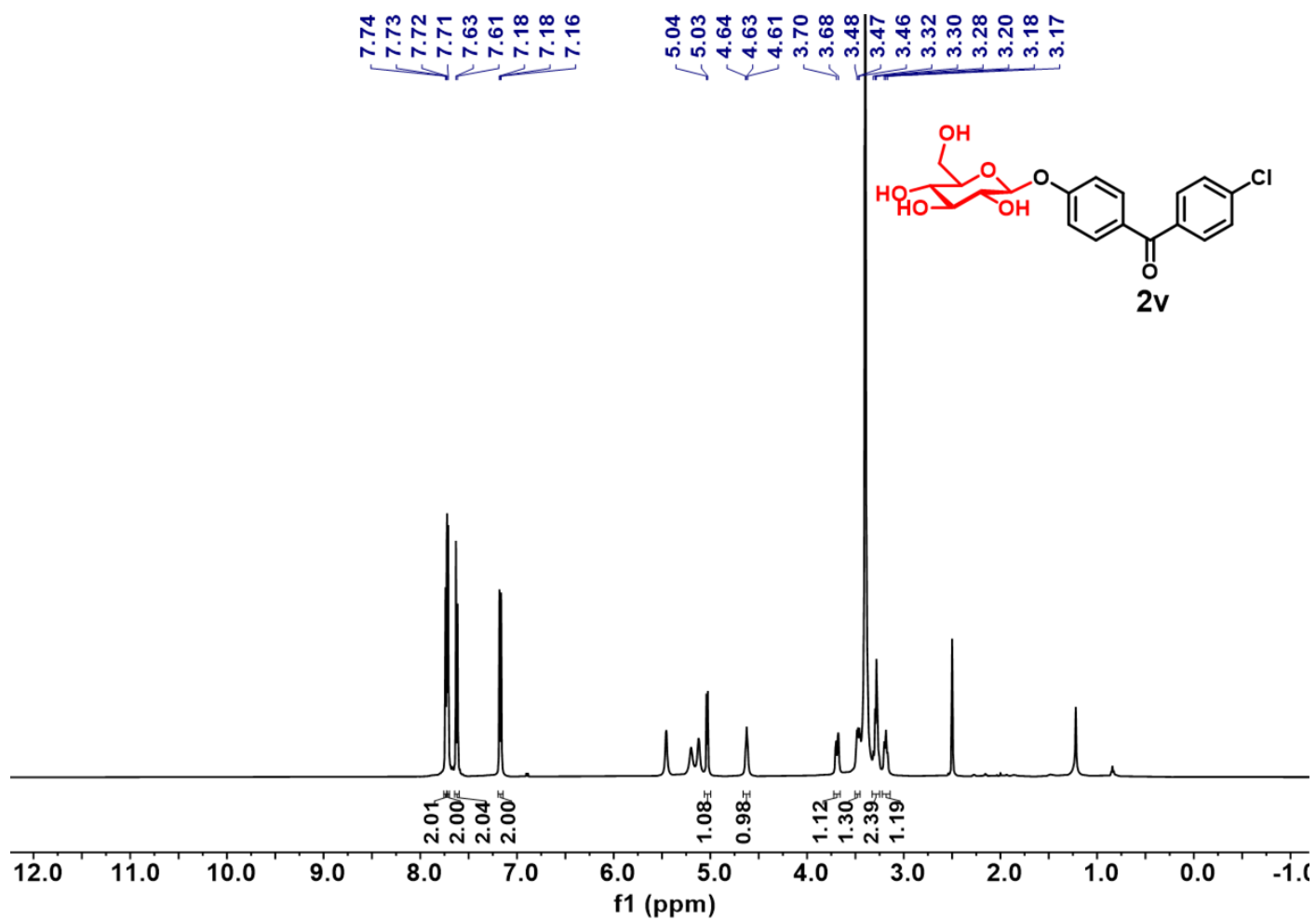
Supplementary Fig. 109 | ^{13}C NMR spectrum of **2t** (126 MHz, $\text{DMSO}-d_6$).



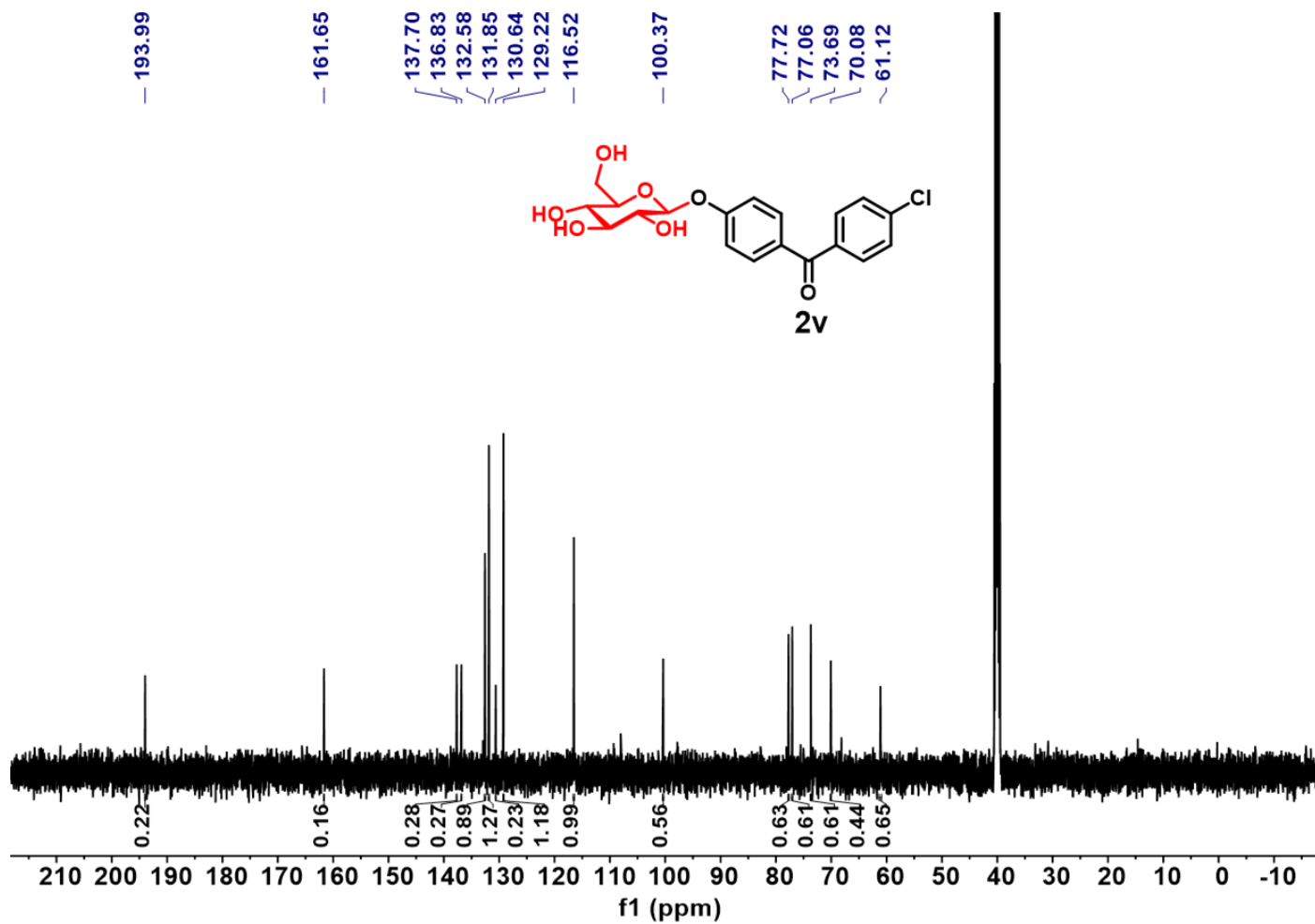
Supplementary Fig. 110 | ^1H NMR spectrum of **2u** (500 MHz, $\text{DMSO-}d_6$).



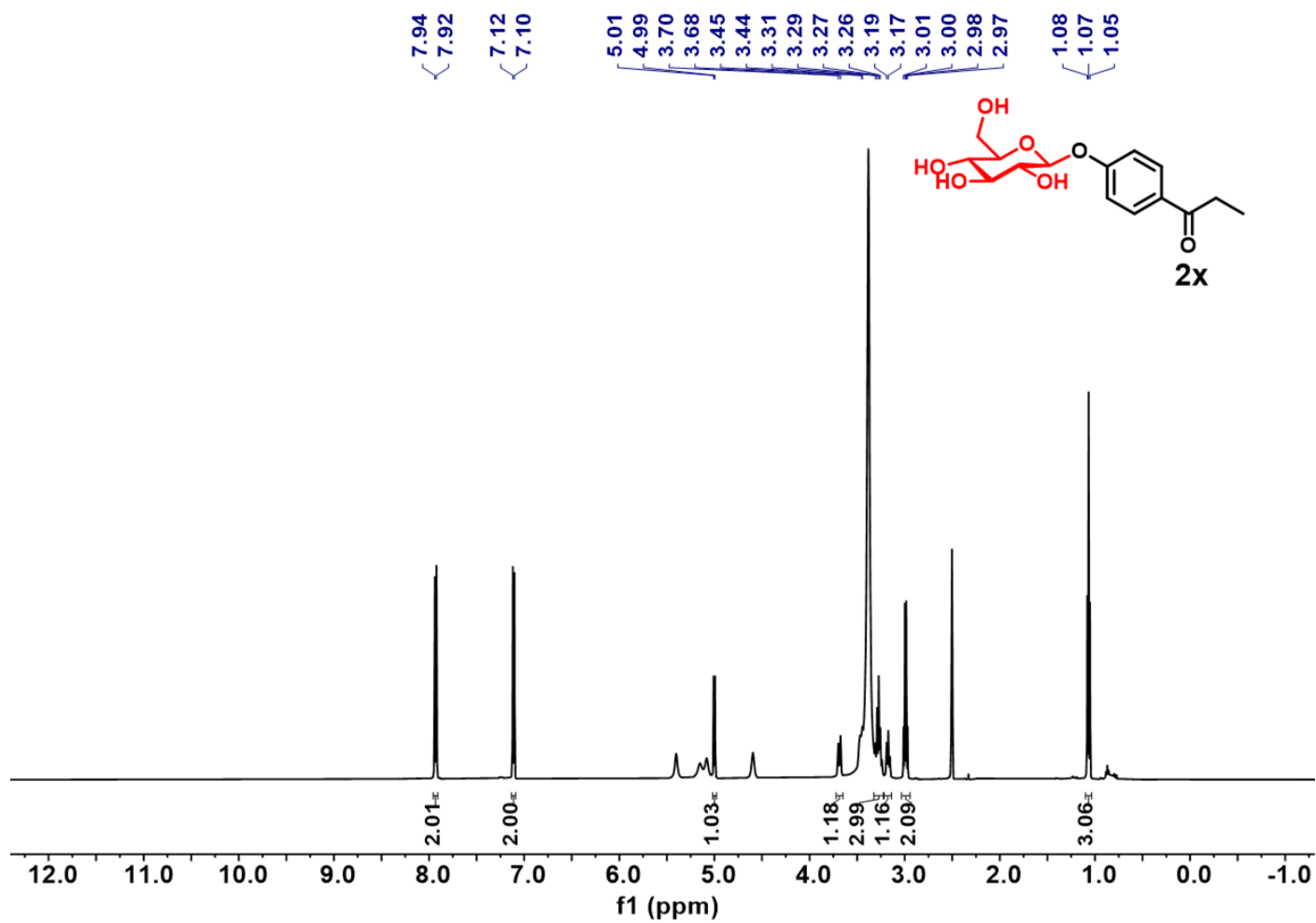
Supplementary Fig. 111 | ^{13}C NMR spectrum of **2u** (126 MHz, $\text{DMSO}-d_6$).



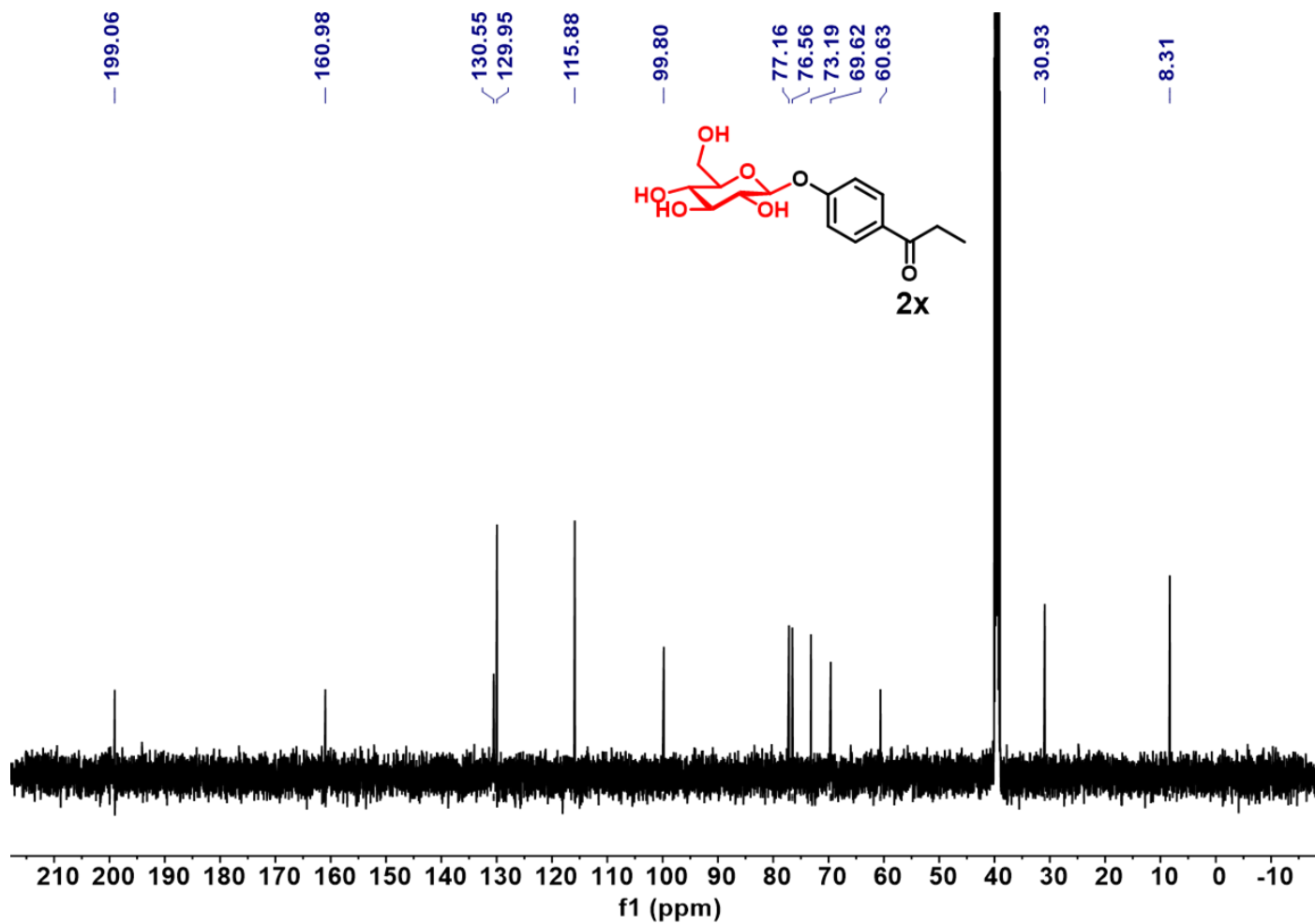
Supplementary Fig. 112 | ^1H NMR spectrum of **2v** (500 MHz, $\text{DMSO-}d_6$).



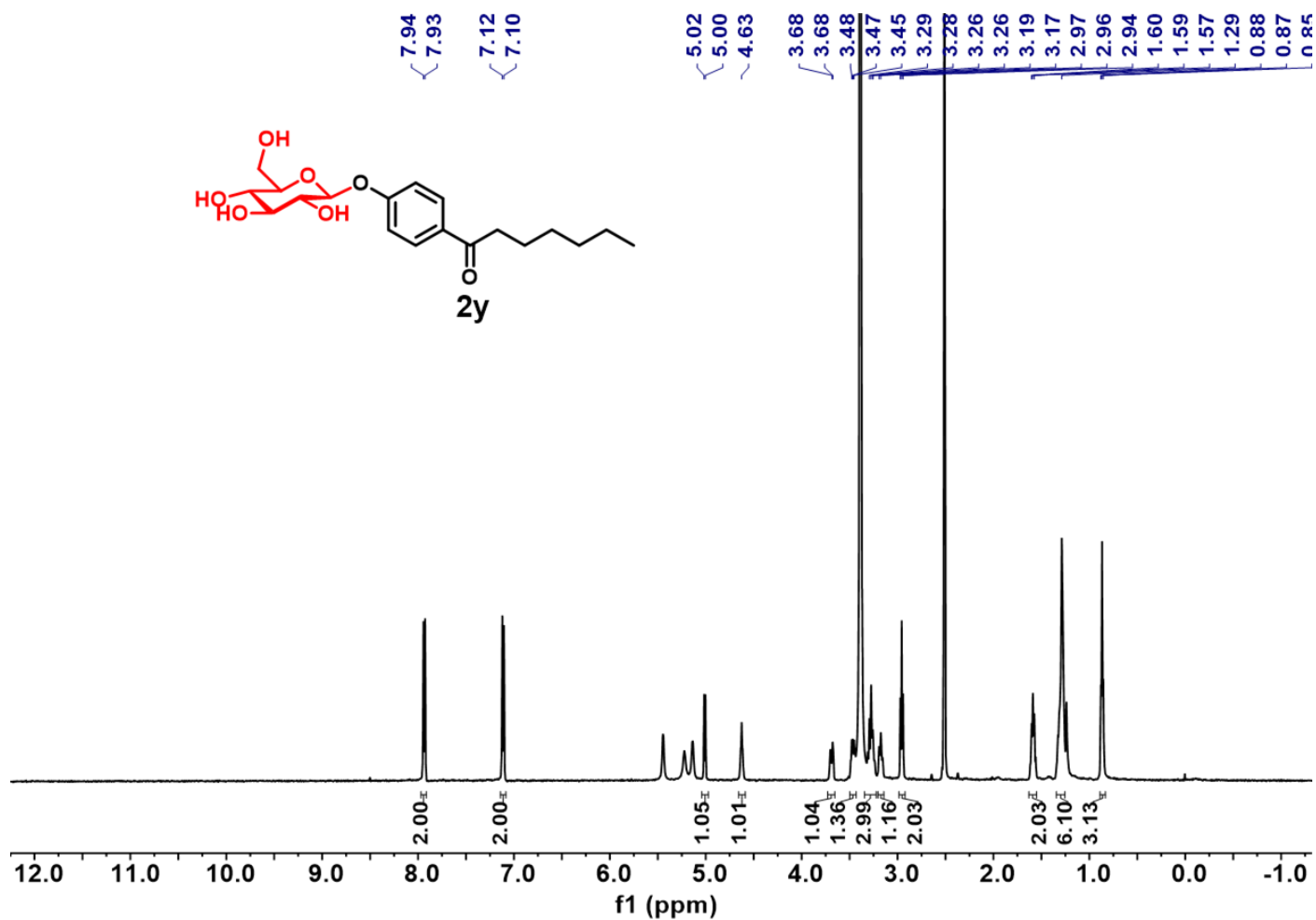
Supplementary Fig. 113 | ¹³C NMR spectrum of **2v** (126 MHz, DMSO-*d*₆).



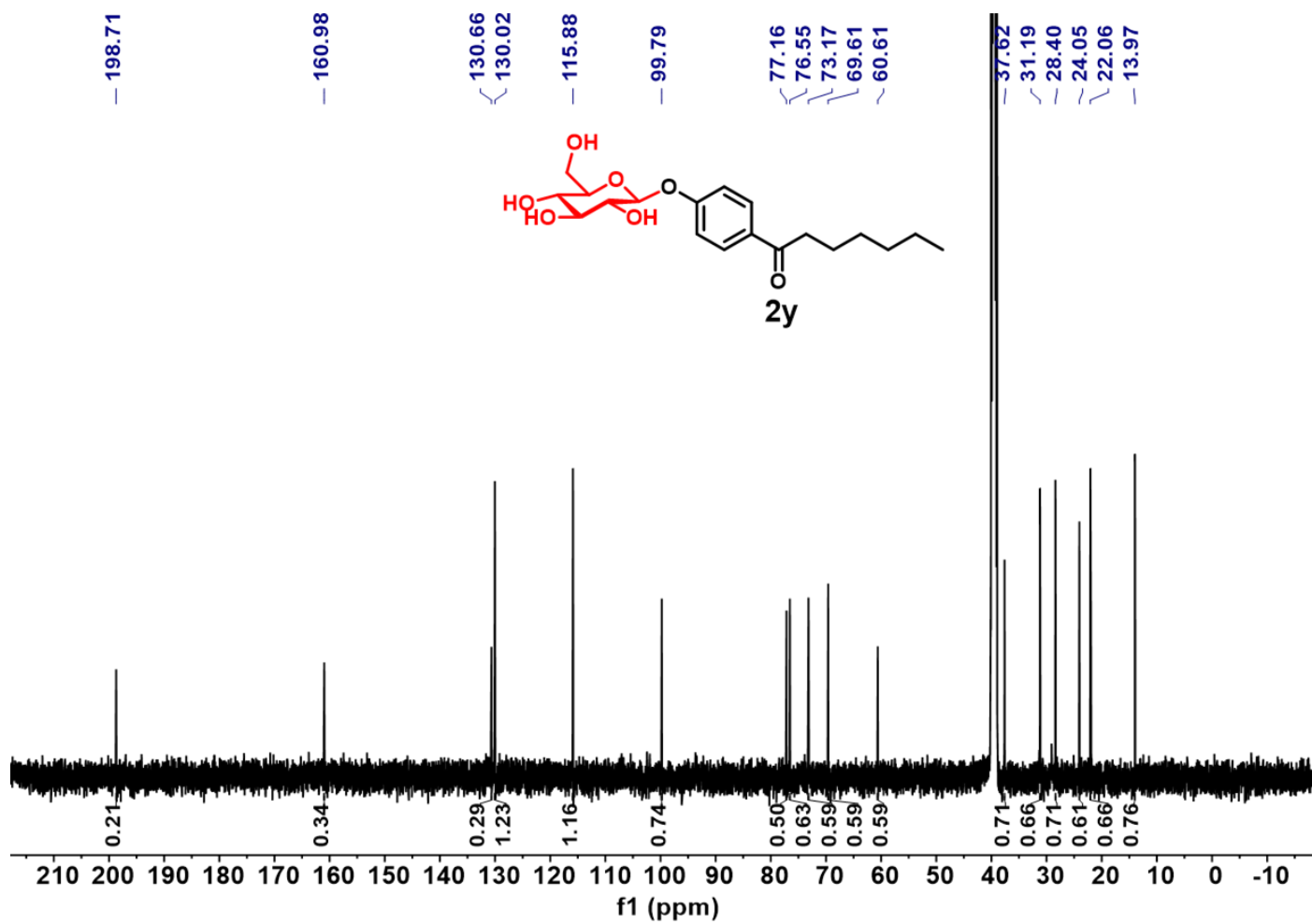
Supplementary Fig. 114 | ¹H NMR spectrum of 2x (500 MHz, DMSO-d₆).



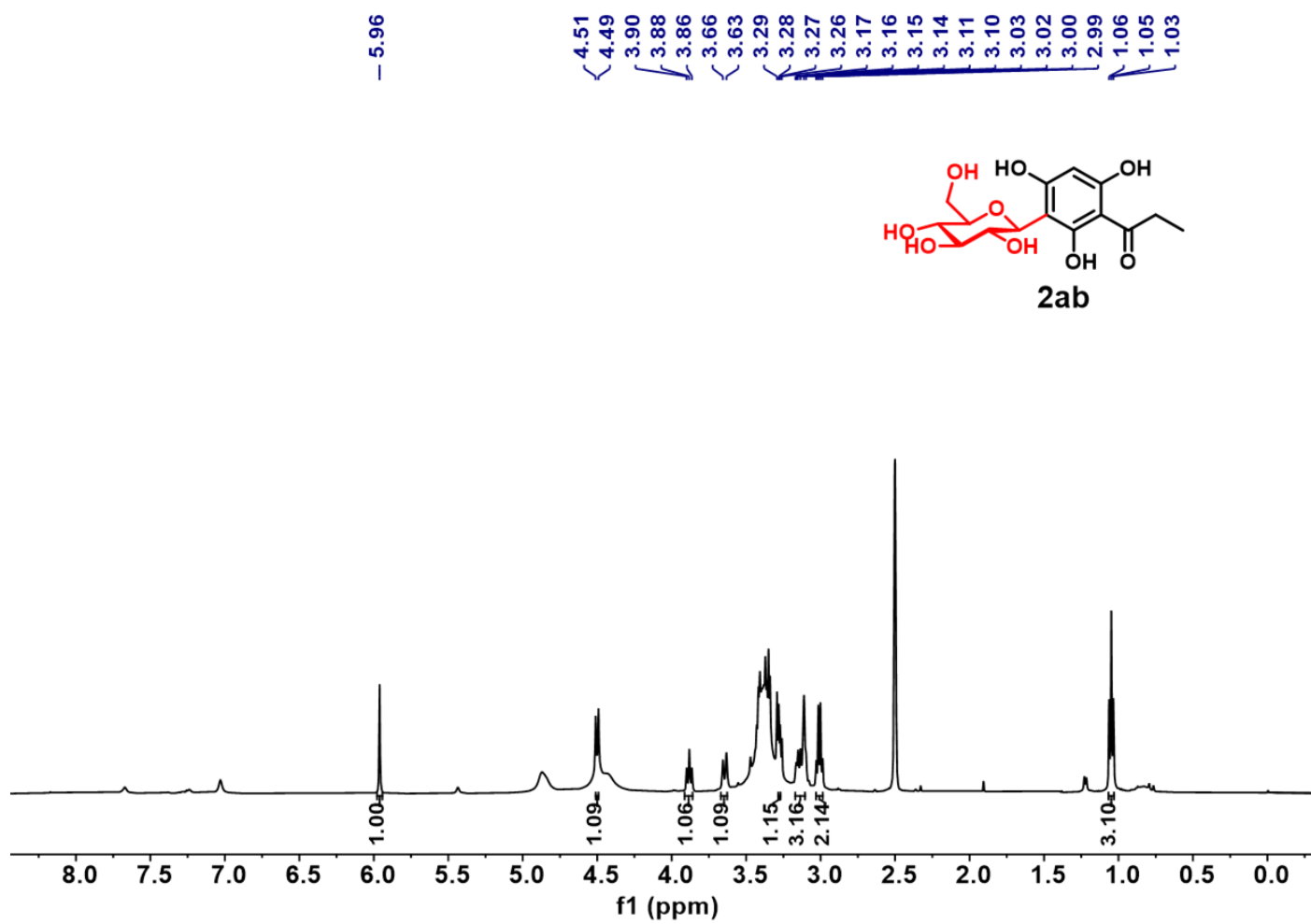
Supplementary Fig. 115 | ¹³C NMR spectrum of 2x (126 MHz, DMSO-*d*₆).



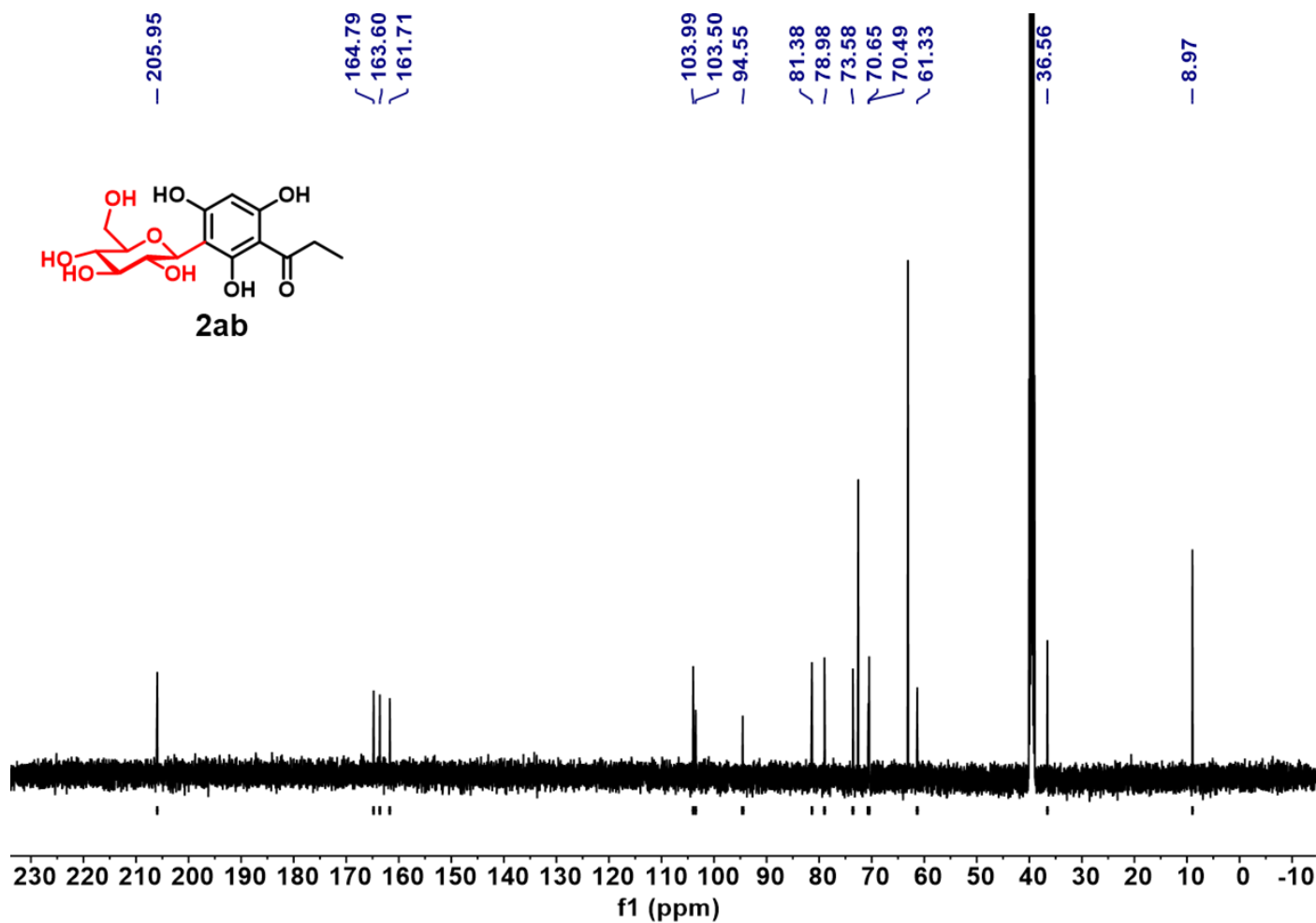
Supplementary Fig. 116 | ¹H NMR spectrum of **2y** (500 MHz, DMSO-*d*₆).



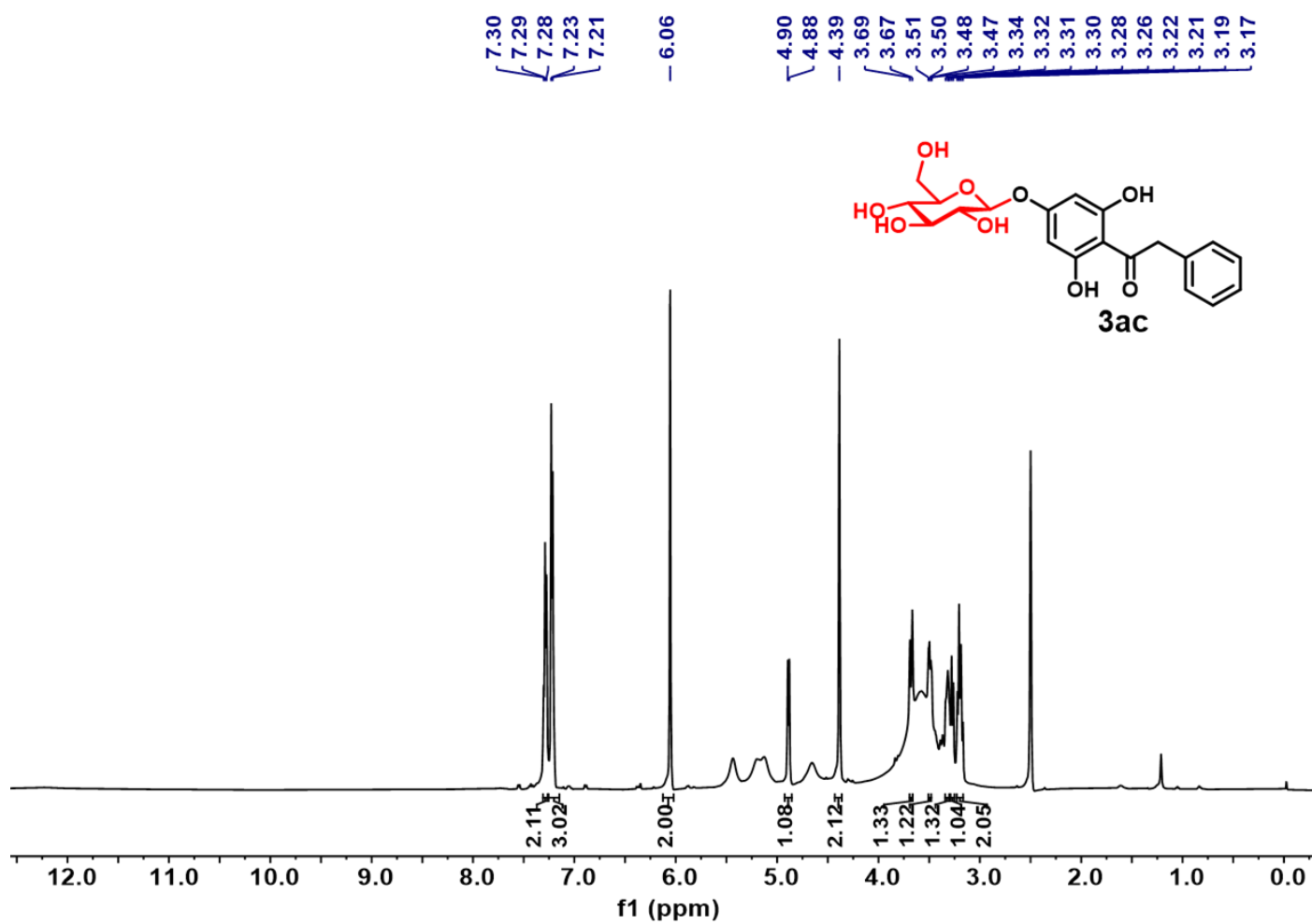
Supplementary Fig. 117 | ^{13}C NMR spectrum of **2y** (126 MHz, $\text{DMSO-}d_6$).



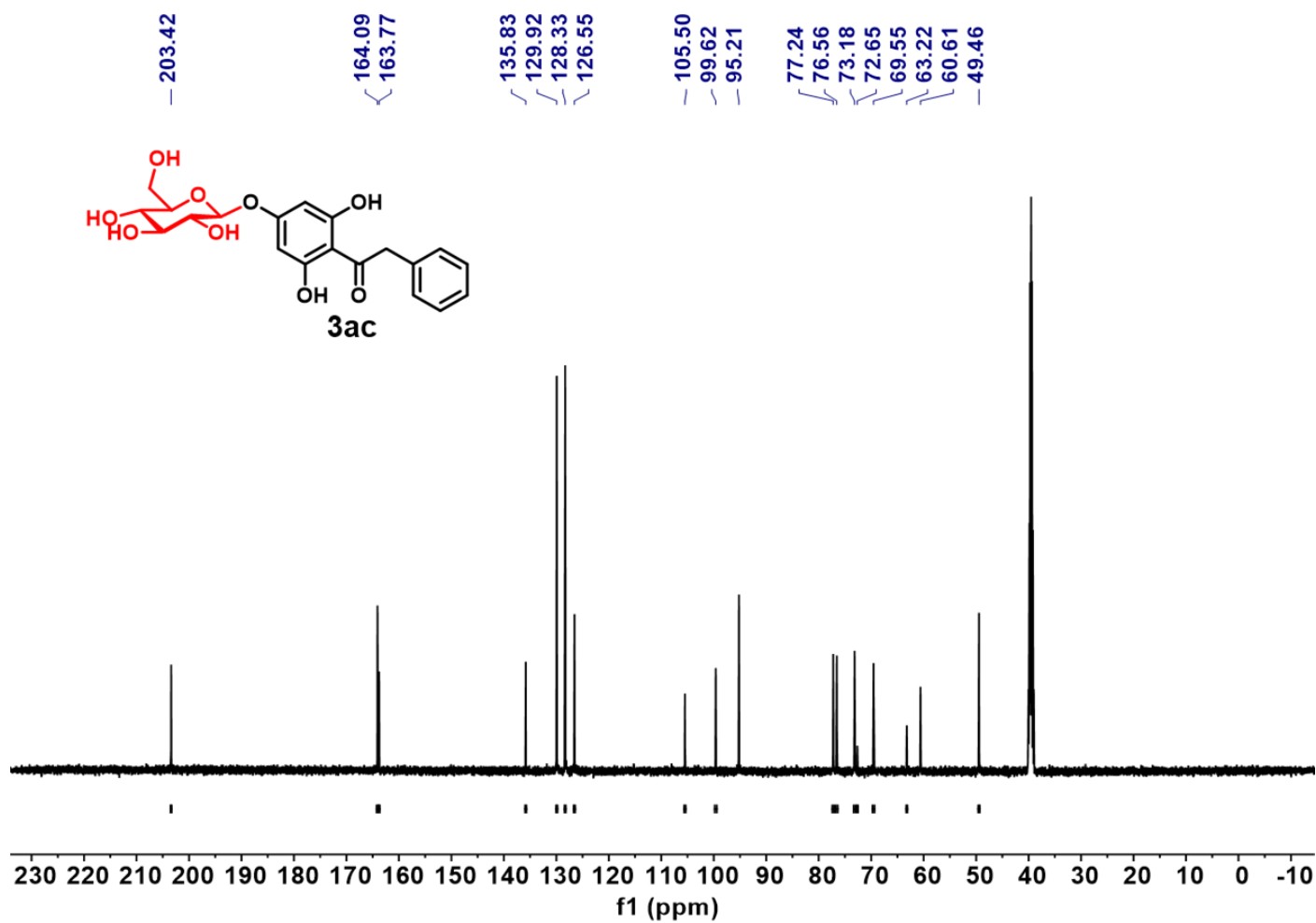
Supplementary Fig. 118 | ^1H NMR spectrum of **2ab** (500 MHz, $\text{DMSO-}d_6$).



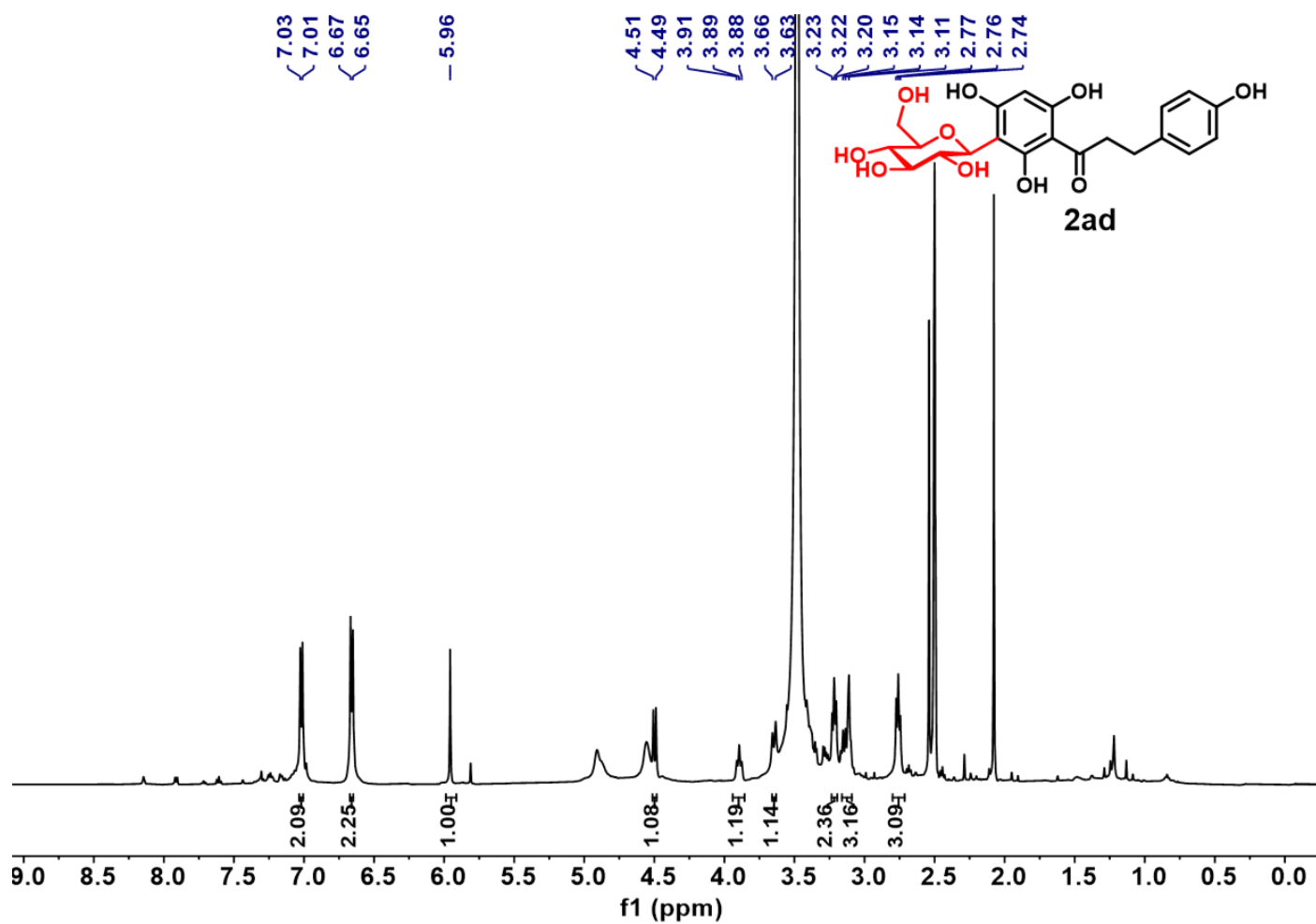
Supplementary Fig. 119 | ^{13}C NMR spectrum of **2ab** (126 MHz, $\text{DMSO-}d_6$). Glycerol: ^{13}C NMR (126 MHz, $\text{DMSO-}d_6$) δ 72.5, 63.1.



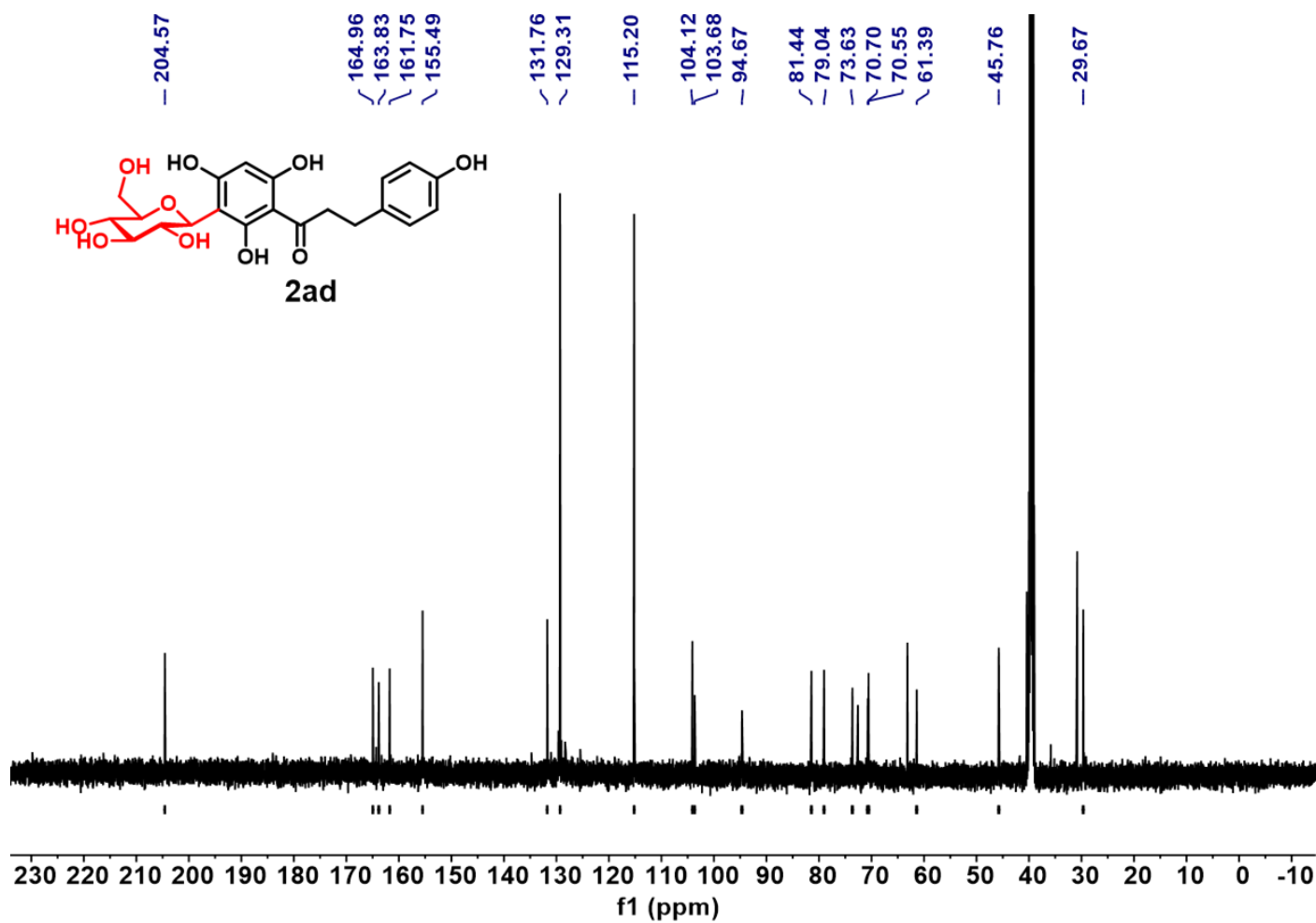
Supplementary Fig. 120 | ^1H NMR spectrum of **3ac** (500 MHz, $\text{DMSO}-d_6$).



Supplementary Fig. 121 | ¹³C NMR spectrum of **3ac** (126 MHz, DMSO-*d*₆).



Supplementary Fig. 122 | ^1H NMR spectrum of **2ad** (500 MHz, $\text{DMSO-}d_6$).



Supplementary Fig. 123 | ^{13}C NMR spectrum of **2ad** (126 MHz, $\text{DMSO-}d_6$).

Supplementary Methods

Protein and DNA Sequences

1. The original sequence of MiCGT

MSASDALNSCPHVALLLSSGMGHLTPCLRFAATLVQHHCRVTIITNYPTVSVAESRAISLLLSDF
PQITEKQFHLLPFDPPSTANTTDPFFLRWEAIRRSAHLLNPLLSSISPPLSALVIDSSLVSSFVPVAAN
LDLPSYVLFTSSTRMCSLEETFPFAVASKTNFDSIQLDDVIEIPGFSPVPVSSVPPVFLNLNHLFTT
MLIQNGQSFARKANGILINTFEALEGGILPGINDKRAADGLPPYCSVGPLLPCKEFEKTECSAPVKW
LDDQPEGSVVYVSFGSRFALSSEQIKELGDGLIRSGCRFLWVVKCKKVDQEDEESLDELLGRDV
LEKIKKYGFVIKNWVNQQEILDHRAVGGFVTHGGWNSSMEAVWHGVPMLVWPQFGDQKINA
EVIERSGLGMWVKRWGWGTQQLVKGEEIGERIKDLMGNNPLRVRAKTLREEARKAIEVGGSSSE
KTLKELIENWKTSRKT*

2. The original sequence of FcCGT

MSDSGGFDSHPHVALIPSAGMGHLTPFLRLAASLVQHHCRVTLITTYPTVSLAETQHVSHFLSA
YPQVTEKRFHLLPFDPPNSANATDPFFLRWEAIRRSAHLLAPLLSPPLSALITDVTLSAVLPVTINL
HLPNYVLFTASARMFSLTASFPFAIVASKSTSSGSVEFDDDFIEIPGLPPIPLSSVPPAVMDSKSLFA
TSFLENGNSFVKSNGVLINSFDADLEADTLVALNGRRVAVAGLPPVYAVGPLLPCFEFEKRDDPSTSL
ILKWLDDQPEGSVVYVSFGSRLALSMEQTKELGNLLSSGCRFLWVVKGKTVDKEDDEESLKNV
LGHELMKIKDQGLVVKNWVDQDKVLSHRAVGGFVSHGGWNSLVEAARHGVPVLVWPQFG
DQKINAEAVESAGLGMWVRSWGWGTTELRAKGDEIGLKIKDLMANDFLREQAKRIEEEEARKAI
GVGGSSERTFKELIDKWKCNNNTH*

3. The original sequence of NnGT

MEMSSSSDLQILPPHVALLPSSGMGHLVPFVRLAAALARRNCLVTFITTHPTVSLSESRLVSRLLS
AFPHVTPLEFHLLPLDHSTANSKDPFFLQFEAIRRSAHLLSPLLSSCDPPLSALITDVSLASAFISI
TDELRLPNYILFTSSAWMLSLCLNFPTFVVNTSTNLSGGSATASDDIEIPGVPPVPKSWLPPLLLD
LSNLFTTQFMANGQELIKSNGILINTFGNAEQATVAALNEGKVVNGLPPVTIKIGPLAPCEFEKGS
SVEWLDGQPAGSVLYVSFGSRTAMSREQIRELGDGLVRSGCRFLWVVKDKKVDREDEEELGGI
VGQELMERMKDNGLVVKNWVDQGVVLAHPAVGGFVSHCGWNSVTEAAWNGVPVLAWPQH
GDQSINAQVMEKSGIGMWVKSWSWGRTEVVKGEEIGQKIGEMMGNDRLKAQAANIREESRR
AVEVGGGSYKELEGIIEKWKISRI*

4. The original sequence of GjGTa

MENTLVLYPAPGIGHMISMLELAKLILRHYSNKFSRIHILINTGFRDMKSTYLDHISSTNPSIVVH
QFPFIQADLSSSLSPPAIGFKFIRKNAPNVHHALQEISKTSIRALIIDFFCTSAMPYSNNLGIPVYY
FFTSGAAAVALFLYFPTIHKQTSESFKDLVQTKFDVPGLPPIPATQMPEPVLDRDDPAYDDILYY
SVHLPKSSGIIVNTFDELEPIALKAITDGLCVPDAPTPLYNIGPLIADADSRPAIDGDKGIDLDQS
DCFSWLDROPDQCVVFLCFGSRGTFSEVQIKEIAKGLERSGKRFLWVVKPLRNNKSKQVEGS
GGFEIDSILPERFLEKTKGIGLVVKSZIPQLQVLRHPAVGGFVTHCGWNSTLEAVVAGVPLVAW
PLHAEQHVNMALVQDMKMAIPVEQGGDGIVRGEEVEKRVRELMDSERGRELRKLSQKTRDI
AAESGVHLGSSSTALASLIHVVFNGNLEHHHHHHH*

5. The original sequence of GjGTc

MKNQELVFIPSAVMSHLVSTVELAKLLIDRNEHLSITVLIMKLPYEKNVGNYTYPQTEASDSRIR
 FLELKKDESASQTVSPILFIYQFVEDHKNSARDVLTEISNSASSDLVGVVVDMFCSSMIDVANEF
 GVPSYVFYTSGAAMLGLMLHLQSLRDDFGEDVTNYENSKVELAVPTYINPVPVKVLP SRLFDM
 EGGGNMFLNLTKRFRETKGIVINSFFELESHAIQALSNDKTIPPVYPVGPILDLKESNGQNQETEM
 ITKWLDIQPDSSVFLCFGSRGCFDGGQVKEIACALESSGYRFLWSLRRPPPKGKFESPGDYENL
 EEALPEGFLQRTAEVGKVIGWAPQAAILSHPAVGCFVSHCGWNSTLESVWFGVPMATWPLYAE
 QQVNAFLLLKDLGMAVDIKMDFKSTSFEPSTEIVAADLIEKAIKHLMDPENIEIRKKVKEKKEKS
 RLSLSEGGPSSASLGRFLDALIDNIP*

Characterization of synthetic substrate and glycosides products

1h: ^1H NMR (500 MHz, DMSO- d_6) δ 11.95 (s, 1H), 10.70 (s, 1H), 7.71 (dd, J = 8.5, 5.6 Hz, 2H), 7.39 – 7.36 (m, 2H), 7.35 (s, 1H), 6.39 (dd, J = 8.6, 2.3 Hz, 1H), 6.36 (s, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 197.05, 164.98, 164.70, 163.94, 162.99, 135.03, 134.64, 134.61, 131.57, 131.50, 115.52, 115.34, 112.84, 108.22, 102.75. ^{19}F NMR (376 MHz, DMSO- d_6) δ -108.16.

1i: ^1H NMR (500 MHz, DMSO- d_6) δ 11.95 (s, 1H), 10.76 (s, 1H), 7.58 (q, J = 7.7 Hz, 1H), 7.45 (q, J = 8.7, 6.6 Hz, 3H), 7.36 (d, J = 8.7 Hz, 1H), 6.39 (d, J = 8.7 Hz, 1H), 6.36 (s, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 196.92, 165.02, 164.18, 162.70, 160.75, 140.33, 140.27, 135.20, 130.62, 130.56, 124.78, 124.75, 118.44, 118.27, 115.32, 115.14, 112.59, 108.40, 102.74. ^{19}F NMR (376 MHz, DMSO- d_6) δ -112.43.

1j: ^1H NMR (500 MHz, DMSO- d_6) δ 11.92 (s, 1H), 10.72 (s, 1H), 7.65 (d, J = 8.3 Hz, 2H), 7.59 (d, J = 8.4 Hz, 2H), 7.35 (d, J = 8.7 Hz, 1H), 6.38 (d, J = 8.7 Hz, 1H), 6.36 (s, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 197.14, 164.84, 163.99, 136.83, 136.43, 135.05, 130.60, 128.49, 112.80, 108.30, 102.76.

1k: ^1H NMR (500 MHz, DMSO- d_6) δ 11.87 (s, 1H), 10.75 (s, 1H), 7.68 (dt, J = 5.4, 2.6 Hz, 1H), 7.64 (s, 1H), 7.56 (d, J = 6.0 Hz, 2H), 7.34 (d, J = 8.8 Hz, 1H), 6.40 (dd, J = 8.8, 2.3 Hz, 1H), 6.36 (d, J = 2.3 Hz, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 196.74, 165.00, 164.02, 140.11, 135.12, 133.19, 131.27, 130.34, 128.08, 127.24, 112.73, 108.42, 102.75.

1l: ^1H NMR (500 MHz, DMSO- d_6) δ 7.74 (s, 1H), 7.72 (s, 1H), 7.56 (s, 1H), 7.55 (s, 1H), 7.31 (d, J = 8.7 Hz, 1H), 6.34 (d, J = 9.0 Hz, 1H), 6.31 (s, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 196.98, 164.39, 137.39, 135.06, 131.48, 130.77, 125.27, 112.17, 108.98, 102.92.

1m: ^1H NMR (500 MHz, DMSO- d_6) δ 7.81 (d, J = 7.9 Hz, 1H), 7.76 (s, 1H), 7.61 (d, J = 7.7 Hz, 1H), 7.49 (t, J = 7.8 Hz, 1H), 7.33 (d, J = 8.8 Hz, 1H), 6.39 (dd, J = 8.8, 2.3 Hz, 1H), 6.36 (d, J = 2.2 Hz, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 196.65, 165.24, 164.08, 140.39, 135.13, 134.21, 130.94, 130.64, 127.66, 121.69, 112.71, 108.58, 102.83.

1n: ^1H NMR (500 MHz, DMSO- d_6) δ 11.95 (s, 1H), 10.76 (s, 1H), 7.62 – 7.55 (m, 1H), 7.53 – 7.39 (m, 3H), 7.36 (d, J = 8.7 Hz, 1H), 6.45 – 6.34 (m, 2H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 197.32, 165.34, 164.26, 142.00, 135.34, 131.08, 129.36, 125.50, 125.47, 112.79, 108.67, 102.88. ^{19}F NMR (376 MHz, DMSO- d_6) δ -61.43.

1o: ^1H NMR (500 MHz, DMSO- d_6) δ 12.22 (s, 1H), 10.69 (s, 1H), 7.53 (d, J = 7.6 Hz, 2H), 7.39 (d, J = 8.6 Hz, 1H), 7.34 (d, J = 7.7 Hz, 2H), 6.37 (d, J = 11.2 Hz, 2H), 2.39 (s, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 198.49, 164.68, 164.40, 141.85, 135.30, 135.22, 128.92, 112.64, 108.10, 102.76, 21.10.

1p: ^1H NMR (500 MHz, DMSO- d_6) δ 12.26 (s, 1H), 10.73 (s, 1H), 7.45 – 7.40 (m, 3H), 7.40 (s, 1H), 7.37 (d, J = 8.7 Hz, 1H), 6.38 (d, J = 8.6 Hz, 1H), 6.36 (d, J = 1.9 Hz, 1H), 2.38 (s, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 199.00, 164.89, 164.61, 138.07, 137.83, 135.40, 132.17, 128.93, 128.24, 125.84, 112.50, 108.22, 102.75, 20.90.

1q: ^1H NMR (500 MHz, DMSO- d_6) δ 12.16 (d, J = 2.8 Hz, 1H), 10.74 (d, J = 2.8 Hz, 1H), 8.23 (s, 1H), 8.07 (dd, J = 17.4, 8.5 Hz, 2H), 8.02 (d, J = 8.0 Hz, 1H), 7.73 (d, J = 8.4 Hz, 1H), 7.69 – 7.58 (m, 2H), 7.47 (d, J = 9.4 Hz, 1H), 6.41 (s, 2H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 198.53, 164.82, 164.36, 135.46, 135.38, 134.20, 131.89, 129.55, 129.19, 128.12, 128.10, 127.71, 126.95, 125.17, 112.95, 108.31, 102.82.

1r: ^1H NMR (500 MHz, DMSO- d_6) δ 12.11 (s, 1H), 10.66 (s, 1H), 8.13 (d, J = 2.0 Hz, 1H), 7.97 (s, 1H), 7.75 (d, J = 8.6 Hz, 1H), 7.61 (d, J = 8.5 Hz, 1H), 7.42 (d, J = 8.6 Hz, 1H), 7.09 (s, 1H), 6.39 (d, J = 11.2 Hz, 2H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 198.37, 164.51, 164.08, 155.91, 147.52, 135.32, 133.23, 127.06, 125.46, 122.90, 113.04, 111.33, 108.08, 107.37, 102.76.

1s: ^1H NMR (500 MHz, DMSO- d_6) δ 12.27 (s, 1H), 10.75 (s, 1H), 8.17 (dd, J = 2.9, 1.3 Hz, 1H), 7.69 (dd, J = 5.0, 2.9 Hz, 1H), 7.63 (d, J = 8.8 Hz, 1H), 7.44 (dd, J = 5.0, 1.3 Hz, 1H), 6.41 (dd, J = 8.8, 2.4 Hz, 1H), 6.34 (d, J = 2.3 Hz, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 192.13, 164.81, 164.50, 140.03, 134.77, 132.70, 128.15, 127.27, 113.07, 108.35, 102.81.

2a: ^1H NMR (500 MHz, DMSO- d_6) δ 7.17 (d, J = 8.8 Hz, 1H), 7.11 (d, J = 8.3 Hz, 1H), 6.41 – 6.32 (m, 2H), 6.30 (d, J = 2.2 Hz, 1H), 4.67 (d, J = 9.8 Hz, 1H), 4.01 (t, J = 9.1 Hz, 1H), 3.67 (d, J = 11.5 Hz, 1H), 3.45 (dd, J = 11.6, 4.5 Hz, 1H), 3.20 (d, J = 19.4 Hz, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 199.04, 162.73, 162.30, 158.59, 134.12, 132.01, 116.36, 112.06, 108.38, 107.32, 102.73, cc0.43, 61.31.

3a: ^1H NMR (500 MHz, DMSO- d_6) δ 7.24 (d, J = 8.6 Hz, 1H), 7.21 (d, J = 8.4 Hz, 1H), 6.59 (d, J = 7.9 Hz, 2H), 6.33 (d, J = 8.0 Hz, 2H), 4.92 (d, J = 7.5 Hz, 1H), 3.71 (d, J = 10.5 Hz, 1H), 3.50 (dd, J = 11.8, 5.4 Hz, 2H), 3.30 – 3.17 (m, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 199.43, 164.06, 162.49, 161.16, 158.43, 134.70, 131.93, 118.66, 114.44, 107.92, 107.34, 103.72, 102.47, 100.11, 77.18, 76.60, 73.22, 69.61, 60.65.

2c: ^1H NMR (500 MHz, DMSO- d_6) δ 7.73 (d, J = 9.0 Hz, 1H), 6.42 (d, J = 8.9 Hz, 1H), 4.59 (d, J = 9.8 Hz, 1H), 4.00 (t, J = 9.3 Hz, 1H), 3.27 (dd, J = 10.8, 5.8 Hz, 2H), 3.15 (dd, J = 17.7, 8.0 Hz, 3H), 2.96 (q, J = 7.2 Hz, 2H), 1.08 (t, J = 7.3 Hz, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 205.91, 164.69, 163.51, 131.91, 112.24, 81.65, 79.12, 73.35, 70.63, 70.43, 61.51, 30.64, 8.87.

3d: ^1H NMR (500 MHz, DMSO- d_6) δ 7.66 (d, J = 6.8 Hz, 2H), 7.63 (d, J = 7.4 Hz, 1H), 7.55 (t, J = 7.7 Hz, 2H), 7.42 (d, J = 8.8 Hz, 1H), 6.66 (d, J = 2.4 Hz, 1H), 6.62 (dd, J = 8.8, 2.4 Hz, 1H), 5.00 (d, J = 7.4 Hz, 1H), 3.70 (d, J = 11.8 Hz, 1H), 3.49 (dt, J = 11.2, 4.7 Hz, 1H), 3.28 (dd, J = 14.9, 8.3 Hz, 3H), 3.19 (t, J = 9.2 Hz, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 198.50, 162.97, 162.66, 137.84, 134.19, 132.13, 128.93, 128.51, 115.61, 108.16, 103.75, 99.79, 77.20, 76.50, 73.14, 69.55, 60.60.

2e: ^1H NMR (500 MHz, DMSO- d_6) δ 7.52 (d, J = 8.7 Hz, 2H), 7.40 (d, J = 8.8 Hz, 1H), 6.89 (d, J = 8.6 Hz, 2H), 6.41 (d, J = 8.8 Hz, 1H), 4.66 (d, J = 9.8 Hz, 1H), 4.05 (t, J = 9.2 Hz, 1H), 3.68 (d, J = 11.5 Hz, 1H),

3.42 (dd, $J = 11.8, 5.8$ Hz, 1H), 3.23 – 3.14 (m, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 198.14, 164.29, 164.02, 161.18, 134.41, 131.70, 128.84, 115.24, 112.59, 111.79, 108.15, 81.69, 79.14, 73.55, 70.64, 70.49, 61.52.

3e: ^1H NMR (500 MHz, DMSO- d_6) δ 7.59 (d, $J = 8.3$ Hz, 2H), 7.44 (d, $J = 8.8$ Hz, 1H), 6.89 (d, $J = 8.3$ Hz, 2H), 6.69 – 6.58 (m, 2H), 4.99 (d, $J = 7.5$ Hz, 1H), 3.71 (d, $J = 11.8$ Hz, 1H), 3.51 (dd, $J = 12.2, 5.3$ Hz, 1H), 3.40 (d, $J = 7.8$ Hz, 1H), 3.30 (dd, $J = 15.8, 8.4$ Hz, 2H), 3.23 (d, $J = 9.2$ Hz, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 196.87, 162.33, 162.03, 161.76, 133.71, 132.08, 128.79, 116.38, 115.35, 107.86, 103.89, 100.04, 99.92, 77.26, 76.63, 73.28, 72.65, 69.68, 63.23, 60.75.

3f: ^1H NMR (500 MHz, DMSO- d_6) δ 7.44 (d, $J = 8.5$ Hz, 1H), 7.15 (d, $J = 2.1$ Hz, 1H), 7.06 (dd, $J = 8.2, 2.1$ Hz, 1H), 6.85 (d, $J = 8.2$ Hz, 1H), 6.60 (d, $J = 8.8$ Hz, 2H), 4.97 (d, $J = 7.4$ Hz, 1H), 3.70 (dd, $J = 12.0, 2.1$ Hz, 1H), 3.49 (dd, $J = 11.9, 5.5$ Hz, 1H), 3.32 – 3.16 (m, 4H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 196.68, 162.05, 161.69, 150.33, 145.15, 133.39, 128.96, 122.66, 116.62, 116.48, 115.13, 107.55, 103.78, 99.92, 77.18, 76.53, 73.17, 69.59, 60.64.

2i: ^1H NMR (500 MHz, DMSO- d_6) δ 7.64 – 7.54 (m, 1H), 7.47 (td, $J = 8.6, 8.2, 2.5$ Hz, 1H), 7.44 – 7.38 (m, 2H), 7.32 (d, $J = 8.8$ Hz, 1H), 6.46 (d, $J = 8.8$ Hz, 1H), 4.66 (d, $J = 9.8$ Hz, 1H), 4.05 (t, $J = 9.2$ Hz, 1H), 3.68 (d, $J = 11.6$ Hz, 1H), 3.22 – 3.06 (m, 4H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 197.96, 164.59, 162.75, 160.80, 140.22, 140.17, 134.71, 130.80, 130.73, 124.76, 124.74, 118.41, 118.24, 115.39, 115.21, 112.71, 111.59, 108.54, 81.72, 79.05, 73.26, 70.62, 70.33, 61.51. ^{19}F NMR (376 MHz, DMSO- d_6) δ -112.18.

2t: ^1H NMR (500 MHz, DMSO- d_6) δ 7.67 (d, $J = 8.8$ Hz, 2H), 7.64 (d, $J = 8.7$ Hz, 2H), 7.15 (d, $J = 8.8$ Hz, 2H), 6.92 – 6.87 (m, 2H), 5.02 (d, $J = 7.1$ Hz, 1H), 3.71 (d, $J = 11.7$ Hz, 1H), 3.50 (d, $J = 4.8$ Hz, 1H), 3.29 (t, $J = 6.8$ Hz, 2H), 3.20 (t, $J = 8.9$ Hz, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 193.29, 161.72, 160.41, 132.40, 131.58, 131.48, 128.43, 115.80, 115.26, 99.98, 77.22, 76.61, 73.25, 69.63, 60.67.

2u: ^1H NMR (500 MHz, DMSO- d_6) δ 7.73 (d, $J = 8.8$ Hz, 2H), 7.70 (d, $J = 6.8$ Hz, 2H), 7.66 (t, $J = 7.4$ Hz, 1H), 7.55 (t, $J = 7.6$ Hz, 2H), 7.20 – 7.15 (m, 2H), 5.03 (d, $J = 7.1$ Hz, 1H), 4.61 (s, 1H), 3.70 (ddd, $J = 11.9, 4.7, 2.1$ Hz, 1H), 3.48 (dt, $J = 11.3, 5.4$ Hz, 1H), 3.30 (d, $J = 7.1$ Hz, 2H), 3.20 (t, $J = 8.3$ Hz, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 194.60, 160.99, 137.65, 132.31, 132.02, 130.46, 129.39, 128.56, 115.95, 99.89, 77.21, 76.58, 73.20, 69.60, 60.63.

2v: ^1H NMR (500 MHz, DMSO- d_6) δ 7.73 (d, $J = 6.2$ Hz, 2H), 7.72 (d, $J = 6.0$ Hz, 2H), 7.62 (d, $J = 8.4$ Hz, 2H), 7.17 (d, $J = 8.6$ Hz, 2H), 5.03 (d, $J = 7.0$ Hz, 1H), 4.62 (d, $J = 5.9$ Hz, 1H), 3.69 (d, $J = 11.1$ Hz, 1H), 3.50 – 3.45 (m, 1H), 3.29 (d, $J = 7.1$ Hz, 2H), 3.18 (t, $J = 8.9$ Hz, 1H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 193.99, 161.65, 137.70, 136.83, 132.58, 131.85, 130.64, 129.22, 116.52, 100.37, 77.72, 77.06, 73.69, 70.08, 61.12.

2x: ^1H NMR (500 MHz, DMSO- d_6) δ 7.93 (d, $J = 8.8$ Hz, 2H), 7.11 (d, $J = 8.9$ Hz, 2H), 5.00 (d, $J = 7.3$ Hz, 1H), 3.69 (d, $J = 11.7$ Hz, 1H), 3.28 (q, $J = 8.6$ Hz, 3H), 3.18 (d, $J = 9.0$ Hz, 1H), 2.99 (q, $J = 7.2$ Hz, 2H), 1.07 (t, $J = 7.2$ Hz, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 199.06, 160.98, 130.55, 129.95, 115.88, 99.80, 77.16, 76.56, 73.19, 69.62, 60.63, 30.93, 8.31.

2y: ^1H NMR (500 MHz, DMSO- d_6) δ 7.94 (d, $J = 8.4$ Hz, 2H), 7.11 (d, $J = 8.4$ Hz, 2H), 5.01 (d, $J = 7.2$ Hz, 1H), 4.63 (s, 1H), 3.73 – 3.66 (m, 1H), 3.46 (dd, $J = 11.8, 5.8$ Hz, 1H), 3.34 – 3.22 (m, 3H), 3.17 (t, $J = 9.1$ Hz, 1H), 2.96 (t, $J = 7.2$ Hz, 2H), 1.59 (t, $J = 7.3$ Hz, 2H), 1.29 (s, 6H), 0.89 – 0.83 (m, 3H). ^{13}C NMR (126

MHz, DMSO- d_6) δ 198.71, 160.98, 130.66, 130.02, 115.88, 99.79, 77.16, 76.55, 73.17, 69.61, 60.61, 37.62, 31.19, 28.40, 24.05, 22.06, 13.97.

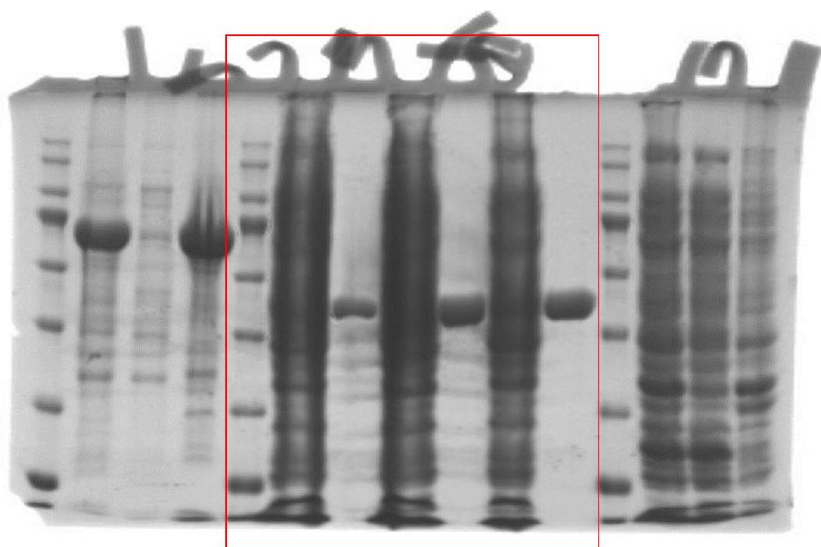
2ab: ^1H NMR (500 MHz, DMSO- d_6) δ 5.96 (s, 1H), 4.49 (s, 1H), 3.88 (t, J = 9.1 Hz, 1H), 3.64 (d, J = 11.7 Hz, 1H), 3.28 (d, J = 5.1 Hz, 1H), 3.17 – 3.10 (m, 3H), 3.01 (q, J = 7.2 Hz, 2H), 1.05 (t, J = 7.2 Hz, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 205.95, 164.79, 163.60, 161.71, 103.99, 103.50, 94.55, 81.38, 78.98, 73.58, 70.65, 70.49, 61.33, 36.56, 8.97.

3ac: ^1H NMR (500 MHz, DMSO- d_6) δ 7.29 (t, J = 7.0 Hz, 2H), 7.22 (d, J = 7.5 Hz, 3H), 6.06 (s, 2H), 4.89 (d, J = 7.7 Hz, 1H), 4.39 (s, 2H), 3.68 (d, J = 11.8 Hz, 1H), 3.51 – 3.48 (m, 1H), 3.34 – 3.30 (m, 1H), 3.27 (d, J = 8.9 Hz, 1H), 3.20 (q, J = 9.1 Hz, 2H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 203.42, 164.09, 163.77, 135.83, 129.92, 128.33, 126.55, 105.50, 99.62, 77.24, 76.56, 73.18, 72.65, 69.55, 63.22, 49.46.

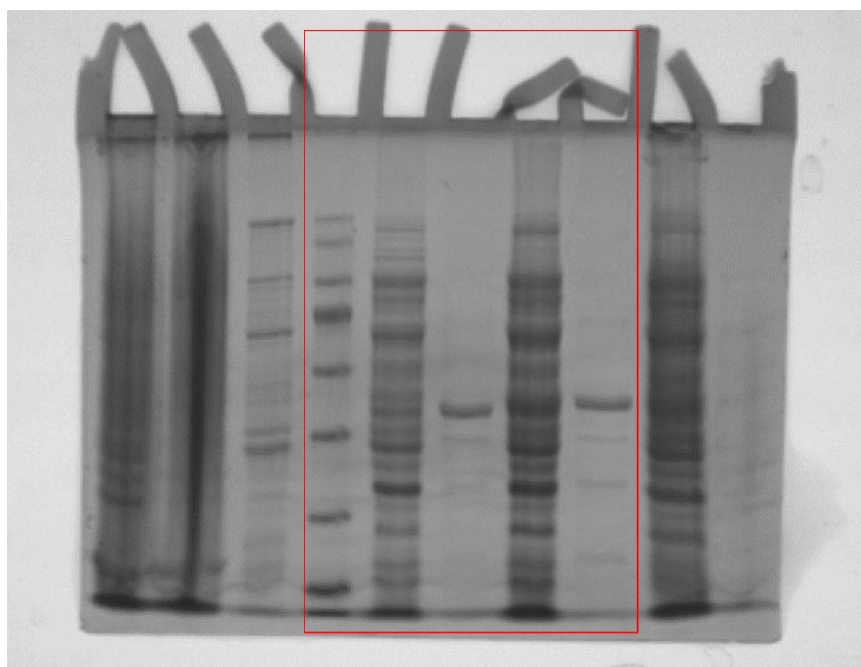
2ad: ^1H NMR (500 MHz, DMSO- d_6) δ 7.02 (d, J = 8.3 Hz, 2H), 6.66 (d, J = 8.0 Hz, 2H), 5.96 (s, 1H), 4.50 (d, J = 9.8 Hz, 1H), 3.89 (t, J = 9.2 Hz, 1H), 3.65 (d, J = 11.6 Hz, 1H), 3.22 (t, J = 7.8 Hz, 2H), 3.16 – 3.09 (m, 3H), 2.76 (t, J = 7.8 Hz, 3H). ^{13}C NMR (126 MHz, DMSO- d_6) δ 204.57, 164.96, 163.83, 161.75, 155.49, 131.76, 129.31, 115.20, 104.12, 103.68, 94.67, 81.44, 79.04, 73.63, 70.70, 70.55, 61.39, 45.76, 29.67.

Uncropped Gels from Supplementary Figures

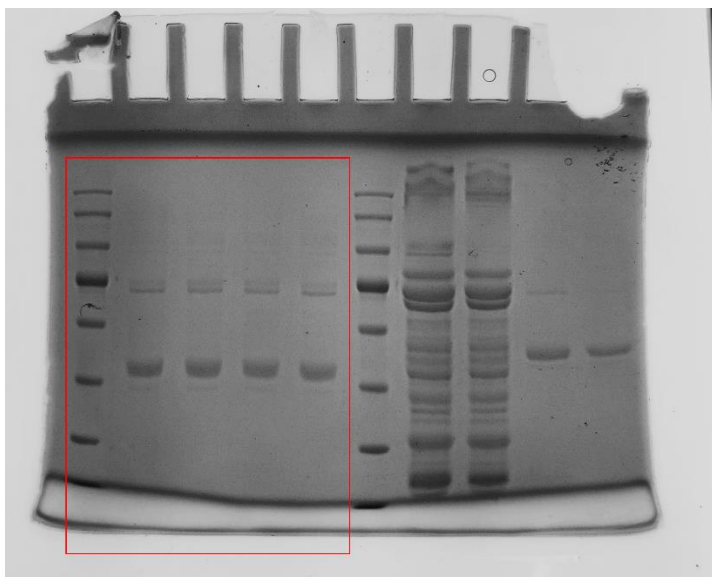
Supplementary Fig. 9a:



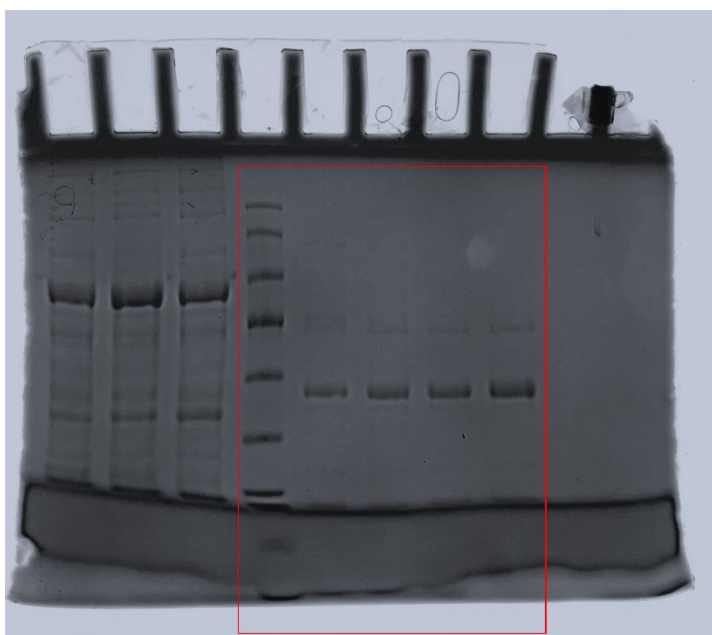
Supplementary Fig. 9b:



Supplementary Fig. 9c:



Supplementary Fig. 9d:



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